The Value Momentum Investing (VMI) Master Document

Complete Notes from Adam Khoo's *Whale Investor / Value Momentum Investor* Course

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About This Document

This document is a complete, exhaustive written record of **every core-methodology lesson video** in the course — **55 videos totalling 23 hours 7 minutes** of instruction.

Every video was transcribed in full (OpenAI Whisper `medium`, 236,275 words of raw transcript), then every transcript was read end-to-end and distilled into structured notes. Nothing was skipped or sampled. Concepts, formulas, numeric thresholds, worked examples, company names, tickers, tool navigation steps, and rules are preserved as taught.

Provenance note: all content below comes from the course videos themselves. No outside material has been added. Where the instructor gives a specific number, that number is reproduced exactly as stated in the lesson.

Coverage note: this document covers the full core investing methodology — stock basics, trend following and strategy, fundamental analysis, valuation, technical analysis, the 7-step VMI formula, worked case studies, portfolio construction, finding/monitoring/selling, and investor psychology.

Deliberately **excluded** from this document (not part of the core stock-picking method): the ETF lessons, the broker-selection and IBKR platform walkthroughs, and the options / leverage lessons, along with the bonus webinar recordings.

The Core Idea in One Page

When you buy a share of stock, you are buying a piece of a business. If the business makes more money over time, the shares become worth more and the price follows. Everything else — elections, the Fed, geopolitics, yield curves, daily headlines — is noise.

The whole method answers four questions, which the instructor abbreviates **WWHW**:

Key	Question	Answered by
W	**What** to buy	Fundamental analysis → only the top ~1% of businesses
W	**When** to buy	Valuation (buy below intrinsic value) + technical analysis for timing
H	**How many** shares	Position sizing and portfolio construction
W	**When** to sell	Explicit exit rules — and knowing the many invalid reasons to sell

Two engines drive returns, and the name of the method comes from combining them:

- **Value** — buy great businesses only when price sits below calculated intrinsic value.

- **Momentum** — use trend and chart structure so you buy into strength and avoid catching falling knives, and so exits are disciplined rather than emotional.

And one factor sits above the method itself: **temperament**. The instructor's own weighting is that **70% of an investor's success is mindset** — the ability to follow the rules when fear or greed says otherwise. That is why Part X exists, and why he tells students to watch that lesson more than any other.

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- E. Growth rates are triangulated from independent sources, never taken from one
- F. Step 3 (moat) is the one step with no number — and the verdicts genuinely differ
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Part I — Stock Market Basics

Lesson 1: Stock Market Basics — Master Notes (Whale Investor / VMI Course)

Source videos: (06) VMI Course Introduction, (07) Stock Basics Part 1 — Parts of a Business, (08) Stock Basics Part 2 — P/E & PEG, (10) Stock Basics Part 4 — What Drives Stock Prices, (11) Stock Basics Part 5 — How to Make Money in Stocks, (12) Stock Basics Part 6 — Stock Index Performance, (14) Shocking Market Facts.

Course Introduction (Video 06 — Course Structure, Rules, Community)

- Course also called "The Whale Investor" or "Value Momentum Investor (VMI)" course.
- Goal: consistent profits, stress-free, low-risk, sustainable investing; build net worth toward "financial freedom."
- Course explicitly cuts out financial-media "noise/bullshit" and "impractical theories" from business school/ financial academia.
- **The Four Keys — "W, W, H, W":**
  1. **What** to buy — only the top 1% of the strongest stocks in the market.
  2. **When** to buy — only when strong stocks are attractively priced/undervalued.
  3. **How** many shares to buy — position sizing, building a diversified, balanced portfolio.
  4. **When** to sell — exit rules.
- Following the methodology strictly → can beat the market, achieve double-digit returns over the long run, outperform most retail investors and most professional money managers.
- You CAN fail — but only by failing to take action, failing to apply what's taught, or breaking the rules. Rules exist to keep you safe.
- **Course structure — 12 core lessons, each ~2–3 hours:**
  1. Stock Market Basics (this lesson)
  2. Exchange Traded Funds (ETFs) — simplest way to start investing
  3. Keys to Profitable Investing — overview of the methodology
  4. Fundamental Analysis — analyzing a business inside out, identifying great companies
  5. Stock Valuation — how to value businesses (expensive vs. cheap)
  6. Technical Analysis — optimizing entries/exits via chart patterns
  7. The Seven-Step VMI Formula (Value Momentum Investment Formula) — everything combined into a checklist/ methodology
  8. Building a bulletproof, diversified portfolio that performs in all economic conditions
  9. Finding great stocks/opportunities + exit rules (when to say goodbye to a stock and replace it)
  10. **Psychology** — Adam calls this one of the most important lessons even though numbered 10th. "70% of an investor's success is their mindset" — ability to manage temperament and emotions. Recommended to rewatch multiple times.
  11. Choosing the right stockbroker for your objectives
  12. Turbocharging your portfolio using options (generating extra income)
- Plus additional sections: how to use TradingView (Adam's charting platform of choice) and how to use Interactive Brokers (his brokerage); any reputable broker will do since broker mechanics are similar once you learn one.
- Bonus section: past webinars for students.
- **Recommended learning approach:**

- Complete every lesson, don't skip any — even experienced/professional investors should redo the basics.
- Learn actively: take your own notes (even though slides/notes are provided), pause and repeat videos.
- Each lesson ends with a quiz — attempt it before seeing answers.
- Downloadable course notes + resources (Excel templates for managing investments, an intrinsic value calculator) available under "Course Notes and Resources."
- Time to finish: intensive pace (12–13 hrs/day) = ~2 days; moderate pace (~2 hrs/day) = ~13 days (~1.5 weeks); relaxed pace (~1 hr/day) = ~26 days (~1 month). "Repetition is the mother of skill."
- **Community (Discord):**
- Active chat 7 days/week, including weekends/holidays, ~15–16 hours/day.
- Members can tag "Adam Koo" directly with questions.
- New members only granted chat-group access **7 days after enrolling**, so they first watch lessons (most questions get answered by the content itself).
- Members encouraged to share their own stock analyses/ideas — but should NOT ask "Adam, what do you think of this stock?" (Adam won't spoon-feed picks; he guides members to do their own analysis, and will give feedback on analysis already done).
- Security warning: turn off direct messages (DMs) in the group — scammers impersonate Adam or others via DM. Adam states he will **never** DM anyone directly; all his communication happens in the main chat group (under "Investing Costs" → main chat stock group).
- Separate "Friendly Chit Chat" channel exists for off-topic/social conversation.

What Is a Share of Stock (Parts of a Business)

- A share of stock is **not** a lottery ticket / price-prediction bet. Buying a share = **becoming a part owner (shareholder) of an underlying business.**
- A business (McDonald's, Microsoft, Apple, etc.) is fundamentally **a money-making machine**: it sells products/services to make money.
- Core logic chain: more money the business makes → more the business is worth → more valuable the shares → share price rises over time. As an owner, you have a claim over the business's profits.
- Adam uses "stocks" and "businesses" interchangeably — deliberately.
- **Core rule:** "If the business does well, the shares will do very well… Everything else is noise and distraction." Ignore financial-media sensationalism (geopolitics, elections, war, yield curve, inflation, what the Fed is doing) — the only real question is: *will this business make more money over time?*
- **Example — Microsoft:** Adam is a shareholder because regardless of who's US President, the yield curve, inflation, or wars, people keep using Microsoft, more people use it every year, Microsoft keeps raising prices, and profits keep growing → as a shareholder his claim on those profits grows → share value rises over time.

Shares Outstanding (the "pizza" concept)

- Before buying into a business, you need to know how many total "pieces" (shares) exist.
- **Example: "Success Company"** — has **100 million shares outstanding**. Think of the company as a pizza with 100 million slices; you can buy as many slices as you want.
- Companies can **issue more shares** (increasing shares outstanding) or **buy back and cancel shares** (decreasing shares outstanding) — the number is not fixed over time.

- If you buy 1 million shares out of 100 million total, you own **1% of the business**, entitling you to 1% of the profits and 1% of voting rights (voting rights not a focus of this course — "we're just interested in the money").

Minimum Shares / Fractional Shares by Market

- **US market:** biggest market in the world; **no minimum number of shares** — you can buy as little as 1 share. Some brokers (e.g., **Interactive Brokers**, Adam's broker) even allow **fractional shares**.
- Example: **Meta Platforms (owns Facebook & Instagram)** — one share priced at **~$600**. If you can't afford a full share, you could buy **0.1 of a share for $60**.
- **Asian exchanges (e.g., Singapore):** typically require a **minimum of 100 shares, in multiples of 100**.
- Example: **DBS Bank** on the Singapore Exchange — 1 share = **$40**, so minimum investment = $40 × 100 = **$4,000**.
- Conclusion: Asian markets are "less flexible" than US markets for position sizing.

Yahoo Finance — First Tool Introduced

- Website: **finance.yahoo.com**. Search by company name or **ticker symbol** in the search bar.
- Every stock has a ticker symbol (does not need to be memorized): Meta = **META**, Apple = **AAPL**, Microsoft = **MSFT**.

Net Income & Implied Shares Outstanding

- The first basic question when considering owning a piece of a business: **how much money is the business making?** → this is the **net income**.
- **Net income = net profits = earnings** — these three terms are used interchangeably.
- Net income specifically means **net income after tax** — what's left after deducting everything.
- Found on Yahoo Finance under **Statistics** → "Net income available to common shareholders" (shares you buy as a retail investor are "common shares").
- **TTM** = **Trailing Twelve Months** — the last 12 months from the date you're looking at the data (e.g., recording in Dec 2024 → TTM covers Dec 2023–Dec 2024).
- **Example — Meta Platforms:** Net income (TTM) = **$55.54 billion** = 55,540 million (1 billion = 1,000 million).

Shares Outstanding vs. Implied Shares Outstanding

- Yahoo Finance Statistics page shows two figures for Meta:
- **Shares Outstanding:** 2.18 billion (current pieces of the "pizza" actually in the market).
- **Implied Shares Outstanding:** 2.52 billion.
- **Why the difference:** companies grant employees free shares via **options**, which employees can convert into actual shares in the future. Implied shares outstanding accounts for what happens if all employees convert their options — i.e., a larger, more conservative share count.
- **Rule: always use the implied shares outstanding figure** (to be conservative). For Meta: **2.52 billion = 2,520 million shares.**

Earnings Per Share (EPS)

- **Formula:** EPS = Total net income ÷ Total (implied) shares outstanding.
- Calculation for Meta: $55.54 billion ÷ 2,520 million.
- Pre-calculated on Yahoo Finance's **Summary** page as "Earnings Per Share (TTM)."

- **Meta EPS (TTM) = $21.17** — meaning every share entitles the owner to $21.17 of profit per year (based on trailing 12 months). This figure is *entitled* profit, not necessarily cash received.

The Four Ways a Company Returns Profit to Shareholders

Adam poses the question: as a shareholder "entitled" to $21.17/share, how do you actually get that money? Four mechanisms:

1. **Dividends** — company pays out cash from profits.
   - Not all companies pay dividends; some keep 100% of profit ("like a savings account"), some pay a little, some pay a lot.
   - **"Forward" dividend** = expected dividend over the *next* 12 months if you buy now.
   - Meta example: forward dividend = **$2.00 per share** (i.e., out of the ~$21 earned, the company pays $2 cash).
   - **Dividend taxation:** depends where the stock is listed and where the investor resides.
     - US-listed stocks: dividends are taxed. A US resident pays tax per their income bracket. A resident of a country with **no tax treaty with the US** (e.g., Singapore) pays a **flat 30% withholding tax** — i.e., only 70% of each $1 of US dividend is received.
     - Singapore or Hong Kong-listed stocks: dividends are generally **tax-free** for the investor — though **certain Hong Kong stocks carry a 10% withholding tax**.
     - Adam's personal preference: doesn't like buying **US stocks for dividends** (30% tax hit); prefers US stocks for **capital gains** (zero capital gains tax for a Singaporean). For dividend income specifically, he prefers **Singapore-listed stocks** (no dividend tax).
2. **Share buybacks** — company uses profit to repurchase its own shares from the market and cancel them.
   - Good when done while shares are undervalued and the business is healthy/growing profits.
   - Mechanism: fewer shares outstanding → each remaining share represents a larger ownership slice → EPS rises (same total profit split among fewer shares) → over time this should push the share price higher.
   - **Extreme illustrative example:** if a company buys back and cancels **half** of all its shares outstanding — Meta's 2,520 million shares ÷ 2 = **1,260 million shares** remaining. Recomputing EPS: $55.54 billion ÷ 1.26 billion = EPS **doubles from ~$21 to ~$42 per share**.
   - **Pizza analogy:** a pizza has 4 slices, you own 1 slice (25% of pizza). If the company buys back and cancels 2 slices, only 2 slices remain, and you still own 1 — now you own **50%** of the pizza, without spending any additional money yourself. "The company buys shares on your behalf."
3. **Reinvest profits in the business** — via R&D, entering new markets, building factories, new products — aiming to generate more future revenue/profit, which should raise intrinsic value and share price.
   - Key metric to evaluate this option: **Return on Invested Capital (ROIC)**.
   - **Not available on Yahoo Finance** — Adam uses **Stock Oracle** (**app.stockoracle.com**) instead, found under "Company Metrics" (as a chart or as a standalone favorited metric).
   - **Meta ROIC (TTM) = 22.81%.**
   - **Benchmarks:** ROIC of **12% and above** = good business; **15%+** = excellent business.
   - **ROIC does not apply to:** financial-services companies (banks, insurers) — use **Return on Equity (ROE)** instead for those; also does not apply to **REITs** (Real Estate Investment Trusts). Applies to all other company types.
   - Interpretation: Meta grows reinvested profits internally at ~22.81%/year on average — an investor must weigh whether they could personally beat that return if paid out as a dividend instead.
   - **Warren Buffett / Berkshire Hathaway example:** Berkshire has never paid a dividend in **60 years**, despite Buffett being the largest shareholder — he prefers reinvesting internally at a high ROIC, compounding his wealth tax-free.

4. **Retain the cash (retained earnings)** — company simply keeps the profit without returning it via any of the above three methods.
   - Shareholders generally dislike this: "why are you keeping my cash?" — not returned as dividend, buyback, or productive reinvestment.
   - Adam's preference: companies that return capital via the **first three** methods.

Market Capitalization

- **Formula:** Market Cap = Share price × Total shares outstanding (theoretical cost to buy 100% of the company).
- No need to calculate manually — available on any finance website.
- **Meta example:** Share price ~$599.81 × shares outstanding → **Market cap = $1.51 trillion** (i.e., **$1,514 billion**; 1 trillion = 1,000 billion).
- **Company size classifications by market cap:**
- **Large-cap:** $10 billion or more.
- **Mid-cap:** $2 billion – $10 billion.
- **Small-cap:** less than $2 billion.
- It doesn't matter which size category a company falls into as long as it meets the course's investment criteria — but Adam personally buys **mostly large-cap companies**, believing they are generally the "strongest and safest" in the markets.

P/E and PEG Ratio

Price-to-Earnings (P/E) Ratio

- **Formula:** P/E = Share price ÷ EPS.
- **Meta example:** $599 ÷ $21.70 (approx.) = **28.31×** (Yahoo Finance's own precomputed figure showed **28.33×**, a slight rounding difference). Based on trailing twelve months (TTM) earnings.
- P/E changes constantly — every second while the market is open, since price fluctuates continuously.
- **Interpretation:** P/E of 28× means you pay $28 for every $1 of annual earnings per share — theoretically it would take 28 years of flat $21/year earnings to "get back" your $599 investment.
- **But** this ignores earnings growth. Illustrative extreme example: if EPS were to **double every year** (100% growth) starting from $21 → $42 → $84 → $168 → $336, cumulative profit exceeds $650 (the cost of the share) in under 5 years, not 28.
- **Core teaching:** "P/E ratio by itself is quite useless / meaningless" unless you know the earnings growth rate. A common investor mistake: assuming low P/E = cheap and high P/E = expensive. This is false — a high-P/E stock can be cheap if growth is high, and a low-P/E stock can be expensive if growth is low.

Finding Earnings Growth Projections

- Previously available free on Yahoo Finance and GuruFocus, but both removed this data. Two current free sources:
- **finviz.com** — shows past 5-year growth and projected next-5-year growth.
  - Meta: past 5-year EPS growth = **14%**; projected next 5-year growth = **17%**.
- **zacks.com** — a second opinion source (Adam recommends getting more than one growth estimate).
  - Meta: projected earnings growth = **20.05%**.

- Combined range used for Meta: **17%–20%** projected growth.

PEG (or "PEC") Ratio

- Theoretical rule: for a **fairly valued company**, the P/E ratio should roughly equal the earnings growth rate (e.g., 15% growth → P/E of 15; 5% growth → P/E of 5; 20% growth → P/E of 20).
- **Formula:** PEG = P/E ratio ÷ Earnings growth rate (%).
- **Meta example:** P/E 28.31 ÷ 20 (using 20% growth) = **PEG 1.41**.
- **Textbook/theoretical interpretation:** PEG > 1 = expensive; PEG < 1 = undervalued.
- **Adam's practical/real-world rule (differs from pure theory):** Strictly requiring PEG < 1 finds very cheap stocks but causes you to **miss many great companies**. Adam's threshold: PEG up to **1.5** is acceptable; **above 1.5** starts to look expensive.
- Meta's PEG of 1.41 (on 20% growth assumption) = "reasonably priced."

Worked Comparison — NVIDIA vs. Ford Motor Company

- **NVIDIA:** price ~**$137**/share, P/E = **53.97×**, past-5-year EPS growth = **48%**, projected next-5-year growth = **35%**.
- PEG = 53.97 ÷ 35 = **1.54** (slightly above the 1.5 threshold — "not cheap, but not super expensive... slightly expensive").
- **Ford Motor Company:** price ~**$10**/share, P/E = **11.44×**, projected next-5-year growth = **3%** (stated as 3.06%).
- PEG = 11.44 ÷ 3.06 = **3.74** — "very expensive" by the PEG framework.
- **Conclusion:** despite Nvidia's much higher headline P/E (53.97 vs. 11.44) and much higher share price, **Nvidia is actually the cheaper stock on a PEG basis** — Ford is more expensive. This directly rebuts the "low P/E = cheap" fallacy.
- **Trailing vs. Forward P/E clarified:** P/E = Price ÷ Earnings, but "Earnings" can be last-12-months (trailing) earnings or next-12-months (projected) earnings.
- Trailing P/E uses last 12 months of earnings.
- **Forward P/E** uses projected earnings for the next 12 months. In Nvidia's case, the video shows the forward P/E figure at **30.95×**, lower than the trailing 53.97× because next year's earnings are projected to be much larger than the trailing-12-months figure (bigger expected future earnings → lower forward P/E for the same price). *(Note: transcript audio says "Ford PE" here in context of discussing forward P/E for Nvidia — this is understood as "forward P/E," not a reference to Ford Motor Company, based on the surrounding explanation of trailing vs. projected earnings.)*

Historical Case Study — Why P/E Alone Misleads (Adam's Own Trades)

- **NVIDIA:** Adam first bought in **October 2022**. At that time, many avoided it because the P/E ratio looked high — approximately **39.9× (~40×)** on TradingView's P/E indicator overlay at the time of purchase.
- **Result:** From October 2022 to the time of recording (~3 years later, 2022–2024/25), Nvidia's share price rose **1,133%** (roughly **12×**).
- **Ford Motor Company (contrast):** Many investors bought Ford around the same period (Oct 2022) because its P/E looked very low — **3.8×** at that time.
- **Result:** Holding Ford for 3 years, "the price basically went nowhere" — no meaningful gain.
- **Takeaway (stated emphatically):** "P/E ratio itself is totally useless, is totally meaningless. Don't look at it." Only the **PEG ratio** (P/E relative to earnings growth) should be considered — and even PEG "is also not the best way to value a stock... only a rough guide." The most comprehensive valuation method taught later in the course is the

Discounted Cash Flow (DCF) method, which Adam uses primarily because it gives "the most accurate intrinsic value of the stock."

Stock Oracle Full Walkthrough — Microsoft (MSFT) Example (recorded 2026)

Using **app.stockoracle.com**, searching MSFT:
- Share price: **$397** per share.
- Market capitalization: **$2.9 trillion** (stated precisely as **$2,958 billion**).
- Net profit (annual): **$119 billion**.
- Shares outstanding: **7.46 billion**.
- EPS = net profit ÷ shares outstanding (derivable from the above).
- **P/E ratio (trailing 12 months): 24.93×** (~24 stated first, then refined to 24.93).
- **Forward P/E ratio (next year's projected earnings): 20.95×.**
- **Projected earnings growth rate (5-year): 19.62%.**
- **PEG ratio = P/E ÷ growth rate = 24.93 ÷ 19.62% (or forward P/E basis) ≈ 1.27.**
- **Long-term growth rate beyond 5 years: 8.9%** — Adam notes **Stock Oracle is the only website that also shows long-term growth projections beyond the standard next-5-years window.**
- **ROIC: 26.71%.**
- **ROE (Return on Equity): 34.39%.**
- Stock Oracle's "Favorites" feature lets you customize which metrics display (out of **~479–497 available metrics** on the platform). Adam's **personal "10 favorite metrics"** to check on any stock at a glance: trailing P/E, forward P/E, 5-year projected growth rate, long-term growth rate (beyond 5 yrs), shares outstanding, PEG ratio, ROIC, ROE (plus other items rounding out the ten).

What Drives Stock Prices

Short-Term Price Movement: Demand, Supply, and Emotion

- A common beginner surprise: prices are highly volatile day-to-day (e.g., up 5% one day, down 5% the next, up 8%, down 3%, etc.).
- **Key mistake to avoid:** believing price always reflects business worth. A 5% two-day price swing does NOT mean the business made 5% more or fewer profits, or that usage changed 5% — "it's the exact same business."
- **Stock market = an auction-driven market** (compared explicitly to a live auction: "do I get $25… sold to the man for $29").
- **Short-term price driver = demand and supply.** More buyers than sellers on a given day → price rises (same company); more sellers than buyers → price falls (same company).
- **What drives demand/supply short-term = emotions**, specifically **fear and greed**.
- Fear → panic selling.
- Greed → excited buying.
- Fear can also cause buying — **FOMO (fear of missing out)**: fear that a rising stock will keep rising without you, so people buy, which itself pushes price higher.
- **Emotions are triggered by news** — lockdowns, recession/depression fears, war headlines → panic/selling; good earnings announcements, rate news → buying/price rises.
- Rule for investors: **ignore the news noise and disassociate emotionally from the market**; "if you get emotional in the market… you're going to lose money, I guarantee you that." The way to make money is to go opposite of the crowd.

Case Study — Microsoft, 2019–2020 (COVID Crash)

- **2019:** Microsoft share price ~**$190** ("said $109" appears to be a transcription slip immediately corrected/contextualized as $190 throughout); Adam's calculated **intrinsic value = $230** at that time (intrinsic value calculation covered in the later Stock Valuation lesson). Conclusion: the stock was **already undervalued** at $190.
- **2020 COVID-19 pandemic / global lockdown announced** → panic selling across the market ("whether it's McDonald's or Microsoft, every stock came down").
- Microsoft dropped from **$190 to $133 in four weeks** (roughly a 30% drop as characterized in the discussion — later stated as "35% less").
- **Reality check:** the business itself did not deteriorate — Microsoft's revenue and profits actually **increased** during the pandemic (whether working from home or in-office, people still needed Microsoft subscriptions).
- **Why the price fell anyway:** short-term market price reflects emotion and manipulation, not underlying value. Adam also implicates **market makers/institutions** deliberately fueling fear to cause retail investors to panic-sell at the bottom, so institutions can buy the cheap shares — "that's how they make a lot of money, by scaring the retail investors."
- **Margin selling mechanic explained:** some investors get greedy during rallies and borrow money from their broker to buy more shares — **"buying on margin."** Existing shares serve as **collateral**. When the market drops suddenly, collateral value falls, and the broker demands more cash ("margin call"); investors without spare cash get **forcibly liquidated** by the broker (shares sold at very low prices). Mass forced liquidations create a feedback loop of extra selling pressure, accelerating the price's "free fall" during a panic.
- **Investor response:** rational investors benefit from this mispricing by buying shares of great companies while they're 30–40% (or more) below intrinsic value due to panic and forced liquidations.
- **On timing the bottom:** "no one knows where is the bottom... there's no way to predict the bottom." As long as a purchase is made while the stock is undervalued, timing precision doesn't matter — you don't have to catch the exact bottom. Adam's method: **never buy all at once**; instead **"nibble"** — buy in small tranches as the price moves.

Quotes Cited

- **Benjamin Graham** (mentor to Warren Buffett, known as "the father of value investing," author of *The Intelligent Investor*, which Adam highly recommends reading):

> "The market is a pendulum that forever swings between unsustainable optimism that makes stocks too expensive at times and unjustified pessimism which makes them too cheap at times. And the intelligent investor is a realist who sells to optimists when stocks are too expensive, and buys from pessimists when stocks are too cheap."

- **Peter Lynch** (described as one of the very few fund managers who actually beat the market by a wide margin over his career; Adam has studied his work extensively):

> (paraphrased per Adam) "Often there's no correlation between the success of a company's operations and the success of the stock price over a few months or even sometimes a year or so" — but Adam adds that over the **long term** (defined here as **at least two years or more**), "there's a hundred percent correlation between the success of a company and the success of its stock price."

- Adam's synthesis: "this disparity is the key to making money. It pays to be patient and to own successful companies."

Long-Term Price Drivers (5 Factors)

Over the long run (the S&P 500 specifically referenced as having risen for over 120 years), five factors drive stock prices higher:

1. **Earnings** — companies (in aggregate, the 500 biggest US companies) make more profit over time. **Corporate earnings grow at an average of ~8% per year over the long run** (explicitly called out to "write this down"). Higher EPS → higher share value → higher share price.
   - Beyond EPS, **free cash flow per share** is highlighted as even more important than reported accounting earnings — "cash flow is more tangible" than accounting profit (expanded further in the Fundamental Analysis lesson).
2. **Share buybacks** — reduce share count, so each remaining share is worth more (same mechanism as described earlier).
3. **Dividends** — more dividends paid (e.g., by the S&P 500 in aggregate) increases attractiveness/value, since investors are willing to pay more for dividend income.
4. **Future earnings growth (anticipation)** — the market often prices in *future*, not just current, earnings. Some companies with current losses still see rising share prices because investors anticipate future profitability/cash flow growth.
5. **Money supply** — controlled largely by the **Federal Reserve**. When the Fed lowers interest rates and buys bonds, money supply increases, inflating asset prices (including stocks). Conversely, raising rates / halting bond purchases reduces money supply.
   - **Example: 2022–2023** — the Fed **reduced money supply by ~40%** (raised interest rates, stopped buying bonds), yet the **stock market still rose in 2023** — because even with lower money supply, growing corporate earnings were enough to push the market higher. When all five factors move favorably together (money supply up, earnings up, dividends up), the market rises even more strongly.
   - Supporting chart description: S&P 500 share price (red line) tracks alongside forward earnings/projected company earnings (blue line) over the long run — they move together over time despite short-term divergence.

Summary Framework: Short-Term vs. Long-Term Drivers

- **Short term (next day/week/month/few months):** price driven mainly by **emotion** (fear/greed) and **market sentiment**, triggered by news. The **study of these short-term emotional/price patterns is called Technical Analysis.**
- **Long term (2+ years):** price driven by the underlying **intrinsic value** of the business — determined by revenue, profit, cash flow, assets, and liabilities. **The study of the business itself is called Fundamental Analysis.**
- **Weighting rule for investing decisions (explicitly told to "write this down"): 90% of buy/sell decisions should be based on fundamentals**, and only **10%** on technical analysis (used to optimize entry/exit timing). This 90/10 split applies to *investing*; short-term speculative trading would weight technicals more heavily.
- Two things needed to maximize investing profit: (1) identify which are the great companies, and (2) determine the stock's **intrinsic value**.

Trend/Pattern Vocabulary Introduced

- **Uptrend pattern:** price makes a repeating "wave up, wave down" sequence trending higher overall.
- **Downtrend pattern:** price makes a repeating wave sequence trending lower overall.
- **Consolidation:** price moves sideways in waves without a clear directional trend.
- These patterns are explored in depth in the later Technical Analysis lesson; described here as "the cherry on the cake," with fundamentals being "the main cake."

Illustrative Example — McDonald's

- McDonald's: a long-standing, well-known company. Revenue, profits, and cash increase globally over time → **intrinsic value** (shown as a black dotted line in Adam's chart) rises steadily over time.
- **Share price**, however, does **not** track intrinsic value smoothly day-to-day/month-to-month — it moves "like a yo-yo" due to short-term volatility, emotion, and manipulation, cycling through consolidation → uptrend → downtrend → uptrend → downtrend → consolidation → uptrend patterns, etc.
- Zooming into a downtrend segment: Adam narrates the emotional swings driving it ("Oh my God, we're going to die, no one's going to eat burgers anymore, I hate Ronald McDonald" during downtrends, vs. excitement during uptrends), likening the market to "a bipolar, manic, depressive person."
- **Key conclusion:** "the majority of the time, the price movements… are caused by reasons that have nothing to do with the company's long-term business potential… fundamentals and intrinsic value."

How to Make Money in Stocks

Don't Ask "Why Did the Market Move?"

- Using the S&P 500 chart as illustration: prices move in repeated wave-up/wave-down patterns driven by cycling excitement/fear.
- **"Why did the market go up/down?" is called "the most stupid question to ask."** Reasoning: analysts always retroactively construct a plausible-sounding explanation for any price move (e.g., "market went up because GDP beat expectations," "market went down because of the war in Gaza") — there are always both positive and negative narratives available in the world to cherry-pick from; these justifications are a waste of time and not useful for investment decisions.
- **Simple operating rule:** when the market/a favored company goes down and becomes undervalued, buy. When it goes up and becomes expensive, don't buy (and it may be a time to sell certain positions).

Procter & Gamble (P&G) Example — Overvalued vs. Undervalued

- P&G: a long-standing company (household products, e.g. shampoos) whose sales/profits/value keep rising over time → intrinsic value line (green) rises steadily.
- Share price oscillates above and below the intrinsic-value (green) line repeatedly, "like a pendulum."
- **Definitions:**
- **Overpriced/overvalued** = share price above the intrinsic value line.
- **Underpriced/undervalued** = share price below the intrinsic value line.
- **The investor's job, simply stated:** add shares of great companies when they are undervalued (below the green line). "Why the price went down, why the price went up is totally irrelevant."
- **You do not need to buy the exact bottom.** Nobody can predict a wave's bottom. Adam's personal tranche-buying ("nibbling") method:
- Once price drops **~10% below intrinsic value**, nibble (buy a small amount).
- If it drops another **5–10%** further, nibble more.
- If it drops another **5–10%**, nibble again.
- This averages down the cost basis over time; doesn't require catching the exact bottom, since the goal is simply an average entry price below where intrinsic value (rising over time) eventually reconnects with price.
- **Key aphorism: "Price is not value… Price is what you pay, value is what you get."** (Price = the fluctuating line; Value = the green intrinsic-value line.)

- Adam states he specifically *loves* when a great company he holds gets hit by bad news, because panic selling creates buying opportunities.

Meta/Facebook — Long-Term Holding Through Repeated "Bad News" Buying Opportunities

- Adam has held Facebook/Meta for many years. A few years prior to recording it traded between **$120 and $360 per share**; by the time of recording it was near **$600**.
- Throughout the holding period, intrinsic value (green line) rose steadily as profits grew, while a string of negative headlines periodically crashed the price below intrinsic value — each cited by Adam as a buying opportunity:
- **April 2018** — Facebook data-selling scandal (accused of selling user data to other providers) → panic, price fell below intrinsic value → buying opportunity.
- Facebook fined over the **Cambridge Analytica scandal** → more panic → another buying opportunity.
- Headline: *"Facebook shares extend slide as Starbucks joins major global brands in social media ads spending boycott"* → advertiser boycott news → opportunity.
- Headline: *"Tech stocks closed sharply lower as bond yields jumped"* — long-term interest rates rose, causing broad panic and a share price drop → opportunity.
- **Whistleblower complaint** alleging Facebook's platform was used to promote human trafficking and domestic servitude (Adam clarifies he is not condoning the underlying issue — merely noting the market reaction) → opportunity.
- Headline: *"Facebook's outage frustrates advertisers heading into holiday season"* → an outage-driven scare → opportunity.
- **General lesson:** every company (McDonald's, Visa, MasterCard, Meta, etc.) periodically faces scandals/bad news. Rational investors who evaluate fundamentals (is the company still making more money? what is intrinsic value?) and buy the dip rather than panic-sell tend to profit; those who "buy high, sell low" out of panic tend to lose money repeatedly.

Two Ways Stocks Build Wealth

1. **Capital appreciation** — profit from the share price rising over time between purchase and sale.
   - You should not expect gains "next day, next week, or even next month" — short-term price movement is essentially random/unpredictable.
   - **Worked example — Apple:** bought at **$320**/share (bullish thesis: great company, widely used products, undervalued); later sold at approximately **$502**/share (the video also momentarily states "$520" during narration, but the profit calculation that follows — $182 profit per share — is only consistent with a $502 sale price: $320 + $182 = $502). **Profit = $182 per share.**
   - Adam's own biggest wins came from **not** selling too early — holding long enough for a position to compound **5×, 10×, 20×, or even 50×** the original cost. "That's how you get really rich."
   - **Capital gains tax:** depends on investor's home country. Adam (Singapore resident) pays **zero capital gains tax** investing in either the US or Singapore markets. By contrast, US citizens investing in US shares owe capital gains tax per their income bracket. (Viewers told to check with their own tax advisor.)
2. **Collecting cash dividends** — a share of company profits paid out periodically (commonly quarterly, some companies pay twice yearly).
   - Not all companies pay dividends. Dividend payers tend to be **mature, large, established companies with excess cash** beyond what's needed for reinvestment.
   - **US dividend-payer examples cited:** PepsiCo, Johnson & Johnson.
   - **Adam's personal dividend preference: Singapore-listed banks — DBS Bank, OCBC, UOB** — for high, tax-free dividend income.

- **Non-dividend / high-growth company examples:** NVIDIA, Palantir (which Adam owns) — these reinvest all profits into the business or buy back shares rather than pay dividends; investors profit via capital appreciation instead.
- **Investor decision framework:** dividend-heavy stocks (Pepsi, J&J, OCBC) tend to have slower-rising share prices; growth stocks (Nvidia, Meta, Apple) tend to pay little/no dividend but appreciate faster. Choose based on whether you want regular cash income or growth of wealth.
- Adam personally runs **multiple portfolios**: two focused on capital gains (own companies with fast-rising price, low dividends) and a separate **dividend portfolio** (UOB, OCBC, DBS — slower price appreciation, high dividend income). Full detail in the later "building winning portfolios" lesson.

Dividend Yield

- **Formula:** Dividend yield = Dividend per share (paid over last 12 months) ÷ Current share price.
- **Simple example:** a $20 stock paying $2/year in dividends → yield = **10%**. Framed as compared against bank interest or bond yields; described as "pretty good."
- **"Attractive" yield benchmarks by market:**
- **US market:** ~**3%** dividend yield considered attractive.
- **Singapore market:** ~**5%–6%** dividend yield considered attractive.
- Adam prefers Singapore for dividend investing specifically because yields run higher (5–6% vs. US's typical 2–3%) **and** are tax-free, whereas US dividends are taxed. The US market is framed as better suited for **capital gains** investing instead.
- **PepsiCo (Yahoo Finance) example:** last closing price **$152.89**/share; forward dividend (expected over next 12 months) = **$5.42**/share → dividend yield = $5.42 ÷ $152.89 = **3.55%**.
- **UOB Bank (Singapore) example:** share price **$36.02** (minimum purchase 100 shares, as with most Asian markets); dividend **$1.76**/share → yield = $1.76 ÷ $36.02 = **4.89%**. Confirms Singapore stocks generally carry higher dividend yields than US stocks, and Singapore dividends are tax-free.

Stock Oracle — Dividend Data Deep Dive (DBS Bank Example)

(Shown via DBS Bank's US OTC-market listing on Stock Oracle at time of recording; Adam notes Stock Oracle plans to add native Singapore/China/Hong Kong-listed data by end of 2026.)

- **Current dividend yield: 3.86%.**
- **Dividend payout ratio: 63.86%** — meaning DBS pays out ~63% of its profits as dividends, retaining the other ~37% as retained earnings.
- **Dividend growth rate:** last **5 years**: **26% per year** average growth; last **10 years**: **16% per year**; last **3 years**: **23% per year** — described as "amazing," consistent dividend growth.
- Dividends from DBS (Singapore-listed) are tax-free.
- **Dividend yield history (10 years):** yield has cycled up and down over time; generally, when share price falls, yield rises (and vice versa). At time of recording, DBS's share price had risen a lot recently, pushing the yield **down from ~5.8% to under 4%** — Adam characterizes this as **DBS currently being "a bit overvalued"** for dividend-buying purposes; he'd prefer to buy back above a ~4% yield, and would find it "very attractive" if yield approached or exceeded **5%**.
- **Dividend-per-share chart (10 years):** shows a steadily increasing dividend per share over time, **except during COVID**, when DBS was forced to cut dividends **"because of MAS"** (Monetary Authority of Singapore) — dividends resumed increasing afterward.

- **Most recent financial year (FY2025) figures:** EPS (profit per share) = **$2.95**; dividends paid out of that = **$1.88** per share; free cash flow generated ≈ **$3.7x per share** (transcript audio here is garbled — heard as "three dollars into a seven since it's free cash flow" — figure is approximate, likely intended as roughly $3.7–3.79 per share).
- **Ex-date / pay-date mechanic:** if you own shares before the stock's "ex-date," you are entitled to receive the dividend, which is then distributed on the "pay date." DBS pays dividends **quarterly (4 times per year)**.

Stock Exchanges

- The **US market** is Adam's primary focus — "world's biggest market," home to "the best companies in the world." Comprises about **24 stock exchanges**, of which the **three biggest** are:
- **New York Stock Exchange (NYSE)**
- **American Stock Exchange (AMEX)**
- **NASDAQ Stock Exchange**
- Example: Coca-Cola is listed on the NYSE; Apple is listed on NASDAQ. Knowing which exchange a stock trades on is **not important** for investors — your broker automatically routes the order to the correct exchange.
- **China:** three main exchanges — **Shanghai Stock Exchange**, **Shenzhen Stock Exchange** (both mainland China), and the **Hong Kong Stock Exchange**.
- **UK:** the **London Stock Exchange**.
- **Europe:** several exchanges, biggest include **Euronext** (based in Amsterdam) and the **Frankfurt Stock Exchange**.
- **Asia (other): Singapore Exchange**, **Tokyo Stock Exchange**, **Bombay Stock Exchange**, **Jakarta Stock Exchange**, and others.

Stock Index Performance

What Is a Stock Index and Why It Matters

- Every stock market has at least one index measuring its overall performance — used by investors as a **benchmark to beat**.
- **How an index is built:** a curated "basket" of the biggest/most representative companies stands in for the entire market — compared to elected representatives (members of parliament/congress) representing a whole population.

Major US Indexes

- The US market has roughly **10,000 total listed stocks**.
- **S&P 500** = the **500 biggest** US companies, averaged together — the index most commonly used to represent the overall US market, and the one **most followed by retail investors** (because it captures far more companies than the Dow).
- **Dow Jones Industrial Average (DJIA)** = the **30 biggest** US companies (a subset that is also within the S&P 500).
- **NASDAQ 100** = the **100 biggest** companies listed specifically on the NASDAQ exchange, many of them technology companies.
- These three indexes only include **large-cap** companies (market cap ≥ **$10 billion**).
- **Russell 2000** = a basket of **2,000 small- and mid-size companies** (fills the gap for smaller companies not captured by the above three).

- Important: when headlines say "the market went up 20% this year," this typically refers to the **S&P 500** specifically, **not** all ~10,000 US-listed stocks.

Singapore & Other Global Indexes

- **Singapore market:** only **~600 total stocks** (a "very small market"). The top **30 biggest** are combined into the **Straits Times Index (STI)**.
- **Global index reference list provided:**
- US: Dow Jones (30), S&P 500 (500), NASDAQ 100 (100), Russell 2000 (2,000 small/mid).
- Singapore: Straits Times Index (STI) — weighted average of 30 biggest companies.
- UK: **FTSE 100** — 100 biggest companies on the London Stock Exchange.
- Europe: **Euro Stoxx 50** — 50 biggest companies across 11 European countries; **DAX** — weighted average of the 40 biggest companies on the Frankfurt Stock Exchange; **CAC 40** — 40 stocks on the Paris Stock Exchange.
- China: **Shanghai Composite Index**, **Shenzhen Composite Index**, **Hang Seng Index** (represents Hong Kong), **CSI 300** (the 300 biggest mainland Chinese companies — generally used to represent mainland China).
- Japan: **Nikkei 225** — the 225 biggest Japanese companies on the Tokyo Stock Exchange.
- Australia: **ASX 200**.
- Malaysia: **KLCI (Kuala Lumpur Composite Index)** — the 30 largest companies listed in Malaysia.

Index Membership Examples & Overlap

- **Apple:** IS in the S&P 500, IS in the Dow Jones, IS in the NASDAQ 100, is **NOT** in the Russell 2000 (too large for that index).
- **Coca-Cola:** IS in the S&P 500, IS in the Dow Jones, is **NOT** in the NASDAQ 100.
- Some stocks appear across multiple indexes; others appear in just one.

Index Composition Changes Over Time

- Index members are **not fixed** — a company losing profitability/relevance gets removed and replaced with a stronger/more representative company.
- **DJIA historical replacements cited: Citigroup, AIG, and General Electric** were previously in the Dow Jones but have all been removed, replaced over time by companies including **Nike, Salesforce, and Amazon**. More recently, **Intel** was removed from the Dow Jones and replaced by **Nvidia** "just a few months ago" (relative to recording).
- **Current DJIA member examples cited (partial list):** Amazon, American Express, Apple, Boeing, Cisco, Nvidia, Visa, Disney, Walmart, 3M, Nike, and others.
- Observation: DJIA's 30 companies sell not just to US consumers but **globally** — hence Adam calls them "global companies" and states the **Dow Jones "in a way represents the global economy."** When asked "what is the safest thing I can invest in," Adam's answer is **the Dow Jones**, reasoning that people worldwide consistently consume McDonald's, Apple products, etc.

Index Weighting

- The 500 companies in the S&P 500 do **not** carry equal weight — bigger **market cap** = bigger weighting/influence on the index's overall movement.
- Example: **NVIDIA has a much bigger weighting than Texas Instruments** because Nvidia's market cap is larger — a 1% move in NVIDIA affects the index more than a 1% move in Texas Instruments.
- Reminder: market cap = share price × total shares outstanding.

FinViz "Maps" Feature — Visualizing the S&P 500 by Sector

- On **finviz.com**, the "Maps" (referenced initially as "groups") feature visually displays all 500 S&P components, colored by year-to-date performance (green = price increased YTD, red = price decreased YTD; example date context: YTD from January to December 2024).
- Components are grouped into sectors, e.g.:
- **Technology:** Microsoft, Apple, Oracle, Nvidia, Broadcom, Salesforce, ServiceNow.
- **Consumer Cyclical / Consumer Discretionary:** Amazon, Tesla, Home Depot.
- **Financials:** Visa, JPMorgan, American Express, MasterCard.
- **Healthcare:** Eli Lilly, Johnson & Johnson, Merck, UnitedHealth.
- **Consumer Defensive:** Pepsi, Walmart.
- (Sectors covered in more depth in the next lesson.)

Historical S&P 500 Performance (data through end of 2024, as presented in Adam's chart tool)

- **20-year period (Jan 2005 – end 2024):** index moved from ~**80 points** to **595 points** → total gain **613%**; equivalent to a **compounded annual growth rate (CAGR) of 10.32%**.
- **10-year period (Jan 2015 – end 2024):** index moved from ~**170 points** to **595 points** → total gain **246%**; CAGR **~13%**.
- **5-year period:** index started at **300 points** → total gain **100%**; CAGR **14.9%**.
- All figures noted to include **dividends reinvested** into the index.
- Major crashes/corrections referenced along the 20-year timeline: **2008–2009 Great Financial Crisis**, **2018 trade-war crash**, **2020 COVID-19 pandemic crash**, **2022 bear market** (driven by high inflation and high interest rates).
- Lesson: despite multiple crashes, the long-term trajectory is upward because companies make more money over time and the index composition upgrades itself (weak companies replaced by strong ones).

Global Market Comparison (via TradingView, 10-year lookback, priced in local currency first, then currency-adjusted)

- Tool: **TradingView.com** (free version usable; paid not required); comparison built by charting the **S&P 500 (symbol SPX)** against other indexes added via the "+" button.
- **10-year returns (local currency, unadjusted), ranked:**
  1. **US (S&P 500): 191%**
  2. **India (Nifty): 179%**
  3. **Europe (Euro Stoxx 50): 60%**
  4. **Australia (ASX): 53%**
  5. **UK (FTSE 100): 24%**
  6. **Singapore (Straits Times Index): 14%**
  7. **Hong Kong (Hang Seng): –16%** (a loss; described as "very sad")
- (Also referenced: China market generally lands in last place.)
- **Currency-adjustment point:** indexes are quoted in different currencies (e.g., Nifty in Indian rupees). When converted/normalized to a single currency (e.g., USD), the US market's outperformance over India **widens further**, because the Indian rupee has been depreciating against other currencies — so India's 179% local-currency gain looks smaller once currency loss is factored in.
- **Japan (Nikkei 225), once added to the comparison:** performed **better than India** over the 10-year currency-adjusted view — landing in **second place behind the US**.

- **5-year ranking (currency-adjusted):** US (best) → Japan → India → Europe → Singapore → Australia → UK → China (worst).
- **Takeaway:** if the objective is maximizing capital gains, the data supports focusing on the **US market**, which delivered the best returns of all markets compared.

Lesson 1 Quiz — Questions & Adam's Answers (9 Questions)

1. **Which statement is FALSE?**
   - (A) In the US market you can buy fractional shares (less than 1 share) — **TRUE** (via brokers like Interactive Brokers).
   - (B) In the Hong Kong market you must always buy a minimum of 100 shares — **FALSE** (flagged as the "tricky" answer: it actually depends on the specific stock/counter — some require 100-share minimums, others require 500-share minimums).
   - (C) In Singapore and Malaysia markets, minimum purchase is 100 shares — **TRUE.**
   - (D) In the US market, there is no maximum number of shares you need to buy — **TRUE.**
   - **Answer: B.**
2. **Success Company** earned **$350 million** net income (last 12 months) with **100 million shares outstanding**. EPS?
   - EPS = $350M ÷ 100M = **$3.50**. **Answer: B.**
3. Success Company's share price is **$50**. P/E ratio?
   - P/E = $50 ÷ $3.50 = **14.3×**. **Answer: B.**
4. Success Company's share price is **$50**, with **100 million shares outstanding**. Market capitalization?
   - Market cap = $50 × 100M = **5,000 million = $5 billion**. *(The transcript's stated answer letter here is garbled/unclear — likely a transcription artifact.)*
5. What is the PEG ratio formula — P/E ratio multiplied by growth rate, or P/E ratio divided by growth rate?
   - **Answer: divided by (B).** Reminder: PEG of **1.5 or less** is considered reasonable; below 1 is considered cheap.
6. Success Company's share price is **$50** and it paid **$5** in dividends (last 12 months). Dividend yield?
   - Yield = $5 ÷ $50 = **10%**. **Answer: A.**
7. If a stock generates EPS of **$5**, how can the company return this capital to shareholders? (Multi-part answer — all four mechanisms restated:)
   - **Dividends** (taxable in US markets, not taxable in markets like Singapore).
   - **Share buybacks** (indirectly returns capital — with fewer shares outstanding, you own a bigger percentage of the business).
   - **Retain the earnings** (like a savings account).
   - **Reinvest the capital back into the business.**
8. **Company A:** P/E = 10, projected earnings growth = 5%. **Company B:** P/E = 40, projected earnings growth = 100%. Which is cheaper?
   - Company A PEG = 10 ÷ 5 = **2**. Company B PEG = 40 ÷ 100 = **0.4**.
   - Lower PEG = cheaper → **Answer: B (Company B is cheaper).**
9. **True/False set:**
   - "Great investors can predict where the share price will go in the short term." → **FALSE** — "no one can predict where prices go in the short term… too many moving parts," only probability-based guessing is possible.
   - "In the short term, share prices are driven by emotions, manipulation, and forced selling." → **TRUE.**
   - "In the long run, share prices are driven by the fundamental value of the business." → **TRUE.**
   - "Short-term trading/speculation has a much higher win rate than long-term investing." → **FALSE.** Best win rates

in short-term trading run **50–60%**, whereas long-term investing in great companies wins **90%+** of the time (companies make money over time).

Shocking Market Facts

(Note: this video's opening lines introduce Lesson 2 on ETFs, but the bulk of the content — as transcribed — covers statistical "facts" about long-run market behavior, cycles, and probabilities; captured here as delivered.)

Fact 1: The US Index Always Goes Up Over the Long Run

- Illustrated via a **Dow Jones Industrial Average, 121-year chart** (January **1904** to December **2024**) — chosen over the S&P 500 because the S&P 500's history only goes back about 60 years, while the Dow Jones is the oldest/longest-history US index.
- Despite an extensive list of historical crises across the 121 years — **World War I, World War II, the Great Depression, the Vietnam War, the assassination of President JFK, the Cuban Missile Crisis, the Black Monday stock crash, the Gulf War, a 21-day US government shutdown, the Asian Financial Crisis, the dot-com crash, the 2008–09 Great Financial Crisis, the European sovereign debt crisis, the US credit-rating downgrade (from excessive debt), the 2018 US–China trade war, the 2020 COVID-19 pandemic, and the 2022 "Fed hyperinflation war"** — the market kept climbing.
- **Total gain over 121 years: 11,188,000%.** Equivalent **CAGR: 10.04%.**
- **Why the index always goes up long-run (two reasons given):**
  1. Underlying driver = **earnings/profits**. Example: McDonald's — a Big Mac costs more today than 20 years ago (price increases) *and* more burgers are sold globally today than 20–30 years ago (volume increases) → rising prices × rising volume → rising profits → rising share value.
  2. **Index composition self-upgrades.** Weak/unprofitable companies get removed and replaced by stronger ones (again citing **AIG, Citigroup, General Electric** removed from the Dow Jones, replaced by better companies like **Apple, Salesforce, Goldman Sachs**) — the index is effectively "engineered to always go up."

Fact 2: The Market Doesn't Go Up Every Year, But Goes Up Most Years

- Looking at the **last 74 years (1951–2024):**
- Sample year-by-year figures cited: 1951 +23%, 1952 +18%, 1953 −1%, etc.
- **16 of 74 years ended down (21.6% of the time).**
- **58 of 74 years ended up (78.4% of the time).**
- Conclusion: buying US stocks in January and holding to December gives roughly a **78% probability of making money** in any given year — "better than the casino" (casino games generally offer worse-than-50/50 odds; investing in the US market offers better-than-70% odds).
- **Holding for 5-year periods:** out of **70 total 5-year periods** (every overlapping 5-year combination across the 74-year span), **6 periods ended in a loss (8.6%)**, **64 periods ended in a gain (91.4%)**.
- **Holding for 10-year periods:** out of **65 total 10-year periods**, only **2** ended in a loss (both spanning the **dot-com crash**) → **loss probability 3.1%**, **gain probability 96.9%**.
- **How long to hold for a historically "100% guaranteed" gain?** According to the historical data shown, **about 15 years** — even if you had bought at the exact top of the dot-com bubble or the Great Financial Crisis, holding 15 years historically always produced a gain. (Adam notes the course will still teach how to avoid blindly buying at market tops, aiming instead to buy when undervalued or near fair value.)
- **Distribution of annual return outcomes (over the 74 years):**

- +10% to +20% in a year: happened **23%** of the time.
- +20% to +30%: happened **21.6%** of the time.
- +30% to +40%: happened **13.5%** of the time.
- +40% to +50%: happened **once** (in **1958**) = **1.4%** of the time.
- +50% to +60%: happened **once** (in **1954**).
- 0% to −10% (down): happened **12.2%** of the time.
- −10% to −20%: happened **5.4%** of the time.
- −20% to −30%: happened **2.7%** of the time (twice in 74 years).
- −30% to −40%: happened **once** (in **2008**, the Great Financial Crisis) = **1.4%** of the time.
- Comparative takeaway: a +20–30% year (21.6% probability) is roughly **~10× more likely** than a −20–30% year (2.7% probability) — pushing back against financial-media pessimism about crash predictions.
- Overall: of 74 years, **58 were bullish, 16 were bearish** — "it pays to stay invested… it pays to stay bullish."

Singapore Market Comparison (Straits Times Index, via STI ETF, ticker ES3)

- Because most Singapore market gains historically come from **dividends** (unlike the US, where gains come mostly from price appreciation), Adam uses the **STI ETF (ticker: ES3)**, which only has data back to **2003** (started in 2003), giving **22 years of data** (vs. 74 for the US) — "better than nothing."
- **Annual performance (22 years, dividends included):** e.g., 2003 +36% (incl. dividends), 2004 +18.96%, etc.
- **5 of 22 years ended in a loss (22.73% probability).**
- **17 of 22 years ended in a gain (77.27% probability).**
- Note: **without** counting dividends, the annual gain-probability drops to roughly **56%** — dividends are a major driver of Singapore market total returns.
- **5-year holding periods: 18 total 5-year periods**; only **1** period (**2011–2015**) ended in a loss even after dividends → **gain probability 94.44%**, **loss probability 5.56%**.
- **10-year holding periods: 13 total 10-year periods**, **all 13** ended in a gain (**100%** gain probability, 0% loss) — *(the transcript momentarily also states "zero out of 30 periods," which appears to be an inconsistency/speech slip relative to the "13 total periods" figure stated just before it)*.

Fact 3: Stock Prices Move in Cycles, Not Straight Lines

- **Bull market (uptrend):** a series of **higher highs and higher lows** — generally associated with economic growth and market optimism. Most of the time, the market is in a bull-market phase.
- **Bear market (downtrend):** a series of **lower highs and lower lows** — generally (though not always) associated with recessions or broad pessimism.
- On an actual long-run S&P 500 chart, Adam demonstrates visually marking off alternating bull-market and bear-market segments over time.

Fact 4: Bull Markets Are Opportunities for Investors

- The stock market spends roughly **80% of the time in a bull-market phase**.
- Even within a bull market, prices still regularly decline — but "every time they go down, they go even higher" afterward, producing a pattern of **higher highs and higher lows** made up of repeated **"wave up, wave down"** cycles.
- **Breathing analogy:** price behavior compared to breathing — "breathe out, breathe in" repeatedly; a market "can't go up every single day… it's got to come down before going even higher."

- **Pullback:** a wave-down decline of **1% to 9%**. Occurs on average **~3 times per year**. Timing is unpredictable ("no one knows… it will happen when it happens").
- **Correction:** a bigger wave-down decline of **10% to 19%**. Occurs on average **~once per year** (essentially, one of the roughly three pullbacks in a given year tends to deepen into a correction).
- **Duration:** pullbacks and corrections typically **last 1 to 4 months**.
- Both pullbacks and corrections are framed as **buying opportunities** — the investor's aim is to buy during wave-downs (small ones = pullbacks, big ones = corrections).
- **Estimating possible bottoms — Moving Averages:** since no one can predict the exact bottom of a wave-down, **moving averages** are used as a probabilistic guide (full detail deferred to the later Technical Analysis lesson).
- **50-day moving average** — shown as the **blue line**.
- **150-day moving average** — shown as the **green line**.
- **200-day moving average** — shown as the **red line**.
- These are auto-calculated/auto-plotted by charting software.
- Moving averages help identify **support levels** — price levels where a decline tends to stall and bounce back up (not guaranteed, but "a good probability"). Example given: the S&P 500 repeatedly using its 50-day (blue) moving average as a support level, bouncing off it multiple times in succession.
- Practical application: once price falls to a moving-average support level, that's a **possible bottom** and a point to start adding shares — but because price could still fall further to the *next* moving average down, positions should always be built in **tranches at successive support levels**, never bought "at one go."

Tools & Websites Referenced (Summary)

- **Yahoo Finance** (finance.yahoo.com) — ticker search; "Statistics" page (net income, shares outstanding, implied shares outstanding); "Summary" page (EPS TTM, P/E TTM, forward dividend/yield, market cap, current share price).
- **Stock Oracle** (app.stockoracle.com) — subscription platform; "Company Metrics" page (ROIC, ROE, P/E trailing & forward, PEG, 5-year and beyond-5-year growth projections, dividend yield/history/growth-rate/payout ratio/ex-date-pay-date schedule); customizable "Favorites" from a pool of ~479–497 available metrics; Adam's primary go-to tool for metrics not available on Yahoo Finance (e.g., ROIC).
- **FinViz** (finviz.com) — ticker search; past-5-year and projected-next-5-year earnings growth; "Maps" feature visualizing the entire S&P 500 by sector/performance, color-coded green (up YTD) / red (down YTD).
- **Zacks** (zacks.com) — alternative source for earnings growth projections, used to cross-check FinViz's numbers (Adam recommends getting more than one source/opinion).
- **TradingView** (tradingview.com) — charting software (free version sufficient); used for P/E-ratio overlays on price charts, moving averages, and cross-market/currency-adjusted index comparisons (index symbols such as SPX for the S&P 500). Course includes dedicated tutorials on setting up TradingView charts.
- **Interactive Brokers** — Adam's personal brokerage; notable for allowing fractional share purchases in the US market. (Any reputable broker is acceptable for the course; broker mechanics are broadly similar once learned.)

Key Takeaways

- A share is ownership in a real business, not a lottery ticket — the business's growing profitability, not news headlines, is what ultimately makes a share worth more.

- Learn to find and use **net income**, **implied shares outstanding**, and **EPS** — and understand the four ways a company can return that profit to you: dividends, buybacks, reinvestment (ROIC), or retained earnings.
- **P/E ratio alone is meaningless** — it must always be evaluated against the **earnings growth rate** via the **PEG ratio** (PEG = P/E ÷ growth%; roughly ≤1.5 reasonable, <1 cheap, >1.5 getting expensive), and even PEG is only a rough guide — full valuation comes later via discounted cash flow.
- Short-term price action is driven by **demand/supply, emotion (fear and greed), news, and manipulation/forced selling** — largely unpredictable and not correlated to actual business performance.
- Long-term price action (2+ years) is driven by **fundamentals**: earnings growth (~8%/year long-run average), share buybacks, dividends, future earnings anticipation, and money supply — with a **100% long-term correlation** between business success and stock price success.
- The investing decision mix should be **90% fundamental analysis, 10% technical analysis**.
- Making money requires buying great businesses **when undervalued** (below intrinsic value) and being willing to buy into bad news/panic, rather than asking "why did the market move" (an unproductive question).
- Wealth is built two ways: **capital appreciation** and **dividends** — know which one your objective favors, and choose stocks/markets accordingly (US market: better for capital gains, low-taxed; Singapore market: better dividend yields, tax-free).
- Stock indexes (S&P 500, Dow Jones, NASDAQ 100, etc.) are curated, self-upgrading baskets of top companies used as **performance benchmarks** — the course's implicit goal is to beat them.
- Historically, the **US market has the best long-run and 10-year track record** among major global markets, with a high (~78–96%+) probability of gains the longer the holding period, especially past 10–15 years.
- Markets move in **bull/bear cycles** made of repeated wave patterns; expect regular **pullbacks (1–9% drops, ~3x/year)** and **corrections (10–19% drops, ~1x/year)** even within a bull market — these are normal and are buying opportunities, not reasons to panic.

Part II — Trend Following, Strategy Execution & The Keys to Profitable Investing

Section 02 — Trend Following, Strategy Execution & The Three Steps to Successful Investing

Source videos

File	Actual content
20 — Trend Following Part 1 (20:09)	Moving-average trend-following theory + charting setup
21 — Trend Following Part 2 (36:36)	Live 20-year SPY back-test, trade-by-trade; parabolic drops; bear-market staged buying
22 — Strategy Execution (30:31)	200 MA rule, combined signal, three method summary, buy-and-hold vs trend-following back-test, timing studies, macro vs price action, 2020 case study
25 — Part 1 (27:34)	**Lesson 3: The Three Steps to Successful Investing — Part 1** (great businesses, intrinsic value, five criteria, reasons 1–3 a stock goes on sale)
26 — Part 2 (23:03)	**Lesson 3: The Three Steps to Successful Investing — Part 2** (reasons 4–5 a stock goes on sale, GOOGL/J&J case studies, when to sell)

PART A — TREND FOLLOWING (Videos 20 & 21)

A1. The Two Methods of Investing — Where Trend Following Fits

- **Method 1 (previous lesson): Buy and hold.**
- Buy consistently every couple of months regardless of the trend — through bull markets and bear markets.
- Hold through the ups and downs.
- Long-run average return: **about 10% per year**.
- **Method 2: Trend following.**
- We attempt to **time the market by following the trend**.
- Goal: identify uptrends (bull markets) and identify bear markets.
- Far more **active** than buy and hold.
- Procedure:
  1. Buy **only during the uptrend / bull market**.
  2. Once we identify that it is turning into a bear market, **sell all our shares**.
  3. Only once the bear market has ended and a new bull market starts, **get back in** to ride the new bull market.
  4. Repeat the process indefinitely.
- The tool that makes this possible: **moving averages**.

A2. What a Moving Average Is

- A moving average is a mathematical calculation: **the average of the stock price over a certain period of time**.

- **50-day MA** — software takes the last **50 days of the stock's closing price**, adds them together, divides by 50, repeats every day, and plots the line.
- **150-day MA** — software adds the last **150 days of closing price**, divides by 150, repeats every day, plots the line.
- **Colour convention used throughout the course (Adam's own convention — "because Adam says so"):**
- **50 MA = BLUE**
- **150 MA = GREEN**
- **200 MA = RED** (introduced later, in video 22)

A3. The Hulk-and-Smurf Mnemonic

- **Green (150) = the Incredible Hulk. Blue (50) = a Smurf.**
- Alternative framing: a **150 kg man** and a **50 kg man**.
- **When the Hulk sits on the Smurf → it goes DOWN.** So when the 150 green is *above* the 50 blue → **confirmed downtrend / bear market**.
- **When the Smurf sits on the Hulk → the Hulk carries him UP.** So when the 50 blue is *above* the 150 green → **uptrend / bull market**.

A4. RULE — Identifying a Change in Trend (50/150 Crossover Method)

State definitions

- **Downtrend / bear market:** 150 green is above 50 blue, and **both moving averages are sloping down**.
- **Uptrend / bull market:** 50 blue is above 150 green, and **both moving averages are sloping up**.
- As long as blue stays below green, the downtrend continues.

RULE — Downtrend reverses into an Uptrend (TWO conditions, both must be met)

1. **Both moving averages must change their slope.** From sloping down, both lines must **flatten or slope upwards**.
   - *If either line is still sloping down, it is still a downtrend. It is NOT a change in trend.*
   - Both lines have to be flat or sloping up for the change in trend to be confirmed.
2. **The 50 MA (blue) must cross ABOVE the 150 MA (green).** There must be a crossover.
   - When both are satisfied → **uptrend confirmation / bull-market confirmation → BUY to get back into the market.**

RULE — Uptrend reverses into a Downtrend (TWO conditions, both must be met)

1. **Both moving averages must change their slope.** From sloping up, both must **flatten or slope downwards**.
   - *If one moving average is still sloping up, it is NOT a change in trend.*
2. **The 50 MA (blue) must cross BELOW the 150 MA (green).**
   - When both are satisfied → confirmed bear market → **SELL**.

The single most important refinement

- **The slope is the most important thing — you cannot just look at the 50/150 crossover.**
- The classic mistake: people see the 50 above the 150 and buy immediately — and it was a **fake-out** / false reversal.
- **Watch the GREEN line.** You have to wait for the green line to flatten before confirming the uptrend, and wait for the green line to flatten/slope down before confirming the downtrend.

- **If you are not sure whether the green has flattened, wait a few more days.** (Adam does exactly this repeatedly in the back-test.)

A5. RULE — Timeframe Restriction

- **The 50/150 crossover change-in-trend method ONLY works on DAILY candles.**
- **Do NOT use it on weekly or monthly candles** for the change-in-trend method. (Repeated for emphasis in both video 20 and video 22.)

A6. Adding Shares Inside an Uptrend — "Buy the Dip on the Uptrend"

- Price moves in **wave up, wave down, wave up, wave down** = **impulse, correction, impulse, correction** ("breathe out, breathe in").
- **We buy on the wave down.** When price waves down and hits one of the moving averages, we can add more shares.
- Valid as long as the blue line remains above the green line (still an uptrend).
- **Write this down: "Buy the dip on the uptrend."**
- **Dip = correction = retracement** — all the same thing.
- The moving averages **act like support levels** — that's where you know you can add more shares.
- Frequency of adding is up to you: **once in three months, once in six months, once a year.**
- You can't tell how long an uptrend will last — it could last many months or many years. Keep adding along the way.

A7. Worked Example — 2006 to 2011 (Financial Crisis Window)

- If you bought the Dow Jones or the S&P at the start of this ~five-year window and simply held, you would end up with **a very small return**.
- Using trend following you exit before the bear market and buy back to catch the bull market. Three trades:
- Trade 1: **+22%**
- Trade 2: **+31%** (after avoiding the entire bear market)
- Trade 3: **+19%**
- **Total: 72% over the five-year period** — better than buying and holding across the same window.

A8. Charting Setup — Tools and Exact Steps

- **Any charting software will work** — all charting software lets you add moving averages. Named options: **TradingView**, or **the charting software from your broker**, whichever broker you use.
- Adam demonstrates on **Think or Swim (TD Ameritrade)** charting software.
- Instrument used: **SPY** (the S&P 500 ETF).

Steps to install the moving averages (Think or Swim)

1. Set candles to **daily**.
2. To back-test, set the chart range to **20 years** — the software automatically switches to monthly candles, so go to **Time Frame** and **change it back to daily candles** (20 years, daily candles).
3. Click the **"magic potion" (studies) icon** → type **"moving averages"**.

4. Two kinds exist: **Simple Moving Average** and **Exponential Moving Average**. **For the 50/150 method, use the SIMPLE moving average.**
5. Double-click Simple Moving Average → it defaults to a **9-day** MA.
6. Click **Edit**, change **9 → 50**, set colour **blue**, width **2 or 3** (doesn't matter).
7. Click again to add a second study → **Edit** → change to **150**, colour **green**, width **2**.
8. Click OK. Optionally **Save as "50 150"** so the preset is reusable.
9. (Later: add the **200 MA in red**.)

A9. The 20-Year SPY Back-Test — Trade-by-Trade Walkthrough (Video 21)

Back-test protocol

- Chart starts at **2001**. Move forward day by day.
- **Pretend you don't know the future** — make decisions only on what is visible at that moment.
- **Execution rule: buy or sell on the OPENING PRICE of the day AFTER the crossover is confirmed.**
- Chart legend used to read the price: **O = opening price, H = high, L = low, C = closing price.** We trade the **O**.
- Prices recorded in an **Excel spreadsheet** to calculate profit/loss.

Signals and fake-outs encountered

- Starting state 2001: **downtrend** (blue below green) → do not buy.
- **First crossover: NOT a buy** — the blue had sloped up but the **green was still sloping down**. It was a fake-out; the 50 fell back below the 150 and the downtrend continued.
- **Second crossover: confirmed** — both sloping up + crossover → **BUY**.

Complete trade log (SPY, opening prices)

#	Action	Price	Note
1	BUY	94.89	Day after confirmed crossover
2	SELL	109.62	Crossover came first, waited for green to flatten
3	BUY	113.12	Prior downtrend was very short-lived
4	SELL	125.20	Blue sloping down, green flat
5	BUY	132.12	Downtrend again short-lived, reversed back up
6	SELL	147.93	Confirmed by price action too: higher highs turned into **lower highs**
7	BUY	93.67	After the crash; waited for green to flatten
8	SELL	106.02	Blue sloping down, green flat
9	BUY	118.10	
10	SELL	124.42	Sold into a **gap down** after green flattened
11	BUY	128.73	Waited extra days for green to flatten
12	SELL	187.49	**Parabolic drop** — sold only "to follow the rules of the game"
13	BUY	210.12	After a **double dip**
14	SELL	266.52	Took several extra days of waiting to confirm green had flattened

#	Action	Price	Note
15	BUY	284.70	
16	SELL	249.52	**COVID / pandemic crash, February 2020 — a parabolic drop**
17	BUY	320.13	Blue sloping up, green flattened
—	Mark-to-market	**390.25**	Last closing price at time of recording

Back-test results and observations

- Over the 20 years there were **nine buys and nine sells** — nine round-trip transactions.
- **Most trades made money.** There was **one loss — caused by the February 2020 plunge** — but it was made back very quickly on the next trade.
- **Overall: yes, you make money.** As with any method there is no 100% win rate — you win some and lose some, but you win more than you lose.
- Biggest win of the method: **it got us out before the financial-crisis bear market — a 50% plunge — and kept us out of the market for a year**, then put us back in near the start of the new bull market.

Discipline principles stated during the back-test

- **We are not predicting the future. No one can predict the market.** "I'm not a fortune teller. All I'm doing is following the trend."
- **River analogy:** if the river flows north, I swim north; if it flows south, I get off the boat. Follow the market — don't outsmart the market.
- **Ignore the news and the newspapers — everything is noise.** Look only at price action and the trend.
- We can identify **when** a trend changes, but **not how long the new trend will last**. Some downtrends are very short-lived and reverse straight back up.

A10. The Limitation — Parabolic Drops

- **Definition: a parabolic drop is when the market drops 90 degrees — straight down.**
- Normal trend change is roughly **45 degrees up / 45 degrees down**. A parabolic drop is a 90-degree collapse.
- **This does happen in the market once in a while.** In the 20-year back-test it happened twice — including the **February 2020 COVID crash**.
- **The failure mode:** when the market drops that fast, **by the time the moving averages cross and you sell, you are selling far too low — after the crash has already happened.** There is no point selling at that stage.
- **RULE (Adam's real-life behaviour): when you see a 90-degree drop, DO NOT SELL. Just hold it.**
- He states explicitly that in reality he did **not** sell in February 2020; he ignored the moving averages completely.
- In the back-test he sold anyway, only to follow the mechanical rules of the exercise.
- **Rubber-ball theory:** the harder you throw the ball down, the faster it bounces back up. If you hold great companies/good ETFs, they bounce back just as fast.
- **The problem this creates:** where do you buy more shares to take advantage of the discount? Waiting for the 50 to cross back above the 150 means buying **only after the big recovery — too high**.

A11. RULE — Staged Buying Into a Crash (Bear-Market Ladder)

The historical facts underpinning the rule

- Adam has studied the markets for **28 years** and lived through **six bear markets**; the statistics are drawn from **100 years** of market history.
- **A bear market is a drop of more than 20% from the high.** Anything more than 20% is a bear market.
- **The average bear market drops 36%.**
- **The worst drop in history was 55%** (the financial crisis).

The staged-entry rule (using a target of 100 shares of the ETF as the illustration)

- **At −25% from the top:** start adding — buy **25 shares (a quarter position)**. Do *not* deploy the full 100 at this level.
- **At −35%/−36% (the historical average):** add another **quarter position** → now half the planned position is filled.
- **At another −10% down (i.e. −45%):** add the **third quarter position**.
- **At −55% (the biggest drop the market has ever seen):** you know you are definitely at the bottom → add the **last quarter position** — "all in maybe."
- **Rationale:** you can never know where the exact bottom is — the market can reverse at −25%, at −35%, anywhere. Buying in stages means that if it reverses at −35% you still captured at least half your intended position. Once the moving average crosses back and the uptrend is confirmed, **add more shares later on**.

A12. Supplementary Resources Referenced

- YouTube: search **"Adam Khoo bull market"**.
- **"Has the stock market bottomed?" — 3 April 2020** — the video where he called the bottom (or at least "near the bottom").
- **"The bull market is back" — 10 April 2020** — where he called the new bull market and told viewers to get back into the markets.
- These videos show the additional parameters and indicators he uses.

PART B — STRATEGY EXECUTION (Video 22)

B1. RULE — The 200-Day Simple Moving Average (Red Line)

- Install the **200 simple moving average** into the chart; Adam's preference is to colour it **red**.
- **The 200 works ALONE** — it is a single moving average, so the rules are stricter.

RULE — Uptrend by the 200 MA (two criteria)

1. **The price of the stock/ETF must be ABOVE the 200 moving average**, and
2. **The 200 must be SLOPING UP.**
   - **It cannot be flat.** It must slope up.
   - *Contrast with 50/150:* for the 50/150 the lines may be **flat or sloping up** for an uptrend, and **flat or sloping down** for a downtrend. Because the 200 is only one moving average, **we have to be more strict — only sloping up counts as an uptrend.**

RULE — Downtrend by the 200 MA (two criteria)

1. **The price must cross BELOW the 200 moving average**, and
2. **The 200 must slope downwards.**

Interpretation notes

- Price dipping below the 200 while the 200 is **still sloping up** is **not** a downtrend.
- Price rising above the 200 while the 200 is **still sloping down** is **not** an uptrend.

B2. RULE — Which Signal Do I Use? Combine All Three and Take Whichever Comes First

- **Answer: combine all three moving averages (50, 150, 200) to determine the trend.**
- **RULE: follow the signal that comes FIRST.** Whichever fires earlier is the change-in-trend signal.
- Worked illustration:
- On an uptrend turning down, the 50/150 exit fired earlier than the 200 exit (the 200 exit only came once the red line was sloping down with price below it) → take the **50/150** signal.
- On a downtrend turning up, the 50/150 entry fired earlier than the 200 entry → take the **50/150** signal.
- **The 50/150 does not always come first. There are times the 200 comes first.** Always take the earlier signal.
- **Re-emphasised: this trend-reversal method only works on daily candles.**

B3. RULE — Complete ETF Trend-Following Strategy (Summary)

BUY when the index/ETF is making a dip on an uptrend. Uptrend is defined as EITHER:
- The **50 MA is above the 150 MA and both moving averages are flat or sloping up** (they must not be sloping down), **OR**
- The **price of the ETF is above the 200 MA and the 200 is sloping up.**
- Either condition satisfies the uptrend condition.
- As long as it is an uptrend, **buy when the price retraces or dips to the moving average.**

SELL / exit the ETF when the index reverses into a downtrend (a potential bear market). Downtrend is EITHER:
- The **50 MA crosses below the 150 MA and both moving averages have flattened or are sloping down**, **OR**
- The **ETF price crosses below the 200 MA and the 200 is sloping down.**
- **Whichever comes first.**

EXCEPTION TO THE SELL RULE: if the price drops **90 degrees (parabolic drop)**, as in the 2020 pandemic — **do not sell.** By then it is too late; you would be selling near the bottom, which makes no sense. Hold it. Good ETFs and good companies bounce back like a rubber ball — the faster it drops, the faster it bounces back.
- On such a crash, rather than waiting for the 50 to get back above the 150 (by which time price has risen a lot), **add at −25%, add more at −35%, more at −45%, and at −55% that's the bottom — all in.**

B4. The Three Methods Compared

Method 1 — Dollar-Cost Averaging (passive)

- The simplest method. **No market timing, no need to look at the trend.**
- **Invest a fixed dollar amount at regular intervals**, regardless of uptrend or downtrend. The only requirement is **consistency**.
- Effect: on average you **buy more shares at a low price and fewer at a high price**, averaging your cost to a reasonable level.

- You always make money over time.
- Can be done on a portfolio of **equity ETFs** or a portfolio of **multi-asset ETFs**.
- Passive approach — buy and hold through the ups and downs.
- *"If you're colourblind, just do this."*

Method 2 — Trend Following

- Requires seeing the colours of the moving averages.
- Identify uptrend vs downtrend using the **50, 150 and 200 moving averages**.
- Buy when the index/ETF starts an uptrend; **add more shares on every dip on the uptrend** (wave down toward a moving average).
- You can't pick the exact bottom, but near the moving averages you know you're near the bottom.
- **Yes, you do get out of the market** — when the uptrend reverses into a downtrend/bear market.
- There is no telling how long bear markets last — one month, six months, one year.

Method 3 — The Hybrid (the method Adam actually uses)

- A hybrid of methods 1 and 2.
- **Dollar-cost average — buy shares with a fixed amount of money every six months — but do NOT buy blindly.**
- **Look at the trend of the market.** Only buy when:
- the market is making a **dip on the uptrend**, **or**
- the market is at the **support level of a consolidation**.
- **If the market is in a downtrend, avoid dollar-cost averaging.**

B5. BACK-TEST — Buy-and-Hold vs Trend Following over 26 Years

- Comparison run over the **last 26 years**:
- **Buy and hold** (dollar-cost averaging consistently regardless of trend, holding through the ups and downs): **total return 644%**
- **Trend following** (buy on the uptrend, exit at the start of the downtrend, buy back in): **total return 696%**
- **Conclusion: in the long run the two are about the same.** Trend following gives slightly better results, but passive buying and holding would have done pretty well as well.
- **Whatever strategy you use, the most important thing is to GET STARTED as early as possible.** The worst thing a person can do is procrastinate and never start.

B6. Cash Rules and the Risk-Over-Time Inversion

- **Only invest with cash you don't need in the short term.** If you need the cash for emergencies or expenses in the next one to two years, **don't invest it** — keep it in cash or time deposits.
- If you don't need the cash for the next one, two, three, four years and you want to compound your wealth, **start as early as possible**.
- **Short term:** holding cash *is* safer than stocks. Example: in the **2022 bear market the stock market dropped up to 20% in that one year**. Cash didn't drop nominally — though cash still loses value to **inflation**, so there is a drop in real terms.
- **Long term (5 to 10 years) the relationship inverts:**

- **The longer you hold cash, the more risky it becomes** — cash hardly grows, you only get very minute interest from time deposits, it doesn't build wealth, and in the long run you won't have enough for retirement or to beat inflation.
- **The longer you hold stocks, the safer it becomes** — despite going through repeated bear markets and ups and downs, over time you are much better off than someone who stayed in cash for 5–10 years.
- **On the fear "what if I buy and it crashes next week?":**
- First, **you can never predict when the next crash will come.**
- Second, **even if you are unlucky enough to buy right at the top before a crash, if you hold on over the long run you are STILL better off than the person who stayed in cash the whole time.**

B7. STUDY — Bad Timing Still Beats Procrastination (2001–2020, $2,000/year into the S&P 500 ETF)

Five scenarios, each investing **$2,000 annually into the S&P 500 ETF from 2001 to 2020** (20 years):

Scenario	Behaviour	Ending value
1. Perfect timing ("like God")	Invests $2,000 at the exact **bottom of the market every year** (impossible in practice)	**$151,000**
2. Invest immediately	Puts in $2,000 on the **first trading day of every year** (2nd/3rd January)	**$135,000**
3. Dollar-cost averaging	Splits $2,000 into 12 months = **$167 per month**, every month	**$134,856**
4. Worst timing ("Charlie Brown")	Invests $2,000 at the **top of the market every year**	**$121,000**
5. Stayed in cash	Kept $2,000/yr in **time deposits** for 20 years, always waiting for a lower price	**$44,000**

- **Lesson: even bad timing beats procrastination.** The worst outcome by far belongs to the person who never started.
- Note how close scenario 3 (DCA, $134,856) is to scenario 2 (lump sum at the start of each year, $135,000).

B8. RULE — Ignore Macroeconomics; The Stock Market Is Not The Economy

- **Adam does not use macroeconomic factors — interest rates, GDP, employment data — in making investment decisions at all.** He reads them "for entertainment purposes."
- **Looking at macro data will not only fail to help — it will make your decisions WORSE.** It will adversely affect your returns: you'll buy at the worst time and sell at the worst time.
- **The stock market is not the economy. They are two totally separate animals.**
- **The stock market is a LEADING INDICATOR of the economy — it leads the economy by about SIX TO NINE MONTHS.**
- When the stock market turns from uptrend to downtrend, **six to nine months later** the economy will show a recession.
- When the stock market turns from downtrend to uptrend, **six to nine months later** the economy will show a recovery in GDP.
- **If you focus on economic data you are six to nine months late and you will always make late decisions.**
- **Focus only on the price action / the trend of the market.**

The economy-vs-market cycle chart (red line = stock price, green line = GDP)

- **At the market top:** the market reverses into a downtrend (50 crossing below the 150 and sloping down = the trend-follower's exit signal) — but **the economy is still showing growth, in full recovery**. Investors reading the news see a strong economy and BUY right at the top, then wonder why it goes down.
- **When the recession finally appears:** those investors sell — and by then **they are selling near the bottom of the stock market cycle**. This is why most retail investors **buy high and sell low**.
- **At the market bottom:** a new uptrend forms (50 crossing above the 150 sloping up, or price above the 200 sloping up). Smart investors buy here. Retail investors won't — "it's a recession, GDP is still dropping."
- **They wait for the green line (early recovery) and end up buying near the top again.** It goes up a bit, then collapses.
- **Instruction: ignore the economy, ignore GDP, ignore inflation, ignore interest rates, ignore employment. All you have to look at is the price action.**

B9. CASE STUDY — March/April 2020: Buying While the News Screamed Depression

The headlines that scared retail investors out (late March / early April 2020)

- Jobless claims soar to surpass **3 million**, a record high.
- **30 March:** coronavirus job losses could total **47 million**; the unemployment rate may hit **32%**.
- **6 April:** Janet Yellen (Federal Reserve Chairperson) says second-quarter GDP could **decline by 30%** and unemployment is already at **12 to 30%**.
- **9 April:** death toll reaches a third straight record, **799**; the IMF chief says the pandemic will unleash **the worst recession since the Great Depression**.
- **9 April:** the bull market started — **the Dow jumped 500 points, up 13% on the week.**

What actually happened

- **The market bottomed on 23 March 2020**, precisely when the media was most pessimistic and predicting the worst depression in history.
- Most people were selling like crazy; **many banks were recommending their clients sell everything** (Adam notes he has many banker friends).
- **Adam started buying on 25 March 2020 — two days after the bottom** — and was buying shares every single day on 4, 6 and 9 April, along with his entire investment community.

The Ultimate Investors Playbook (UIP)

- A subscription service where Adam sends a **notification within a couple of minutes** the moment he buys or sells something.
- At the time it had **1,200 members**; a few thousand more have joined since.
- Actual notifications from the crash:
- **24 March:** heads-up — looking to add **MasterCard** to the portfolio as it's at the **150 MA support**.
- **25 March:** bought **MasterCard at $236.40**, with the intention of adding more if it dropped to **$210**.
- **30 March:** added **Facebook (now Meta Platforms) at $159.43**.
- **The buying rationale (important):** the decision to buy is **not** a prediction that it's the bottom or that it will go up. **No one can predict the bottom.** The criteria are simply: it's a **great company**, it is **undervalued**, and it's **at a support level** → just buy, and you will do very well.

The results

- By the end of 2020, when most people made no money:
- **Portfolio 1 grew 44%**
- **Portfolio 2 grew 41.37%**
- **Actual dollar gain: $791,000 in profits** during that recession/bear-market year.
- **These returns required more than index ETFs** — he picked individual companies that were **super high quality and very, very undervalued**. That is the subject of the next stage of the course.

PART C — LESSON 3: THE THREE STEPS TO SUCCESSFUL INVESTING (Videos 25 & 26)

> *Content identification:* Video 25 ("Part 1") and video 26 ("Part 2") are the two halves of **Lesson 3 — The Three Steps to Successful Investing**. Video 25 opens with "welcome to lesson 3, the three steps to successful investing" and video 26 closes with "that concludes lesson number three... in the next lesson we're gonna dive deeper into analyzing businesses... lesson four, fundamental analysis."

C1. Framing

- The previous lesson taught the simplest way to invest: buy an ETF tracking an index, owning the 30 or 500 biggest companies. **But that only gets you market returns.**
- **The purpose of this course is to BEAT the market** — to get much higher returns than the index provides — by picking individual stocks that outperform.
- Lesson 3 is deliberately shorter: it delivers the broad concept; the details come in the subsequent lessons.

C2. STEP 1 — Identify Great Businesses

- When you buy a share of stock **you are not buying a lottery ticket — you are buying a piece of a business. You are owning a business.**
- **Only buy great businesses — best of breed.** Not mediocre businesses, not lousy businesses.
- **In the US market there are over 10,000 businesses** and you can buy any one you want — so why waste time with low-quality businesses?
- **Adam focuses only on the top 1% of great businesses and ignores the other 99%.**

C3. STEP 2 — Buy Only at the Right Price (Below Intrinsic Value)

- **A great business can be a lousy investment if you pay too high a price** — if you buy it when it's overvalued.
- **A great business becomes a great investment only when you buy it for less than what it is worth.**
- **Property analogy:**
- The bank values an apartment at **$1 million**. You fall in love with it and pay **$1.5 million** — **50% above valuation**. Bad investment: it will be very hard to profit, and any eventual return will be low because you overpaid.
- Alternatively the seller is desperate (a divorce, or he lost his business and needs cash) and sells at **$700,000** — **30% below intrinsic value**. Great investment.

- **In real estate, a great property with a desperate motivated seller is really hard to find.** In the stock market it happens all the time.
- **Frequency:** not every day, but **at least once a year, or once every few years**, a great business goes on sale at a **20%, 30% or 50% discount** because there are desperate, panicking sellers reacting to bad news.
- "That's how I got really rich, and that's how people like Warren Buffett got really rich."
- **Ideally the price has waved down to a SUPPORT LEVEL.**
- **Support level** = a technical-analysis term (taught in **lesson 6**): a price level where price drops to that level and **there is a high potential for it to reverse and bounce back up.**

C4. STEP 3 — Sell (Or Rather, Usually, Don't)

- **If you have done steps 1 and 2 very well, you shouldn't ever have to sell** — until you need the cash to pay for something.
- Reasoning:
- A great business keeps growing and compounding your wealth.
- If you sell it, you must reinvest the cash into another great business — **which is not easy, because there aren't that many great businesses around.**
- "Once you latch on to a great business — like having the perfect wife — you're going to hold on for the rest of your life."

Holding-period return evidence (as presented)

Company	1 year	5 years	10 years	20 years
Apple	~16% to 35% (depending where you sell)	**191%**	**792%**	**22,830%**
MasterCard	~20%	**68%**	**563%** (despite COVID crash, 2022 bear market, trade war)	**6,000%** (despite recessions, crashes, the 08/09 great financial crisis)
NVIDIA (leader in the AI revolution, one of his biggest holdings)	**144%**	**2,200%**	**29,615%** (from 2015)	—
American Airlines (a lousy business)	**+33%**	**−33%**	**−60%**	—

- **The big money is in staying invested in a great business and letting it compound massively. That's how you make millions and build generational wealth.**
- **You do NOT build generational wealth by taking profit at 20%, 30% or 50%. You want a 10x, a 15x, a 20x.**
- **With American Airlines, the longer you hold, the worse it gets.** Only great businesses compound your wealth over time; mediocre and lousy businesses stagnate or reduce your wealth.

RULE — Whole industries to avoid

- **Write this down: ALL airlines are lousy businesses.** It is **not the fault of the management — it is the nature of the industry.**
- **The property development industry** is likewise not a great business by the nature of the industry.
- Certain industries are simply not very profitable.

The legitimate reasons to sell (broad concept; full detail in the exit-rules lesson)

1. **The fundamentals of the business have deteriorated.**
   - This does **NOT** mean a short-term drop in profits or sales — that is cyclical and very normal.
   - It means **the long-term business model is disrupted, or the company loses its competitive advantage.**
   - Examples of businesses that were once great and no longer are: **Intel** and **General Electric** — over time they lost their competitive advantage and their business got disrupted.
   - The moment you see that, get out and reinvest into a great business.
2. **You found an even better business** → sell and swap into the greater business.
3. **The share price has gone crazy / way overvalued / way too expensive** → nothing wrong with selling part or all, holding cash and waiting for it to get cheap again, or reinvesting into a more attractively priced great business.

C5. What Makes a Business "Great"

- **Definition:** a great business is one where you are **confident that over the next three, five, ten, twenty years the business will consistently increase in value.**
- The **green line** on the charts = **the value of the business = the intrinsic value** of the business / of its shares.
- **Mechanism:** as the business increases in value over time and is worth more and more, people are willing to pay more for it — and hence the share price rises over time.
- **A business is a money-making machine. The more money it makes, the more it is worth.**
- **What makes a business more valuable: consistently making more money** — generating **increasing sales revenue, increasing net income (also known as earnings), and increasing cash flow from operations**. When these rise, the intrinsic value keeps rising.

THE FIVE CRITERIA of a great business (taught in full in **lesson 7**)

1. **Track record** — at least the **last five to ten years** showing the ability to increase **revenue, net income and cash flow from operations consistently.**
   - *But history does not predict the future — this alone is not good enough.*
2. **A sustainable competitive advantage that protects it from competition — an ECONOMIC MOAT.**
3. **High return on equity and high return on capital** — i.e. the management is extremely efficient.
4. **A strong balance sheet** — **low, conservative debt structure and high levels of cash.**
5. **Strong growth catalysts and drivers.**

- **Fewer than 1% of businesses in the market meet all five criteria.** When a business does meet all five it is highly likely to be a great business that will keep growing in value over time.
- **Fails the criteria:** American Airlines.
- **Meets the criteria:** Microsoft, NVIDIA, Apple, MasterCard.

C6. Why Price Diverges From Value (Intrinsic Value vs Share Price)

- The intrinsic value line rises steadily for a great business (examples shown: **Microsoft**, **Visa**) — but **the share price does not go up in a straight line and does not always reflect the value of the business.**
- **Why:** the share price is determined by **demand and supply in an auction-driven market**. In the short term the market price **is not driven by logic — it is driven by news and emotions** (and manipulation).
- Result: the price behaves like **a pendulum swinging from overvalued to undervalued and back**, repeating endlessly.
- Bad news / pessimism → price drops **below** intrinsic value → **undervalued**.

- Good news / euphoria / excitement → price runs **above** intrinsic value → **overvalued**.
- **When the share price gets very far from the intrinsic value, we have MISPRICING — and taking advantage of the mispricing is how you get really rich in the markets.**

The man-and-dog analogy

- A man walks in a straight line from his house to his office. He has a dog on a **very long leash**.
- The dog runs all over the park — far to the right, far to the left — because it's excited by a cat or a tree. **You have no idea where the dog is going.**
- That's the market in the short term: **very random; you don't know if it's up or down tomorrow.**
- **But eventually the dog meets back up with the man when he reaches his office — because the dog is on the leash.** Price eventually converges to value.

C7. RULE — How to Buy: Tranching at Support Levels

- **Job: patiently wait for the stock price to go BELOW the intrinsic value (the green line), then start to add — ideally at support levels** (support-level drawing is taught in **lesson 6**, technical analysis; valuation methods in **lesson 5**, and different businesses are valued a bit differently).
- **You do not have to buy at the exact bottom of every correction or wave down.** No one can predict the exact bottom. **As long as you buy below the intrinsic value, you've done your job.**
- **RULE: never buy at one go. Always buy in three to four — or even five — tranches.**
- Once the price goes below intrinsic value by a **margin of safety of about 10–15%**, start adding some shares **at the support level**.
- If it goes even lower — say **20% below intrinsic value** — add more.
- If it goes lower still, add even more.
- You **slowly average into the position**. It's like dollar-cost averaging, **except you are adding at support levels**.
- Your average cost ends up somewhere in the middle — **not at the bottom, and it doesn't have to be.**
- **Why it works:** as long as you buy below intrinsic value, over time the price gets back above intrinsic value — **plus the intrinsic value itself is increasing.** That's how you see growth in the value of your shares.
- **Warren Buffett (as quoted in the lesson):** the time to be greedy is when others are fearful, and the time to be fearful is when others are greedy.

C8. THE FIVE REASONS A GREAT BUSINESS SELLS BELOW INTRINSIC VALUE

> Reasons 1–3 have **nothing to do with the company itself** — the stock is just collateral damage in the short term. Reasons 4–5 are company-specific. Adam explicitly says he does **not care why** sellers are desperate (panic, emotions, or being force-sold by a broker because they were on margin) — the job is simply to take advantage of the discount.

Reason 1 — Overall stock market correction

- **On average, once a year the whole market drops by at least 10%.**
- There is **always a trigger** but you can **never predict** it: a war, a disaster, geopolitical tensions, tariffs, weak economic numbers — it doesn't matter.
- When the market drops, **most companies go down together — the good, the bad and the ugly.**

- **That is a great time to buy the good companies**, because the market drop has nothing to do with the company; it's overall market sentiment.

Reason 2 — Bear market

- **Definitions:**
- **Correction:** the market drops **more than 10% but less than 20%** — happens more frequently.
- **Bear market:** the market drops **more than 20% from the high** — usually due to a recession, an economic crisis or a financial crisis.
- **Frequency:** looking at the last 100 years, **on average once every six years** — but it is **not** a regular occurrence:
- **From 2009 to 2019 — ten years — there was no bear market at all.**
- **Then a bear market in 2020, and another bear market in 2022.**
- Sometimes they come close together, sometimes very far apart. We don't know.
- In a bear market **everyone panics and loses their mind; good companies and bad companies all fall 20, 30, 40%** — that's how you buy a great company at a discount when there's maximum fear.
- **"I love bear markets."** After every bear market his portfolio grows exponentially because he gets to add shares at huge discounts.

Reason 3 — The sector or industry is unloved / "out of rotation"

- There are **11 sectors** in the market (technology, financial, healthcare, and so on). Within each sector there are **industries** — e.g. within technology: semiconductor, hardware, software.
- **Sectors take turns being loved and unloved by investors.**
- When a sector is loved, money (especially institutional money) flows in — that sector is **"in rotation"** and **most or all of its stocks go up — the good, the bad and the ugly. A rising tide lifts all boats.**
- **The money has to come out of somewhere** — the sector it leaves is **"out of rotation" / unloved**, and **all its stocks drop**, good, bad and ugly.
- **It's almost impossible to predict exactly when a rotation will happen or reverse.** "My job is not to predict the rotation. My job is to take advantage of the rotation."

Worked example — Healthcare, 2023 and especially 2024
- The whole healthcare sector was out of rotation and unloved; money flowed out. Reasons given:
- **Politics:** usually in an **election year** money flows out of healthcare because of **uncertainty over new political regulations**.
- **Post-COVID surgery backlog:** COVID caused people to delay non-essential surgeries; afterwards they all went for them, **increasing insurance companies' claim costs → costs up, profits down**.
- Healthcare stocks got very, very cheap → **that's when he bought them**.
- Long-run case: **everyone needs healthcare; healthcare is a recession-proof sector; and the aging population is a long-term growth driver.** When people get excited about healthcare again and money flows back, that's how you get **20, 30, 40% returns**.

Worked example — Consumer Staples, 2024
- **Consumer staples** = companies that sell **low-growth essential goods** — e.g. **Coca-Cola, Pepsi, Procter & Gamble, Colgate**.
- They are very **defensive** stocks and **recession-proof**.
- **When the economy is doing very well, people don't want these boring companies, so they sell them** — that's why they tend to drop during good economic growth. And when they drop, they're cheap.
- **In 2024 he bought a lot of consumer staple stocks — for example Pepsi — and a lot of healthcare stocks.**

- Whenever a fear of recession appears (we can't know when), everyone gets scared, they buy healthcare and consumer staples again, and those stocks fly.

Reason 4 — The company is hit by short-term bad news

- **When you buy a great business that meets the five criteria, 99.9% of the time any bad news that hits the company is SHORT TERM.** They will sort it out, solve the issue, and the company will do very well again.

(a) Lawsuits. Great companies get sued all the time — by **consumers**, by **investors** (class-action lawsuits after a price drop), and by **the government** (e.g. the **Department of Justice** suing them for being a monopoly). **Every time a great company gets sued and the share price drops on the bad news, Adam buys** — it's always temporary and rebounds very fast.
- **2024 — three great companies sued by the US DOJ: Google, Apple and Visa.** Google actually lost its lawsuit. Each caused a temporary drop that made the stock undervalued — **and by the end of the year all three were back at all-time highs.**
- **March 2024:** the Justice Department sues **Apple**, alleging it illegally monopolised the smartphone market → **Apple shares dropped about 10%** → back to an all-time high by the end of the year.
- **5 August: Google loses its landmark US antitrust case over search dominance** → share price plunged → back to all-time highs by the end of the year / the following year.
- **24 September:** the US DOJ **sues Visa over the debit card market monopoly** → Visa's price dropped and was undervalued very briefly → all-time highs again.

(b) The Buffett / American Express archetype (1963).
- American Express was **not yet listed on the stock market** — it traded **over the counter as a pink sheet**.
- Buffett wanted the company: it was a great business, but **too expensive at about $60 per share**, so **he waited patiently for bad news.**
- In **1963 the "Salad Oil Scandal"** hit: American Express was **sued for negligence** and lost roughly **$30–50 million** because of the lawsuit, causing a short-term drop in profits.
- Buffett's insight: the short-term loss didn't matter because **the brand was still very strong** — at the time, credit cards and travel both meant American Express; the advertisement said "don't leave home without it."
- **The scandal cut the share price by about 50%, from $60 to $30, and Buffett pounced and bought shares.**
- Once the scandal was resolved the share price went back to **$60 and a lot higher**. He bought in **1964 and has never sold** — **he has made over US$20 billion on that one great business.**

(c) Earnings misses. A great company will have good years and bad years, good quarters and bad quarters. Sometimes when it reports earnings, **earnings or revenue come in lower than expected**. People panic and sell, the **share price drops 10, 20, 30%** — a **great time to buy**, because it will rebound very strongly.

(d) Cyclical slowdown. A business can be in a long-term growth industry yet still have a temporary slowdown.
- **Digital advertising** is a long-term growth industry — five or ten years from now more companies will be spending on online advertising. **The biggest players are Google, Amazon and Meta.** (**Meta owns Facebook, Instagram and WhatsApp.**)
- **In 2022 there was a short-term cyclical slowdown in advertising spend.** Google's and Meta's revenue slowed, and **the share prices dropped roughly 15–20%.** Adam **bought a lot more in 2022**, and it has rebounded significantly since as advertising spend re-accelerated.
- **Cybersecurity** is the same: in the long run more and more companies will spend on cybersecurity systems. **His favourite cybersecurity companies are Palo Alto and Fortinet.** In **2022 and early 2023 there was a short-term slowdown in corporate cybersecurity spending**, and **Palo Alto and Fortinet dropped 10%, 20%, 30% — way below intrinsic value.** He **added a lot of shares** and it has produced very nice gains.

(e) Product recalls. Usually affects auto companies — **Adam does not buy auto companies** (reasons explained later in the course).

(f) Diseases. Years ago, **mad cow disease** meant no one dared eat beef. **McDonald's**, which sells primarily beef, saw profits fall and **the share price drop 30%.** He bought it undervalued knowing it was a short-term problem — "the cow won't be mad forever… one day the cow will be happy again, people eat beef again." **McDonald's went back to all-time highs.**

Reason 5 — Great companies making stupid mistakes

- Sometimes even great companies make **stupid mistakes that hit their short-term profits**, while their **competitive advantage, brand, economies of scale and economic moat remain intact.**

Example — American Express, 1963 (as a self-inflicted mistake). They made the mistake of **certifying salad oil for a commodities trader who had falsified the records**, and were **sued for negligence**. The **share price dropped 50%.** Buffett recognised the brand was still very strong — people used AmEx credit cards and travellers' cheques every day — **bought shares at half price and made billions.**

Example — Nike. Nike is a great brand with the **biggest marketing budget among all the athletic footwear companies.** But in **2019 they hired a new CEO who made very bad management decisions:**
- He **cut out all the wholesalers**, wanting to sell **Nike direct to consumers** — a bad mistake.
- He **killed a lot of the innovation**: instead of having a **football department and a running department** that built the most innovative shoes, he reorganised Nike into only **men's, women's and children's departments.**
- Result: **innovation fell, they lost shelf space**, and **for the first time Nike's revenue growth decelerated and profit decelerated. The share price dropped 40%.**
- **That was when Adam accumulated shares of Nike** — the brand is still very strong, there's a long runway, they **fired the old CEO** and brought in a CEO to repair everything, and it's on the path of recovery.
- **Caveat stated at the time of recording: Nike shares had NOT yet rebounded to their prior glory — it is in the process of rebounding.**

C9. CASE STUDY — Alphabet / Google (GOOGL), two-year chart

- **Share classes:** Alphabet has two classes you can buy, **GOOGL and GOOG**. **There is no difference — you can buy either one.** The only difference: **the one with the L has voting rights.**
- Google is a great business: **revenue and profits keep growing year after year, so intrinsic value keeps rising** — but the share price "goes up and down like a crazy yo-yo" between overvalued and undervalued because of the news flow:

Date	News event	Effect
February 2023	Their new **AI chatbot made a mistake**	People panicked and sold; shares became very undervalued, then recovered once people realised everyone still uses Google
24 October 2023	Earnings not as good as expected — **cloud business revenue missed expectations, profit missed expectations**	Share price dropped, then went back to new highs
March 2024	**Not company-specific** — the entire market dropped on **fears of inflation coming back**; the **Dow Jones dropped 475 points** and the **S&P 500 suffered its worst day since January**	Google fell with everything ("the good, the bad and the ugly"), became really undervalued, then back to all-time highs
September 2024	Google **lost its fight against a $2.7 billion EU antitrust fine**	Share price really undervalued — **Adam added more shares** — then rebounded

Date	News event	Effect
October 2024	**Google faces a US government attempt to break it up**	Panic, then recovery

- At the time of speaking, **Google is near intrinsic value — slightly below it**; be patient and you'll get a chance to buy soon.

C10. CASE STUDY — Johnson & Johnson

- **He no longer owns J&J.** He owned it **before they spun off their consumer health division**, when J&J was a very diversified healthcare company. After the spin-off they became a **pure pharma stock, which he does not invest in.**
- While he owned it: **revenue and profit grew year after year, so intrinsic value kept growing**, while the share price swung on litigation news.

Event	Effect
Failed to overturn a $2.12 billion baby powder verdict, planned a Supreme Court appeal (the baby powder was alleged to contain carcinogenic, cancer-causing material)	Price dropped a lot, got really undervalued — **Adam was adding more shares here** — then recovered
23 February — faces another **25,000 baby powder lawsuits**; assigns **$4 billion in litigation expenses and lawyer's fees**	Price dropped, then rose again
The Supreme Court rejects J&J's appeal against the **$2 billion penalty** in the baby powder case	Price dropped again, then rose again

C11. CRITICAL CAVEAT — Only Do This On Great Businesses

- **All of the buy-the-bad-news examples apply ONLY to great businesses / high-quality companies.**
- Great businesses: bad news + pessimism → share price drops → **great time to buy — it will rebound higher, because the company makes way more money over time.**
- **DO NOT do this on low-quality companies.**
- Low-quality = companies that **don't make profits**, or whose **profits are inconsistent**, or whose **revenue, profits and cash flow are not growing consistently.**
- **For those companies, when bad news hits and the share price drops, it may NEVER come back.**
- "Please remember that."

C12. THE FINAL STEP — Selling to Take Profits

- Full exit rules come in **lesson 9** (portfolio management and exit rules). The broad concept:
- **If you really did a good job of buying a great business that keeps growing year after year, there is no reason to sell it — unless you need the money to buy something** (a new car, a new house). Otherwise, **stay invested.**

Reason to sell 1 — Long-term fundamentals / competitive advantage have deteriorated

- Stocks he actually sold for this reason: **Estée Lauder** and **Disney**.
- They were great businesses but he doesn't think they're great anymore — **they lost their competitive advantage; the long-term moat is not as strong as it once was.**
- He **sold and reinvested into another company with a very, very strong moat.**

Reason to sell 2 — He found an even better business

- Nothing wrong with the business, but a **greater business exists → do a swap**: sell this, reinvest in that one.

Reason to sell 3 — The share price is overvalued (way above intrinsic value)

- **How much is "way above" depends on the KIND of stock:**

(i) Mid-to-lower-growth, very predictable companies
- Examples: **Pepsi, Procter & Gamble, McDonald's, Yum Brands, Lockheed Martin.**
- They are great companies and they grow, **but not super high growth — they grow at maybe 10% or 12%.**
- **RULE: once the share price gets about 20% or more above intrinsic value, he may start to sell some of it, or even all of it.**

Worked example — McDonald's
- **Intrinsic value at the time of recording: $273.**
- He **added shares at about $252** — a pretty good discount.
- The share price then made a **parabolic move** — going up 90 degrees. (**Parabolic = a 90-degree move; covered further in the technical analysis lesson, lesson 6.**)
- Price reached about **$300**, roughly **15–20% overvalued**. He felt it was **overextended — "like a rubber band, once it goes up too much it tends to snap back"** — and he needed cash to buy something else, so **he sold near the top (not at the top).**
- It has since come back down, **but he has NOT bought it back — because it is still not below his intrinsic value; it's still above the valuation.**
- **If it drops back to about $265 or $259, he'll add it back.**

(ii) High-growth companies
- Examples: **NVIDIA, Palantir, Amazon, Meta, Microsoft, Salesforce** — these **make up the bulk of his portfolio.**
- **In the past he bought them undervalued, sold them when overvalued — and that was one of the biggest mistakes he made.** After he sold, they'd dip a bit, but **before he could buy back they zoomed even higher and he missed out on a lot of returns.**
- **Apple and Microsoft:** he bought them **over 15, 16, 18 years ago** and jumped in and out many, many times, making "okay money." **Looking back, if he had simply bought and stayed invested through the ups and downs, he would have compounded his wealth probably FIVE TO TEN TIMES MORE.** "It's a lesson I learned as I became more experienced."
- **RULE TODAY: for great high-growth companies (NVIDIA, Palantir, Microsoft), even if it gets overvalued, he WON'T sell it.** Short term it may be overvalued and come down a bit, but long run the intrinsic value continues to compound — it makes more sense to just stay invested.
- **Exception: if it gets CRAZY overvalued — more than 100% above intrinsic value — he may sell PART of it.** Especially if it has taken **too big an allocation in the portfolio.**
- **At 30, 40, 50% overvalued for a high-growth company, he stays invested and does not sell.**

C13. Course Roadmap References Made in Lesson 3

Lesson	Content promised
Lesson 4	Fundamental analysis — reading a business inside out to judge great / mediocre / lousy
Lesson 5	How to value stocks of businesses (different businesses valued differently)
Lesson 6	Technical analysis — support levels, resistance, parabolic moves

Lesson	Content promised
Lesson 7	The five criteria of a great business, in detail
Lesson 9	Portfolio management and exit rules, in detail

KEY TAKEAWAYS

- **Trend following = two moving-average tests, and the SLOPE matters more than the crossover.** Uptrend confirmed when the 50 (blue) crosses above the 150 (green) **and** both lines have flattened or turned up; downtrend confirmed when the 50 crosses below the 150 **and** both have flattened or turned down. A crossover without a slope change is a fake-out. Daily candles only.
- **The 200 MA (red) is a stricter, standalone confirmation:** price above a **rising** 200 = uptrend; price below a **falling** 200 = downtrend. Combine all three MAs and **act on whichever signal comes first**.
- **The method's one failure mode is the 90-degree parabolic drop.** By the time the MAs cross you'd be selling at the bottom — so don't sell; hold, and instead **ladder in at −25%, −35%, −45% and −55%** (bear market = >20% drop; average bear = −36%; worst ever = −55%).
- **Over 26 years, buy-and-hold returned 644% vs trend following's 696% — essentially the same.** Getting started matters far more than which method you choose: the $2,000/year study showed perfect timing $151k, start-of-year $135k, DCA $134,856, worst timing $121k — and staying in cash only $44k. **Bad timing beats procrastination.**
- **Ignore all macroeconomics.** The stock market is not the economy — it **leads** the economy by six to nine months, so trading off GDP, inflation, rates or employment guarantees you buy near tops and sell near bottoms. Price action only. Adam's proof: buying MasterCard at $236.40 and Facebook at $159.43 in late March 2020 while headlines predicted 32% unemployment — producing 44% and 41.37% portfolio gains and **$791,000 in profit** in a bear-market year.
- **Three steps to successful investing: (1) identify great businesses — only the top 1%; (2) buy only below intrinsic value, ideally at a support level; (3) don't sell** unless fundamentals deteriorate, you find a better business, or it becomes extremely overvalued.
- **Five criteria for a great business:** 5–10 year track record of rising revenue/net income/operating cash flow; a sustainable competitive advantage (economic moat); high ROE and return on capital; a strong balance sheet with low debt and high cash; strong growth catalysts. **Fewer than 1% of companies qualify.**
- **Value compounds in a straight line; price yo-yos on news and emotion** (the man walking his dog on a long leash). **Mispricing is the source of outsized returns** — and great businesses go on sale for five reasons: market corrections (~10% drop once a year), bear markets (>20%, roughly once every six years), sector rotation, short-term company bad news, and self-inflicted management mistakes.
- **Never buy in one go — always tranche into 3–5 buys at support levels**, starting at roughly 10–15% below intrinsic value and adding more at ~20% below and lower.
- **Sell discipline scales with growth rate:** predictable 10–12% growers (Pepsi, P&G, McDonald's, Yum, Lockheed) get sold at ~20%+ overvaluation; high-growth compounders (NVIDIA, Palantir, Amazon, Meta, Microsoft, Salesforce) are held even when 30–50% overvalued, trimmed only above **100% overvalued** — because selling Apple and Microsoft in and out cost Adam an estimated 5–10x of compounded wealth.

Part III — Fundamental Analysis: Reading the Financial Statements

Lesson 4 — Fundamental Analysis (Financial Statements)

> Adam Khoo, Value Momentum Investing (Whale Investor). Videos 28–34.
> Adam's own framing: *"This is probably the heaviest, most technical lesson in the entire course."*
> Everything below is drawn only from the lesson transcripts.

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0. Introduction — Why Fundamental Analysis Matters

0.1 The core claim

- Fundamental analysis = **"the art of learning how to analyze a business"** — reading a business inside out to decide whether it is a **good**, **mediocre**, or **lousy** business.
- **70% of your investment decisions are based on the fundamentals of the business.**
- When you buy a stock, **you are not buying a ticker that goes up and down — you are buying a piece of an underlying business.** If the business does well, the stock will do well over time.

0.2 Why great businesses give you "no worries"

- If you own a great business and the price falls 20%, 30%, 40% on emotion or manipulation, you don't sweat it — you know it will eventually bounce back higher. In fact you **use the correction to add more shares**.
- If you own a lousy or average business and the price drops, you have **every reason to panic**, because you don't know if it will come back.

0.3 The real purpose: avoiding disasters

- Fundamental analysis is **not just about finding great companies — more importantly it is about identifying and avoiding lousy businesses.**
- **The main reason investors lose money is that they buy stocks of lousy businesses.** They get seduced by short-term speculation/hype, buy in, the bubble bursts, the price crashes and **never comes back** because the underlying business was never great.

0.4 Our job as investors

- **Not** to predict where the market goes next week / next month / next year — short-term price movement is impossible to predict with certainty.
- **Our job is to invest into a portfolio of great businesses that will thrive and grow under all economic conditions.**
- We cannot predict interest rates, inflation or GDP — **and we don't have to.** Rain or shine, recession or no recession, high or low inflation, high or low rates, a portfolio of great businesses will grow consistently and grow our wealth over time.

0.5 "Blue chip can get blue black" — the collapse case files

Company	Market / Sector	What happened
Noble Group	SGX blue chip, component of the **Straits Times Index**, one of the biggest companies in Singapore and Asia	Collapsed. Investors said "I was cheated, all my money's gone." **Many analysts had recommended the stock.**
Hyflux	Singapore water treatment company, blue chip	Collapsed. **Both share investors and bondholders lost most of their money.** (Full case study in §7.)
China Evergrande	One of the biggest real estate developers in China, blue chip	On the verge of collapse, **defaulting on its debt.**

- Adam: *"Blue chip can get blue black if you invest in the wrong ones."* Don't be deceived by the word "blue chip."
- Adam has **never invested in any of these companies**, because he studies the financial statements and does a fundamental deep-dive. **All of these companies failed his fundamental criteria.**

0.6 Great businesses can decay — but slowly

- Does a great business become a lousy business? **Yes, it happens** — but it does **not** happen overnight, over one quarter or one year. **It takes years.**
- **Intel** — was a great business with a strong economic moat, but over time **lost its competitive advantage to Nvidia and to AMD**.
- **Boeing** — was a great business, but **management missteps** destroyed a lot of its reputation.
- Therefore: ignore short-term performance; look at the **long-term year-on-year trends** of revenue, profits and cash flow.

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1. Reporting Cadence, Earnings Dates, and Where to Get the Data

1.1 How often companies report

- Companies report **four times a year**: a **quarterly report** every quarter, and an **annual report** at the end of every financial year.
- They report **three things**:
  1. **Income statement** — how much money they made, how much they spent, what's the profit at the end of the day.
  2. **Balance sheet** — how financially strong the business is: assets minus liabilities = shareholders' equity.
  3. **Statement of cash flows** — **the most important** — how much cash was generated from the business, how much was paid out, how much was borrowed or repaid.

1.2 You don't need to read the raw reports

- You *can* download quarterly and annual reports from the company's website, but modern financial websites already **compile and collate the data and pre-calculate most of the ratios**.

1.3 Finding the next earnings date

TradingView (charting software)
- Enter the ticker; an **"E" icon** appears on the chart marking earnings.
- Example shown: **Apple — earnings to be announced Thursday, 30 January 2025** (recording date 28 January, so two days away).

- Companies announce either **before the market opens ("BFO")** or **after the market closes** — Apple's example showed *after market close, 4pm Eastern Time*.
- Historical earnings markers show beats/misses. Example quarter shown: analysts expected **$1.60 EPS**, company reported **$0.97 — 39% below expectations**; **revenue was above expectations by 0.44%**. Stock dropped a bit, then eventually went up again. Another marker showed earnings and revenue **both better than expected**, and the stock **initially dropped, then went up again**.

Yahoo Finance earnings calendar
- `finance.yahoo.com/calendar/earnings`
- Shows all companies reporting in a chosen period; search a ticker (e.g. **MSFT**) to get its next date. Example shown: **next earnings 25 April 2022**.

1.4 Should you buy before or after earnings?

- **"It is totally unimportant."**
- Reason: **a great business will have good quarters and bad quarters, good years and bad years. There is no great company that has a good quarter every quarter.**
- In the short term the market is **often manipulated / illogical** — a company can beat on earnings *and* revenue and the share price still drops; it can report bad results and the price goes up. **On an earnings announcement, anything can happen — it's a coin flip.**
- What matters: over the years, **revenue, profits and cash flow from operations trend upward**.
- Once you *own* a company it *is* useful to read the quarterly results and the **CEO conference call transcript** to confirm the company is following its plan and roughly delivering on projected revenues and profits.

1.5 Where to pull financial statements

Website 1 — StockOracle (preferred; presents data very visually)
- Enter the ticker in the **top right** search (e.g. Apple = **AAPL**).
- The company page shows the **next earnings date** immediately. (Example: Apple next earnings **30 April 2026**; recording made **February 2026**.)
- Go to **Financials → financial statements**.
- Data can be displayed as a **line chart or bar chart**. Charts show, in one view: **sales revenue**, **operating income (orange)**, **net income**, and on the right the cash flow data — **operating cash flow (orange)** and **free cash flow (dark green)**.
- Scroll down for the three statements in order: **income statement → balance sheet → cash flow statement (statement of cash flows)**.
- **Free / basic account = last 5 years of data. Paid account = 10 years of data.** (Visual trend charts of margins and ROE/ROIC require the paid/Pro account.)
- Toggle **annual** or **quarterly**.
- Additional locations for ratios: **Company Metrics** page (profitability ratios, debt ratios, efficiency ratios) — the basic free account also carries roughly the last 5 years there.
- **Coverage limitation:** StockOracle covers **all US-listed stocks** only. Stocks listed in India, South Korea, Malaysia, Singapore, China are **not yet available**. **By end of 2026, Singapore and China–Hong Kong listed stocks will be added.**

Website 2 — Yahoo Finance (for non-US listings)
- Example walkthrough: search **DBS Bank → ticker D05 → Financials**. Same three statements: income statement, balance sheet, statement of cash flows.
- Free version shows **4 years of data plus the last 12 months**; toggle annual/quarterly.

Website 3 — Morningstar.com (fallback for foreign stocks; free section)
- Normally a paid site, but a free section carries **ROE and ROIC**.
- Example: **Ping An Insurance** (China–Hong Kong), choose the **H shares listed on the Hong Kong market → Key Metrics → Profit and Efficiency Ratios** → ROE is shown. **ROIC is not provided** (and in any case ROIC does not apply to banks or insurance companies — Ping An is both).

1.6 ⚠️ Two formatting traps to watch

	Yahoo Finance	StockOracle (and most other sites)
Column order	Data runs **right to left** (2022, 2023, 2024, 2025 reading right to left)	Data runs **left to right** (2021, 22, 23, 24, 25, then TTM)
Denomination	Figures reported **in thousands**	Figures reported **in millions** (US dollars, adjusted to USD)

- Reading millions: **1,000 million = 1 billion.** So Apple's `435,617` = **435,617 million = $435.617 billion**.

1.7 TTM (trailing twelve months)

- The last column on StockOracle is **TTM = the last 12 months = the last four quarters added together**.
- **The last 12 months may or may not match the last financial year** — they don't always coincide. (For Alphabet in the example they *did* coincide, so the TTM figure equalled the last financial year figure.)
- You can verify TTM yourself by clicking **Quarterly** and summing the last four quarters.
- **Not all companies end their financial year in December** — Apple's financial year **ends in September**.

1.8 Which period to use for which statement

Statement	Measures	Use
Income statement	A **period of time** (how much was made over a year)	**Annual** figures, compared year to year
Statement of cash flows	A **period of time** (cash in/out over 12 months)	**Annual** figures
Balance sheet	A **snapshot at a point in time** (how much cash/debt right now)	**Latest quarterly** report

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2. The Income Statement

> *"The income statement tells you how good the company is at making money."* Sales generated, all costs and expenses, and the profit left for shareholders.

2.1 The line items in order (Adam's whiteboard shoe-company example)

Worked example: a shoe company sells shoes at **$500 each**, sells **1,000 shoes a year**.

#	Line item	Also called	Example figure	Note
1	**Sales revenue**	Revenue	**$500,000**	$500 × 1,000 shoes
2	**− Cost of goods sold**	**COGS**	**−$250,000**	$250 to make each shoe × 1,000
3	**= Gross profit**		**$250,000**	Sales − COGS
4	**− Operating expenses**	**OPEX** / overheads	**−$100,000**	Costs of running the business

#	Line item	Also called	Example figure	Note
5	**= Operating income**		**$150,000**	**Profit the company makes from its core business**
6	**− Non-operating expenses**		**−$50,000**	Not really related to the core business
7	**+ Other (non-operating) income**		**+$60,000**	Extra income not from the core business
8	**= Net income / net profit after tax**	Net profit	**$160,000**	The bottom line

What sits inside operating expenses (OPEX / overheads):
- Staff salary
- Rental
- Utilities
- **Depreciation** — a **non-cash expense**. Cash does not actually go out; you already paid for the equipment years ago. Adam's example: you bought equipment five years ago (cash left five years ago) but you **depreciate it over ten years**, deducting a certain amount each year — non-cash.
- **R&D** (research and development)
- Marketing costs

What sits inside non-operating expenses:
- **Lawsuit / lawyer fees** (company got sued)
- **Taxes**
- **Restructuring costs**
- **Interest payments** on borrowed money
- (Elsewhere Adam adds: **inventory write-offs**)

What sits inside other / non-operating income:
- **Dividends** from a company it has invested in
- **Capital gain** on selling a property or an asset
- **Interest income**
- **Forex gains**

2.2 What actually matters in the income statement

Adam: *"The only thing you've got to pay attention to in the income statement"* — **three items**:

1. **Revenue (sales revenue)**
2. **Operating income**
3. **Net income after tax**

All other line items are **not so important** unless you are digging for specific information.

2.3 The tests

- **Test 1 — Sales revenue must be increasing consistently** over **at least the last 5 years**, **10 years even better**.
- **Test 2 — Net profit (net income after tax) must also be increasing consistently** over the last 5 years.
- **"Increasing consistently" does not mean it must rise every single year.** One or two down years are fine **as long as the general trend is upwards.**
- **Test 3 — the escape hatch:** If **net income is *not* consistent**, that's **OK — provided operating income is increasing consistently.**

Why the escape hatch exists: Net profit is frequently **skewed or distorted by non-operating income and expenses** — a big tax bill, a lawsuit, restructuring, a merger/acquisition cost, or a windfall gain on an investment or on selling a building. These are **one-off / non-recurring gains and losses**, **exceptional items** that are **not part of the core business**, happen once, will not recur, and can therefore be **disregarded when analyzing the business.**

Operating income *"reports the amount of profit from the business's ongoing operations"* and **excludes**:
- Non-operating **expenses**: taxation, interest costs, lawsuit costs, restructuring costs, inventory write-offs
- Non-operating **income**: dividend income, capital gains on investments, interest income, forex gains

Worked distortion examples:
- **Pepsi** — profits rising, but one or two years had a big drop. **Nothing to do with the business** — it was **taxation changes** or **forex losses/gains** from cross-border transactions. Ignore.
- **Salesforce** — recently spent a lot of money **acquiring other companies**; the acquisition cost caused a **big drop in profit last year**. Non-operating. Ignore.

2.4 Speculative growth companies (the exception to "must be profitable")

- Companies named: **Sea ("C"/SE), Roku, Palantir, Airbnb** — revenues growing but **not profitable yet**.
- Adam calls these **speculative growth companies** — **more risky** because they are not yet profitable.
- **He does buy them once in a while, but with a very small allocation.**
- **His requirement:** although net profit is not yet positive, **the company must already have positive free cash flow.**
- Example given: **Palantir** — revenues growing, not yet profitable, but **free cash flow has already turned positive**.

2.5 Profit margins

Gross profit margin

```
Gross profit margin = Gross profit ÷ Sales revenue × 100%
Example: $250,000 ÷ $500,000 = 50%
```

- **Meaning:** for every **$1 in sales**, the company keeps **50 cents** to pay all its other expenses.

Net profit margin

```
Net profit margin = Net profit after tax ÷ Sales revenue × 100%
Example: $160,000 ÷ $500,000 = 32%
```

- **Meaning:** for every **$1 in sales**, the company keeps **32 cents** in net profit, which it can use to **pay off debt, pay dividends, buy back shares, or reinvest**.

How to judge a margin — there is no absolute number:
1. **Comparative test:** compare the company's margins to its **closest competitors**. You want the company to have **the same or higher** margins than its closest competition. If competitors have *higher* margins, that is **not good**.
2. **Trend test:** look at the margin over the **last 5 years**. You want it **increasing or consistent**. **Falling gross margins over 5 years is a bad sign** — it means the cost of making the products is rising and eroding the margin.
3. **Escape hatch again:** if **net profit margin** is inconsistent because it's skewed by non-operating items, look at the **operating profit margin** instead. **As long as gross profit margin and operating margin are both consistent or growing, that's fine.**

Where to find margins on StockOracle: Company Metrics → scroll down (available on the free basic account, roughly last 5 years) → **gross profit margin, operating profit margin, net profit margin**, plus the **10-year average** for comparison. Visual margin charts require the **paid account** (scroll up on the same page).

2.6 Worked example — Apple's income statement (StockOracle)

Line-by-line for the year shown:

Line item	Figure
Sales revenue (last financial year)	**$416.161 billion**
Sales revenue (TTM / last 12 months)	**435,617 million = $435.617 billion**
− Cost of goods sold	**~$220,000 million**
= **Gross profit**	**$195,201 million**
− Total operating expenses	**$62.151 billion**
• of which SG&A (selling, general & administrative)	**$27.6 billion**
• of which R&D	**$34.55 billion**
= **Operating income**	**$114 billion**
− Interest expense on debt (FY2023 detail)	**$3.9 billion**
+ Interest income earned on its cash pile	**$3.75 billion**
= **Net interest expense**	**$183 million** — *"considered very low for a company"*
− Other non-operating income & expenses	**$382 million**
= **Earnings before taxes**	**$113.736 billion**
− Income tax expense	—
= **Net income**	**$96.995 billion**

- Note: **Apple takes on debt for taxation reasons**, and simultaneously holds a lot of cash earning bank interest — hence both an interest expense and interest income line. In the two most recent financial years these lines were **not shown / treated as zero**.

Apple's trend check (Adam reads them off the table):
- **Revenue:** 2020 **$274B** → **$365B** → **$394B** → slight drop to **$383B** → **$391B** → **$416B** → TTM **$435.617B**. Verdict: **"pretty consistent"** — yes.
- **Operating income: $65B** → **$108B** → **$119B** → **$114B** (slight drop) → **$113B** → TTM **$141B**. Verdict: generally **uptrend** — yes.
- **Net income: $57B** → **$94B** → **$99B** → **$96B** → **$93B** (two dip years) → **$112B** → **$117B**. Verdict: generally **uptrend**.
- Easiest method: **just look at the StockOracle chart** set to **annual** — revenue uptrend? net income (green) uptrend? if not, is operating income (orange) going up? **"That's all you've got to know about the income statement."**

Apple's margins (last five years):
- **Gross profit margin:** 43% → 44% → 46% → 46.9% → 47%. **Increasing — they are getting more and more profitable**, and current gross margins are **higher than the 10-year average**. Good sign.
- **Operating margin:** 30.29% → 29.82% (slight drop, *"no need to panic"*) → 31.51% → 31.97% → 32%. Generally rising and **above the 10-year average**.
- **Net profit margin:** 25.31% → 25.31% → 23% (dipped) → 26% → 27%. *"Everything looks fantastic."*

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3. The Balance Sheet

> *"The income statement tells us how profitable the business is. The balance sheet tells us how financially strong the business is."*

3.1 The master equation

```
Total Assets – Total Liabilities = Shareholders' Equity
```

- **Assets** = what the company **owns**
- **Liabilities** = what the company **owes**
- **Shareholders' equity** = what is **left behind for the shareholders — you and me**

Always use the LATEST QUARTERLY report for the balance sheet, because it is a **snapshot at a point in time**, not a period.

3.2 ASSETS

Current assets (short-term assets) — things the company owns that it **can convert to cash easily within three months**:
1. **Cash and equivalents** — cash in the bank plus **short-term bonds, commercial paper, fixed deposits** convertible to cash within three months.
2. **Short-term investments** — investments the company can sell and convert to cash easily.
- *(1 + 2 are shown combined on StockOracle as "cash and short-term investments".)*
3. **Inventory**
4. **Accounts receivable** (and **net accounts receivable** after deducting bad debt) — money customers owe the company that it has yet to collect.

Non-current assets (long-term assets) — things the company owns that it **can't convert to cash easily**:
- **Physical / tangible assets** (assets you can actually touch): **real estate, machinery, plant, equipment (PP&E)**
- **Long-term investments**
- **Intangible assets**: **goodwill, patents, the brand**

> ⭐ **The one asset figure Adam actually uses: "cash and short-term investments."** *"This figure is important because later when you learn to calculate the intrinsic value of a business, this is one of the things we're going to input."*

3.3 LIABILITIES

Current liabilities (short-term liabilities) — things the company owes that it must pay **within three months** (Adam also describes them in the Pepsi walkthrough as payable **within a year**):
- **Accounts payable** — owed to suppliers, owed to employees as unpaid salary. Must be paid within three months or the company may get sued.
- **Taxes payable**
- ⭐ **Short-term debt** — borrowed money repayable within three months. **This is what matters.**
- **Deferred revenue** (see §3.8) — subscription money collected upfront for services not yet delivered.
- Other current liabilities

Non-current liabilities (long-term liabilities):
- ⭐ **Long-term debt** — the two most common forms:
1. **Term loans** the company has taken
2. **Bonds the company has issued** (corporate bonds)
- Long-term leases, provisions for risks and charges, etc.

> ⭐ **The one liability figure Adam actually uses: TOTAL DEBT.**
> `Total Debt = Short-term debt + Long-term debt`
> **Total debt is NOT total liabilities** — total liabilities include a lot of other stuff besides debt. Total debt is used in **the debt ratios** and in **the intrinsic value calculation**.

How much debt is too much? It depends entirely on how much the company earns.
- Company earning **$1 billion/year** with **$1 billion of debt** → fine, it can repay all debt in one year.
- Company earning **$1 billion/year** with **$20 billion of debt** → **it will take 20 years to repay. That is too much debt.**

3.4 SHAREHOLDERS' EQUITY — three main parts

1. **Common stock + additional paid-up capital**
   - The money the company raised when it **first sold shares to the public, usually at the IPO** (initial public offering).
   - **Is it important? "Not really."** General knowledge only.
   - **Key insight:** when you and I buy Apple stock today, **the money does NOT go into the company** — it goes to the shareholder who sold you the shares. Money only enters the company when the company **issues new shares**. That is **very rare for US companies like Apple and Microsoft** (they already have so much cash they don't need to raise any) but **very common for REITs**.
2. **Retained earnings** — *"this is more important."*
   - Profits made over the years that the company **did not pay out as dividends** and kept inside the business — **accumulated savings from the profits**, which all belong to the shareholders.
   - **A good business shows a lot of retained earnings accumulated over the years.**
3. **Treasury stock** — **always a negative; it always reduces shareholders' equity.**
   - Created when a company **buys back its own shares** and holds them.
   - Bought-back shares held in treasury **no longer receive dividends and have no voting rights — they are kept in cold storage.**
   - **Why buybacks are generally good:** fewer shares in the market → **each remaining share becomes worth more**.
   - **But buybacks are only good if:** (a) the company buys back **when the shares are cheap / undervalued**, and (b) the company is **already profitable**.
   - **Why the reduction in equity is good, not bad:** with less shareholders' equity, the company is **making profits using less equity**, and you are **sharing with fewer shareholders**.
4. *(Also on the statement — general knowledge only)* **Accumulated other comprehensive income** — any **unrealized loss or profit** from the company's investments or from **foreign exchange**. Example: Pepsi buys other companies whose value falls but Pepsi doesn't sell → unrealized loss recorded here; or Pepsi holds a lot of cash overseas and the **US dollar strengthens against foreign currencies** → unrealized forex loss recorded here. *"Is this really important for you as an investor? No, it's really not important."*

3.5 ⚠️ Negative shareholders' equity is NOT automatically bad

- Normally you want equity **positive** (assets > liabilities). Negative equity normally means liabilities exceed assets — **generally not good**.

- **THE EXCEPTION:** companies like **Domino's Pizza, McDonald's, and Yum Brands** have **negative shareholders' equity** — **not** because they are losing money. On the contrary, **they are making a lot of money**. The negative equity comes from **years of aggressive share buybacks creating a huge negative treasury stock line.**

Worked example — McDonald's (latest quarterly balance sheet on StockOracle):
- **Shareholders' equity = −790** (negative)
- **Retained earnings = $70 billion** — they have made a lot of money
- **Treasury stock = $79 billion** — shares bought back from the market
- $70B retained earnings − $79B treasury stock → **negative equity**
- Scrolling up confirms: **revenue increasing, net income increasing, free cash flow increasing. Very, very good business.**

> **Rule:** negative shareholders' equity is **fine — even a good thing — as long as** (a) the company is making a lot of money, (b) **profits are increasing consistently**, (c) **free cash flow is increasing consistently**, and (d) the cause is **a large treasury stock balance**.

3.6 Worked example — Pepsi's balance sheet (StockOracle, latest quarterly, US$ millions)

Line	Figure
Total assets	**107,399**
Total liabilities	**86,852**
= Total shareholders' equity	107,399 − 86,852 (transcript reads the figure as "24,06", partially garbled)
Cash and short-term investments	**9,530** (= **$9.53 billion**)
Net accounts receivable	**9,035**
Total current assets	**27,901**
Total current liabilities	**32,764**
Total non-current assets	**79,498**
Short-term debt	**6,861**
Long-term debt	**42,321**
Total debt	6,861 + 42,321
Common stock	**23**
Additional paid-in capital	**4,451**
Retained earnings	**72,788**
Treasury stock	*(deducted — always negative)*
Accumulated other comprehensive income	**−15…**

- Current ratio for Pepsi = **27,901 ÷ 32,764** (all pre-calculated on the websites).
- Adam's repeated warning: **"The figure by itself is meaningless. All these figures are meaningless unless you compare them with other figures. It's all relative."**

3.7 The THREE debt/safety ratios Adam uses

> *"There are of course many other debt ratios. You don't need to look at all of them. These are the three main ones that are sufficient."*

① Current Ratio

```
Current Ratio = Current Assets ÷ Current Liabilities
```

- **Threshold: greater than 1 is safe.** You want more assets than liabilities — the top bigger than the bottom.
- Pre-calculated on all financial websites.

② Debt to EBITDA Ratio

```
Debt / EBITDA = Total Debt ÷ EBITDA
where Total Debt = Short-term debt + Long-term debt
```

- **EBITDA** = **Earnings Before Interest, Taxes, Depreciation and Amortization** — another kind of earnings, before deducting interest expenses, taxes, depreciation and amortization.
- **Threshold: less than 3 is safe.**
- **Worked example:** total debt **$3 billion** ÷ EBITDA **$1.5 billion** = **1.5×** → the company could **repay all its debt within 1½ years** of profits.
- If the ratio is **3**, it would take **three years** of annual profits to repay the debt.
- **Why taxes are excluded:** *"because if it's making losses, there's no taxes"* — so taxation is not included in the measure.

③ Debt Servicing Ratio

```
Debt Servicing Ratio = Net Interest Expense ÷ Cash Flow from Operations × 100%
```

- **Net interest expense** = interest paid on debt **net of** interest earned on investments.
- **Threshold: less than 30% is safe. More than 30% would be very dangerous.**
- **Worked example:** company pays **$1 million** of interest a year and generates **$10 million** of cash flow from operations → **10% → safe.**
- **Personal analogy:** you earn **$5,000 a month take-home** (your "cash flow from operations") and pay **$1,000 a month interest** on bank loans → **20% → safe**. But if you paid **$4,000 in interest out of $5,000 earned → too high.** *"Same thing for an individual, same thing for a company."*

Pass requirement: *"Must all three pass for it to be a safe investment? Ideally, yes. But if the company can pass two out of three, I'm also okay in investing in that kind of company."*

3.8 ⚠️ When a current ratio below 1 is FINE

Reason A — deliberate supplier squeeze (Pepsi)
- Pepsi's current ratio is **below 1 — about 0.85**. **They do it intentionally.**
- Because Pepsi is so big it has **enormous leverage over its suppliers**, so it **deliberately drags out payments** — instead of paying suppliers in **30 days**, it pays in **60 to 90 days**.
- This **reduces the working capital Pepsi needs (a good thing)** but **inflates accounts payable → inflates current liabilities → pushes the current ratio below 1.**

- Check the rest of the picture and Pepsi is clearly safe:
- **Debt servicing ratio 9.79%** (way below 30%)
- **Interest coverage ratio 12.52** (way above 3)
- **Debt/EBITDA 2.59** (below 3)
- Revenue growing strongly; free cash flow very consistent and positive; net profits very positive
- **Cash conversion cycle very low** (a good thing) and **accounts receivables grow slower than revenue growth**

Reason B — subscription businesses and deferred revenue (ServiceNow)
- Software/subscription companies (**ServiceNow, Adobe, Salesforce**) collect **one year of service upfront**. Because the service has not yet been provided, this is booked as **deferred revenue**, which is recorded as a **current liability**.
- Lots of deferred revenue **expands current liabilities → pushes the current ratio below 1**.
- **ServiceNow: current ratio 0.95**, but **debt servicing ratio 0**, **interest coverage 90**, **debt/EBITDA 0.55**. Very profitable, very high quality.

> **Conclusion: you can't look at one ratio in isolation. You have to compare it with the other ratios and look at everything in totality.** *"Current ratio is important, but if it's below one, it doesn't matter if the other ratios are very, very strong."*

3.9 Two optional ratios Adam does NOT personally use

Interest Coverage Ratio *(Adam: "I don't use it myself because it is an overlap")*

```
Interest Coverage = EBITDA ÷ Net Interest Expense
```

- **Worked example:** EBIT/EBITDA of **$100 million**, interest expense of **$10 million** → **10×** → the company can cover its interest expense **10 times over from one year of profits — very, very safe.**
- **Thresholds: 3 or more = safe. 5 or more = very, very safe. Below 3 = getting risky, especially below 2.**
- Meta Platforms example: **71** — very, very, very safe.

Cash Ratio *(Adam: "I think it's not necessary")*

```
Cash Ratio = Cash and Cash Equivalents ÷ Current Liabilities
```

- Same as the current ratio except the numerator **excludes accounts receivable and inventory** — cash and equivalents only.
- **Thresholds: 0.5 and above = very, very safe. Below 0.3 = getting risky.**
- Meta Platforms example: **1.97** — very, very safe.

3.10 Worked example — Meta Platforms debt ratios (StockOracle → Company Metrics → scroll down)

Ratio	Latest	5-year average	Threshold	Verdict
Current ratio	**2.6** (last quarter)	**2.72**	> 1	Way above 1
Debt servicing ratio	**0**	—	< 30%	*"Very, very conservative"*
Debt / EBITDA	**0.59**	**0.3**	≤ 3	Pass
Interest coverage	**71**	—	≥ 3 (≥5 very safe)	Very, very safe
Cash ratio	**1.97**	—	≥ 0.5	Very safe

3.11 ⚠️ Where the three debt ratios DO NOT APPLY

The current ratio / debt-to-EBITDA / debt servicing trio **applies to all companies EXCEPT**:
- **Banks**
- **Financial companies, e.g. insurance companies**
- **Property developers**
- **REITs**
- **Commodity companies**

Why: *"by the nature of their business, they are highly leveraged — they take on a lot of debt as compared to their equity. So if you use these three ratios, they will all fail."*

Adam's own position:
- **He personally avoids property developers and commodity companies** because of their high leverage.
- **He does invest in banks, insurance companies and REITs** — using **different measures of debt** (below).

3.12 Bank-specific ratios

① Common Equity Tier 1 Ratio (CET1)

```
CET1 Ratio = Common Equity Tier 1 ÷ Risk-Weighted Assets
where Common Equity Tier 1 = cash + common stock + retained earnings
```

- Part of the **Basel III requirements** — the **international standard for banks**.
- **International standard minimum: at least 4.5%.**
- **Adam's own more conservative minimum: more than 10%.**
- The calculation is *"actually very complicated"* — **don't calculate it yourself; just search it online** (e.g. Google "DBS Bank common equity tier one ratio").

Bank	CET1 ratio	Verdict
DBS Bank (biggest and most profitable bank in Singapore)	**14.1%**	Above Adam's 10% minimum, way above the 4.5% international standard. *"Singapore banks are the safest in the world, based on this ratio."*
JP Morgan (US)	**12.4%**	Very safe, above 10%
Bank of China	**11.28%**	Above 10%

② NPL Ratio (Non-Performing Loan Ratio)

```
NPL Ratio = Non-Performing Loans ÷ Total Loans
```

- A **non-performing loan** is a loan not repaid within a certain period of time, which **could eventually become a bad debt**.
- **Threshold: less than 5% is safe.**
- **DBS Bank NPL ratio = 1.5%** — *"which is great"* — only 1.5% of DBS's loans are non-performing.

3.13 REIT-specific ratio

Gearing Ratio (a.k.a. Aggregate Leverage)

```
Gearing Ratio = Total Debt ÷ Total Assets
```

- A **REIT (real estate investment trust)** means **buying a portfolio of real estate properties**. REITs **take bank loans to finance the properties**; too much loan is risky, especially since they pay a lot of interest and **if interest rates rise they could be in trouble.**
- **Threshold: less than 45%.** Meaning **less than 45% of the properties are financed by debt**, the rest financed by equity.
- **Where to find it:** go to the **REIT's own website → download the annual report / investor presentations →** the gearing ratio is stated.
- Example: **Ascendas REIT — aggregate leverage 37.6%**, less than 45% → **safe**.

---

4. The Statement of Cash Flows — the MOST IMPORTANT statement

> *"Out of these three parts, which do you think is the most important one? If we could only look at one, which would it be? The answer is the statement of cash flows. This is the most important of all."*

4.1 Why cash beats profit

Cash flow tells you how much cash is **actually** coming into and going out of the business. **A company can report high profits without the cash ever arriving.**

Illustration: a company reports **$5 million in profits** but its **cash flow from operations is only $2 million.** How?

1. **Uncollected sales / bad debt** — the company sells products to customers and **records the profit, but the customer never pays.** It becomes a **bad debt**. The profit was recorded; the cash never came in.
2. **Asset revaluation gains** — Adam's real example: a **palm oil company** owned palm trees worth about **$50 million**. The next year the palm trees grew bigger, the company **re-valued the assets at $100 million**, declared a **$50 million gain on assets**, and **added it into other income** — **inflating net profit by $50 million with zero cash actually coming in**, because nothing was sold.

> **"Many times companies could creatively inflate their profits, where the company looks profitable but they are not really generating cash. And this is what causes a lot of companies that look profitable to go bankrupt."**
> **"Which is more important, recording the profit or actually collecting cash? It's the cash."**

4.2 The three sections

① Cash Flow from Operations (net operating cash flow, OCF) — ⭐ the important one
- How much cash came in or went out **as a result of the company's operations**. For McDonald's: **selling burgers and fries**. For Google: **selling advertisement through search and YouTube**.
- **This is the section used later to calculate the intrinsic value of the business and of the shares.**
- **RULE: OCF must be POSITIVE and INCREASING CONSISTENTLY over the last 5 years.**
- If OCF is **negative**, *"something's wrong — the company, while doing business, is losing more cash than earning cash."*
- **Low-quality business signature:** OCF that is **not consistently positive** (positive, negative, negative, positive, negative) **or** that **goes up, down, up, down, up, down**. *"Those are businesses I don't like to invest in."*
- ⚠️ **Exception:** by the nature of their business, **banks, commodity companies and property developers** tend to

have **inconsistent operating cash flows**. **For these three types of company this rule doesn't apply.** For all other companies it does.

② **Cash Flow from Investing (net investing cash flow)**
- Cash in/out as a result of the company's **investment activities**.
- **Cash OUTFLOW (negative):** the company **buys equipment, buys land, builds a plant** — this is **capex (capital expenditures)**.
- **Cash INFLOW (positive):** the company **sells a machine, sells equipment, sells an asset, sells shares it owns of other companies**.
- **Positive does NOT mean good and negative does NOT mean bad. It just tells you the company's investment activities.**

③ **Cash Flow from Financing (net financing cash flow)**
- Cash in/out as a result of the company's **financing activities** — i.e. **raising money**. A company can raise money two ways: **borrowing** or **selling equity**.
- **Cash INFLOW (positive):** borrowing money from the bank, **selling bonds**, or **selling new shares through a rights issue**. *"Generally, positive means the company's raising money, either through debt or through equity — and that normally happens for new companies."*
- **Cash OUTFLOW (negative): paying back bank loans**, **buying back shares**, **paying dividends**.
- **Again, positive is not good and negative is not bad.** Mature, big, profitable companies (**Microsoft, Google, Nvidia**) don't need to raise money — so their financing line is **negative because they are buying back shares and paying dividends**, which **is good for us as shareholders.**

> **Summary of signs:** *"For the second and third section, positive and negative is not good or bad — it just tells you how money's moving. But the first one has to be positive, and it has to be increasing over the last five years."*

4.3 Free Cash Flow

```
Free Cash Flow (FCF) = Net Operating Cash Flow (OCF) – Capital Expenditures (Capex)
```

- Capex sits inside the **cash flow from investing** section. FCF is already **pre-calculated at the bottom of the cash flow statement** on StockOracle.
- **RULE: free cash flow must be CONSISTENTLY POSITIVE.** Ideally **also increasing consistently.**
- **When FCF is positive, the company can:**
  1. **Pay dividends** (which we like as shareholders)
  2. **Buy back shares** (which we like as shareholders)
  3. **Keep the money as retained earnings**
  4. **Pay off debt**
- **Negative FCF for one year or two years, once in a while, is OK.**
- **But consistently negative FCF for two or three years in a row is very dangerous:** more cash is going out than coming in, **the company will eventually run out of cash**, and it is then **forced to raise cash — sell more shares (diluting shareholders), take more bank loans, or sell more bonds (increasing debt).** *"That's a lousy business."*

4.4 ⚠️ The AI capex exception (recorded 2026)

- **Microsoft, Google, Meta Platforms and Nvidia** have been **investing very heavily into AI data centers**, spending **billions and billions** to build AI capability.

- Since **FCF = OCF − capex**, the **capex spike has caused their free cash flow to fall. Their FCF is still positive**, just lower.
- **Adam is not concerned, for two reasons:**
  1. **The increased capex will lead to higher future growth and will further strengthen their economic moat.**
  2. **The increase in capex is temporary** — it is a burst of massive AI investment that **will take maybe two, three, four years**, after which **capex drops and free cash flow goes back up.**
- **The diagnostic rule:**
- **FCF fell because OPERATING cash flow fell → that IS a cause for concern.**
- **FCF fell because CAPEX rose temporarily while operating cash flow is still going up (investing for future growth) → that is fine.**

4.5 Worked examples

Alphabet / Google — HIGH QUALITY (StockOracle → Financials → cash flow statement, annual)
- Last financial year ended **31 December 2025**; TTM happened to coincide exactly with the last financial year.
- **Net operating cash flow (TTM): 164,713 million.**
- OCF series: **$54 billion → $65B → $91B → $91B (slight drop) → $101B → $125B → $164B.** *"Do you generally see an uptrend? … The answer is yes. That's a high quality business."*
- **Net cash flow from investing:** negative every year; last 12 months **−120,000 million (−$120 billion)** — simply means the company **is making investments**, not bad.
- **Net cash flow from financing:** negative every year — **Google is buying back shares and paying dividends.** Good for shareholders.
- **Free cash flow:** positive, positive, positive — and rising. On the Pro-account chart: **operating cash flow (orange) rising consistently; free cash flow (dark green) rising consistently and positive.** *"That is what we want to see. That is a high quality business."*

Intel (INTC) — NOT high quality
- Semiconductor company. *"Not all semiconductor companies are high quality."*
- **Operating cash flow increased from 2017 to 2020, then dropped and went sideways after 2020.** *"This is not what I like to see."*
- **Free cash flow rose, then decreased, then turned negative — five years of negative free cash flow.** *"For me, I won't touch this kind of business."*

Singapore Airlines — NOT investable (despite being a great airline)
- *"Singapore Airlines to me is one of the best airlines in the world. But would I buy the shares? No."*
- **Why all airlines are avoided:**
- Very, very **cyclical** — depends heavily on **tourism and the economy**
- **Very high maintenance capex** — refurbishing and repairing planes, buying new planes
- **High fuel costs**
- **Very tough and very competitive business**
- Result: **all airlines have very inconsistent operating cash flow and very inconsistent free cash flow.**
- *"It's not the fault of the management. It's the nature of the industry."*
- Chart reading: OCF **flat, going down, turned negative during COVID, positive, up, then down again** = **inconsistent cash operations**. FCF **negative for five straight years**; then after **2022** very high FCF, but **declining ever since**.
- Consequence of persistent cash losses: **keep issuing bonds, keep raising debt, keep selling shares → dilutes shareholders and loads the company with debt.**

MasterCard — the contrast
- A **payment technology** business Adam has invested in: **operating cash flow very, very consistent; free cash flow**

very, very consistent.
- *"There are many ways to make money, and to me I'd rather make money in an easy way by buying businesses where the cash flow generation is very, very consistent, very, very predictable."*

---

5. Financial Metrics — Profitability, Fraud Screens and Efficiency

5.1 ROE — Return on Equity

```
ROE = Net Profit After Tax (Net Income) ÷ Shareholders' Equity × 100%
```

- *"This is one of Warren Buffett's favorite criteria in investing in a company."*
- **What it measures:** how good the company is at making money **based on the shareholders' stake in the business**.
- **Worked example:** net income **$126 million** ÷ shareholders' equity **$420 million** = **30% ROE**.
- **Thresholds:**
- **ROE of 12% → a good business**
- **ROE above 15% → an excellent investment**
- *"The majority of companies that I'm investing in, you find that the ROE is often above 15%."*
- **Look at ROE over the last 5 years** to ensure it is **consistently above 12–15%**.

⚠️ **ROE when shareholders' equity is negative (McDonald's, Yum Brands):**
- When equity is negative because of large treasury stock, **ROE becomes undefined.**
- **Some websites report McDonald's ROE as negative — "that is actually not accurate,"** because profit generated from negative equity is **effectively infinity**. **ROE is infinite, which is very good.**
- *"It basically means that company makes profit without having any equity."*

5.2 ROIC — Return on Invested Capital

```
Simplified:  ROIC = Profit ÷ (Equity + Debt)

Full formula:
ROIC = [ EBIT × (1 − tax rate) ] ÷ ( Shareholders' Equity + Total Debt − Cash and
Equivalents )
where EBIT = Earnings Before Interest and Taxes
```

- **Why it exists:** ROE alone only tells you **one part of the story**. **Some companies artificially inflate their ROE by taking on a lot of debt and using the debt to buy back shares. ROIC penalizes companies that do that.**
- **What it measures:** how efficient a company is at using its **total capital (equity AND debt)** to generate a profit.
- **Thresholds (same as ROE):**
- **Over 12% → good**
- **Over 15% → excellent**
- ⚠️ **ROIC does NOT apply to banks** — banks are **highly leveraged by the nature of their business**, which makes **ROIC artificially very low**. **For banks, look at ROE only and ignore ROIC.**
- ⚠️ **ROIC also does NOT apply to insurance companies.** (Ping An Insurance is both an insurance company and a bank → **use ROE only**.)
- You never calculate this by hand — it is pre-calculated on the websites.

Where to find ROE and ROIC:
- **StockOracle → Company Metrics → scroll down → Profitability Ratios.** Shows the **latest / last 12 months**, the **5-year average** and the **10-year average**. Pro account also gives multi-year numbers and **trending charts** (line or bar).
- **Fallback for non-US stocks: Morningstar.com free section → Key Metrics → Profit and Efficiency Ratios.**

Worked example — MasterCard (StockOracle):
- **ROE (last 12 months) = 210%** — *"really, really good"*
- **ROIC (last 12 months) = 58.4% (58.41%)** — *"really, really high"*
- On the trend chart: ROE increasing, ROIC pretty consistent. *"We don't need it to increase. As long as it is fairly consistent, it's 12 and above, that's a pretty good business — but of course if it's increasing that's even more powerful."*

5.3 Revenue growth vs Accounts Receivable growth — the fraud screen

RULE: Sales revenue must grow FASTER than accounts receivable. Accounts receivable must NOT grow faster than revenue.

- **Accounts receivable** = the company sold a product (usually to another company) and **has not been paid yet** — revenue that has **not been converted to cash**.
- **Too much accounts receivable is not good** — it means **the company is not collecting its debt quickly.**
- **If accounts receivable grow FASTER than revenue, that is very dangerous.** It means one of three things:
  1. **The customers are in trouble** (can't pay)
  2. **The company is bad at collecting payments**
  3. **The revenue is fictitious** — *"Over the years, I've seen companies that have generated fictitious revenue, fake revenue. And when they record fake revenue, there's no money coming in. So to fool the public, they actually show it under accounts receivables."*
- **"A higher/increasing accounts receivable is a major red flag."**

Where to check:
- **StockOracle → Financials →** one of the **preset default charts** plots sales revenue against accounts receivable. Example shown: **MasterCard** — *"the sales revenue is growing a lot faster than the accounts receivables. That's what I'm looking for."*
- **TradingView (manual):** ticker → choose the **5-year timeframe** → **Indicators** → add **Revenue** and plot it against **Accounts Receivable**.
- **TradingView (Piranha Profits custom indicator — recommended):**

1. Go to the **Piranha Profits lesson page → Lesson 4 → "Course Notes and Resources"** at the end of the lesson.
2. Below the course notes, find **"Total Revenue vs Accounts Receivables" custom indicator for TradingView** and click the link.
3. Scroll down and click **"Add to Favorites"**. (You need a **TradingView account — a free account is fine**; register and log in first.)
4. In TradingView (web charts or the desktop app) → **Indicators → Favorites → "Revenue vs Accounts Receivables"**.
5. It auto-plots **revenue in green** vs **accounts receivable in red**.

- Example shown: **Alphabet** — revenue growing faster than accounts receivable. *"So that's great."*

5.4 CCC — Cash Conversion Cycle (management effectiveness)

```
CCC = DIO + DSO − DPO

DIO (Days Inventory Outstanding)  = Average Inventory ÷ Cost of Goods Sold × 365
DSO (Days Sales Outstanding)      = Accounts Receivable ÷ Net Credit Sales
DPO (Days Payable Outstanding)
```

- **What it measures:** *"the length of time in days that it takes for a company to convert its investments in inventory and other resources into cash flows from sales."*
- ⚠️ **APPLIES ONLY to retail companies that sell physical goods** — e.g. **Nike, Apple, Walmart, Costco, Pepsi**. **Does NOT apply** to companies that do not sell a physical product — e.g. **Google, YouTube, Microsoft**.

Intuition examples:
- **A restaurant:** take out cash → buy ingredients → cook → sell → customer pays. **CCC ≈ 1 to 3 days.**
- **A real estate company:** take out cash → buy land → spend on marketing and materials → buyers pay progressively as construction proceeds. **CCC ≈ 3 to 5 years / "two or three years."**

The rule:
- **It does not matter how long or short the CCC is — what matters is that it is CONSISTENT.**
- **Consistent or getting shorter = good** — the company is getting its money back faster and is **very efficient at turning around its inventory.**
- **Getting longer and longer = RED FLAG** — the company **could be in trouble because it's not able to sell the inventory fast enough to get back the cash.**
- **A NEGATIVE cash conversion cycle is the best** — it means **the business collects cash upfront BEFORE it pays its suppliers. Pepsi's CCC is negative all the way** — *"that's very, very good."*

Where to find it:
- **TradingView:** ticker → 5-year timeframe → **Financials → type "cash conversion cycle"** → choose **quarterly** (more data points) or annual. Example: **Nike** — trend is flat, one bump up then coming down again → *"it looks fine."*
- **StockOracle: Financials → scroll down → cash conversion cycle chart** (hover for numbers); **or Company Metrics → Efficiency Ratios** → shows current, **5-year average** and **10-year average** (history depth depends on your package).

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6. The 12-Point Fundamental Checklist (Adam's Full Summary)

> *"If you can ensure that the company you invest in checks all the 12 points, you know it's a great company: it's very profitable, it's very safe, and it's well managed."*

Group A — Is the company PROFITABLE? (metrics 1–6)

#	Check	Threshold / rule
1	**Sales revenue AND net income increasing consistently**	Over the **last 5 years** (10 better). If net income is not consistent, **operating income** must be growing consistently.
2	**Cash flow from operations increasing consistently**	Over the **last 5 years**
3	**Free cash flow consistently positive**	FCF = OCF − capex

#	Check	Threshold / rule
4	**Gross profit margin and net profit margin consistent or increasing**	Not decreasing; compare vs closest competitors too
5	**ROE**	**12–15% or higher**
6	**ROIC**	**12–15% or higher**

Group B — Is the company FINANCIALLY STRONG? (metrics 7–10)

> *"It's not going to go bankrupt easily. In fact, it's something you can buy and you can sleep soundly at night."*

#	Check	Threshold
7	**Current ratio**	**1 or more**
8	**Debt to EBITDA ratio**	**3 or less**
9	**Debt servicing ratio**	**Less than 30%**
10	**Special cases**	**Banks: Common Equity Tier 1 ratio > 10%. REITs: gearing ratio < 45%.**

Group C — Is MANAGEMENT EFFECTIVE? (metrics 11–12)

#	Check	Threshold
11	**Cash conversion cycle**	**Consistent or declining, NOT increasing.** *Applies only to retail companies selling physical goods.*
12	**Accounts receivable vs sales revenue**	**Accounts receivable must NOT be growing faster than sales revenue.**

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7. Case Study — HYFLUX (how the 12 metrics would have saved you)

7.1 What Hyflux was

- A **water treatment company**, a **blue chip in Singapore**, and *"once upon a time the darling of the Singapore Stock Exchange."*
- **Widely promoted by analysts and stockbrokers.**
- On the surface it looked good: **profits were going up, revenue was going up.**
- **Temasek Holdings** — the **Singapore government sovereign wealth fund** — was a big backer of the stock, which gave people a lot of confidence.

7.2 What happened

- Share price collapsed from **about $2 / $2.50 a share → 4 cents → eventually zero.**
- **Not only stock investors lost everything — bond investors lost hope of recovering any money at all.**
- Adam **never bought it and told his students to stay away** because **the fundamentals failed many of the metrics.**

7.3 The warning signs, in the financial statements, years in advance

> *"It didn't go bust overnight. Over the years, a lot of warning signs could be seen in the financial statements." "On the surface, revenue was growing, profit was growing — but if you dig deeper into the statement of cash flows, that's where you find problems. That's why I keep saying the statement of cash flows is the most important statement of all."*

🚩 **Red flag 1 — Negative and inconsistent operating cash flow (as far back as 2006)**
- **2006 through 2011** operating cash flow figures cited: **−26.8 million, 74 million, 29 million, 60 million, −49.5 million** — *"extremely inconsistent, negative, negative, negative."*
- The rule is that operating cash flow must be **increasing consistently** — Hyflux's was neither consistent nor reliably positive.

🚩 **Red flag 2 — Negative free cash flow**
- FCF = operating cash flow − **purchase of PPE (property, plant and equipment) = capex**.
- **−26 million OCF − 19 million capex → negative free cash flow.**
- **29 million − 18 million → still positive**, but **it turns negative again** in following years.

🚩 **Red flag 3 — Paying dividends while free cash flow was negative**
- *"If your company has negative free cash flow, should you be paying out dividends? No. You should be conserving cash… You should only pay out dividends if your free cash flow is positive and you have got excess cash."*

🚩 **Red flag 4 — Accounts receivable exploding vs revenue**

Year sequence	Revenue	Trade / accounts receivable
	142 million	**30 million**
	192 million	**46 million**
	554 million	**41 million**
	524 million	**27 million**
	569 million	**182 million**

- **In 2010, revenue grew 8% while accounts receivable grew 575%.**
- *"When accounts receivables suddenly jumps a lot higher than revenue, something's wrong with the business."*

🚩 **Red flag 5 — Unsustainable debt levels (Debt/EBITDA fail)**
- Short-term debt: **"loans and borrowings" of 95 million in 2010** (under current liabilities) **+ long-term debt**
- **Total debt = $599 million**
- **EBITDA = 106 million**
- **Debt / EBITDA = 5.64** vs the threshold of **3 or less** → **"a big fail."**

7.4 Why so many investors still bought it

> *"The trouble is most investors are lazy, or they're incompetent — they don't do a deep-dive research into the fundamentals of the business. They buy simply because of what they read in the media, because of what they are told by their bankers or their analysts or their brokers."*

1. **"Temasek Holdings is a major investor, so it must be safe."** — *"a very, very wrong belief system."*
   - **What people didn't realize: Temasek invested in Hyflux very early on and had already sold most of its shares for a big profit. They were already out of the stock by the time retail investors were buying.**
2. **The celebrity status of the CEO, Olivia Lum** — featured in **Forbes Magazine**, **Entrepreneur of the Year**, a **Nominated Member of Parliament**. *"I'm not saying she's not a good businesswoman — in fact I think she's a very good businesswoman. But no matter how great you are, you make mistakes as well, so do I. Don't just be seduced by how good someone is."*
3. **Blindly following analyst and bank recommendations** — e.g. **Daiwa Capital Markets** said it believed Hyflux's share price had **reached an inflection point on 7 March**, expected the share price **to continue to rise as sentiment improves**, issued a **BUY rating with a target price of $2.40.** The stock then collapsed to **4 cents and then zero.**
4. **"It's a water treatment business, so it must be a strategic investment — everyone needs water."**

> **Lesson:** *"Do not be fooled by all these surface reasons for investing. You can always dig deep into the financial metrics of any investment."*

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8. Great Business Models — The Pillars of a Great Business

> *"We only want to invest in the best businesses in the world — businesses with great business models that can survive and thrive through all economic conditions."*

Adam opens by listing **six pillars** and then walks through them (the walkthrough order differs slightly from the opening list, and at the very end he says "seven pillars" — the substance is the six below):

Opening list: ① reduce input costs over time ② increase output of products and services ③ self-finance its growth ④ weak suppliers ⑤ growing customers / huge customer base ⑥ weak competition.

Walkthrough order: ① input costs ② output & pricing power ③ customers ④ weak suppliers ⑤ protection from competition / economic moat *(described as "the most important one")* ⑥ self-financing.

PILLAR 1 — Ability to REDUCE INPUT COSTS over time

There are **three main input costs**:

1a. Infrastructure and capital equipment
- You want a business that can **grow without constantly spending more and more on new infrastructure and new equipment — spend one time, then scale on that same infrastructure.**
- ❌ **Businesses Adam avoids on this test:**
- **Airlines** — to grow they must **constantly buy new planes**, causing large cash outflow, **constant new capex to replace old planes**, plus **continuous maintenance and repair spending → reduces profitability over time**.
- **Shipping companies** — must **buy new ships to replace old ships and repair the ships**.

- **Auto manufacturers** — must **constantly invest in new factories and new equipment** to build cars more efficiently. *"All this sucks up their cash flow and creates a very low free cash flow in the business."*
- ✅ **Businesses Adam likes — invest once, then scale without adding cost:**
- **Credit card / payment technology companies — Visa, MasterCard.** They invested heavily in payment infrastructure once; now they **dominate the industry and can expand to all countries and many platforms without incurring additional cost.**
- **Software companies** — once the software is made, they can **scale production without investing a lot more in capital equipment.**
- **The general principle: "I like companies that have very little physical assets, physical equipment. I like companies with more INTELLECTUAL assets."**
- **Nike** — doesn't make its own shoes. It **designs the shoes, owns the intellectual property, and outsources manufacturing to China or Vietnam**, where **the manufacturers** carry the capital equipment burden, not Nike.
- **Apple** — doesn't make its own iPhones or laptops. It **designs them, creates the software, owns the IP and the brand**; manufacturing is outsourced.
- **Coca-Cola** — doesn't bottle its own drinks. It **owns the syrup and the brand "Coke"** and **outsources the bottling and production to its suppliers**.
- **McDonald's and Yum Brands (owner of KFC and Pizza Hut)** — a **franchise model**. They don't really own their own restaurants or invest in capital equipment; they **own the brand and the recipes and sell the franchise.**
- **Inflation benefit:** these companies **are not affected by inflation because they don't have to keep spending more and more money on input cost.**

1b. Raw materials

- Ask: **could raw material costs go up?**
- **Starbucks** — raw materials are **coffee and sugar**. If **sugar prices** or **cocoa bean prices** go up, **input costs rise and profits could drop.**
- **The test:** if raw material costs rise and the company **can raise the price of its drinks and pass the cost to customers, that's OK — they are hedged against inflation.** But if the company **cannot pass on the cost and has to absorb it, that really screws up profitability and profit margins.**

1c. Labour costs

- **Key man risk** — *"I tend to avoid investing in companies where the success of the company is based on the brilliance of one man."*
- Example: **Elon Musk.** *"One reason why I'm very nervous to invest in Tesla or SpaceX, because a lot is driven by his personality, his vision. Now what if he dies? What if he gets hit by a truck?"*
- **Old saying (Adam attributes it to Warren Buffett): "It's best to invest in a company where any idiot can run the business — because one day an idiot may run the business."** You want a business to be great **because of the business model itself**, not because of one man, so **anyone can run it**.
- **Unique talent requirement** — *"Is unique talent required to maintain the competitive advantage?"* If special talent is needed to maintain the company's advantage, **that's dangerous — once you lose the talent, the whole business is gone.**
- **Strong unions** — Adam **avoids companies with very strong unions**. Another reason he avoids **airline companies, especially in the US and European countries**: if the **pilots strike, the cabin crew strike, it brings the entire company to its knees.**

PILLAR 2 — Ability to INCREASE OUTPUT (products and services)

More output → **more revenue, more profits.** Four sub-tests:

1. **Pricing power** — *"the business must be unique enough to command pricing power."* You want companies **able to raise their price every year**, because then **revenue rises even if the quantity sold stays the same**, and the company is **hedged against inflation** — if labour and raw material costs rise, they simply raise prices.
2. **High switching costs** — once a customer buys, it must be **very expensive or very difficult to change to another brand**, so customers are **stuck to that brand for the rest of their life**.
   - **Microsoft** — once people are familiar with **Office, PowerPoint and Word**, it's very difficult to change to another word processor or spreadsheet app. **It locks in the customers.**
3. **Recurring income** — *"once customers buy one time, they keep paying year after year after year. I don't like businesses where they sell the product once and that's it."*
   - Adam invests in **Adobe, Microsoft, Salesforce** because of recurring subscription revenue.
4. **Multiple product revenue streams**
   - ❌ **Zoom** — *"a one trick pony"* with only the Zoom video-sharing software. If users get sick of it or jump to **Microsoft Teams** or a competitor, **they're done.** This is why Adam doesn't invest in Zoom.
   - ✅ **Alphabet** — multiple product lines: **YouTube, Google Search, Google Maps** — diversified revenue streams.
   - ✅ **Disney** — **Disney Parks / Recreation / Experiences, Disney+, ESPN, Hulu, Network TV.** *"Very diversified in their revenue streams — that's a lot safer."*

> **How to verify Pillars 1 and 2 quantitatively:** *"Very simple, look at the profit margins."* **As long as gross profit margins and net profit margins are consistent or increasing over the last five years, that's a reflection of the first two pillars.**

PILLAR 3 — Growing, strengthening CUSTOMERS

We want the customer base to be **growing** and also **able to buy more products and services over time** (rising purchasing power). Four questions:

1. **Is the customer base sizable and growing?**
2. **Are the customers' wages rising?** If customers earn more, they can **buy more from your business** or **pay more as you raise prices.**
3. **Is there customer concentration risk?**
   - Adam avoids companies with very few customers.
   - **Example: Upstart (ticker UPST, given in the transcript as "UPSD")** — a **fintech company providing credit-rating software services to banks and property developers**, rating the credit of customers so a bank knows whether to give a loan. The business model is interesting, **but 80% of Upstart's revenue comes from just three customers.** *"That's what I call customer concentration risk. If they lose one customer, or lose all three, they're done."*
   - **What you want instead:** *"a business that has got thousands or hundreds of customers — so if they lose one or two or three or four, it doesn't matter."*
4. **Can customers switch to substitutes or competitors easily? Are they loyal to the brand?**
   - Easy switching + no brand loyalty = **a lousy business.**
   - ✅ **Apple** — even though **Samsung or Huawei** offer cheaper phones with more features, people still buy iPhone because they are **in the ecosystem where everything works seamlessly** — **locked into the Apple ecosystem** = **high switching cost + brand loyalty**. Apple customers also **tend to have higher disposable income**, are **willing to pay more for a premium product**, and **the customer base is growing all around the world.**

- ✅ **Costco** — people **pay membership fees every year in the US**, so **they keep coming back**; strong brand loyalty.

PILLAR 4 — WEAK SUPPLIERS

- The opposite of Pillar 3: while you want **customers strong and growing**, you want the **suppliers to your business to be WEAK relative to the business.**
- You want your business to have **stronger bargaining power than its suppliers**, so it can **squeeze the suppliers for lower input costs.**
- **If suppliers are stronger than your business, they can easily raise their price and increase your business's costs.**
- **Test: does the business have many alternative suppliers?** If one supplier raises price, the company can go to another, and another. **You want suppliers to have weak bargaining power.**

PILLAR 5 — PROTECTION FROM COMPETITION / STRONG ECONOMIC MOAT ("the most important one")

- *"The one thing that can kill your business or reduce its revenue and profits would be competition wanting to take away your customers."*
- **A great business is one that's able to protect itself against potential competition so competitors cannot take away its customers and market share.** This only happens if the business has a **strong economic moat.** (Introduced in the previous lesson under the three steps of investing; covered in detail in **Lesson 7 — the seven-step Value Momentum Investing formula.**)
- **Sources of a strong moat / competitive advantage:**
  1. **Brand monopoly** — e.g. **Google** in search; **Facebook / Instagram** in social media.
  2. **High switching costs** that lock in the customer for life.
  3. **Huge economies of scale** that competitors cannot compete with.
  4. **High barriers to entry** — difficult for new players to enter the industry because of **regulations** or **high cost of entry**.
- **What makes competition WEAK:**
- **Few competitors** (many competitors is bad)
- Competitors with **very small market share**
- Competitors with a **weaker competitive position**
- **Adam's practice:** *"Before investing in a business I always look at all the closest competitors, to ensure the closest competitors are always weaker than the business I want to invest in. If the competitor is stronger, then I'd rather invest in the competitor."*
- **"You always want to invest in the top dog — the strongest business within the industry. And if there's no one strong business, forget it, don't even invest in that. You want to invest in an industry where there's clearly one leader, and you want to be with that leader."**
- **How this shows up in the numbers:** when a company has a strong competitive advantage and weak competition, it is **able to increase revenue and income consistently, has consistently growing cash flow from operations, and its free cash flow is positive.** *"All this is reflected in those financial metrics you have learned — but now you understand what goes behind creating those metrics."*

PILLAR 6 — Ability to SELF-FINANCE growth

- As a business grows it needs money to grow. **We want businesses that grow ORGANICALLY — generating enough cash to grow themselves, without borrowing more money and without selling more shares.**

- **Questions to ask:**
  1. **Is the company able to grow without issuing new shares or new debt?** If yes, **that's a great business.**
  2. **Does the company have a conservative debt structure?**
  3. **Is the company buying back shares?**
- **"Conservative debt structure" = exactly the metrics already taught:**
- **Current ratio more than 1**
- **Debt/EBITDA ratio 3 or less**
- **Debt servicing ratio less than 30%**
- **ROE and ROIC above 12% or above 15%**
- **On buybacks:** *"In general, we like companies to buy back their own shares — provided they are profitable and provided the shares are undervalued in the first place."*

> **Closing:** *"If it is able to meet these pillars and it is reflected in all the financial metrics, then I know I've got the confidence to invest in this business, and over time it will grow under all economic conditions and thrive."*

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9. Key Takeaways

1. **You are buying a business, not a ticker.** 70% of the investment decision rests on fundamentals. If the business does well, the stock does well over time.
2. **The purpose of fundamental analysis is defensive first.** Finding great companies matters, but **avoiding lousy businesses is what saves you** — investors lose money mainly by owning bad businesses whose prices crash and never recover.
3. **"Blue chip" is not a safety label.** Noble Group, Hyflux and China Evergrande were all blue chips. **"Blue chip can get blue black."**
4. **The statement of cash flows is the most important of the three statements.** Profits can be manufactured (uncollected sales booked as revenue; asset revaluation gains). **Cash cannot be faked.**
5. **Income statement: only three numbers matter** — revenue, operating income, net income after tax. All three should trend up over 5 (ideally 10) years. If net income is lumpy because of one-off items, **operating income is the clean read.**
6. **Balance sheet: only two numbers matter for Adam's work** — **cash + short-term investments** and **total debt (short-term + long-term)**. Both feed the intrinsic value calculation later.
7. **Balance sheet uses the LATEST QUARTERLY report** (a point in time). **Income statement and cash flow statement use ANNUAL figures** (a period of time).
8. **Negative shareholders' equity is not automatically bad** — check whether it is caused by large treasury stock from years of buybacks (McDonald's, Domino's, Yum Brands). In that case ROE is effectively infinite, not negative.
9. **A current ratio below 1 is not automatically bad** — Pepsi does it deliberately by stretching supplier payments; subscription companies like ServiceNow do it because deferred revenue inflates current liabilities. **Never read one ratio in isolation.**
10. **Operating cash flow must be positive and rising; free cash flow must be consistently positive.** Two-to-three consecutive years of negative FCF is dangerous — it forces dilution or more debt.
11. **A falling FCF is only alarming if operating cash flow is falling.** If FCF fell because of a temporary capex surge (the AI data-center build-out at Microsoft, Google, Meta, Nvidia) while OCF keeps rising, that is fine and even strengthens the future moat.

12. **Certain industries are structurally excluded from the standard debt tests** — banks, insurers, property developers, REITs, commodity companies (all naturally highly leveraged). Banks use **CET1 > 10%** and **NPL < 5%**; REITs use **gearing < 45%**. Adam avoids property developers and commodity companies entirely.
13. **Avoid structurally hard industries** — airlines (cyclical, huge maintenance capex, fuel costs, strong unions, brutal competition), shipping, auto manufacturing. Singapore Airlines is a great airline and still not investable. *"It's not the fault of the management. It's the nature of the industry."*
14. **Accounts receivable growing faster than revenue is a major red flag** — it means customers in trouble, poor collection, or **fake revenue**.
15. **Hyflux showed every warning sign for over a decade** — negative/inconsistent OCF from 2006, negative FCF, dividends paid despite negative FCF, receivables +575% against revenue +8% in 2010, and Debt/EBITDA of 5.64 versus the threshold of 3. **None of it required insider knowledge — just reading the cash flow statement.**
16. **Never outsource conviction.** Not to a sovereign wealth fund's presence (Temasek had already sold most of its Hyflux stake at a profit), not to a celebrated CEO, and not to an analyst's buy rating with a $2.40 target on a stock heading to zero.
17. **Great business models share six traits:** falling/low input costs (asset-light, IP-heavy, outsourced manufacturing), the ability to increase output with pricing power / high switching costs / recurring income / multiple revenue streams, a large and growing customer base with no concentration risk, weak suppliers, a strong economic moat against weak competition, and the ability to self-finance growth without issuing shares or debt.
18. **"It's best to invest in a company where any idiot can run the business — because one day an idiot may run the business."** Avoid key-man risk.
19. **All ratios are pre-calculated for you.** You never need to compute ROIC, CCC, debt/EBITDA or CET1 by hand — you only need to know **what they mean, where to find them, and what the threshold is.**
20. **Pass 12 of 12 and you own a great company. Two out of three on the debt ratios is acceptable to Adam.**

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10. Quick Reference — All Metrics and Thresholds

Profitability

Metric	Formula	Threshold / Rule	Exceptions & notes
Sales revenue	—	**Increasing consistently** over 5 yrs (10 better)	One or two down years fine if the trend is up
Net income (net profit after tax)	Revenue − all costs & taxes	**Increasing consistently** over 5 yrs	If distorted by one-off items → use **operating income** instead
Operating income	Gross profit − OPEX	**Increasing consistently**	Excludes tax, interest, lawsuits, restructuring, write-offs, dividend/capital-gain/interest/ forex income
Gross profit margin	Gross profit ÷ Sales revenue	**Consistent or increasing** over 5 yrs; **≥ closest competitors**	Falling gross margin = rising production cost = bad sign
Net profit margin	Net profit after tax ÷ Sales revenue	**Consistent or increasing**; **≥ closest competitors**	If lumpy, use **operating profit margin**
Operating profit margin	Operating income ÷ Sales revenue	**Consistent or increasing**	Backup for a distorted net margin

Metric	Formula	Threshold / Rule	Exceptions & notes
ROE (Return on Equity)	Net profit after tax ÷ Shareholders' equity	**≥ 12% good; > 15% excellent**; consistent over 5 yrs	Warren Buffett's favourite. **Undefined/ infinite (good) when equity is negative from treasury stock**
ROIC (Return on Invested Capital)	EBIT × (1 − tax rate) ÷ (Equity + Total debt − Cash & equivalents)	**≥ 12% good; > 15% excellent**	**Does NOT apply to banks or insurance companies** (leverage makes it artificially low)

Cash Flow

Metric	Formula	Threshold / Rule	Exceptions & notes
Cash flow from operations (OCF)	—	**Positive AND increasing consistently over 5 yrs**	**Does NOT apply to banks, commodity companies, property developers** (naturally inconsistent)
Free cash flow (FCF)	**OCF − Capex**	**Consistently positive** (ideally increasing)	1–2 negative years OK; **2–3 consecutive negative years = dangerous** → dilution/debt. Falling FCF is fine if caused by temporary capex while OCF rises
Cash flow from investing	—	**No good/bad sign.** Negative = buying assets (capex); positive = selling assets	—
Cash flow from financing	—	**No good/bad sign.** Positive = raising money (loans, bonds, new shares); negative = repaying debt, buybacks, dividends	Negative is typical of mature, cash-rich companies — good for shareholders

Financial Strength / Debt

Metric	Formula	Threshold	Exceptions & notes
Current ratio	Current assets ÷ Current liabilities	**> 1**	Below 1 OK if other ratios strong — e.g. **Pepsi 0.85** (supplier payment drag), **ServiceNow 0.95** (deferred revenue)
Debt / EBITDA	(Short-term debt + Long-term debt) ÷ EBITDA	**< 3**	= years of pre-interest/tax/D&A profit needed to repay all debt. **Hyflux was 5.64 = big fail**
Debt servicing ratio	Net interest expense ÷ Cash flow from operations × 100%	**< 30%**	> 30% = very dangerous
Interest coverage ratio *(optional)*	EBITDA ÷ Net interest expense	**≥ 3 safe; ≥ 5 very safe; < 3 risky; < 2 dangerous**	Adam doesn't use it — overlaps with the others
Cash ratio *(optional)*	Cash & equivalents ÷ Current liabilities	**≥ 0.5 very safe; < 0.3 risky**	Excludes receivables and inventory. Adam doesn't use it
CET1 ratio — BANKS ONLY	Common Equity Tier 1 (cash + common stock + retained earnings) ÷ Risk-weighted assets	**Basel III minimum 4.5%; Adam requires > 10%**	DBS **14.1%**, JP Morgan **12.4%**, Bank of China **11.28%**. Just Google it
NPL ratio — BANKS ONLY	Non-performing loans ÷ Total loans	**< 5%**	DBS **1.5%**

Metric	Formula	Threshold	Exceptions & notes
Gearing ratio / Aggregate leverage — REITS ONLY	Total debt ÷ Total assets	**< 45%**	Ascendas REIT **37.6%**. Find it in the REIT's own annual report / presentations

*The three core debt ratios (current, debt/EBITDA, debt servicing) do not apply to: banks, insurance/financial companies, property developers, REITs, commodity companies. Passing 2 out of 3 is acceptable.***

Management Effectiveness

Metric	Formula	Threshold	Exceptions & notes
Cash conversion cycle (CCC)	DIO + DSO − DPO — where DIO = Avg inventory ÷ COGS × 365; DSO = Accounts receivable ÷ Net credit sales; DPO = Days payable outstanding	**Consistent or DECLINING. Increasing = red flag. Negative CCC is best** (collects cash before paying suppliers, e.g. Pepsi)	**Only for retail companies selling physical goods** (Nike, Apple, Walmart, Costco, Pepsi). **Not** for Google, YouTube, Microsoft
Revenue growth vs Accounts receivable growth	Compare the two growth rates	**Revenue must grow faster.** Receivables growing faster = **major red flag**	Signals customers in trouble, poor collection, or **fictitious revenue**. Hyflux 2010: revenue +8%, receivables +575%

Data Sources Cheat Sheet

Need	Go to
US financial statements, ratios, visual charts	**StockOracle** → Financials / Company Metrics (free = 5 yrs, paid = 10 yrs)
Non-US financial statements (Singapore, HK, etc.)	**Yahoo Finance** → ticker (e.g. DBS = **D05**) → Financials (data right-to-left, **in thousands**)
ROE / ROIC for foreign stocks	**Morningstar.com** free section → Key Metrics → Profit and Efficiency Ratios
Next earnings date	**TradingView** "E" icon, or **finance.yahoo.com/calendar/earnings**, or the StockOracle company page
Revenue vs accounts receivable chart	**StockOracle** preset chart, or **TradingView** with the **Piranha Profits custom indicator** (Lesson 4 → Course Notes and Resources → Add to Favorites)
Cash conversion cycle	**TradingView** → Financials → search "cash conversion cycle", or **StockOracle** → Financials chart / Company Metrics → Efficiency Ratios
Bank CET1 and NPL ratios	**Google search** the bank name + ratio name
REIT gearing ratio	The **REIT's own website** → annual report / investor presentations
Raw source documents	Company website: **quarterly reports, annual reports**; also read the **CEO conference call transcript** after results

Part IV — Stock Valuation: Calculating Intrinsic Value

Lesson 5 — Stock Valuation

Source videos: 36 (Introduction & PEG Ratio), 37 (DCF Part 1), 38 (DCF Part 2), 39 (Discounted Net Income), 40 (Price/Book & Banks), 41 (REITs & Price/Sales)

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1. Why Valuation Matters

1.1 The core premise

- Previous lessons taught how to identify a **great business**. This lesson teaches how to know whether the **shares** of that business are expensive or cheap.
- A great business can be a lousy investment if you pay too high a price. A great business becomes an **excellent investment only when bought below intrinsic value**.
- **Property analogy:**
- Great apartment, bank values it at $1,000,000, seller asks $2,000,000. If you pay $2M for a $1M asset, it is very hard to make money on resale unless you wait many years.
- If the seller is desperate and you buy at $800,000 — 20% below valuation — that is a big deal.
- In the stock market **no bank does the valuation for you**. You must do your own.

1.2 Buffett's priority ordering

- Buffett: it is far better to buy a wonderful company at a fair price than a fair company at a wonderful price.
- Adam's restatement: **"cheap crap is still crap."** Only buy an excellent business when it is cheap.
- **Priority: great business FIRST, valuation SECOND.** But valuation is still essential.
- Better to buy an excellent business that is slightly cheap than a mediocre business that is very, very cheap.

1.3 Two things that tell you NOTHING about value

1. **Absolute share price.**
   - A $1,000 stock is cheap if intrinsic value > $1,000.
   - A $5 stock is expensive if intrinsic value is only $1.
2. **How far the price has moved recently.**
   - NVIDIA: up **120%** over the last year, yet on calculation it is **still not expensive — fairly priced**.
   - AMD (NVIDIA's competitor): **fell 50%** over the last year and is **still expensive — still above intrinsic value**.
   - What matters is only: intrinsic value today vs. share price today.

1.4 What drives value

- When you buy shares you buy a piece of a business. A business is a **money-making machine**. The more money it makes, the more it is worth.
- How much a company makes is measured by:
- Cash flow from operations (CFO)
- Free cash flow (FCF)

- Net income

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2. The Five Valuation Methods — Overview

#	Method	When used	Key precondition
1	**PEG ratio**	Quick check / stock screening only	Earnings positive and consistently increasing
2	**Discounted Cash Flow (DCF)** — Adam's default, used **90% of the time**	Most companies; most comprehensive & accurate	CFO consistently increasing; FCF positive
3	**Discounted Net Income**	Companies whose business model makes cash flow inconsistent/negative (finance, insurance, brokers, asset managers)	Net income consistently increasing
4	**Price to Book (P/B)**	Banks; REITs; property developers; loss-making / distressed / near-bankrupt companies	Meaningful balance sheet / book value
5	**Price to Sales (P/S)** and **Price to Sales Growth (PSG)**	Revenue growing but not yet profitable; highly cyclical companies with inconsistent profits	Revenue is measurable and meaningful

- Why DCF is the default: a business is a money-making machine, and the best measure of how much it makes is the **cash it generates for shareholders**. Value of a business = **present value of all expected future cash flows**. Adam projects **20 years** and discounts to present value.
- Why net income is second-best: net income is an **accounting entry that can be manipulated/distorted** by non-operating income and expenses. Cash is more real than accounting profits.

2.1 Tools mentioned

- **Intrinsic Value Calculator (Excel template)** — download under **Lesson 5 → Course Notes and Resources**, scroll to the bottom. Once opened, the **tabs at the bottom** hold a separate calculator for each method (discounted FCF, discounted CFO, discounted net income, price-to-book, etc.).
- **Only key into the WHITE boxes.** Coloured boxes contain the formulas — editing them breaks the calculator.
- The discount-rate table inside the calculator is **updated every year** and re-sent to students; a link inside the calculator points to the source where you can compute the risk-free rate and average market risk premium yourself.
- **StockOracle (Pro subscription)** — auto-calculates intrinsic value under the **Intrinsic Value** tab, showing each method side by side, then an algorithm picks the most appropriate one and displays it as the **"Oracle Value."** Pro also has a **built-in intrinsic value calculator** with all inputs pre-pulled from the balance sheet and cash-flow statement, which you can override with your own assumptions.
- Data provider for growth estimates: **FactSet**.
- Currently **US-listed stocks only**; Singapore and Hong Kong listings expected **by end of 2026**.
- Free StockOracle account is enough to read the financial statements.
- **FinViz** — PE, PEG, EPS growth next 5 years, plus a **stock screener** (~9,900 stocks in the database).
- **Zacks (zacks.com)** — third opinion on projected growth rate.
- **GuruFocus** and **SimplyWallStreet** — also carry PEG.
- **Yahoo Finance** — required for **non-US listed stocks** (financials, analyst growth estimates, statistics, book value per share, dividend yield).

- **TradingView** — charting; add the **"Price to Book Ratio" indicator** to read historical P/B for any market (Singapore, Hong Kong, India, Malaysia, US).
- **Ultimate Investors Playbook** (subscription) — Adam's monthly workings, buy levels and fair values.

2.2 StockOracle worked illustration (NVIDIA)

- Current share price: **$184**
- Discounted cash flow (latest FCF): intrinsic value **$258**
- Discounted net income (latest net income): **$370**
- Price to sales ratio method: **$226**
- **Oracle Value (algorithm blend): $208** → at $184 the stock is **slightly undervalued**
- Lesson: different methods give different numbers. **Intrinsic value is a range, never one number.**

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3. Method 1 — PEG Ratio

3.1 Purpose and limits

- Use as a **very quick check** on value and, mainly, as a **screening criterion** to shortlist stocks for deeper research (there are >10,000 stocks; you cannot examine them all). Screening covered fully in Lesson 9.
- Adam does **not** use PEG to actually value a stock — it is not comprehensive.
- **Precondition:** earnings must be **positive AND increasing consistently**. Otherwise PEG is not accurate at all.
- **Limitation:** based on reported net income, which can be distorted by non-operating expenses and non-operating income. That is why a full DCF is needed.

3.2 Formula and thresholds

```
PEG ratio = PE ratio ÷ projected earnings growth rate (%)
```

- Higher PEG = more expensive. Lower PEG = cheaper.
- **Textbook / CFA theory:** PEG > 1 → overvalued (PE above growth rate); PEG < 1 → undervalued.
- **Problem with the textbook rule in practice:** if you only buy PEG < 1 you get lots of cheap stocks, but many of them are not great companies — and **many great companies trade above PEG 1**, so you miss them.
- **Adam's rule of thumb: look for PEG ≤ 1.5.**
- Way above 1.5 → probably expensive.
- 1.5 or less → worth a further look.

3.3 Worked examples (from FinViz)

NVIDIA

1. PE ratio = **54.69** (~54×)
2. Projected earnings growth, next 5 years = **61.17%**
3. PEG = 54.69 ÷ 61.17 = **0.89**
4. Below 1.0 and below 1.5 → **looks cheap**

AMD (share price down >50% over the year)

1. PE ratio = **112** (despite the 50% fall)
2. Projected earnings growth, next 5 years = **33%**

3. PEG = 112 ÷ 33 = **3.4**
4. Way above 1.5 → **expensive**

Conclusion: NVIDIA is the cheaper stock, even though it rose 120% and AMD fell 50%.

3.4 Screening use (FinViz)

- FinViz screener database: **9,900 stocks** — impossible to review individually.
- FinViz does not offer a 1.5 cut-off, so Adam sets **PEG < 2**, which narrows ~10,000 stocks down to about **700**.
- Other criteria are then layered on to narrow further (Lesson 9).

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4. Method 2 — Discounted Cash Flow (DCF)

Adam's primary method — used ~90% of the time.

4.1 The concept

- A business is a machine that will print cash for the next 10, 20, 30 years. To value the machine, **project the future cash it will generate and discount it to present value**. Divide by shares outstanding → **intrinsic value per share**.
- Formal definition: *the intrinsic value of a stock is the present value of its future expected cash flows* (Adam prefers cash flow over earnings because cash is more tangible than accounting profits).
- **Default projection horizon: 20 years.**

4.2 The magic-machine teaching sequence (discount rate = 5%)

Scenario	Cash timing	Present value calculation	Value today
1	$100 today (t=0)	none needed	**$100**
2	$100 in 1 year	100 ÷ 1.05	**$95**
3	$100 in 2 years	$100 \div 1.05^2$	**$91**
4	$100 today + $100 in yr 1 + $100 in yr 2	100 + 95 + 91	**$286**

- **Discount rate = the opportunity cost of your money** — what you could earn elsewhere at the same risk (5% in the example).
- Reverse check: $95 invested at 5% for 1 year → $100. $91 invested at 5% for 2 years → $100. This is the **time value of money**.
- Valuing a business is identical, except the machine prints cash for **20 years**, not 3. In addition, the calculator brings in the company's **total debt** and its **cash and equivalents** to arrive at the value of the business today.

4.3 Preconditions for using DCF

- The company must generate **positive cash flow**.
- **Cash flow from operations must be increasing consistently.**
- You must be able to confidently expect that consistent increase to continue for the next 20 years.
- "Consistently increasing" does **not** mean up every single year — one or two down years are fine as long as the general trend is up. (MasterCard's free cash flow chart is the illustration of what a clean uptrend looks like.)

4.4 Which cash flow to use

- **Default: Free Cash Flow (FCF).**
- `FCF = Cash flow from operations − Capital expenditure`
- It shows how much cash the business generates from operations after capex — what is actually **left for shareholders** (to pay dividends, buy back shares, or retain). This is the **true earning power** of the business.
- Alternatives when FCF is distorted — see §5.

4.5 Input-by-input procedure (Intrinsic Value Calculator)

Step	Input	Where to get it
1	Stock name	—
2	Ticker symbol	Any financial site (StockOracle, Yahoo Finance)
3	**Valuation method** = Discounted Free Cash Flow	Dropdown in the calculator
4	**Latest free cash flow (last 12 months)**	Financials → **Cash Flow Statement → Annual → first column = last 12 months**
5	**Total debt** = short-term debt + long-term debt (**NOT total liabilities**)	**Balance Sheet → Quarterly (latest point in time) → Liabilities**
6	**Cash and short-term investments**	**Balance Sheet → Quarterly → Assets**
7	**Growth rate years 1–5**	Consensus analyst estimates (see §4.6)
8	**Growth rate years 6–10**	Base case = same as years 1–5, **capped at a maximum of 15%**
9	**Growth rate years 11–20**	**Default 4%**
10	**Shares outstanding**	Overview → shares outstanding → use the **diluted average**
11	**Discount rate**	From the beta table in the calculator (see §4.7)
12	Currency of financial statements / currency of share price / exchange rate	Needed for non-US stocks (§6.1)

- **Denomination warning:** StockOracle reports in **millions**; Yahoo Finance reports in **thousands**. Always check and convert before keying in — the calculator expects millions.

4.6 Growth rate — sourcing and rules

- Use **consensus analyst estimates**. Hundreds of analysts study the company, and most US companies give forward guidance.
- Three sources, take an **average of the three** (Adam's illustration: "like going to different doctors, you get different opinions"):
  1. **StockOracle → Company Metrics → scroll to Growth Rates → Projected** (data from FactSet)
     - Use the **projected 3–5 year EPS growth rate** as the proxy for the cash-flow growth rate, because **over the long run earnings and cash flow grow in line**.
     - A **projected long-term growth rate** (beyond 5 years) is also shown — useful for conservative cases.
  2. **FinViz** → "EPS next 5Y" / earnings growth next 5 years
  3. **Zacks** → projected growth rate
- **Years 6–10 rule (Adam's own rule): cap at a maximum of 15%.**
- If years 1–5 = 15.53% → years 6–10 = **15%**

- If years 1–5 = 8% → years 6–10 = **8%**
- If years 1–5 = 20% → years 6–10 = **15%**
- **Years 11–20 rule: 4% by default**, on the assumption the business then grows at roughly **nominal GDP + 2%**. You may use a lower figure (e.g. 2%) if you are more conservative — this is a style choice.

4.7 Discount rate — why Adam's number

- **Not WACC.** Weighted average cost of capital is what CFA/finance-degree curricula use, and it is on StockOracle, but WACC belongs to a **perpetuity (DCF Terminal)** valuation. Adam is not assuming the company lasts forever, so he uses a **modified discount rate**.
- **Buffett's approach (mentioned, not used):** simply the **10-year Treasury bond yield / 10-year Treasury rate** — quoted in the lesson as **3.6%**.
- **Adam's formula:**
  `Discount rate = risk-free rate + (beta × market risk premium)`
- Uses the **5-year average risk-free rate** and the **5-year average market risk premium**.
- You do **not** compute this yourself — the calculator contains a **pre-built lookup table keyed to the stock's beta**, updated annually.
- **What beta means (volatility measure):**
- Beta 1.0 (e.g. Apple): S&P 500 up 1% → stock up 1%; S&P down 1% → stock down 1%.
- Beta 3.0 (e.g. Tesla): S&P up 1% → stock up 3%; S&P down 1% → stock down 3%.
- Beta 0.5 (e.g. Coca-Cola): S&P up 1% → stock up 0.5%; down 1% → down 0.5%.
- **Higher beta → higher discount rate**, i.e. the stock is "punished" for volatility (financial theory). Buffett rejects this and just uses the risk-free rate.
- **Beta → discount rate pairs observed in the worked examples:**

Beta	Discount rate used	Example
0.28	**5.8%** (the lowest rate in the table)	Singapore Exchange (S68)
0.83	**6.1%**	MasterCard (MA)
1.1	**6.6%**	Apple (AAPL)
1.4	**7.4%**	Amazon (AMZN)
1.49	**7.7%**	BlackRock (BLK)

- Where beta falls between table rows, pick the **closest** rate.
- Adam applies the same (US-derived) table to Singapore stocks "to keep it simple."

4.8 Why 20 years and not perpetuity

- Finance theory / CFA assumes the company lasts **forever** (to perpetuity) and applies a **terminal value**. Adam considers that unrealistic and **more aggressive than he is comfortable with**.
- He values the business on **20 years of cash generation only** — anything beyond 20 years is a **bonus**.
- Reality check that 20 years is conservative — company ages cited:
- **Coca-Cola: 106 years**
- **Walmart: 55 years**
- **Apple: 45 years**
- **Microsoft: 39 years**

- A **"DCF Terminal"** (perpetuity) valuation does exist and is auto-calculated on StockOracle, but Adam does not use or teach it.

4.9 WORKED EXAMPLE — MasterCard (MA), Discounted Free Cash Flow

Precondition check: MasterCard's free cash flow chart (dark green) shows a **consistent uptrend** → DCF valid.

Step	Input	Value
1	Stock / ticker	MasterCard / **MA**
2	Method	Discounted **free cash flow**
3	Latest FCF (last 12 months, cash flow statement, annual)	**16,186** (millions USD — StockOracle is already in millions)
4	Total debt = short-term **749** + long-term **18,251**	**19,000** million ($19B)
5	Cash & short-term investments (latest quarter, assets)	**11,459** million
6a	Growth yrs 1–5, StockOracle/FactSet (3–5 yr EPS)	14.94%
6b	Growth yrs 1–5, FinViz	15.68%
6c	Growth yrs 1–5, Zacks	15.98%
6d	**Average of three** = (14.94 + 15.68 + 15.98) ÷ 3	**15.53%**
7	Growth yrs 6–10 — base case = same rate, but capped at 15%	**15%**
8	Growth yrs 11–20	**4%**
9	Shares outstanding (diluted average)	**898** million
10	Beta **0.83** → discount rate	**6.1%**

Results:
- **Base case intrinsic value = $655.97**
- Current share price = **$514.72**
- → **21.5% undervalued**

Sensitivity / conservatism ladder for MasterCard:

Scenario	Yrs 1–5 growth	Yrs 6–10 growth	Yrs 11–20	Intrinsic value	Discount at $514.72
Base case	15.53% (avg of 3 sites)	15% (capped)	4%	**$655.97**	21.5% undervalued
Conservative	14.94% (lowest of the 3)	13.02% (StockOracle **long-term** growth rate)	4%	**$600**	~14% discount
Doomsday	14.94%	9.97% (= 14.94 ÷ 2, growth halves after yr 5)	4%	**$545.50**	~5% below IV

- Interpretation: even under the doomsday case the stock is below intrinsic value → *"these are the kind of stocks I like to buy because it's pretty safe even if things don't work out as I planned."*

4.10 Intrinsic value is a range — margin of safety

- **Buffett & Munger on American Express:** Buffett's calculation → **$60**; Munger (more pessimistic, lower growth rate) → **$50**; the market price was **$35**. The disagreement did not matter because the price was far below both.

- Lesson: **buy when the price is way below intrinsic value so there is a margin of safety** — so that even if your growth assumptions were too optimistic, you are still safe.
- Practical margin-of-safety behaviour shown in the lesson:
- MasterCard at 21.5% below base-case IV and still below the doomsday IV → attractive.
- Apple at $260.83 vs base-case IV $256 → **not bought**; would add "below $190" (well under both the conservative $199.98 and StockOracle's $190).
- Singapore Exchange at $18.37 vs IV $17.67 → **not bought**; interested at $17, $16, $15 or $12.
- BlackRock at $967 vs IV $795 → **not bought**; interested at "$700-plus or even below $700."
- Distressed companies: buy at **P/B ≤ 0.5**, i.e. 50 cents on the dollar of liquidation value.

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5. DCF Part 2 — Handling Inconsistent Cash Flow

5.1 The diagnostic

```
Free cash flow = Cash flow from operations − Capital expenditure
```

Two different causes of inconsistent FCF, with two different answers:

Cause	Verdict
CFO itself is inconsistent	**Cannot use DCF at all.** Use a different method, or forget the stock and find one with consistent cash flow growth.
CFO is consistent but CAPEX is lumpy	**Fine.** Normalize the capex (§5.2).

- **Capex** = money invested in new plants, machinery, equipment. Some companies spend evenly every year; others spend in bursts (data centres, heavy R&D), which crushes FCF in those years and inflates it in quiet years.
- **Hyperscaler illustration:** Microsoft, Meta, Google and Amazon recently spent enormously on **AI data centres** to strengthen their economic moat and future growth. That capex spike knocked their FCF down — **not a concern**, because the underlying business is generating plenty of operating cash.

5.2 Capex normalization rule

```
5-year average capex = (capex yr1 + yr2 + yr3 + yr4 + yr5) ÷ 5

Normalized FCF = latest 12-month cash flow from operations − 5-year average capex
```

Use this normalized FCF as the starting cash flow in the DCF calculator.

5.3 WORKED EXAMPLE — Amazon (AMZN), normalized FCF

Why normalization is required: Amazon's CFO (orange) rises fairly consistently, but reported FCF (dark green) swings wildly — **−$14B, −$16B, +$32B, +$32B, then down to $7B** — because capex tripled (from roughly 50,000 to 150,000 million) building data centres, robots and automation. Using the latest FCF would make intrinsic value artificially low.

Step	Input	Value
1	Stock / ticker / method	Amazon / **AMZN** / discounted free cash flow
2a	Capex, 5 years	61,053 + 63,645 + 52,729 + 82,999 + 131,819

Step	Input	Value
2b	÷ 5 → **5-year average capex**	**78,449** million
2c	Latest cash flow from operations	**139,514** million
2d	**Normalized FCF** = 139,514 − 78,449	**61,065** million
3	Total debt = short-term **2,748** + long-term **65,648**	**68,396** million
4	Cash & investments (latest quarter)	**123,329** million
5a	Growth, StockOracle/FactSet 3–5 yr	18.95% (long-term rate shown as 13.2%)
5b	Growth, FinViz	18.42%
5c	Growth, Zacks	17.15%
5d	**Average → years 1–5**	**18.17%**
6	Years 6–10 (base 18.17% but capped)	**15%**
7	Years 11–20	**4%**
8	Shares outstanding (diluted average)	**10,863** million
9	Beta **1.4** → discount rate	**7.4%**

Results:
- **Intrinsic value = $206**
- Current share price = **$214.33**
- → **3.99% overvalued** (slightly)

5.4 Using Cash Flow from Operations instead of Free Cash Flow

- Permitted **only if** CFO is **very close to net income** — specifically, **CFO must not be more than 20% above net income**.
- Justification cited: **Buffett has said he does not deduct capital expenditure in his valuation if the capex is growth capex rather than maintenance capex.**
- In the calculator, select the method **"Discounted Cash Flow from Operations."**

5.5 WORKED EXAMPLE — Apple (AAPL), Discounted Cash Flow from Operations

Eligibility check: operating cash flow **135,472** vs net income **117,777** (117.78B). 135,472 ÷ 117,777 → less than a 20% difference → **CFO method allowed**.

Step	Input	Value
1	Stock / ticker / method	Apple / **AAPL** / discounted **cash flow from operations**
2	Latest 12-month net operating cash flow	**135,472** million
3	Total debt = short-term **13,824** + long-term **76,685**	**90,509** million
4	Cash & short-term investments (latest)	**66,907** million
5a	Growth, StockOracle (3–5 yr)	12.31%
5b	Growth, FinViz	11.23%
5c	Growth, Zacks	13.33%

Step	Input	Value
5d	**Average → years 1–5**	**12.29%**
6	Years 6–10 (base case = same, under the 15% cap)	**12.29%**
7	Years 11–20	**4%**
8	Shares outstanding	**14,810.36** million
9	Beta **1.1** → discount rate	**6.6%**

Results:
- **Base case intrinsic value = $256**
- Current share price = **$260.83** → **1.8% overvalued**
- **Conservative case:** years 1–5 = **11.23%** (lowest of the three), years 6–10 = **5.61%** (11.23 ÷ 2) → **intrinsic value $199.98 (~$200)**
- StockOracle's own Oracle value = **$190** — lower because it applies a lower growth assumption.
- **Adam's decision:** would **not** buy Apple at $260 even on the optimistic $256 number. Would be attracted to add shares **below $190**.

5.6 When no cash-flow method works — Singapore Airlines

- Singapore Airlines' **operating cash flow is not consistent at all** — sideways, down, up, down. Net income (light green) also alternates up and down.
- Because CFO is inconsistent (not just capex), **DCF cannot be used**.
- Adam would **never buy it in the first place**: it fails a key criterion — **no consistently increasing revenue, net income and cash flow from operations**.
- If forced to value it, the only usable method is the **mean price-to-sales ratio method** (used for very cyclical companies with inconsistent profits).

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6. Non-US Listed Stocks (procedure differences)

6.1 Data sourcing and currency handling

- StockOracle is **US-only for now** (Singapore + Hong Kong expected by end of 2026). For non-US stocks use **Yahoo Finance**.
- **Ticker conventions:** Singapore tickers mix **letters and numbers** (e.g. S68, U11); Hong Kong tickers are **numbers** (e.g. 9988).
- **Reading direction:** StockOracle presents financials **left → right** (newest first); Yahoo Finance presents **right → left** (2022, 2023, 2024, 2025, then LTM). Yahoo typically gives only ~4 years of data.
- **Denomination:** Yahoo reports in **thousands**; convert to **millions** for the calculator.
- **Currency:** the calculator asks for (a) currency of the financial statements, (b) currency of the share price, and (c) the exchange rate between them.
- Singapore Exchange: statements in SGD, price in SGD → exchange rate = **1**.
- **Alibaba illustration:** financial statements are in **Chinese renminbi**; the share price depends on the listing — **BABA** in the US is priced in **USD**, while **9988** in Hong Kong is priced in **HKD**. Enter the appropriate exchange rate.

6.2 WORKED EXAMPLE — Singapore Exchange (S68), Discounted Free Cash Flow

Precondition checks from Yahoo Finance (SGD thousands):
- Operating cash flow: **583 → 446 → 615 → 841 → 826** — not up every year but a general uptrend → acceptable.
- Free cash flow: **539 → 392 → 551 → 773** — general uptrend → acceptable.
- Capex: **44 → 54 → 64 → 68 → 71** — rising smoothly, **no sudden spike** → **no capex normalization needed**, use the standard discounted FCF method.

Step	Input	Value
1	Stock / ticker / method	Singapore Exchange / **S68** / discounted free cash flow
2	Currencies	Statements SGD, price SGD, exchange rate **1**
3	Latest FCF = 755,255 thousands → convert	**755.255** million
4	Debt: short-term/current **326,360** + long-term **299,667** (thousands) → keyed in as	**332.303** million *(figure as stated in the lesson; note the stated sum does not reconcile arithmetically with the two components — likely a transcription artefact)*
5	Cash & equivalents = 2,046,023 thousands → convert	**2,046.023** million
6	Growth, years 1–5	**8.22%**
7	Growth, years 6–10 (assume it continues)	**8.22%**
8	Growth, years 11–20	**4%**
9	Shares outstanding: Yahoo → Statistics → **implied shares outstanding** 1.07 billion (1 billion = 1,000 million)	**1,070** million
10	Beta **0.28** (Yahoo → Summary) → discount rate (lowest in table)	**5.8%**

Growth-rate limitation for non-US stocks:
- There is generally **no 5-year or long-term growth estimate** for Singapore, India or Australian listings — those companies do not give long-range guidance, and FinViz/Zacks do not cover them.
- **Best available:** Yahoo Finance → **Analysis / Analyst → Growth Estimates → next-year growth**, plus the historical growth rate. For S68 that was **8.22%**.

Results:
- **Intrinsic value = $17.67**
- Current share price = **$18.37** → **~3.9% overvalued**
- Decision: **would not buy**. Interested if it falls well below $17.67 — at $17, $16, $15 or $12.

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7. Method 3 — Discounted Net Income

7.1 When to use

- Preference is always DCF, because **cash is more real than accounting profits**.
- But certain business models make cash flow **inconsistent or not always positive by nature**:
- **Finance companies**
- **Insurance companies**

- **Brokers**
- **Asset management companies**
- Reason: their cash flow is distorted by **loans issued and loans redeemed**, and by **customers putting cash into or taking cash out of the funds they manage** — this masks the underlying operations.
- **Precondition: net income must be increasing consistently.** Rule of thumb applied in the lesson: **use whichever series is the more consistent** — if operating cash flow is volatile but net income trends up steadily, use net income.

7.2 Procedure (differences from DCF only)

- Calculator method = **Discounted Net Income**.
- Step 1 input becomes **latest 12-month net income** from the **Income Statement → Annual**, instead of free cash flow from the cash flow statement.
- Every other input is identical: total debt (short + long-term, latest quarter), cash & investments, growth rates for years 1–5 / 6–10 / 11–20, shares outstanding, beta-based discount rate, currencies.

7.3 WORKED EXAMPLE — BlackRock (BLK), Discounted Net Income

Company note: BlackRock = world's biggest **asset management** company (ETFs and public assets). Not to be confused with **Blackstone**, which is private equity.

Precondition check: on the financials chart, net income (green) rises consistently (not every year, but a clear uptrend) and is **far more consistent than operating cash flow**, which is volatile → use net income.

Step	Input	Value
1	Stock / ticker / method	BlackRock / **BLK** / discounted **net income**
2	Latest 12-month net income (income statement, annual)	**5,553** million
3	Total debt: long-term **12,768**, short-term **none**	**12,768** million
4	Cash & investments (latest quarter)	**11,468** million
5	Growth, years 1–5 — StockOracle/FactSet 3–5 yr projected	**11.53%** (could also average with Zacks and FinViz)
6	Growth, years 6–10	**11.53%** (same rate)
7	Growth, years 11–20	**4%**
8	Shares outstanding	**165.4** million
9	Beta **1.49** → discount rate (closest row)	**7.7%**
10	Currencies	Statements USD, price USD

Results:
- **Intrinsic value = $795**
- Current share price = **$967 → 21% overvalued**
- **Why StockOracle shows a higher number ($960):** StockOracle uses the **projected longer-term growth rate of ~16% (16.07%)** rather than the 11.53% 3–5 year figure. Substituting 16.07% into the years 6–10 box pushes intrinsic value **above $900**.
- **There is no right or wrong** — intrinsic value is a range determined by how optimistic or conservative your growth assumption is.
- **Decision:** at $967 this is not a good deal. Interested only at roughly **$700-plus or below $700**.

8. Method 4 — Price to Book Ratio

8.1 The three use cases

1. **Companies with no profits / making losses** — especially **temporary** losses:
   - Banks during the **Great Financial Crisis**
   - Retailers and other companies during **COVID-19**
   - **Distressed businesses**, usually cyclical companies temporarily unprofitable because of an **industry recession or crisis**
   - Also used when a company is **deeply undervalued relative to liquidation value**
2. **Banks** — even highly profitable ones. *"Banks are a very unique animal."* Analysts (Adam included) do **not** use DCF or discounted net income for banks.
3. **Property developers and REITs** — buying a REIT means buying a portfolio of properties.

- Why you have no alternative for loss-makers: **no earnings → no PE**; **negative cash flow → no DCF**; **negative income → no discounted net income**. The only thing left to compare price to is **book value**.

8.2 Book value and liquidation value

- **Liquidation value:** if the company closes down, sells all its assets (property, machinery, equipment), collects the cash and pays off all its debts, **what is left over for shareholders** is the liquidation value = **book value**.

```
Book value = Total assets − intangible assets (goodwill, brand) − total liabilities
Book value per share = Book value ÷ total shares outstanding
Price to Book ratio (P/B) = Share price ÷ Book value per share
```

- Intangibles are deducted because **goodwill and brand disappear when the company closes down**.

8.3 Simple illustration

1. Liquidation value of the whole company = **$100 million**
2. Shares outstanding = **100 million**
3. Book value per share = 100M ÷ 100M = **$1.00**
4. Share price = **$0.50**
5. P/B = 0.50 ÷ 1.00 = **0.5** → you are **paying 50 cents for a dollar** of liquidation value
   - If the company goes bust and liquidates, you get back **more than you paid**.

8.4 Thresholds

Situation	Threshold
Loss-making / distressed company	Ideally **P/B ≤ 0.5** (buying at 50 cents on the dollar)
Banks, general rule	**P/B 1.2 – 1.4 = fairly valued**
Banks, cheap	**Below 1.2**, and especially **below 1.0**, = really undervalued
Banks, more accurate method	Compare **current P/B to the bank's own 5-year average P/B**: above average = overvalued, below = undervalued
REITs, general rule	Do not pay **more than 1.2× P/B (price to NAV)**

Situation	Threshold
REITs with high double-digit DPU growth	Up to **1.5×** is still acceptable

- **Historical example:** in the Great Financial Crisis (2009) Adam bought US banks including **Bank of America at a P/B of 0.4** — 40 cents on the dollar; even in liquidation he would theoretically get $1 back for every 40 cents paid.

8.5 Fair value formula for banks / REITs

```
Fair value (intrinsic value) = (5-year average P/B) × (current book value per share)
```

8.6 WORKED EXAMPLE — Bank of America (BAC)

1. Current share price = **$48.56**
2. Book value per share (StockOracle → Overview) = **$38.44**
3. Current P/B = 48.56 ÷ 38.44 = **1.26** (StockOracle's own "Valuation Ratios → P/B, most recent quarter" shows **1.25** — rounding difference)
4. 5-year P/B history (StockOracle → Company Metrics → Valuation/Price Ratios → P/B): **1.08, 1.01, 1.24, 1.43, 1.26**
5. **5-year average P/B = 1.24**
6. Current 1.26 > average 1.24 → **slightly overvalued**
7. **Fair value = 1.24 × 38.44 = $47.67**
8. Share price $48.56 vs fair value $47.67 → **slightly overvalued**

8.7 Singapore banks — the two-filter approach

Adam buys the three Singapore banks **primarily for dividends** (though they also give good capital gains, especially **DBS**, which is a relatively high-growth bank). Two checks must both be considered:

Filter 1 — Dividend yield
- Target (as of the recording, 26 March, and dependent on the interest-rate environment): **at least 4.5% to 5%**. Below that, he will not buy.

```
Maximum price to pay = Dividend per share (last 12 months) ÷ target dividend yield
```

- **UOB (ticker U11 on SGX), via Yahoo Finance → Statistics:**
  1. Current dividend yield = **4.3%** → below the 4.5–5% requirement
  2. Trailing 12-month dividend per share = **$1.56**
  3. Required price for a 4.5% yield = 1.56 ÷ 0.045 = **$34.67**
  4. Current price = **$36** → too expensive; needs to fall to **$34 or lower**

Filter 2 — Fair value from P/B
- **UOB fair value calculation:**
1. Current book value per share (Yahoo Finance → Statistics → Balance Sheet section; also on StockOracle) = **$29.35**
2. 5-year P/B history read from **TradingView** (add the "Price to Book Ratio" indicator, set 5-year view, read the value on the **first trading day of January** for each of the last 5 years): **2026 = 1.14**, **2025 = 1.30**, plus **1.09** among the earlier years
3. Sum the five readings and divide by 5 → **5-year average P/B = 1.18**
4. **Fair value = 1.18 × 29.35 = $34.63**
5. Current price **$36** → slightly overvalued

- **Adam's stance:** for Singapore banks he is willing to buy **slightly above fair value** as long as the dividend yield is at least **4.5–5%** — but **not too far above**.

8.8 The Price-to-Book Ratio Calculator (Excel tab)

- Located in one of the tabs of the **Intrinsic Value Calculator** workbook, called the **Price to Book Ratio calculator**.
- **Inputs:** current **book value per share** + the **P/B ratio for each of the last 5 years** (five numbers).
- **Outputs:** the fair value **plus the +1 and −1 standard deviation levels** — giving a buy range rather than a single point.
- **UOB application:** fair value ≈ $34, +1 SD ≈ $37 → willing to buy in that band, but **only if the dividend yield is at least 4.5–5%**.

8.9 Update frequency and a full applied example (OCBC)

- **How often to update:** once a **quarter** is fine; **Adam updates monthly**. (Subscribers to the Ultimate Investors Playbook receive his monthly workings, buy levels and fair values.)
- **OCBC (latest calculation shown):**
- **Fair value = $16.39**
- Current share price = **$20** → **overvalued**, with quite a bit to fall before it is attractive
- **+1 standard deviation price = $19.25** — the maximum he would pay above fair value (current price is above even this → far too expensive)
- **5% yield line = $16.40** — only at $16.40 or lower does OCBC yield 5%
- **Four identified buy levels: $17.87, $16.35, $14.40, $13.48**
  - Derived from **technical support levels** (taught in the next lesson) **combined with** the yield requirement
  - Example: buying at **$17.87** is within one standard deviation and gives roughly a **4.5% yield**
- Method: **combine fundamentals (fair value, yield) with technicals (support levels)** to set buy levels.

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9. REITs — Price to Book / Price to NAV + Yield

9.1 Framework

- A REIT (Real Estate Investment Trust) is effectively **a portfolio of properties**. The best ways to value it are **price-to-book ratio** and **dividend yield**.
- Adam specializes in **Singapore-listed REITs** because they pay very high dividend yields and those dividends are **tax-free**.
- **NAV = Net Asset Value = estimated market value of the REIT's total assets − total liabilities.** NAV per share = NAV ÷ number of units.
- For Singapore-listed REITs, **NAV and book value are the same thing** and **price-to-NAV and price-to-book are used interchangeably**, because the book value of a REIT is **marked to market**. (US REITs are slightly different; the lesson focuses on Singapore REITs.)

9.2 Interpreting P/NAV

- **P/NAV = 1.0** → the REIT trades at exactly the market value of its properties less debt.
- **P/NAV = 1.2** → the REIT trades **20% above** the market value of its properties less debt.
- **Rule of thumb: do not pay more than 1.2×.**
- **Exception:** REITs with **high double-digit growth in DPU (distributions per unit)** can justify **up to 1.5×**.

- **Best practice:** compute the individual REIT's own **average P/B** (5-year or 10-year) and derive a fair value from that.

9.3 The two REIT checks

Check 1 — Dividend yield
- Want **at least 5%** in today's interest-rate environment (may change as rates change).

```
Maximum price to pay = DPU (distributions per unit) ÷ 5%
```

Check 2 — Price to NAV / Price to Book

```
Fair value = (10-year or 5-year average P/B) × (current book value / NAV per share)
```

9.4 WORKED EXAMPLE — CapitaLand Ascendas REIT

1. Pull the REIT up on **TradingView**, add the **Price to Book Ratio indicator**, set the **10-year** view.
2. **Proper method:** read the P/B for each year (2025, 2024, 2023, ...), sum all 10, divide by 10.
3. **Shortcut method** (Adam's admitted lazy version, gives a roughly similar number):
   - 10-year **high** P/B = **1.61**
   - 10-year **low** P/B = **1.03**
   - (1.61 + 1.03) ÷ 2 = **1.32** ≈ the 10-year average
4. **Average P/B = 1.32** → below 1.32 = cheap, above 1.32 = expensive. This alone is a very quick valuation test.
5. Current book value per share (Yahoo Finance) = **$2.27**
6. **Fair value = 1.32 × 2.27 = $2.99**
7. Current price = **$2.64** → **undervalued** (below the $2.99 fair-value line)
8. **Yield check:** current dividend yield = **5.759%**, better than the 5% requirement → also attractive
9. **Conclusion:** attractive on both the price-to-book and the yield test.

---

10. Method 5 — Price to Sales (P/S) and Price to Sales Growth (PSG)

10.1 When to use

Values a company on **revenue** — not profits, not cash flow. Two circumstances:
1. **Speculative growth stocks that are not yet profitable** — revenue is growing but the company has not reached economies of scale. One day it could be profitable, so you might take a small position first.
2. **Highly cyclical companies** whose profits swing up and down with the economic cycle:
- **Property developers**
- **Construction companies**
- **Shipping companies**
- **Commodity companies**
- **Utility companies**
- (Also the only method available for something like Singapore Airlines — see §5.6, the "mean price to sales ratio method.")
- In these cases PE ratio and discounted net income / discounted cash flow **do not work**.

10.2 Formulas

```
Price to Sales ratio = Share price ÷ Revenue per share
Fair value = (5-year average P/S ratio) × (current revenue per share)
```

- No need to calculate P/S manually — available on any financial website including StockOracle (**Company Metrics → Price Ratios**).
- Quick read: **current P/S above the 5-year average → overvalued; below the 5-year average → undervalued.**

10.3 WORKED EXAMPLE — Constellation Energy (CEG)

Why P/S: on the financials bar chart, net profit (green) goes up, down, down, up, down — **very inconsistent**, unlike Microsoft or Apple. PE ratio, discounted income and discounted cash flow do not work.

1. P/S history (StockOracle → Company Metrics → Price Ratios):
   - 2022 = **1.16**
   - 2023 = **1.79**
   - 2024 = **2.99**
   - 2025 = **4.13**
   - Most recent = **3.71**
2. **5-year average P/S = 2.52**
3. Current P/S **3.71 > 2.52** → **already tells you it is overvalued**
4. Revenue per share, last 12 months (StockOracle → Overview) = **$85.77**
5. **Fair value = 2.52 × 85.77 ≈ $216**
6. Current share price = **$317** → **overvalued**

10.4 Price to Sales Growth (PSG) ratio

- The **cousin of the PEG ratio** — not commonly used, but useful for **speculative growth companies with very high revenue growth (50%, 100%, 200%) that are not yet profitable**.

```
PSG ratio = Price to Sales ratio ÷ projected revenue growth rate (%)
```

Thresholds:

PSG	Verdict
0.2 – 0.3	Fairly priced
Above 0.3	Overvalued
Below 0.2	Undervalued

10.5 WORKED EXAMPLE — Nebius (NBIS)

Profile: revenue growing "like crazy" but the company is **losing money**.

1. Projected revenue growth rate (StockOracle → Company Metrics → Growth Rates → **Projected Revenue Growth Rate**) = **144.5%** (stated as ~144%)
2. Latest **Price to Sales ratio** (Company Metrics → Price Ratios) = **48.72**
3. **PSG = 48.72 ÷ 144.5 = 0.34** (StockOracle also calculates this for you)
4. 0.34 is **above 0.3** → **slightly overvalued**

5. **Buy trigger:** when PSG falls **below 0.3**, that would be at least a safe price to buy.

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11. Adam's Full Decision Sequence (as summarised in the lesson)

1. **Primary/default: DCF method.** Take the **free cash flow of the last 12 months**, project it **20 years**, discount to present value.
2. **If FCF is inconsistent because capex is inconsistent:** use **latest 12-month CFO − 5-year average capex** = normalized FCF.
3. **Optionally use cash flow from operations directly** instead of FCF — **only if CFO is very close to net income (not more than 20% above it)**.
4. **Precondition for all of the above: cash flow from operations must be increasing consistently.**
5. **If CFO is not consistently increasing → Discounted Net Income method**, provided net income is consistently increasing. Typically **financial stocks: insurance companies, asset managers, brokers, finance companies**.
6. **Banks (special case) → Price to Book.** Fair value = **5-year average P/B × current book value per share**.
7. **REITs → Price to NAV/Book.** Fair value = **5-year or 10-year average P/NAV × current NAV per share**, plus a **dividend yield check (target ≥ ~5%)**.
8. **Companies with inconsistent profits (cyclicals) → Price to Sales ratio.**
9. **Speculative growth stocks, high sales growth, unprofitable → Price to Sales Growth (PSG) ratio.**
10. **Loss-making / distressed companies → Price to Book, ideally P/B ≤ 0.5.**
11. **PEG ratio** sits outside this sequence as a **screening/quick-check tool only** (target ≤ 1.5).

- If you would rather not do it manually, the **StockOracle Pro Intrinsic Value tab** runs every method and its algorithm picks the most appropriate one, published as the **Oracle Value**.
- Example given: **Microsoft — Oracle Value $557 vs share price $405 → undervalued.**

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12. End-of-Lesson Quiz (with answers as given)

1. **What type of companies tend to have inconsistent cash flow from operations?**
   → **All of the above** — property developers, banks and commodity companies.
2. **I want to value a company with inconsistently increasing cash flow from operations but consistently increasing net income. What method?**
   → **(B) Discounted Net Income method.**
3. **I want to value a bank and a property developer. Which method?**
   → **Price to Book ratio method.**
4. **I want to value a speculative growth company with high sales growth but still unprofitable. Which method?**
   → **Price to Sales Growth ratio method.**

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13. Key Takeaways

- **Great business first, valuation second** — but never overpay. "Cheap crap is still crap."
- **Absolute price and recent price moves say nothing about value.** NVIDIA rose 120% and is still fairly priced; AMD fell 50% and is still expensive.

- **A business is a money-making machine; its value is the present value of the cash it will generate.** Adam projects **20 years** (conservative vs. the CFA/textbook perpetuity + terminal-value approach) — anything beyond 20 years is a bonus.
- **DCF on free cash flow is the default, used ~90% of the time**, but only when **CFO is consistently increasing**. One or two down years are fine; a broken trend is not.
- **Growth-rate discipline:** average the **StockOracle (FactSet) 3–5 yr EPS growth**, **FinViz**, and **Zacks** estimates for years 1–5; **cap years 6–10 at a maximum of 15%**; use **4% for years 11–20** (nominal GDP + 2%).
- **Discount rate = risk-free rate + beta × market risk premium**, using 5-year averages, read off the calculator's beta table (observed: beta 0.28 → 5.8%; 0.83 → 6.1%; 1.1 → 6.6%; 1.4 → 7.4%; 1.49 → 7.7%). Not WACC (that belongs to perpetuity DCF). Buffett instead just uses the **10-year Treasury yield (3.6%)**.
- **Lumpy capex ≠ a broken business.** Normalize with a **5-year average capex** (Amazon: $78,449M average capex → normalized FCF $61,065M). Only inconsistent **operating** cash flow disqualifies a stock.
- **CFO may substitute for FCF only if CFO is within 20% of net income** (Apple: CFO 135,472 vs NI 117,777).
- **Intrinsic value is a range, not a number.** Run base / conservative / doomsday growth cases (MasterCard: $655.97 / $600 / $545.50). Buffett said American Express was worth $60, Munger said $50, the market said $35 — the gap is the **margin of safety**.
- **Discipline shown by example:** Adam declined Apple at $260.83 (IV $256, conservative $199.98) and BlackRock at $967 (IV $795). Undervaluation, not a good story, triggers a buy.
- **Banks are valued on P/B, never DCF.** Fair value = 5-yr avg P/B × current BVPS. 1.2–1.4 is fair; below 1.2 (especially below 1.0) is cheap; 0.4 was the GFC bargain in Bank of America.
- **Singapore banks and REITs get a double filter:** fair value from P/B **and** a dividend-yield hurdle (**4.5–5% for banks, ≥5% for REITs**), with buy levels set at technical supports inside the ±1 standard-deviation band (OCBC: fair value $16.39, +1 SD $19.25, 5% yield line $16.40, buy levels $17.87 / $16.35 / $14.40 / $13.48).
- **REIT rule of thumb: don't pay above 1.2× P/NAV**, up to 1.5× only for high double-digit DPU growth. Quick 10-year average shortcut: (10-yr high P/B + 10-yr low P/B) ÷ 2.
- **Loss-making/distressed: P/B ≤ 0.5** — 50 cents on the dollar of liquidation value, so you profit even in liquidation.
- **Unprofitable growth or cyclical: use revenue.** Fair value = 5-yr avg P/S × revenue per share. For hyper-growth unprofitable names, **PSG 0.2–0.3 = fair, >0.3 overvalued, <0.2 undervalued**.
- **Update your valuations at least quarterly; Adam does it monthly.**

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14. Decision Table — Company Type → Valuation Method

Company type / situation	Method to use	Key inputs & thresholds
Any stock — quick screen only	**PEG ratio**	PE ÷ projected 5-yr EPS growth. Target **≤ 1.5**. Theory says <1. Requires positive, consistently rising earnings. NVDA 54.69/61.17 = 0.89; AMD 112/33 = 3.4
Great business, consistently rising CFO and positive, rising FCF (e.g. MasterCard, Singapore Exchange)	**DCF — Discounted Free Cash Flow** (default, ~90% of cases)	Latest 12-mo FCF; total debt (ST+LT); cash & ST investments; growth yrs 1–5 = avg of StockOracle/FinViz/Zacks; yrs 6–10 capped at 15%; yrs 11–20 = 4%; 20-yr horizon; diluted shares; beta-based discount rate
Consistent CFO but lumpy/spiking capex (Amazon, hyperscalers: MSFT, META, GOOGL)	**DCF with normalized capex**	Normalized FCF = latest CFO − **5-year average capex**; all other inputs unchanged

Company type / situation	Method to use	Key inputs & thresholds
CFO very close to net income (CFO ≤ 20% above NI) (Apple)	**DCF — Discounted Cash Flow from Operations**	Use latest 12-mo CFO in place of FCF. Buffett rationale: no capex deduction when capex is growth, not maintenance
Finance companies, insurance companies, brokers, asset managers — cash flow distorted by loans and fund flows (BlackRock)	**Discounted Net Income**	Latest 12-mo net income; requires **consistently increasing** net income; all other DCF inputs identical
Banks — profitable or not (Bank of America, DBS, UOB, OCBC)	**Price to Book**	Fair value = **5-yr avg P/B × current BVPS**. Fair band **1.2–1.4**; cheap below 1.2 / 1.0. For SG banks add **dividend yield ≥ 4.5–5%** filter
Loss-making, distressed, near-bankruptcy, cyclical firms in an industry recession (GFC banks, COVID retailers)	**Price to Book**	Target **P/B ≤ 0.5**; BAC bought at **0.4** in 2009. Book value = assets − intangibles − total liabilities
REITs and property developers (CapitaLand Ascendas REIT)	**Price to NAV / Price to Book + dividend yield**	Fair value = **5- or 10-yr avg P/NAV × current NAV per share**. Don't exceed **1.2×** (1.5× only for high double-digit DPU growth). Yield target **≥ 5%**; max price = DPU ÷ 5%
Highly cyclical: utilities, shipping, commodities, construction, property developers, airlines (Constellation Energy, Singapore Airlines)	**Price to Sales (mean P/S)**	Fair value = **5-yr avg P/S × revenue per share**. Current P/S above avg = overvalued
Speculative growth, very high revenue growth (50–200%), not yet profitable (Nebius)	**Price to Sales Growth (PSG)**	PSG = P/S ÷ projected revenue growth %. **0.2–0.3 fair, >0.3 overvalued, <0.2 undervalued**
Company with inconsistent operating cash flow AND inconsistent net income AND no meaningful book value	**Do not buy**	Fails the core criterion of consistently increasing revenue, net income and CFO — find another stock

Part V — Technical Analysis: Timing Entries and Exits

05 — Technical Analysis (Lesson 6: Timing Your Entries with Technical Analysis)

Source transcripts: 43 Reading Charts · 44 Support & Resistance · 45 Moving Averages · 46 Buying on Uptrends · 47 Buy/Sell on Downtrends · 48 Catching Bottoms

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1. Why Technical Analysis in a Value Framework

1.1 The Rationale

- In the **short term the market is not driven by logic — it is driven by emotions (sentiment)**.
- Prices are driven by **demand and supply**, which in turn is driven by **fear and greed**.
- When people are excited → they buy, buy, buy → price keeps going up **even though it's overpriced**. "High can get even higher."
- When people are scared → they sell → price keeps going down **even though it's a good company and undervalued**. "Low can keep going lower."
- Technical analysis identifies **potential reversal points** where price tends to reverse back up — the ideal points to enter.
- We do **not** want to buy a great, undervalued company and then watch the price keep sliding. We want to buy **at or near the point of reversal**.

1.2 Core Caveats Adam Repeats Constantly

- **There is no 100% certainty in technical analysis.** We cannot predict the market.
- TA is about **probability**, not prediction: "Are prices more likely to keep going up, or more likely to keep going down?"
- We look for **high-probability reversal points** — a high chance the price reverses up at that level. It could still go lower.
- Because of this uncertainty: **NEVER go all-in at once. Always buy in tranches — three to four tranches — at these reversal levels.**
- Adam's phrase used throughout: we buy at "**potential bottoms**", never "the bottom".

1.3 Definition Adam Uses

> **Technical analysis is the study of emotional patterns that are captured on price charts.**

- Analogy: reading a chart is like a **psychologist reading a patient** — facial expression and body language reveal emotion, and from emotion you anticipate behaviour.
- A chart tells you whether investors are **bullish** (increasingly optimistic), **bearish** (increasingly pessimistic), or **confused**, and what the most probable short-term move is.

1.4 The Contrarian Note

- Warren Buffett and Peter Lynch **do not look at charts at all**. If it's a good company and undervalued, they just buy — they hold 5/10/15 years, so a 6-month or 1-year decline doesn't matter to them.
- Adam: "There's nothing wrong with that. You can ignore all these charts as well."
- **Adam's position:** combining chart patterns with great, undervalued companies gives you **better entry points** — not perfect entry points, but better.

1.5 Trend Probability Principle

> **An established trend in price is statistically more likely to continue from one interval to the next than it is to reverse.**

- **River-current analogy:** if the current flows upward, jumping in carries you up. If the current flows downward, no matter how hard you swim you get swept downstream.
- Downtrend → probability is price keeps going lower **until the trend changes**.
- Uptrend → probability is price keeps going higher.

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2. Reading Charts: Candlesticks and OHLC

2.1 The Four Daily Prices

- US market **opens 9:30 a.m. Eastern (New York) time** every weekday. First transacted price = **opening price**.
- Market **closes 4:30 p.m. Eastern time** *(as spoken in the transcript)*. Last transacted price = **closing price**.
- Every trading day has four prices: **Open, High, Low, Close (OHLC)**.
- **Worked example (McDonald's):**
- Open = **$99.70**
- High of day = **$100.30**
- Low of day = **$99.45**
- Close = **$99.99**

2.2 Candlestick Anatomy

- Candlesticks represent all four prices in one bar.
- **Bullish / up day** = close **higher** than open → **green candle** (some platforms use white — same meaning).
- Body **bottom** = opening price
- Body **top** = closing price
- Worked example: opens $10, closes $20, high $25, low $8
- **Bearish / down day** = close **lower** than open → **red candle**.
- Body **top** = opening price
- Body **bottom** = closing price
- Worked example: opens $20, closes $10, high $25, low $7
- **Mnemonic:** bullish candle — open below, close on top. Bearish candle — the **opposite**: open on top, close below.
- **Body** = the rectangle (open→close range).
- **Wick / shadow / tail** = the thin lines (all three words mean the same thing).

- Upper wick = high of the period
- Lower wick = low of the period
- Bottom line for the course: **green = bullish, red = bearish.**

2.3 The Pin Bar (the one candlestick pattern Adam teaches)

- Adam explicitly says: there are many candlestick patterns, but **for investing they are not that important — except the pin bar**.
- **Pin bar = a candle with a very small body and one long tail** (looks like a pin).

Bullish pin bar

1. **Small body**
2. **Long LOWER tail** (tail below the body)
3. Upper tail is short or absent
4. **Colour of the body does not matter** — a red-bodied candle with a long lower tail is still a bullish pin bar
5. **Meaning:** higher likelihood price moves UP after this candle
6. **Critical qualifier:** the candle alone is not good enough — it **must appear at a strong support level or at a moving-average support**

Bearish pin bar

1. **Small body**
2. **Long UPPER tail** (tail above the body)
3. **Colour does not matter** — a green-bodied candle with a long upper tail is still a bearish pin bar
4. **Meaning:** higher chance price goes DOWN
5. **Critical qualifier:** must appear at a **strong resistance level** to be reliable

- **Worked example:** Amazon (AMZN) **daily** candles — multiple red-bodied bullish pin bars with small bodies and long lower tails; price went up after each.

2.4 Gaps

- On shorter timeframes you see **big gaps** between one candle's close and the next candle's open.
- **Cause:** overnight news — most commonly **earnings announcements**.
- If earnings disappoint vs expectations → gap down.
- If earnings beat → often gap up, but **not guaranteed**.
- **Worked example:** Google — a large gap down between Wednesday's close and Thursday's open, caused by earnings.
- **Counter-example given:** McDonald's announced terrible earnings (revenue slowed) and **the price went up** — "in the short term, the market can be totally irrational."
- **Investor interpretation:** short-term drops on news/earnings are **not important** — in fact they are **buying opportunities** if the drop makes the stock undervalued.

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3. Timeframes — The Three-Chart Top-Down Routine

3.1 Candle Periods Available

- **Daily candles** — each candle = one trading day
- **Weekly candles** — each candle = one week

- **Monthly candles** — each candle = one month

3.2 Adam's Fixed Sequence (every new stock, in this order)

Step	Period	Candles	Purpose
1	**10 years**	**Monthly**	Confirm long-term uptrend; is this a long-term performer?
2	**5 years**	**Weekly**	See medium-term cycles (uptrend/downtrend alternation)
3	**1 year** (sometimes **6 months**)	**Daily**	Short-term trend, precise entry timing

> "As an investor, you have to take a look at **all** the timeframes, because every timeframe gives you a different perspective."

3.3 What Each Timeframe Shows

- **10-year monthly:** should be one clear **long-term uptrend**. This is the screening filter.
- **If the 10-year chart is not going up, Adam is not interested — he won't even look at the business.**
- **Rejection examples:**
  - **TransOcean (RIG)** — oil & gas — 10-year chart "not going up" → rejected immediately.
  - **Singapore Airlines (SIA)** — stock did not go up over 10 years → "not long-term performers, forget about it."
- A rising 10-year chart means the company **increases in value over the long run** — makes more and more money, price keeps going up.
- **5-year weekly:** you will **not** see one straight line. You see **big medium-term cycles** — uptrend, downtrend, uptrend.
- Key structural fact: **in a great company, the uptrends will be longer than the downtrends.**
- Worked example: Google 5-year weekly — visible uptrend → downtrend → uptrend sequence.
- **1-year daily:** shorter-term trends, gaps, precise support touches.

3.4 The Unclosed-Candle Rule

1. **Never analyse a candle that has not closed.**
2. Only analyse **closed** candles.
   - Worked example (recorded **11 February 2025**): the February monthly candle is still forming → ignore it. **The last valid candle is the January candle.**
   - Adam then reads the closed monthly candles of Google:
   - **January** — very bullish month, closed way above the open (open at body bottom, close at body top, wicks = month high/low)
   - **December** — very bullish; opened 1 December at body bottom, ran to month high, closed 31 December at body top
   - **November** — bearish month; opened 1 November at body **top**, closed at body **bottom**

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4. Charting Platform Setup — TradingView

4.1 Platform Choice

- Adam uses **TradingView**; you may use **any** charting software (your broker's own charting works too).

- TradingView advantages: **stocks from all around the world** (US, Singapore, India) plus other assets (gold, Bitcoin).
- Access: **tradingview.com → Products → Supercharts**.
- Available as **desktop app (Windows/Mac)** or **web version** — both fine.
- **Paid vs free:** free version limits you to **two indicators**. Adam subscribes to paid for flexibility, but says it is **not a must** — there is a workaround (see §6.6).

4.2 Entering a Symbol

- **Top-left corner** — type the ticker (e.g. `GOOG` or `GOOGL`) and the chart loads.

4.3 Adam's Chart Settings (right-click on chart → Settings; on Mac use **two fingers on trackpad**)

Symbol tab
- Candle **body**: green / red
- **Borders**: green / red
- **Wick** (upper and lower tail): green / red

Status line tab
- Logo
- Title
- Chart values
- Bar change values

Scales and lines tab
- Currency and unit: **always visible**
- Scale modes **A and L**: visible
- On mouse-over scale
- Scale **placement: auto**
- Symbol **name and value**

Canvas tab (background)
- Background: **solid white** (Adam's personal preference — "I don't like a dark background")
- Crosshairs: set colour
- Text: **black**
- Lines: **black**

Events tab (Adam enables all of these)
- **Dividends** (so he sees ex-dividend dates)
- **Splits**
- **Earnings**
- **Latest news**

4.4 Scale Setting

- On the **right-hand side**, enable **log (logarithmic) scale** when viewing the 10-year chart — makes the long-term trend read clearly.

4.5 Period Buttons

- **"All"** → shows the stock from the beginning of its listing and **automatically switches to monthly candles**. Adam then drags the chart back to show roughly the **last 10 years** (in 2025 → back to ~2015).
- **"5Y"** → automatically shows **weekly candles**.

- **"1Y"** → automatically shows **daily candles**.

4.6 Chart Event Icons (bottom of chart)

- `E` **in a square (outlined)** = earnings **not yet reported / upcoming**.
- Worked example: **NVIDIA** — hovering shows earnings on **Wednesday 26 February**; the icon style indicates it reports **after the market closes** that day (some companies report **before the open**).
- **Green** `E` = earnings **already reported**.
- **News icon** = click to see all news related to the stock on that day — useful when price suddenly drops or spikes and you want the cause.
- `D` = **dividend**, usually the **ex-dividend date**. Buy **before** the ex-dividend date to receive the dividend.

4.7 Dividend-Adjustment Toggle (bottom right)

1. **Analysing a single stock on its own → DO NOT adjust for dividends.**
2. **Comparing the performance of two or more stocks → DO adjust for dividends.**
   - **Worked example (5-year comparison):**
   - Unadjusted: **Google +152%** vs **DBS Bank (Singapore) +92%** → Google looks far better.
   - DBS pays large dividends; Google pays none.
   - **Adjusted for dividends: the two are almost neck and neck — only about 4% difference.**
   - Lesson: dividend adjustment is **essential** for fair cross-stock comparison.

4.8 Drawing Tools

- Drawing tools are on the **left-hand side** of the chart.
- **Horizontal Line** → used for support and resistance levels.
- **Horizontal Ray** → used for the **intrinsic value line**.
- To style a drawn line: put mouse on it → **right-click** (or two fingers on Mac trackpad) → **Settings** → set colour, label text, and line style.

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5. Swing Points, Support and Resistance

5.1 Order of Operations

> "The first thing I do when I look at charts is identify **swing high and swing low points**, then **support and resistance levels**. And I do this on **all** the timeframes, **especially on the weekly and on the daily**."

Why: it tells you where price is likely to reverse.
- Price falling → at a **support level** it is more likely to reverse back **up**.
- Price rising → at a **resistance level** it is more likely to reverse back **down**.
- Not a guarantee — a **high probability**.

5.2 Swing Point Definitions

- **Swing high** = price goes **up**, then reverses **down**. The peak is the swing high.
- **Swing low** = price goes **down**, then reverses **up**. The trough is the swing low.

5.3 Support Level — Identification Rules

1. Find a **swing low**.
2. If price comes down again and reverses up at **roughly the same price**, you now have **two swing lows at a similar level**.
3. **Two swing lows at roughly the same price → draw a horizontal line across them → this is a POTENTIAL support level.**
4. If price returns to that same level a **third time** and reverses up again → it becomes a **CONFIRMED support level**.
5. Levels do **not** have to be exact — "roughly the same point, doesn't have to be exact."

Definition: A support level is a price where every time the stock drops to it, it reverses back up.

Why it works (mechanism): At that price there are **a lot of buyers queued waiting to buy** — a **demand zone**. When price hits it, buy orders flood in, **demand exceeds supply**, price reverses up.

5.4 Resistance Level — Identification Rules

1. Find a **swing high**.
2. Price rallies again and reverses down at **a very similar swing-high level**.
3. **Two or more swing highs coinciding at a similar level → draw a horizontal line → resistance level.**
4. Subsequently, when price returns to that level it has a **higher likelihood of reversing back down**.

Why it works (mechanism): At that level there are **many sellers queuing to sell**. When price reaches it, sellers rush in, **supply exceeds demand**, price reverses down.

5.5 THE POLARITY RULE (Adam: "Very important concept. Please write this down.")

> **When a resistance level is broken, that previous resistance becomes a SUPPORT level.**
> **When a support level is broken, that previous support becomes a RESISTANCE level.**

- Practical consequence: **previous swing highs and previous resistance levels are your future support levels — the places where you add shares.**
- Adam: "That is why it's important to look at previously where were the resistance levels, where were the swing highs."
- If it's a good company with good fundamentals and it's undervalued, it will **eventually break out** of a resistance — and then that level flips to support on the way back down.

5.6 Support/Resistance Is a ZONE, Not a Line

- Although you draw a line, **think of it as a zone with a buffer**.
- Price may:
- Overshoot slightly **below** the support line and then reverse up ("like hitting a fence and the fence moves a bit").
- **Fall short** of the line — buyers anticipate the touch, buy early, and price reverses before reaching the line.
- Give the level **a bit of a buffer**.

5.7 Line-of-Best-Fit Rule (where exactly to draw the line)

1. Place the line at the **point of best fit**.
2. Best fit = **the line that touches as MANY swing lows (or swing highs) as possible**.
3. A line touching only one swing low is **not significant**.

4. **The more swing lows it touches, the more significant the level.** (Adam's demonstration line touched **five** swing lows.)

5.8 Timeframe Applicability

- Support and resistance levels appear on **different timeframes at different places** — draw them on **both daily and weekly**.
- **Weekly is usually clearer** than daily.
- **Monthly:** occasionally you will find them, but **usually not** — on monthly the price is normally in a very clear uptrend. S/R normally occurs at the **shorter timeframes (weekly and daily)**.
- Note also: **when a stock is in a very strong trending phase you won't see many support levels** — S/R levels form mostly when price moves into **consolidation**.
- **The market has a long-term memory** — a support level can be respected again **many months later**.

5.9 Worked Support/Resistance Examples

Google (GOOG/GOOGL)
- Support at **$49.90 / ~$50** — three-plus swing lows; price repeatedly bounced there.
- **Polarity:** *before* becoming support, **$49.90 was RESISTANCE** — price hit $49.90 and reversed down, hit $49.90 again and reversed down, was rejected a third time, then **broke out**. After breakout, $49.90 became **support**.
- Multiple additional support levels drawn on the same chart.

McDonald's (MCD)
- Support at **~$230** — price dropped to $230, reversed up; came back to a similar level, reversed up; came back again, reversed up. Many buyers waiting at 230 = demand zone.
- A second, lower support level also identified (swing low, rally, return to the same point, reversal up).

Thermo Fisher (TMO) — weekly and daily
- Support at **~$520** — repeated reversals; demonstrates the **zone** concept (price sometimes goes slightly below $520 before reversing, sometimes reverses just above it).
- Support at **~$439** — repeated reversals; and **many months later** the market "remembered" $439 and reversed there again.
- Resistance at **~$604** — four-plus swing highs rejected; price has been below $604 for **one, two, three years**.
- Adam's forward view: Thermo Fisher is a great company, will earn more over time and increase in value, so **eventually it will break $604 — exactly when, unknown** — and once broken, **that very strong resistance becomes future support**.
- Also noted: the **$520 level was itself a previous resistance** before becoming support.

IWM (Russell 2000 / small-cap ETF)
- Support at **$162.70 / $162.72** — swing low, then repeated bounces "right on the dot".
- Resistance at **$198** — three swing highs rejected at $198.
- Then $198 was **broken**, and on the pullback **$198 acted as support** — repeated bounces, and the stock began to trend up.
- Zooming from weekly to **daily** made the support level "a lot clearer".
- Line-of-best-fit demonstration on this chart: a line touching **five swing lows**; price later reversed again at **~210** *(figure as spoken in the ASR transcript)*.

Mercado Libre (MELI) — daily
- Resistance at **~$288 / $280** formed by two coinciding swing highs → broke out → came back down → **that same line became support**, bounced there repeatedly.

- Another support identified from a matching pair of swing lows → **when broken it became resistance**.
- Adam summarises on this chart: "Support when broken becomes resistance. Resistance when broken becomes support. The same phenomenon."

Mercado Libre (MELI) — weekly
- Resistance at **~$290** — three swing highs rejected; price "not able to break out of it yet."
- Forward view: good company, good fundamentals, undervalued → eventually breaks out → then that resistance becomes support and price bounces off it repeatedly on the way up.

Walmart (WMT) — weekly
- Two swing highs at the same level = **resistance**; once broken, the level became **support** and price bounced off it repeatedly before running higher.
- A second support level from matching swing lows also identified.

5.10 The Investing Purpose of S/R

> "Before we buy a stock — ideally it's a great company, it is undervalued, **and it has dropped to a support level** — we get in right at the support so that we have a high probability of catching the reversal back up."

- Adam: "For me, when I buy shares, **I always buy them at support levels**."
- Corollary: if price is **mid-fall between levels**, he'd **rather not buy there**, knowing there's a high chance it drops to the support first.

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6. Moving Averages

6.1 What They Are and Why

- A **moving average is a mathematical calculation that averages the price of a stock over a certain period of time**; the charting software plots it.
- **Key function: moving averages act as DYNAMIC support levels.**
- "Stocks have a tendency to wave down towards their moving averages and use the moving average as a support where they bounce off."
- Worked illustration: **Meta** — price waves down, hits the blue 50 MA, bounces; repeated six-plus times.
- **Moving averages are areas where Adam likes to buy** — because that's where price tends to bounce and reverse back up after dropping.
- Not every stock bounces off the 50 — some bounce off the 100, some the 150. Hence a **set** of MAs.

6.2 Adam's EXACT Moving-Average Set (6 total)

Simple Moving Averages (SMA) — 4 of them

Period	Colour	Line weight
50 SMA	**Blue**	Thick (2nd thickness)
100 SMA	**Orange**	Thick
150 SMA	**Green**	Thick
200 SMA	**Red**	Thick

Exponential Moving Averages (EMA) — 2 of them

Period	Colour	Line weight
20 EMA	**Red**	**Thin** (to distinguish from the red 200 SMA)
40 EMA	**Blue**	**Thin** (to distinguish from the blue 50 SMA)

- Adam on the colour scheme: "The 50 is always in blue. Why? **Because I said so, so follow it.**" "150 **has to be** green." "200 is **always** red."
- Justification for the periods: "**Over the years I found that these are the most reliable simple moving averages.**"

6.3 SMA vs EMA — The Calculation Difference

- **SMA:** software takes the **last N days of closing prices, adds them together, divides by N**, equally weighted. Repeats every day.
- 50 SMA = sum of last 50 closes ÷ 50.
- 200 SMA = sum of last 200 closes ÷ 200.
- **EMA:** software takes the last N closes but **assigns individual weights** — **recent prices are weighted more heavily** than older prices. It is a **weighted** moving average.
- Adam: "You don't have to worry about the mathematics behind it. It's not important. **Just know when to use which one.**"

6.4 TIMEFRAME RULE — Which MAs Apply Where (critical)

Timeframe	Moving averages that act as support	Ignore
Daily	**50, 100, 150, 200 SMA only**	**20 and 40 EMA**
Weekly	**20 and 40 EMA** (plus SMAs)	—
Monthly	**20 and 40 EMA** (plus SMAs)	—

> "On the **daily** timeframe we pay attention mainly to the simple moving averages — the 50, 100, 150 and 200. **We do not really pay attention to the 20 and 40 EMAs. Please take note of that.** The 20 and 40 EMAs only come into service at the higher timeframes — at the weekly and the monthly candles."

6.5 How to Install Them in TradingView (step-by-step)

1. Click **Indicators**.
2. Type **"moving average"** → select **Moving Average Simple** (or type **EMA** → **Moving Average Exponential**).
3. TradingView defaults to the **9 period** — this must be changed.
4. Click the **gear icon** (settings) on the indicator → **Inputs** tab → change the period to 50 / 100 / 150 / 200 (or 20 / 40 for EMA). **Change nothing else in Inputs.**
5. Go to the **Style** tab → set the colour per the table above and the line thickness (thick for SMAs, thin for EMAs).
6. Repeat for all six.
7. **Save the template:** click the **four-boxes icon ("Indicator templates") → Save** → name it e.g. **"all moving averages"**.
8. For any new stock (e.g. Apple) with a blank chart, just load the saved **"all moving averages"** template instead of re-keying everything.

6.6 Free-Account Workaround — PP Multi MA

- TradingView **free version allows only two indicators**.
- Piranha Profits has built a **single indicator containing all the moving averages**: **PP Multi MA**.
- **How to get it:**
  1. Go to the Piranha Profits website where you watch the video → **Lesson 6** → scroll to **"Course notes and resources"**.
  2. Below the course notes, find **"PP Multi MA"** and click the link.
  3. On the page that opens, **scroll down** and click **"Add to favorites"**. Done.
  4. Back on your chart → **Indicators → Favorites → PP Multi MA** → all moving averages appear at once.
- **Trade-off / only drawback:** with PP Multi MA you must **view all MAs together — you cannot hide individual ones**. Installing them individually lets you hide/show any single MA (e.g. show only the 50).

6.7 Moving Averages and Stock "Personality"

> "Stocks are like people. We are all different. Same with stocks."

- Some stocks like to bounce off the **20**; others ignore the 20 and bounce off the **40**; others use the **50** or **100**.
- **Rule:** *Before buying a stock, look at HISTORICALLY which moving averages it bounced off. Determine the "personality" of the stock. Then wait for it to hit its preferred moving average — the one where it tended to bounce in the past — and add shares there.*
- Adam draws a **horizontal support line at that preferred moving average** on the chart.

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7. Trend Identification

7.1 The Three Trends

1. **Uptrend** — price generally moves upward
2. **Downtrend** — price generally moves downward
3. **Consolidation** — price generally moves sideways

7.2 UPTREND — Definition and Confirmation

Price-structure definition
- **Higher swing highs AND higher swing lows.**
- Prices still drop on an uptrend — but each drop makes a **swing low higher than the previous low**, and each rally makes an **even higher high**.
- Message: **the market is bullish** on this stock; probability is price continues higher.

Moving-average confirmation (two clues)
1. **All moving averages sloping UPWARDS.** "The slope of the moving average tells you the direction of the trend."
2. **Sequence: shorter-term MAs ABOVE longer-term MAs →**
50 above 100 above 150 above 200.

Wave behaviour
- Price moves in **waves**: wave up, wave down, wave up, wave down — "like a breathing pattern: breathe out, breathe in." The pattern continues **until the trend reverses**.
- Normal uptrend advances at roughly a **45-degree angle**.

7.3 DOWNTREND — Definition and Confirmation

Price-structure definition
- **Lower swing highs AND lower swing lows.**
- Prices still go up on a downtrend — each rise is a **short-term rally** — but each rally tops out **lower than the last**, and each drop goes **lower than the last**.
- Message: **the market is bearish / pessimistic**; probability is price continues downward.
- "Even if it's a great company and it's undervalued, on a downtrend, because of the downward momentum — **cheap can get cheaper, low can get lower.**"

Moving-average confirmation (two clues — exact opposite of uptrend)
1. **All moving averages sloping DOWNWARDS** (50, 100, 150, 200 all sloping down).
2. **Sequence reversed: longer-term MAs ABOVE shorter-term MAs →**
200 above 150 above 50.

Timeframe note
- Downtrends can appear on **daily and weekly**.
- On the **monthly** you normally do **not** see a downtrend for a good company — the monthly is the long-term trend and it should always be going up.
- **If the long-term monthly trend is going down → do not touch that company.**
- Daily and weekly downtrends in an otherwise-good company: "**that's fine. I don't sweat it.**"

7.4 TREND-CHANGE RULES (from Lesson 2 — DAILY timeframe specifically)

Uptrend → Downtrend reversal. BOTH conditions required:
1. **The 50 MA (blue) must flatten or slope DOWNWARDS**, AND **the 150 MA (green) must flatten or slope DOWNWARDS.**
- **If EITHER one is still sloping up, it is NOT a change of trend.** Both must be flat or sloping down.
2. **The 50 MA must CROSS BELOW the 150 MA** (blue crosses under green).
- Both satisfied → **the uptrend has reversed into a downtrend.**

Downtrend → Uptrend reversal. The exact opposite, BOTH required:
1. The **50 must flatten and slope UP**, AND the **150 must flatten and slope UP**.
2. The **50 must cross back ABOVE the 150**.
- **Invalidation example given (Mercado Libre):** the blue 50 crossed above the green 150 — **but the green 150 was still sloping down → NOT a valid uptrend confirmation.** You must wait for **both** lines to flatten and slope up.

7.5 CONSOLIDATION / CHANNEL — Definition

- **Same highs, same lows** (contrast: uptrend = higher highs/higher lows; downtrend = lower highs/lower lows).
- **Confirmation rule:**
  1. **Three same highs** → confirms a **resistance**.
  2. **Three same lows** → confirms a **support**.
  3. Price sandwiched between the two = a **CHANNEL** (a range).
- **Moving averages in consolidation:**
- They **move sideways and crisscross each other** back and forth.
- **They do NOT act as support levels in a consolidation.**
- **Therefore: in a consolidation, IGNORE the moving averages entirely.** "The moving averages are useless in a consolidation."
- Look **only** at the drawn support level and the drawn resistance level. "That's it, that's all I look for."

8. BUYING ON UPTRENDS

8.1 The First Golden Rule

> **If the price is on an uptrend, you always BUY THE DIP of the uptrend.**

- Buy **after a wave DOWN**, never after a wave up.
- Logic: if you buy after a wave up, the tendency is that it **waves down right after you buy**. Buy after a wave down and price tends to **wave up right after your entry**, so you're immediately above your entry price.
- Not a guarantee — but **increases returns over time**.

8.2 Dip Size Terminology (from earlier lessons)

- **Pullback** = a wave down of **less than 10%**
- **Correction** = a wave down of **more than 10%**
- **Frequency:** every year you get **at least three pullbacks** and **at least one correction** → these are your buying windows.
- Precondition: **provided it is a good company and it is currently undervalued.**

8.3 Locating the Potential Bottom of the Wave Down

- No one can predict the absolute bottom. But we can make **intelligent probabilistic guesses** and buy at **potential bottoms**.
- **Tool: MOVING AVERAGES.** When a stock is on a clear uptrend and it waves down, it **tends to find support at one of the moving averages**.

8.4 UPTREND ENTRY RULES (numbered)

1. **Confirm long-term uptrend:** 10-year monthly chart is rising (long-term performer). Reject if not.
2. **Confirm the stock is on an uptrend** on the timeframe you're working: higher swing highs + higher swing lows; all MAs sloping up; 50 above 100 above 150 above 200.
3. **Confirm fundamentals:** it is a **great company**.
4. **Confirm valuation:** the price is **BELOW your calculated intrinsic value**. (Adam: "Ultimately remember that **we only buy when the stock price is below the intrinsic value**.")
5. **Wait for a wave DOWN** (a pullback or a correction). Do not chase a wave up.
6. **Identify the support the stock is falling into** — either
   - a **drawn horizontal support** (from swing lows or from a broken resistance), or
   - the **moving average this specific stock historically bounces off** (its "personality").
   - Use the correct MA set for the timeframe: **daily → 50/100/150/200 SMA; weekly & monthly → 20/40 EMA also apply.**
7. **When price hits the FIRST support level, buy only a portion** — "I buy a bit first. I don't buy all the shares."
8. **(Optional, more conservative confirmation)** — Wait for price to touch the support/moving average **and then wait for a GREEN (bullish) candle to print** — i.e. wait until it looks like it's bouncing — before entering.
9. **If price breaks below that support**, do not panic — **watch the NEXT moving average / next support level down**. When it hits that level and bounces, **add more shares**.
10. **Repeat at each successively lower support**, scaling in. **Adam buys over about FOUR TRANCHES.**

11. **Size the tranche by the degree of undervaluation** — if only *slightly* below intrinsic value, "**nibble a small portion**"; buy more heavily only when **significantly undervalued**.
12. **Exit logic:** none required from the chart while it remains a great company — see §9.5 (Adam no longer sells on trend change).

8.5 Worked Uptrend Examples

Microsoft (MSFT) — daily, period A
- Clear uptrend: higher swing highs, higher swing lows, wave up / wave down pattern.
- **Every wave down found support near the 50 SMA (blue)** — hit blue, bounced; five-plus times.
- Once you notice this pattern, **any subsequent drop back near the 50 MA is a potential entry**.

Microsoft (MSFT) — daily, period B (different period)
- The **50 SMA support broke**; price then found support at the **100 SMA (orange)** — hit orange, bounced, four-plus times.
- Later in the same chart, a wave down again found support at the **50** and bounced → another buy point.
- **Illustrates the scale-in rule:** buy a bit at the 50; if it breaks, add more at the 100.

Microsoft (MSFT) — weekly
- On the weekly, **the 20 EMA (thin red line) acted as the strong support** — wave up, wave down, hit the 20 EMA, bounce; repeated many times. "That would be a great place to add shares."

Google (GOOGL) — monthly
- Over a long period, **every wave down found support at the 20 EMA (red) on monthly candles**.
- Every touch of that red line = an area to buy — **provided it is undervalued**.
- Precondition restated: "You've got to calculate the intrinsic value first… always ensure the price is below the intrinsic value **and** it hits a support level where it has tended to bounce in the past."

8.6 FULL WORKED EXAMPLE — Microsoft, Three-Timeframe Build (the model workflow)

Step 1 — 10-year monthly
- Click "All" → monthly by default → drag back ~10 years → enable **log scale**.
- **Q: Is it on a clear uptrend? Yes. ✔**
- **Which MA did it bounce off historically during corrections? The 20 EMA (red line). → Draw a horizontal support line at the 20 EMA.**
- **2022 bear market** (whole market crashed): Microsoft **broke below the red 20 EMA** — but ultimately **bounced off the blue 50 SMA on monthly candles**.
- **→ Draw a second support line at the monthly 50 SMA = $324.**
- Interpretation: "In the event of another big crash, **324 should be the bottom**."
- **Confluence check at 324:** that level was also a **previous swing-high resistance** (two swing highs). Resistance broken → becomes support. So **324 = previous resistance + monthly 50 SMA** → "a very powerful, significant level that you have to watch in the event of a crash."

Step 2 — 5-year weekly
- A **swing low at ~$390** coincides with a line already drawn → **reaffirms 390 as a significant support**.
- The **200 SMA (red) on weekly** was hit and bounced off twice → **and it coincides with the 324 level**.
- Point: "**Everything coincides — all the lines are intersecting to reinforce that these are very strong levels of support.** That's where I want to be a buyer."

Step 3 — 1-year daily
- Short-term picture: **not an uptrend — a choppy sideways CONSOLIDATION pattern.**

- Visually identifiable support at **$407 / $407.80** — price reached it and reversed up repeatedly.
- Two **swing highs** identified and connected with a line — because **swing highs are significant levels that can act as strong future support** if price falls to them.

Step 4 — Draw the intrinsic value line
- Adam has already calculated **Microsoft intrinsic value = $415**.
- Method: **Drawing tools → Horizontal Ray**, drawn at **415**.
- Then right-click (or two-finger click on Mac) → **Settings** → colour **GREEN** ("For intrinsic value, I always use green"), label text **"intrinsic value"**, **line style = dotted**.

Step 5 — The buy decision at the time of recording
- Current price: **$412**.
- Check 1 — **Is it below intrinsic value of $415? Yes, slightly. ✔**
- Check 2 — **Is it at a support level? Yes, at the $407 support. ✔**
- **Decision: Yes, you could add shares here — but only nibble a small portion**, because it's **only slightly undervalued**. Adam buys more only when **more significantly undervalued**.
- **Scale-in ladder for Microsoft:**
1. Nibble at **~$412** (at the $407 support, just under IV $415)
2. If it breaks that support and drops to **$390 → add more**
3. If it breaks down to the third, more significant long-term support at **~$349.50 → add more**
- Rationale: "That is how I build and average into a position — because ultimately a great stock will go up over time. Once it waves down to a support, eventually it will go higher, as the company makes more and more money and its **intrinsic value goes up over time**."

Number of levels to draw

> **"Usually for each stock, I'll draw about FOUR TO FIVE support levels, because this will tell me where to add shares when the price comes down."**

- Support lines may be drawn from **either** (a) swing-low patterns / broken resistance, **or** (b) moving averages. "It could be either or. As long as it's a support level, I would draw a line."

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9. SELLING / THE PARABOLIC MOVE, AND WHAT TO DO IN DOWNTRENDS

9.1 The Worst Time to Buy = The Parabolic Move

- **Normal uptrend advance = ~45-degree angle** (wave up, wave down, wave up, wave down).
- **Parabolic move = price goes up at ~90 degrees** — a very sharp vertical move.
- **Cause: panic buying** — enthusiasm, lots of good news, crazy excitement.
- **Signatures of a parabolic top:**
  1. **90-degree (near-vertical) advance**
  2. **Very long bullish candles** — Adam calls them **"extended range candles"** (exceptionally big green candles)
  3. **Price becomes stretched far away from the moving averages**
- **Rubber-band / pendulum analogy:** the moving average is "home base". Pull price too far above the MA and **it snaps back**. After a parabolic move price **tends to retrace back down to the moving averages**.
- **Rule: a parabolic move is the ABSOLUTE WORST time to buy — and the BEST time to sell (if you are going to sell).**

9.2 IMPORTANT QUALIFIER — Parabolic ≠ Automatic Sell

- A 90-degree move **does NOT mean you must sell**.
- **If it is still a great company, still growing, and not that overvalued → Adam holds. He would not sell.**
- Adam only uses the parabolic move as the **selling trigger** when he has already decided to sell for another reason:
- He thinks it's **really overvalued**, or
- He **needs the money** for something else, or
- He wants cash to **buy another stock**.
- "Even if it drops, I don't really care — I'll just add more shares, because it's going to continue going up eventually anyway."
- **Timing caveat:** "We can't always sell right at the top. Sometimes after you sell, it can still go up a bit more — but it's better to sell nearer the top than near the bottom."

9.3 Covered Calls Note

- **The best time to sell covered call options is when a stock makes a parabolic 90-degree move.**
- Adam sells covered calls **not on growth stocks**, but on **more defensive and moderately cyclical stocks**.
- (Options strategies are covered at the end of the course.)

9.4 Retail Investor Behaviour (the mistake to avoid)

- The **average retail investor does the complete opposite** of what is taught here:
- They **buy when there's good news** — which happens at the top of a parabolic wave up → they always **buy high**.
- They **sell when there's bad news** — which happens after the wave down (rate rises, tariffs, "not as good as before") → they **sell low**.
- Result: buy high / sell low repeatedly → **they lose money even when they are in the right stock**.
- **Therefore: we do NOT buy and sell based on news. We ignore the news. "The news is there to confuse you and to distract you."**
- **We buy and sell purely based on price action** — after we have established it is a great business with great fundamentals and we know the intrinsic value.

Parabolic-top worked examples
- **Broadcom (AVGO):** nice uptrend, wave up/wave down bouncing off the 50 SMA at 45 degrees → then euphoria/great news → **90-degree parabolic move with big green extended-range candles, price far from the MAs** → snapback expected.
- **Lockheed Martin (LMT):** same structure — 45-degree uptrend with wave up/wave down → good news → 90-degree move with big green candles, price stretched far from the moving average → "eventually it will snap back down."
- **Monster Beverage (MNST):** uptrend, nicely bouncing off the **50 SMA** (three identified buy points at the 50, provided undervalued vs intrinsic value) → then good news → **parabolic 90-degree move with green candles** → the point at which to sell covered calls or scale out if raising cash.

9.5 If You ALREADY OWN a Great Company and It Enters a Downtrend — DO NOT SELL

- **What Adam used to do (many years ago):** the moment the downtrend signalled (50 crossing below 150, both sloping down), sell everything, then wait for the new uptrend and buy back.
- **What Adam does now: he does NOT do that anymore.**

- **Reason: WHIPSAW.** Downtrends often don't last long; they reverse fast, so by the time the new uptrend confirms you buy back **at a much higher price** than you sold.
- "Eight or nine out of 10 times you get whipsawed — you end up buying it back 20–30% higher, and all this reduces your returns."
- Conclusion: "**If I had just held on to great companies through these up and down trends, I would have ended up making a lot more** than what I did in the past jumping out and jumping back in."

Amazon (AMZN) whipsaw worked examples

Case	Sell (downtrend signal)	Buy back (uptrend signal)	Result
2022 bear market (Fed raised rates rapidly; 2021 uptrend reversed)	**$165**	**$106**	**Worked** — but this only happens in a bear market
Case 2	**$74**	**$92**	**Lost $8** — whipsawed
Case 3	**$15**	**$19**	**Lost $4 on $15 = 27%** — whipsawed

- **Bear markets happen on average ONCE EVERY SIX YEARS** — so the "sell the downtrend" trade only pays off rarely; **more often than not you get whipsawed.**

9.6 If You DO NOT Yet Own the Stock and It Is in a Downtrend — Two Options

- Buffett-style investors would just buy — they don't care about the downtrend, because downtrends last a couple of months and eventually reverse and go much higher. If you have a 5–10 year horizon, short-term trends don't bother you.
- **Adam personally: "I wouldn't just jump in blindly. I won't do that."**

Universal prohibition on downtrends:

> **NEVER buy after a wave UP in a downtrend.** "If you want to buy on a downtrend, for goodness sake don't buy after a wave up — that's the dumbest time to buy, because after you buy it's going to crash back down again."

The two legitimate options:
- **Option A — Wait for the downtrend to end and the NEW UPTREND to be confirmed, then buy.**
- Trigger: the **50 crosses back above the 150** AND **both the 50 and the 150 have flattened and are sloping up**.
- **Option B — Buy DURING the downtrend, but only at signs of a POTENTIAL BOTTOM** (see §10). "You buy at the bottom of a downtrend anticipating the reversal to an uptrend **before** the uptrend starts."

The cost of Option A (why Option B exists)
- **Main disadvantage of waiting for the new uptrend: you cannot get in near the bottom. You get in after price has already moved typically 30%, 40%, or 50% off the low.**
- **Worked cost examples:**

Stock	Bottom	Uptrend-confirmation entry	Late by
Mercado Libre (MELI), 2022 bear market	**$600**	**~$920**	**~50%**
Apple (AAPL)	**$164**	**$209**	**27%**
Monster Beverage (MNST)	—	uptrend confirmed when 50 crossed above 150	**37%**

10. CATCHING BOTTOMS — Buying During a Downtrend

10.1 The Two Signs of a Bottom

> **There are exactly two signs of a bottom that we use:**
> **1. A PARABOLIC DROP to a strong support level**
> **2. A DOUBLE BOTTOM pattern**

10.2 Why Bottoms Form (the mechanism)

- **A stock can only bottom when there are NO MORE SELLERS LEFT.**
- There are no sellers left **when everyone who wants to sell has already sold.**
- That is induced by **CAPITULATION / PANIC SELLING**.
- Panic selling is induced by a **90-degree drop** — the sharp final plunge makes retail traders "throw in the towel" and get out.
- Roller-coaster analogy: before the ride ends there's usually a big drop at the end.

10.3 SIGN #1 — PARABOLIC DROP TO A STRONG SUPPORT LEVEL

Setup context: in a normal downtrend price falls at a **45-degree angle** (wave down, wave up, wave down, wave up). Before a stock bottoms it typically **drops at 90 degrees** — the parabolic drop.

RULES:
1. Stock is in a confirmed **downtrend** (lower highs, lower lows; MAs sloping down; 200 above 150 above 50).
2. Price makes a **PARABOLIC 90-DEGREE DROP** — a sharp near-vertical plunge.
3. You must see **LONGER-THAN-USUAL RED CANDLES** ("bigger than usual red candles") — the signature of capitulation/panic selling.
4. **The drop must terminate AT a PREVIOUSLY IDENTIFIED support level** — a level you drew earlier, taken from the **monthly, weekly or daily** charts. (This is precisely why you draw 4–5 support levels on the higher timeframes first.)
5. At that point there is a **very high probability** price reverses back up and starts an uptrend.
6. **CONSERVATIVE CONFIRMATION (optional but recommended):** after the big red candles, **wait for a big GREEN bullish candle OR a BULLISH PIN BAR** to appear before entering.
7. **Position sizing:** this is a **potential** bottom, not the bottom — it may still go a bit lower. **Buy in tranches. Never all in at one go.**
8. **What it is worth:** getting in here is "a lot better than buying when the price has already moved up 50%."

10.4 SIGN #2 — THE DOUBLE BOTTOM PATTERN

(Adam notes: Alson Chu, who teaches the Price Action Manipulation course, calls this a BEAR TRAP pattern. "It's a very, very powerful pattern.")

RULES (exact sequence):
1. Stock is in a **downtrend**, falling at ~45 degrees.
2. Price makes a drop to a low → this is the **FIRST SWING LOW**. Mark it with a dotted line. *(Teaching example: $50.)*
3. Price **rallies up** off that first swing low.
4. Price makes a **SECOND sharp drop**. **This second drop must be a 90-DEGREE (parabolic) drop.**
- Both drops can be 90 degrees, but **at minimum the SECOND one must be 90 degrees.**
5. **The second low MUST go BELOW the first swing low.** (If first low was $50, the second must break under it — $49.90, $49.50, $49, whatever.)

6. **WITHIN ABOUT THREE CANDLES of breaking below, price must CLOSE BACK ABOVE the first swing low** (back above $50) **with a BULLISH candle.**
7. That completes the double bottom. **That bullish close back above the first swing low is the BUY trigger** — "that is exactly when I'll get in. I'll buy it right there, and I'll get in right near the bottom before the uptrend."
8. **Caveats:** the pattern does **not always occur** ("sometimes a double bottom doesn't happen"). No guarantee — it could still go lower. **Always buy in tranches.**

10.5 Choosing Between the Two Signs

- If you don't want to buy at the **first** parabolic drop, **wait for the double bottom** — but be aware it **may or may not form**.
- The two signs can also **appear together** on the same chart (see MELI below).

10.6 Worked "Catching Bottoms" Examples

Mercado Libre (MELI), 2022 bear market — BOTH signs present
- Context: nothing wrong with the company — the **whole market was bearish** in the 2022 bear market. Downtrend confirmed: 50 below 150, 150 below 200, all sloping down.
- **Sign #1 (parabolic drop to support):** price falling at 45 degrees → then a **90-degree drop with big red candles → hit a strong support level previously identified on the charts.** Conservative entry: wait for the first green candle → buy.
- **Sign #2 (double bottom):** first swing low at **~$640** (estimated) → rally → second drop **below $640** → then back **above $640** within about three candles with a **bullish candle → buy right there, near the bottom, before the uptrend.**
- **Cost of the alternative:** waiting for uptrend confirmation would have meant entering at **~$920** when the bottom was **$600** — i.e. **~50% off the low**.
- **Uptrend-confirmation invalidation note on this same chart:** blue 50 crossing above green 150 was **NOT valid** while the green 150 was still sloping down — both lines must flatten and slope up.

Apple (AAPL)
- Uptrend → a **short-lived downtrend** → back into uptrend.
- **Waiting for the uptrend:** you'd buy at **$209** when the bottom was **$164 → 27% late** ("209 minus 164… you would have gotten in only after the price had reversed 27% up").
- **The better path — Apple made a DOUBLE BOTTOM:**
1. First bottom = the **first swing low**
2. Price rallied
3. **Parabolic 90-degree drop with big red candles** that went **below the first swing low**
4. **Within 1–2–3 candles it formed the first bullish candle**
5. **Wait for that first bullish candle, then buy** — catching the stock near the bottom.

Monster Beverage (MNST)
- Downtrend → uptrend.
- **Waiting for uptrend confirmation** (50 crossing above 150) meant buying **only after price had already moved 37% up from the bottom.**
- **The chart printed a textbook double bottom:** 90-degree drop, a **double dip**, entry right at the second-bottom reversal → "you caught it very early."

Google (GOOGL), 2022 bear market — PURE PARABOLIC DROP TO SUPPORT
- Daily chart: downtrend → **90-degree parabolic drop with long red candles**.
- The drop terminated exactly at a support level of **$83**.
- **The low: $83.34, hit on 3 NOVEMBER.**

- **Where the $83 support came from:** it was **identified on a HIGHER timeframe — the MONTHLY chart**, where it sat on **the 50 SMA (blue line) of the monthly candles**. Historically Google hit that blue line and bounced.
- Adam: "I bought in there and caught near the bottom."
- **Lesson stated explicitly:** "That's why it's important on the monthly and weekly to draw your support levels that you've identified." Start with the **monthly** timeframe, draw all support levels; go down to the **weekly**, draw the support levels; then finally go to the **daily**.

Mercado Libre (MELI) — second example, pure parabolic drop to support
- Daily: downtrend, moving averages all sloping down → **parabolic 90-degree drop with big red candles**.
- Drop terminated at a strong support at **$666 on 12 MAY**.
- **Where the $666 support came from:** identified on the **MONTHLY** chart, where **$666 had been a swing-high RESISTANCE**. Per the polarity rule — **resistance once broken becomes support**. The line was drawn **in advance**.
- "When it hit my 666 with a 90-degree [drop], I thought: okay, that's the bottom, baby. Got it. And it was beautiful."
- Caveat repeated: "It doesn't always turn out 100%, but **the probabilities are pretty high**."

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11. BUYING IN A CONSOLIDATION / CHANNEL

11.1 Rules

1. Identify the consolidation: **same highs, same lows** (sideways).
2. **Three same highs → confirms the RESISTANCE.** Draw the line.
3. **Three same lows → confirms the SUPPORT.** Draw the line.
4. Price is now **sandwiched between resistance and support = a CHANNEL / range.**
5. **IGNORE the moving averages entirely** — in a consolidation they go sideways, crisscross each other, and **do not act as support.**
6. Look **only** at your drawn support and resistance lines.
7. **BUY WHEN PRICE DROPS NEAR THE SUPPORT LEVEL** — provided it's a good company and it's undervalued.
8. **Keep adding at support** to **build the position until you have a full position.**
9. **Then let it run** — eventually a good company **breaks out of the range and begins a new uptrend.**

11.2 Worked Consolidation Examples

- **Generic channel example:** highs repeatedly at **292** (three same highs → resistance), lows repeatedly at **272** (three same lows → support). **Rule: when price waves down near 272, add.**
- **Bank of America (BAC):** stuck between **resistance $30.84** and **support $26.61**, bouncing within the channel. Moving averages visibly sideways and crisscrossing → ignored.
- **Microsoft daily (from §8.6):** short-term choppy sideways consolidation with support at **$407.80**.

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12. Summary of Trend → Action Mapping

Trend state	Structure	Moving averages	ACTION
Uptrend	Higher highs + higher lows	All sloping **up**; **50 > 100 > 150 > 200**	**BUY THE DIP** — buy after a wave down, at drawn support or at the stock's preferred MA

Trend state	Structure	Moving averages	ACTION
Downtrend	Lower highs + lower lows	All sloping **down**; **200 > 150 > 50**	**Either** wait for uptrend confirmation, **OR** buy at signs of a bottom (parabolic drop to support / double bottom). **NEVER buy after a wave up.** If already holding a great company — **do not sell.**
Consolidation	Same highs + same lows	Sideways, crisscrossing → **useless, ignore**	**BUY NEAR THE SUPPORT** of the channel; build full position; let it break out
Parabolic (90°) move up	Vertical, extended-range green candles, price stretched far from MAs	Price far above MAs — rubber band	**WORST time to buy.** Best time to sell **if** you have already decided to sell (overvalued / need cash / covered calls)

Key Takeaways

1. **Short-term prices are driven by emotion, not logic.** Charts are the record of that emotion — Adam calls TA "the study of emotional patterns captured on price charts."
2. **TA gives probabilities, never certainties.** You look for **high-probability reversal points** and **potential bottoms**, never *the* bottom.
3. **Because you can't pick the exact bottom, you ALWAYS buy in tranches — three to four — never all in at once.**
4. **TA is layered on top of fundamentals and valuation, not instead of them.** The mandatory precondition for every buy: it is a **great company** AND the **price is below your calculated intrinsic value**.
5. **The 10-year monthly chart is a hard screening filter.** If it isn't rising over 10 years, Adam won't even look at the business (RIG, SIA rejected on that basis alone).
6. **Always work top-down: 10-year monthly → 5-year weekly → 1-year daily.** Draw support levels on the higher timeframes first — those are the levels that catch parabolic drops on the daily.
7. **Never analyse an unclosed candle.**
8. **Two swing lows at a similar level = potential support; a third touch confirms it.** Same logic with swing highs for resistance.
9. **The polarity rule is central: broken resistance becomes support; broken support becomes resistance.** Previous swing highs are your future add-on levels.
10. **Support/resistance is a ZONE, not a line** — allow a buffer above and below. Draw the line at the **point of best fit** (touching the most swing points).
11. **Adam's six moving averages: SMA 50 (blue), 100 (orange), 150 (green), 200 (red) — all thick; EMA 20 (red) and 40 (blue) — both thin.**
12. **Timeframe rule for MAs: on the DAILY use only 50/100/150/200 SMA and ignore the 20/40 EMA. The 20 and 40 EMA only work on the WEEKLY and MONTHLY.**
13. **Moving averages are dynamic support.** Every stock has a "personality" — study which MA it historically bounced off, and buy there.
14. **Uptrend = higher highs + higher lows, MAs sloping up, 50>100>150>200.** Downtrend is the exact mirror (200>150>50, all sloping down). Consolidation = same highs, same lows, MAs sideways and crisscrossing.
15. **Trend change (daily) needs BOTH conditions: the 50 and 150 must both stop sloping the old way (flatten or reverse) AND the 50 must cross the 150.** One line still sloping the old way = signal invalid.
16. **On an uptrend, buy the dip — never chase a wave up.** Pullback = <10%, correction = >10%; expect at least 3 pullbacks and 1 correction per year.

17. **A parabolic 90-degree move is the worst possible time to buy** — extended-range candles, price stretched far above the MAs, rubber-band snapback due. It is the *best* time to sell **only if** you had already decided to sell.
18. **Do NOT sell a great company just because it entered a downtrend.** Adam abandoned that approach — whipsaw costs more than it saves (Amazon: sold $74/bought back $92; sold $15/bought back $19 = −27%). Bear markets, where the trade actually works, come only about once every six years.
19. **Waiting for uptrend confirmation costs you 30–50% of the move** (MELI ~50%, AAPL 27%, MNST 37%). That's why bottom-catching exists.
20. **There are exactly two bottom signals: (a) parabolic 90-degree drop with oversized red candles into a pre-drawn support, and (b) a double bottom / bear trap where the second low breaks the first and closes back above it within ~3 candles on a bullish candle.**
21. **Conservative confirmation on any bottom entry: wait for the first big green candle or a bullish pin bar.**
22. **Ignore news for entries and exits.** Retail investors buy on good news at parabolic tops and sell on bad news at bottoms — buy high, sell low, lose money in the right stock. Trade the price action, after the fundamentals are settled.
23. **Draw about 4–5 support levels per stock** — they are your ladder of add-on prices.
24. **Confluence matters.** Adam's strongest levels are where several things intersect at once — e.g. Microsoft $324 = previous swing-high resistance + monthly 50 SMA + weekly 200 SMA.
25. **In a consolidation, the moving averages are useless — trade only the drawn channel support.**

Entry & Exit Rules Cheat Sheet

A. UNIVERSAL PRECONDITIONS (all setups)

1. **10-year monthly chart is in a clear long-term uptrend.** If not → reject the stock outright.
2. **It is a great company** (fundamentals verified in the fundamental-analysis lessons).
3. **Current price is BELOW your calculated intrinsic value** (drawn on the chart as a **green dotted horizontal ray** labelled "intrinsic value").
4. **Support levels are already drawn** on **monthly, then weekly, then daily** — about **4–5 levels per stock**, sourced from swing lows, broken resistances (polarity), and the stock's preferred moving averages.
5. **Never go all in.** Plan for **3–4 tranches**. Tranche size scales with the depth of undervaluation (slightly undervalued → nibble; significantly undervalued → buy more).

B. SETUP 1 — BUY THE DIP ON AN UPTREND

Trigger conditions

1. Higher swing highs **and** higher swing lows.
2. All MAs sloping **up**; **50 > 100 > 150 > 200**.
3. Price has just made a **wave DOWN** (pullback <10% or correction >10%).

Where to buy

4. At the **drawn horizontal support**, **or** at the **moving average this stock historically bounces off** (daily → 50/100/150/200 SMA; weekly/monthly → 20/40 EMA also valid).

Confirmation (optional, conservative)

5. Wait for price to touch the level **and print a GREEN bullish candle** (or a bullish pin bar at that support) before entering.

Scale-in ladder

6. First touch (e.g. 50 SMA) → **buy a small first tranche**.
7. If that level **breaks** → **do not exit**; watch the **next MA / next drawn support below** (e.g. 100 SMA).
8. Bounce at the next level → **add the next tranche**. Repeat down the ladder to about **four tranches**.

What invalidates / what does NOT invalidate

9. Breaking one support level does **not** invalidate the thesis — it simply moves you to the next rung of the ladder.
10. The thesis is invalidated only by **the company ceasing to be a great company** (fundamentals) — **not** by a chart downtrend.

Do NOT buy

11. **Never buy after a wave up.**
12. **Never buy into a parabolic 90-degree move.**
13. Never buy mid-fall between support levels when a clear support sits below.

C. SETUP 2 — BUY IN A CONSOLIDATION / CHANNEL

1. Confirm sideways structure: **same highs, same lows**.
2. **Three same highs** → draw the **resistance**. **Three same lows** → draw the **support**.
3. **Ignore all moving averages** (sideways + crisscrossing = no support value).
4. **Buy when price waves down NEAR the support line.**
5. Keep adding at support until you hold a **full position**.
6. **Hold** and let it eventually break out into a new uptrend.

(Example: support 272 / resistance 292. Bank of America: support $26.61 / resistance $30.84.)

D. SETUP 3 — DOWNTREND, OPTION A: WAIT FOR UPTREND CONFIRMATION

Trigger (daily timeframe, BOTH required)

1. The **50 SMA flattens and slopes UP** *and* the **150 SMA flattens and slopes UP**.
- **Invalid if either line is still sloping down**, even if the crossover has occurred.
2. The **50 SMA crosses back ABOVE the 150 SMA**.

Where to buy

3. On confirmation of both conditions.

Known cost of this method

4. You will typically enter **30–50% above the actual low** (MELI ~50%, MNST 37%, AAPL 27%).

E. SETUP 4 — DOWNTREND, OPTION B1: PARABOLIC DROP TO SUPPORT (catching a bottom)

1. Stock is in a confirmed **downtrend** (lower highs + lower lows; MAs sloping down; **200 > 150 > 50**).
2. Price transitions from a ~45-degree decline to a **90-DEGREE PARABOLIC DROP**.
3. **Red candles are visibly LONGER than usual** (capitulation / panic selling / everyone who wants to sell has sold).
4. **The drop must land ON a support level you had ALREADY drawn** — sourced from the **monthly or weekly** chart (broken resistance, monthly 50 SMA, etc.).
5. **Confirmation (conservative):** wait for the **first big GREEN candle** or a **bullish PIN BAR** at that support.
6. **BUY** — a first tranche.

7. **Accept:** it may still go lower. This is a *potential* bottom. **Keep tranches in reserve** for the next support level down.

(Worked: GOOGL — 90-degree drop into $83 support taken from the monthly 50 SMA; low $83.34 on 3 November. MELI — 90-degree drop into $666 on 12 May, a level drawn earlier as a monthly swing-high resistance that had been broken → became support.)

F. SETUP 5 — DOWNTREND, OPTION B2: DOUBLE BOTTOM / "BEAR TRAP" (catching a bottom)

1. Stock is in a confirmed **downtrend** falling at ~45 degrees.
2. Price makes a low = **FIRST SWING LOW**. **Mark it with a dotted line** (e.g. $50).
3. Price **rallies** off that low.
4. Price makes a **SECOND drop — this one MUST be a 90-degree parabolic drop** with oversized red candles. (Both drops may be 90°, but the second must be.)
5. **The second low MUST break BELOW the first swing low** (e.g. below $50).
6. **Within about THREE candles, price must CLOSE BACK ABOVE the first swing low on a BULLISH candle.**
7. **That bullish close back above the first swing low = the BUY TRIGGER.** Enter there.
8. **Invalidation:** if price breaks below the first swing low and **fails to reclaim it within ~3 candles**, the pattern has not formed — no entry.
9. **Note:** the double bottom **may never form** on a given chart. It is optional confirmation for those unwilling to buy the first parabolic drop.

(Worked: AAPL — first swing low, rally, 90-degree drop below it with big red candles, first bullish candle within 1–2–3 candles → buy, vs. $209 if you waited (bottom $164 = 27% late). MELI — first swing low ~$640, break below, reclaim above $640 within ~3 candles on a bullish candle. MNST — textbook double dip; waiting for the uptrend cost 37%.)

G. EXIT RULES

1. **Default: do NOT sell a great company because the chart turned down.** A daily/weekly downtrend in a fundamentally sound company is not a sell signal. Adam abandoned trend-based exits because of **whipsaw** (8–9 out of 10 attempts lose money; you buy back 20–30% higher).
2. **A parabolic 90-degree move is NOT by itself a sell signal.** If the company is still great, still growing, and not badly overvalued → **hold**.
3. **SELL ONLY IF** you have an independent reason:
   - You judge it **significantly overvalued**, or
   - You **need the cash** (personal reasons), or
   - You want the cash to **buy another stock**.
4. **IF you have decided to sell, the OPTIMAL EXIT TIMING is into a PARABOLIC 90-DEGREE MOVE UP**, identified by:
   - Near-vertical advance (vs. the normal 45-degree uptrend)
   - **Extended-range candles** — exceptionally big green candles
   - Price stretched **far above the moving averages** (rubber band / pendulum → snapback to the MA is due)
5. **Accept imperfect timing:** you cannot sell exactly at the top; it may run further after you sell. Better to sell near the top than near the bottom.
6. **Covered calls:** the **best time to sell covered call options is at a parabolic 90-degree move** — and Adam writes them on **defensive and moderately cyclical stocks**, *not* on growth stocks.

7. **Never sell because of bad news, and never buy because of good news.** That is exactly the retail buy-high/sell-low loop.

H. QUICK REFERENCE — INDICATOR SETTINGS

Item	Exact setting
Platform	**TradingView** (tradingview.com → Products → **Supercharts**); web or desktop app; any charting software acceptable
Timeframe 1	**"All"** button → monthly candles → drag back to **~10 years**; enable **log scale**
Timeframe 2	**"5Y"** → weekly candles
Timeframe 3	**"1Y"** (sometimes 6 months) → daily candles
Background	Solid **white**; text black; lines black
Candles	Body / borders / wicks: **green (bullish), red (bearish)**
Events shown	**Dividends, splits, earnings, latest news**
Dividend adjust	**OFF** for single-stock analysis; **ON** when comparing stocks
SMA 50	**Blue**, thick
SMA 100	**Orange**, thick
SMA 150	**Green**, thick
SMA 200	**Red**, thick
EMA 20	**Red**, **thin**
EMA 40	**Blue**, **thin**
Default trap	TradingView defaults every MA to **9 period** — always change it in **gear icon → Inputs**
MAs on daily	**50/100/150/200 SMA only** — ignore 20/40 EMA
MAs on weekly/ monthly	**20 and 40 EMA also act as support**
Template	Save via **Indicator Templates (four boxes) → Save** as **"all moving averages"**
Free-plan workaround	**PP Multi MA** (Piranha Profits, Lesson 6 course notes & resources) → **Add to favorites** → Indicators → Favorites. Drawback: cannot hide individual MAs
Support/resistance tool	**Horizontal Line** (drawing tools, left panel)
Intrinsic value tool	**Horizontal Ray**, styled **green**, **dotted**, labelled "intrinsic value"
Levels per stock	**4–5 support levels**

Part VI — The 7-Step VMI Formula (Core Methodology)

Section 6 — The VMI 7-Step Formula (Lesson 7)

Source lessons: 50 – VMI Formula Part 1 (30:01) · 51 – VMI Formula Part 2 (30:34) · 52 – VMI Formula Part 3 (21:00) · 53 – VMI Formula Part 4 (28:16)

> This is the core lesson of the course. Everything taught previously — fundamental analysis, financial ratios, technical analysis, stock valuation — is assembled here into a single, ordered checklist.

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0. The Master Framework

The rule

> "Before you make any investment into a stock, you have to put it through the seven steps. If it passes the seven steps, you invest. If it doesn't pass the seven steps, you don't invest. It's as simple as that."

The 7 steps are grouped under **three questions**:

Question	Steps	What it establishes
Q1. Is this a GREAT BUSINESS?	Steps 1–5	Quality filter. Pass → goes on the **watch list** (NOT a buy yet).
Q2. Is this a GREAT PRICE?	Step 6	Share price must be **below intrinsic value** (margin of safety).
Q3. Is this a GREAT POINT OF ENTRY?	Step 7	Chart timing — buy on a wave-down into support, in tranches.

The funnel (Adam's stated pass rates)

Filter	% of all companies in the world that survive
After **Step 1 alone**	**less than 5%** ("just with the first step you eliminate 95% of the companies in the world")
After **Steps 1–5**	**less than 1%** ("it would eliminate 99% of the businesses in the world, and you only invest in the best of the best")

The core logic of the sequence

- **Step 1** proves the business *has* grown consistently (historical record).
- **Step 2** proves the industry/company *is projected to* keep growing (forward catalysts).
- **Step 3** proves nothing can *stop* it growing — competition is the only thing that can, so it must have a moat. **Adam calls Step 3 "the most important step of all."**
- **Steps 4–5** prove it is profitable/efficient and financially unbreakable.
- **Step 6** ensures you don't overpay: *"A great business can be a lousy investment if you pay too high a price for the shares. But a great business becomes a great investment if you're buying the shares at less than what it is worth."*
- **Step 7** optimises the entry price within the undervalued zone.

STEP 1 — Consistent, Predictable Historical Track Record

What it checks

That the company can grow **sales, profits and cash flow in a predictable, consistent manner**, and is **resilient** — "rain or shine, pandemic or no pandemic, recession or economic growth."

Exact criteria

- [ ] **Sales revenue** increasing consistently for **at least the last 5 years, better 10 years**
- [ ] **Net income** increasing consistently over the same 5–10 year window
- [ ] **Cash flow from operations (CFO)** increasing consistently over the same 5–10 year window
- [ ] **Gross profit margin** consistent for the last 5 years, or increasing
- [ ] **Net profit margin** consistent for the last 5 years, or increasing
- [ ] Current margins ≈ **at or above the 5-year average** margin

Priority ranking of the three metrics (Adam's own words)

1. **Sales revenue — most important.** Rising revenue = the company is getting more and more customers, selling more goods and services.
2. **Cash flow from operations — second most important.** The actual cash generated by the business operation.
3. **Net income — also important**, but see the substitution rule below.

Key interpretation rules

- **"Consistent" ≠ up every single year.** *"There could be years that it drops temporarily, but overall if you take a look at the last five years or 10 years, there must be an uptrend."* **One or two years of decline is fine.**
- **Net income substitution rule:** If net income is NOT increasing consistently → **look at operating income instead**, and operating income must be increasing consistently. Rationale: net income can be distorted by items that are not the core business —
- Non-operating **expenses** excluded from operating income: interest costs, taxation, lawsuits, merger & acquisition costs
- Non-operating **income** excluded: forex gains, gains on investment income
- So operating income shows *"the actual income from the business operations."*
- **Recession test:** *"especially during past recessions, during past crisis, I want to see that the revenue continues to grow and the cash flow continues to increase. This tells me the company is resilient, is predictable and it is consistent."*
- **The whole point:** *"When historical earnings and historical sales and cash flow is consistent, then future sales, earnings and cash flows are more predictable."* If the history is choppy, there is no way to project next year or the next five years → avoid.

RED FLAGS / automatic rejections

- ❌ **Profit margins falling for five straight years** → *"that's a warning sign to avoid that company."*
- ❌ Revenue/profit/cash-flow pattern that goes **up, down, up, down** (cyclical roller-coaster) → unpredictable → avoid.
- ❌ Whole industries Adam avoids for failing Step 1: **oil & gas** (profits depend on oil price), **airlines** (affected by oil prices, tourism, "too many things"). *"They are more for short-term trading, not really for long-term compounding of wealth investing."*

EXCEPTIONS to Step 1

Business type	Exception
Banks	Cash flow from operations is naturally inconsistent, sometimes negative. **Look at sales revenue and net income only — ignore CFO.**
Property developers	Same — very inconsistent CFO, ignore it.
Commodity companies	Same — very inconsistent CFO, ignore it.
Speculative / early-stage growth companies	See below.

Speculative growth company exception (the ~5% carve-out)

- Definition: fairly new companies where **revenue is growing 50–100%** (double-digit/triple-digit), but they are **not yet profitable**.
- Requirement: **revenue growing consistently**, AND **net income and cash flow from operations must have turned positive** — Adam states the rule two ways in the lesson:
- overview version: *"turned positive in at least the last two quarters"*
- Palantir case version: *"I only started to invest when they showed at least their first one quarter of positive cash flow"* (checked on the **quarterly** view, not the full-year view)
- Additional gate: he only does it *"if I'm very convinced that they'll be very profitable in the future because they've got a very unique business model and they've got an economic moat that can protect them from potential competition."*
- **Position sizing penalty: half of his normal investment allocation.** *"because they are more risky I don't invest that much — I invest half of my normal investment allocation."*
- Frequency: *"95% of the time I only invest in companies that pass step one, but there will be a 5% of the time where I make an exception."*
- Upgrade rule: once the company has **at least 5 years of consistency**, it is upgraded from "speculative growth" to a normal investment.

Worked examples

Company	Verdict	Numbers / observation
Visa (V) — owned in portfolio	✅ PASS	Revenue, operating income and net income all growing very consistently 5–10 yrs; operating cash flow rising consistently. **Gross margin 78%, operating margin 67%, net profit margin 49%** — "very, very strong."
Microsoft (MSFT) — owned in portfolio	✅ PASS	Sales, net income and operating cash flow all very consistent. **Gross margin 68%, operating margin 46%, net profit margin 39%.**
Johnson & Johnson (JNJ)	✅ PASS (illustrative)	Net income "increasing year after year after year" over the same period as GM.
General Motors (GM)	❌ FAIL	Over a 6-year window: high profits → break-even → profits → more profits → less profits → loss. "Inconsistent… unpredictable."
ExxonMobil (XOM)	❌ FAIL	Revenue up/down like a roller coaster; operating and net income choppy including a loss year; operating cash flow up/down. Profit driven by oil price (2016 war → big profits; COVID → big losses). "Good for speculation, short-term trading, but not for long-term investing."

Company	Verdict	Numbers / observation
Singapore Airlines	❌ FAIL	Revenue cyclical; profits choppy with loss years; net operating cash flow up/down/negative. **Gross profit margin only 16%, net profit margin only 9.6%** — low margins from labour, fuel and maintenance costs. "Although I fly on Singapore Airlines, I won't invest in the business."
Palantir (PLTR)	⚠️ Speculative exception	First invested **2022**. At the time revenue growing very strongly, net income still negative, but **cash flow from operations turned positive on the quarterly view in December 2022** — that quarter triggered the buy. Thesis: only company then using AI inside enterprise software (pre-ChatGPT). ~4 years later revenue still growing, profits rising for 4 years — *"has not yet reached a five-year track record so I still classify it as a speculative growth company."*
Meituan (China)	⚠️ Speculative exception	Bought ~2 years prior, when profit and cash flow had just turned positive.

Where to find the data (tools)

- **StockOracle → Financials → "Financial Trends" chart** — visual of revenue / operating income / net income + operating cash flow.
- **StockOracle → Financials → Income Statement** — revenue growth, operating income growth.
- **StockOracle → Statement of Cash Flows** — cash flow from operations growth.
- **StockOracle → Company Metrics** — margin trend chart plus a table of **current vs 5-year average** gross / operating / net margin.
- **Quarterly toggle** on StockOracle financials — required for the speculative-growth check (full-year may still be negative while quarterly has turned positive).

STEP 2 — Positive Projected Growth Rate (Growth Catalysts)

What it checks

That the business *"is projected to grow in the next 10 to 20 years"* — a **positive growth rate** — driven by real growth catalysts (being in a growth industry, opening more outlets, creating more products, expanding the addressable customer base).

Exact criteria

- [ ] Projected earnings growth rate is **positive** → PASS
- [ ] *"Obviously the higher the growth rate, the better"* (no numeric minimum is given — positive is the bar)
- [ ] **Ideal (not mandatory):** the company sits inside one of the six high-growth secular trends listed below

PASS = positive projected growth. FAIL = zero/negative projected growth. This is the loosest of the seven gates.

Where to find the projected growth rate (tools)

1. **StockOracle → Company Metrics → scroll to the bottom → "Projected Growth"** — gives the **next 3–5 year** projected earnings growth rate AND a **long-term** growth rate.
2. **finviz.com**
3. **Yahoo Finance**
4. **zacks.com**

Where to find the *reasons* for growth (best sources)

- The company's **quarterly reports**
- The company's **annual reports**
- The company's **earnings call transcript** — *"where the entire management team will talk about what are their growth plans in the future... what new products, what new services, what new markets are they growing into."*
- **StockOracle → Earnings tab** — download the quarterly press release and open the earnings call transcript.
- **Shortcut: StockOracle → "AI Insights" → scroll to Growth Drivers.**

Worked example — Microsoft

- Projected earnings growth **19.62% for the next 3–5 years**
- Projected earnings growth **8.9% beyond that (long-term)**
- Verdict: positive → **PASS Step 2.** ("Is that positive? Yes. Is that good? Not bad.")
- **AI Insights growth drivers listed for MSFT:** AI-driven cloud expansion · enterprise SaaS and Copilot monetization · proprietary AI infrastructure · commercial bookings and contract backlog · global cloud market share gains · Windows 11 adoption and OEM growth.
- Bonus: Microsoft sits inside **two** of the six preferred high-growth industries (AI & robotics; cybersecurity).

The six high-growth industries Adam focuses on

1. **AI and robotics**
2. **AI energy infrastructure**
3. **Healthcare** — driven by the world's **aging population**
4. **Digital payments** — shift from physical cash to digital (Mastercard, Visa)
5. **Cybersecurity** — more companies using AI and being online → more hacking and cybercrime
6. **The new space economy** — ⚠️ *"I have not yet invested in it, but it's something I'm watching... because many of the companies involved are not yet profitable."* Applications beyond space tourism: **data centers in space, missile defense shield, low-earth-orbit satellite communications.**

STEP 3 — Sustainable Competitive Advantage (Economic Moat) ⭐ MOST IMPORTANT

What it checks

That the company has a **sustainable competitive advantage that protects it from competition**, so competitors cannot easily take away its customers or market share. Warren Buffett's term: an **economic moat**.

Why it is the most important step: Step 1 shows it *has* grown; Step 2 shows the industry *will* grow. *"But can the business continue to grow its revenue, profits and cash flow in the next 10, 20 years? The only thing that can stop it is competition."*

The consequence of a moat: the company keeps its **pricing power** — *"it's able to raise prices consistently and not lose customers and not lose market share"* — and therefore generates consistent growth in sales, net income and cash flow for the next 10–20 years, which grows intrinsic value.

Adam's preference: *"I like to invest in monopolistic businesses that have got ideally little or no competition."*

The three moat grades

Grade	Definition	Competitors	Pricing power	Financial signature	Adam's action
WIDE	Very strong, sustainable advantage; *"we are pretty certain it can't easily be disrupted for the next 20 years."* Monopolistic.	None or insignificant	Can raise prices without materially affecting demand	Very high profit margins; consistent increase in sales & profits; predictable long-term outlook	✅ **70–80% of all stocks he buys**
NARROW	Some competitive advantage, *"probably sustainable for the next 10 years. But beyond 10 years, we are not so sure."*	Several significant competitors	Limited — raise price too much and customers defect	Average profit margins ("pretty good, but not super normal"); increasing sales & profits over next 10 yrs	✅ Buys some ("I would still invest in it")
NO MOAT	No clear competitive advantage; can be killed by a stronger competitor *"in the next one or two or three years."*	Many competitors selling near-identical / commodity products	None — **price is the ultimate buying decision**; customers just buy the cheapest	Low profit margins; inconsistent sales & profits; unpredictable growth beyond 5 years	❌ **"Avoided at all costs."** Only usable for short-term speculation, never buy-and-close-your-eyes.

⚠️ **Important nuance:** a no-moat stock can still rise in the short term and still grow revenue and be profitable. That is not the point — *"it can be easily disrupted by bigger competitors."*

The FIVE sources of moat

#	Moat source	What it means	Examples given
1	**Strong brand / brand monopoly**	People are loyal enough to keep buying even if the company raises price and competitors are cheaper. A **brand monopoly** = **top-of-mind recall** — the first name people think of for that product category.	Apple, McDonald's, Coca-Cola, Nike, Microsoft. Brand-monopoly test cases: office software → **Microsoft**; search → **Google** ("we say *Google this*, *Google that*, not *Bing this*"); free online video → **YouTube** (owned by Google); paid online video → **Netflix**; food anywhere/anytime when everything is closed → **McDonald's**; social media → **Meta** (WhatsApp, Instagram, Facebook)
2	**High customer switching costs**	Once a customer uses the product it becomes very expensive or very inconvenient to switch; normally subscription-based; locks in the customer for years.	**Apple ecosystem/iCloud** ("Hotel California — you can check out but you can never leave"; Watch + iPhone + MacBook + iPad + home camera/ lighting all synced; a better, cheaper Samsung phone still wouldn't be bought because it can't sync). **Enterprise software:** ServiceNow, Salesforce (embedded into all workflows). **Cybersecurity vendors:** Palo Alto, CrowdStrike, Fortinet (integrated into the entire company; switching is risky and difficult). **Amazon Web Services** (hard to migrate to another cloud).
3	**Network effect**	The more customers use the product, the better it gets and the more others want to use it — a flywheel.	WhatsApp / Instagram / Facebook (Asia & Europe — "you're left out of the loop"); **iMessage** in the US; **WeChat** (Tencent) in China; **Amazon** (sellers and buyers must be there); **Alibaba** and **Pinduoduo** in China; **MercadoLibre** in Latin America; **Google search / Gemini** (more usage → smarter searches → more powerful AI)
4	**High barriers to entry**	Regulatory restrictions, intellectual property, patents, trademarks.	**Pharmaceutical patents — protected for 20 years** (⚠️ but *"it doesn't last forever"*). **Regulatory moats:** Palantir, Lockheed Martin — need very high-level security clearance with the US military/Pentagon, so new competitors can't enter. Airbus.

#	Moat source	What it means	Examples given
5	**Economies of scale**	So big that competing is almost impossible: buy in bulk, run more efficiently, spread fixed costs over a larger base → can temporarily cut price and kill competitors, or keep price and earn far higher margins and out-spend on R&D.	Amazon, Walmart, Costco, Airbus

🔑 THE SCORING RULE

> **"If a company has got at least THREE out of these FIVE moat sources, then I would want to invest in the company. It tells me the company has a strong moat."**

The pricing-power test (the fast qualitative check)

"Can [the company] raise their price every year and not lose market share?"
- **Yes** → wide moat.
- **No / must discount** → narrow moat at best.
- Worked example — **Tesla:** cannot raise car prices annually without losing share; facing heavy EV competition from China; *"not only can they not raise their price, they have to keep cutting their price, they have to keep discounting."* → **narrow moat, not wide.**

The 10-year chart shortcut (visual moat proxy)

- Look at the **long-term chart of the last 10 years**.
- **Clear 10-year uptrend** → *"usually a sign of a strong economic moat."*
- **Generally on a downtrend over the long run** → *"then it doesn't have a competitive advantage."*
- Brand monopolies illustrated this way: Microsoft, Google, Netflix, McDonald's all *"in the long run, it always goes up."*

Tool: StockOracle Pro moat analysis

- Enter the ticker → a moat grade (Wide / Narrow / None) is displayed immediately.
- Click **"Oracle IQ"** → **three perspectives on the moat**:
  1. **Oracle Moat** — assigned by the StockOracle research team based on an algorithm.
  2. **Community Moat** — crowd vote (users vote wide/narrow).
  3. **AI Moat** — AI-generated, scored **out of 10**, with a written explanation and a **breakdown by the five moat sources**, each scored /10.

AI Moat score thresholds (Adam's stated preference)

- **9/10** = "a very, very wide moat."
- **7, 8 or 9 out of 10** = *"that gives me more confidence."*
- **6/10** → *"I don't often invest in 6 out of 10. It's not strong enough."*
- **2/10** = no moat = *"the most dangerous."*

Worked examples — moat scoring

Meta Platforms (META) — WIDE MOAT, AI moat score 9/10

Moat source	Score
Brand loyalty & pricing power	9/10
Barriers to entry	9/10
Switching costs	8/10
Network effect	**10/10**
Economies of scale	9/10

Rationale summarised by the tool: operates the largest global social media and digital advertising ecosystem via Facebook, Instagram, WhatsApp, Messenger and Threads; competes with Alphabet, ByteDance, Amazon and emerging platforms; dominant share of global social media usage and advertising revenue; scale and network effects smaller competitors struggle to match.
Adam's fun fact: ~6 billion people worldwide have internet access; **3.5 billion** use one of Meta's products ≈ **60% of everyone on the internet.** No competitor (Twitter/X, Snapchat, Reddit) comes close.

Zoom Communications (ZM) — NARROW MOAT, AI moat score 6/10 → NOT BOUGHT

Moat source	Score
Brand loyalty & pricing power	6/10
Barriers to entry	5/10 (*"bigger competitors can always replicate a similar platform"*)
Switching costs	6/10 (*"you can easily switch to another platform"*)
Network effect	6/10
Economies of scale	5/10

Adam's reasoning at the time: Zoom was growing well, especially during the pandemic, *"but I felt that although Zoom was growing, it didn't have a very strong moat. It could easily be disrupted by bigger competitors like Microsoft and Google."*

Beyond Meat — NO MOAT, AI moat score 2/10 → AVOIDED, subsequently collapsed
No pricing power, no brand loyalty, no barriers to entry, no switching costs, no network effect, low economies of scale. *"No moat is the most dangerous."*

PayPal — AVOIDED. *"Although PayPal was gaining traction, it was getting popular, it was growing, but I felt that the business itself could be easily disrupted and replaced by much stronger competitors. And good thing I didn't buy it, because after going up initially, these shares have collapsed and I still wouldn't buy it."*

Reference lists given

- **Wide moat examples:** Meta Platforms, Amazon, S&P Global, Alphabet (Google), Netflix, Nvidia, Mastercard, Visa, Apple ("and there are many, many more").
- **Narrow moat examples:** Cisco Systems, FedEx, Expedia, Dow Chemical, Alcoa, Chevron, Schlumberger, Tesla, ExxonMobil, Royal Dutch Shell.
- **No-moat industries (mostly, not all):** commodity companies, **airlines, auto manufacturers, property developers, construction companies, shipping companies, oil & gas companies.** *"It's like a bloody red ocean of competition. So I stay away from those industries."*

Step 3 summary questions to ask before buying

1. Does it have a **strong brand / brand monopoly**? (McDonald's, Visa, Mastercard, Microsoft, Google, Coca-Cola, Pepsi)
2. Does it have **high barriers to entry** — regulatory barriers, IP or patents? (Lockheed Martin, Airbus, Palantir)
3. Does it have **huge economies of scale**? (Amazon, Walmart, Airbus)
4. Does it have a **network effect**? (Google, Meta, Apple, Alibaba, Tencent)
5. Does it have **high switching costs**? (Microsoft, Palo Alto, CrowdStrike, Amazon Web Services)
   → **≥3 of 5 = invest.**

STEP 4 — Highly Profitable and Operationally Efficient

What it checks

That the company is genuinely profitable and runs efficiently — and that its profitability is **not being manufactured by leverage**.

The FOUR metrics and exact thresholds

#	Metric	Threshold	Notes
1	**Return on Equity (ROE)**	**≥ 12% = "a very profitable business" > 15% = "a super exceptional business"**	Measures how well the company makes money from shareholders' stake in the business
2	**Return on Invested Capital (ROIC)**	**≥ 12% = a good business ≥ 15% = an excellent business**	**Why both?** Companies that finance with a lot of **debt** need less equity, which **inflates ROE artificially** — but heavy debt **drags ROIC down**. Requiring high ROE *and* high ROIC ensures *"its profitability is not being pushed up by too much debt."*
3	**Sales revenue vs Accounts Receivable**	**Sales revenue must be growing FASTER than accounts receivable**	Working-capital efficiency check
4	**Cash Conversion Cycle (CCC)**	**Consistent or declining** — must not be increasing too much	*"how many days it takes for every $1 the company spends... to get the money back. The faster the better."* **A NEGATIVE CCC is fantastic** — the company collects cash upfront from customers before it even pays its suppliers. Applies mainly to companies selling **physical goods** (Apple, Nike).

Trend rules

- ROE and ROIC **trend charts**: *"as long as it is consistent or rising, that's a good thing. But if it's consistently declining, then it's something to watch out for."*
- Always compare the **current** figure to the **5-year average** — current at or above the 5-yr average is what you want.

⚠️ ROIC EXCEPTIONS — do NOT apply ROIC to:

- **Banks**
- **Insurance companies**
- **Regulated utilities**

Reason: these are highly leveraged by nature, or have very high fixed capex requiring debt financing. *"We just look at ROE, because if you use ROIC, they will all fail."*

Worked example — NVIDIA (NVDA)

Metric	Value	Verdict
ROE	**101%**	✅ Way over 15%, and above the 5-year average
ROIC	**93%**	✅ Way above 15% and above the 5-year average
Revenue vs Accounts Receivable	Revenue growing faster than AR	✅ Pass
Cash Conversion Cycle	"Generally moderately rising, but nothing to be concerned about"	✅ Acceptable

Singapore Telecom (SingTel) — CCC on TradingView is **negative** and mostly sideways over 5 years. *"Negative cash conversion cycle is a very good thing. That means SingTel collects cash upfront before paying its suppliers. So that is of course the best scenario."*

Where to find the data (tools)

- **StockOracle → Company Metrics → ROE/ROIC trend chart**, then scroll down to **"Profitability Ratios"** for the exact figures and 5-year averages. *Works on the FREE basic account.*
- **StockOracle → Financials → scroll down** for revenue growth vs accounts receivable. *Also on the free site.*
- **Morningstar.com → [ticker] → Key Metrics → Profitability and Efficiency** — alternative source, **shows ROE and ROIC for the last 6 years**; works for **foreign (non-US) stocks not covered by StockOracle**.
- **TradingView → Indicators → search "Cash Conversion Cycle"** — plot CCC over the last 5 years.
- **TradingView custom script — "Total Revenue vs Accounts Receivables"** (labelled **Piranha Profit**): download from the **Whale Investor course resources**, under **Lesson 4 (Fundamental Analysis) → Course Notes and Resources**, then load via TradingView → Indicators. Revenue plots **green**, accounts receivable plots **red**.

STEP 5 — Conservative Debt Structure (Strong Balance Sheet)

What it checks

That the company will not be crippled or bankrupted by debt. *"If you invest in companies with little or no debt, it can't go bankrupt. You can only go bankrupt with high debt."*

Why it matters: companies that can make a lot of money by taking on high debt are risky — *"if interest rates rise too much, these companies would suddenly pay a lot of interest expense... that would lower their profits and the share price would suffer."*
Bankruptcy examples cited: Evergrande (China property) and **Hyflux** (Singapore). *"What did all these companies have in common? They had very high debt."*

The THREE main ratios and exact thresholds

#	Ratio	Threshold	Meaning
1	**Current Ratio**	**> 1**	Company has more current assets than current liabilities
2	**Debt / EBITDA**	**≤ 3**	*"the company can pay back all their debt in three years using their profits"* (EBITDA = earnings before interest, taxes, depreciation and amortization)
3	**Debt Servicing Ratio**	**< 30%**	**Formula: Interest expense ÷ Cash flow from operations × 100%**

Supplementary ratio

Ratio	Threshold
Interest Coverage Ratio	**> 3 = OK; > 5 = very safe**

⚠️ Current-ratio exception

Some companies **intentionally** run a current ratio **below 1** because they **collect sales upfront** — e.g. **subscription companies like ServiceNow**, or **Apple**. Cash collected before the service is delivered is booked as **deferred revenue**, which sits in **current liabilities**, enlarging the denominator. *"In cases where the current ratio is less than one, don't be too alarmed, it's because of this reason."*

⚠️ Sector exceptions — these three ratios do NOT apply to banks or REITs

Sector	Use instead	Threshold
Banks	**Common Equity Tier 1 (CET1) ratio**	**> 10% = pretty safe**
REITs	**Gearing ratio = Total debt ÷ Total assets**	**≤ 45% is the minimum requirement; below 40% is the safest**

- CET1 location: **StockOracle → [bank ticker] → Overview → scroll to the bottom → "Financial Strength."**
- Gearing ratio location: the REIT's **quarterly and annual report**, or the REIT's own **website → Investor Presentation** (Singapore-listed REITs explicitly disclose "gearing ratio").

Worked examples

NVIDIA (NVDA) — "solid balance sheet"

Ratio	Value	Threshold	Verdict
Current ratio	**3.91**	> 1	✅ "way above 1"
Debt servicing ratio	**0**	< 30%	✅ "fantastic"
Debt / EBITDA	**0.06**	≤ 3	✅
Interest coverage	**503**	> 5	✅

Banks — CET1

Bank	CET1	Verdict
JPMorgan (JPM)	**14.1%**	✅ "way above 10%"
DBS Group Holdings	**17%**	✅ *"Singapore banks are very, very, very conservative and very safe."*

Where to find the data (tools)

- **StockOracle → Company Metrics → scroll down → "Debt and Liquidity Ratios"** — current ratio, debt servicing ratio, debt/EBITDA, each shown **current vs 5-year average**. *Available on the free basic account.*
- **StockOracle → Overview → Financial Strength** — CET1 for banks.

✅ CHECKPOINT — After Steps 1–5

> *"If the company passes the first five steps, it's a great business."*
> *"Once it passes the first five steps, does it mean it's a buy? NO — because it may be at the wrong price."*

Action: put it on the **WATCH LIST** and be patient. *"You watch it until the price is right."*

STEP 6 — Great Price: Buy Below Intrinsic Value (Margin of Safety)

What it checks

That the **share price is below the intrinsic / fair value** of the business.

The rule

- [ ] **Share price < intrinsic value** → undervalued → eligible to buy
- [ ] *"The lower the share price is versus the true value, the greater the margin of safety."*
- ❌ **Overvalued great business = a bad investment**: *"even though it's a great business, it may take you a long time to break even on your investment if the share price happens to drop way below the intrinsic value after you bought it."*

Valuation method by business type

Business type	Method	Requirement / formula
Normal companies (primary method)	**Discounted Cash Flow (DCF)**	Project how much cash the business generates for the **next 20 years** and discount to present value. **Assumption: cash flow from operations must be increasing consistently.**
Companies with inconsistent CFO but consistent earnings	**Discounted Net Income method**	Requires **net income increasing consistently for at least the last 5 years**
Both CFO and net income inconsistent	❌ **Do not buy** — *"then it won't pass the first five steps in the first place"*	For a high-quality business, **at least one of cash flow or earnings must be increasing consistently**
Banks (JPMorgan, Bank of America, DBS, OCBC)	**Price-to-Book (P/B) method**	**Fair value = 5-year average P/B ratio × current book value per share**
REITs	**Price-to-NAV method**	**Fair value = average P/NAV ratio over the last 5 or 10 years × current NAV per share.** Buy when price is below fair value.
Speculative / early-stage growth companies (no consistent profits or cash flow)	**Price-to-Sales (P/S) method**, or **Price-to-Sales-Growth ratio**	(a) **Intrinsic value = 5-year average P/S ratio × current revenue per share** (b) **P/S-Growth = P/S ratio ÷ projected revenue growth rate**

Price-to-Sales-Growth ratio thresholds (speculative growth stocks)

Value	Verdict
Below 0.2	Undervalued
0.2 – 0.3	Fair value
Above 0.3	Overvalued

REIT extra requirement

- When buying a REIT, Adam is buying it **for the dividend yield**. For **Singapore-listed REITs he wants a minimum 5% dividend yield.**

Tools

- **StockOracle → "Intrinsic Value" tab** — computes intrinsic value automatically using several different methods.
- **StockOracle Pro → "Oracle Value"** — Adam's preferred output because it **normalizes capex** and **adjusts when recent free cash flow is inconsistent**, giving *"the most appropriate intrinsic value."*
- **Intrinsic value calculators** taught in the previous lesson, for manual calculation.

Worked example — Alphabet / Google

- **Plain discounted free cash flow method: intrinsic value $190.** ⚠️ Caveat: this method **does not normalize capex** — Google's capex recently spiked and free cash flow dropped a lot, *"that's why the number is lower."*
- **Oracle Value (normalizes capex): $302.**
- Verdict at the time of recording: **Google is selling slightly ABOVE intrinsic value** → not a buy yet.

(Note: in the Step 7 chart walkthrough of the same lesson, Alphabet's intrinsic value is quoted as ~"3,008" against support levels of 2,833 / 2,670 / 2,479 / 2,261 — the same relationship, undervalued, but on a different quoted scale in the transcript. The relationship taught is what matters: price below intrinsic value.)

Speculative growth positions Adam has taken under this rule

- **Palantir** and **Celsius** (the energy drink) — bought with a **small allocation** because *"the moat is very strong and there's a huge opportunity."*

STEP 7 — Great Point of Entry: Buy the Wave Down into Support

What it checks

That you enter at an **optimal chart level**, using support structure and tranching, once Steps 1–6 already pass.

The core principle

- Great businesses go up over the long run, **but never in a straight line** — price moves in **wave up / wave down** patterns driven by **greed and fear**.
- **THE GOLDEN RULE: "Never buy during a wave up. Never chase the stock. Wait for the stock to come to you."**
- *"Whenever the price gets too far from the moving average, like a rubber band, it will snap back eventually. So don't get in when it's waving up."*

- You cannot catch the exact bottom: *"As long as we can buy near the bottom, we've done our job. And as long as we don't buy near the top, we're doing the right thing."*

The three chart scenarios

Scenario A — Buy the DIP on an UPTREND

- **Uptrend definition: higher highs and higher lows**; the **50 MA is above the 150 MA** and both **sloping up**; **price above the 200 MA**, which is also sloping up.
- **Buy trigger:** wait for the **wave down** to a support level, then add.
- **Valid support levels = any of the moving averages: 50 MA, 100 MA, 150 MA, 200 MA.** *"Any one of these moving averages are fine, because they act as natural support levels."*
- **Every stock has its own character** — some bounce off the 50, some off the 100, some off the 200. Identify which MA *that specific stock* historically respects.
- **Worked example — Apple:** clear uptrend on daily candles (50 MA above 150 MA, both sloping up, 200 MA sloping up). Entries taken on each wave down into a moving average; when it broke the 50 MA and fell to the 200 MA, that was the second tranche.

Scenario B — Buy at the SUPPORT of a CONSOLIDATION (sideways)

- ⚠️ **"In a consolidation, the moving averages are USELESS. Ignore the moving averages, they tend to be criss-crossing."** MAs only work in a clear uptrend (sloping up = support) or clear downtrend (sloping down = resistance).
- **Instead: draw horizontal support and resistance levels yourself** from the price pattern.
- Every time price drops near the drawn support → add a tranche.
- *"Whenever it goes into consolidation, eventually it will break out into a new uptrend. So you want to accumulate during this period of consolidation."*
- **Worked example — Boeing:** sideways consolidation with clear resistance at **375** and support at **326** (as spoken in the transcript: "$3.75" / "$3.26"). Buy each time price drops near the support.
- **Worked example — TSM (TSMC):** consolidated sideways for about a year after a strong uptrend. Adam identified the support and **added shares every time price came near ~108**, repeatedly through the year; the stock has since broken out of the consolidation resistance into a new uptrend.
- **Worked example — Amazon:** noted as currently in a sideways consolidation pattern.

Scenario C — Buy the REVERSAL of a DOWNTREND into a new uptrend

- **Downtrend definition: lower highs and lower lows**; **50 MA below the 150 MA** and sloping down; **price below the 200 MA**, sloping down.
- **Standard trigger — wait for confirmed trend change:**
- **50 MA (blue) crosses above the 150 MA (green)** and both slope up, **OR**
- **Price goes above the 200 MA (red)** and the 200 MA slopes up, **OR**
- **Price action alone**: a **higher high AND a higher low** appear.
- **Trade-off:** waiting for confirmation means **you buy higher** than if you had bought during the downtrend. In one example, waiting for the 50/150 cross meant buying around **46** versus a much lower current price.
- **Pre-emptive entry (optional, allowed for fundamentally good businesses):** you may enter **before** the confirmed trend change as soon as you see **higher highs and higher lows in the price action**, *"especially the moment it crosses above the 50 moving average, the blue line, and the blue line is kind of flattening — that could warrant an entry."*

- Justification for entering early: *"when you're buying good companies they will go up eventually... if you take a one year, three or five year view they will be a lot higher."*

Buying INTO a downtrend — the special-circumstances rule

Adam will add shares while the **daily** timeframe is still in a downtrend **only if BOTH conditions are met**:

1. [ ] **Price is at the LOW of the downtrend, after a sharp ~90-degree drop.** *"When I see a sharp drop, usually after a sharp drop it tends to reverse back up. Not always, but pretty high probability."*
2. [ ] **Price is at a STRONG SUPPORT on a HIGHER timeframe** — i.e. the **weekly** or **monthly** chart.

Worked example — FICO (Fair Isaac): daily candles clearly in a downtrend (wave up / wave down, lower highs and lows) with a **sharp drop to the low of the downtrend**. Zooming out to **weekly candles** showed a **strong historical support at the 200 MA (red line)** — price had previously plunged into that line and bounced. That combination triggered the buy, and the stock bounced.

Worked example — MercadoLibre (MELI): daily timeframe in a clear downtrend with moving averages sloping down, plus a **sharp drop**. Zooming out to **monthly candles** showed the **40 EMA (blue line)** had **historically been a strong support** — wave up, wave down, hit the blue line and bounce, repeatedly. That gave the confidence to add shares before the bounce.

Multi-timeframe support mapping — use ALL THREE timeframes

> *"Which time frame should I look at? The answer is: I look at all three time frames. Because on each time frame, you will see different levels of support."*

Worked example — Alphabet (GOOGL): building four buy levels

#	Timeframe	Support identified	Level (as quoted)
1	**Daily**	**100 MA (orange line)** — recent retracements repeatedly bounced off it	**2833**
2	**Daily**	**Drawn horizontal support** from two previous swing lows	**2670**
3	**Weekly**	**50 MA (blue line)** — historically bounced off it	**2479**
4	**Monthly**	**20 EMA (red dotted line)** — price hits it and bounces during normal corrections	**2261**

Intrinsic value quoted as **~3008** → price is undervalued at all four levels, so all four are valid buy zones.

- **Support can come from either a moving average OR a level you draw yourself from previous swing-low patterns.**
- *"This is a bit subjective... it really depends on you identifying what makes sense. It doesn't have to be perfect."*
- **How many levels? "Usually I draw like three or four buy levels and that's enough."**

Tranching — the position-building method

> *"When investing in a stock, I never buy a full position all at once. I spread out my purchase over three to four tranches."*

- Decide the **full intended allocation** first (e.g. 100 shares this year).
- Divide into **3 or 4 tranches** (e.g. **25 shares per level**, or 33 shares if using thirds).

- **Buy one tranche each time price hits a support level.** If it breaks that level and hits the next one down, buy the next tranche. *"The lower it goes, the more you buy."*
- **STOP when the full allocation is bought.** *"You buy until you have bought your entire allocation and that's it, leave it, don't add anymore."*
- If price hits only level 1 and then runs up, you simply hold a quarter position — *"don't worry, eventually it will come back down again and then you take your next quarter position."*
- Result: *"You may not pay the exact bottom price, but you get a very good average."*

Averaging DOWN vs averaging UP

	Definition	Adam's verdict
Averaging down	Buying more and more as price goes lower and breaks more support levels	**Better** — *"you get to buy more shares at a lower price, your average cost is much lower, you make a lot more money over time."* But psychologically hard: *"they see red, red, red, red, and then they freak out."* **Adam's preference.**
Averaging up	Price hits support 1, you buy, it rallies, comes back to support later, you buy the next quarter at a higher price	**Also fine** *("although it's still undervalued, it's still fine")* — feels better psychologically but you pay more.

"So you could either average up or average down, depending on how it plays out. To me, I'm fine either way, but I prefer averaging down."

Key Takeaways

1. **The formula is a pass/fail gate, not a scorecard.** All seven steps must pass before you invest. No step can be skipped or traded off against another.
2. **The seven steps answer three questions in order:** (1) Is this a great business? [Steps 1–5] (2) Is this a great price? [Step 6] (3) Is this a great entry? [Step 7].
3. **The formula is brutally exclusionary by design.** Step 1 alone eliminates ~95% of the world's companies; Steps 1–5 together eliminate ~99%. *"Why settle for anything less than the best?"*
4. **Predictability is the whole point of Step 1.** Consistent historical revenue, cash flow and profits are what make future revenue, cash flow and profits *projectable* — and everything in valuation depends on being able to project. Cyclical roller-coaster businesses (oil & gas, airlines) are rejected not because they can't make money but because they are unpredictable.
5. **Revenue > cash flow from operations > net income** in importance; and if net income is lumpy, substitute **operating income**, which strips out interest, taxes, lawsuits, M&A costs, forex gains and investment gains.
6. **Step 3 (moat) is the most important step.** Steps 1 and 2 establish that a business has grown and its market is growing; only competition can stop that, so a sustainable moat is what converts past growth into future growth. **≥3 of the 5 moat sources = invest.**
7. **The single fastest moat test is pricing power:** can the company raise prices every year without losing customers or market share? Tesla fails this test (must discount), hence narrow moat, not wide.
8. **Never buy a no-moat company as an investment.** It may be growing and profitable today and can still be killed within 1–3 years by a bigger competitor. Zoom (6/10) was passed on; Beyond Meat (2/10) collapsed; PayPal was passed on and collapsed.

9. **ROE and ROIC must BOTH be high.** High ROE alone can be a debt illusion — leverage shrinks the equity base and inflates ROE while crushing ROIC. Exception: banks, insurers and regulated utilities are judged on ROE only.
10. **Debt is the only thing that makes a company go bankrupt.** *"If you invest in companies with little or no debt, it can't go bankrupt."* Evergrande and Hyflux are the cautionary cases.
11. **Passing Steps 1–5 buys you a watch-list slot, not a trade.** A great business bought at an overvalued price can take years to break even.
12. **Valuation method must match the business type:** DCF for normal companies, Discounted Net Income if cash flow is lumpy, P/B for banks, P/NAV for REITs, P/S or P/S-Growth for speculative growth names.
13. **Never chase a wave up.** Wait for the wave down into support. If price is far from its moving average it snaps back like a rubber band.
14. **Moving averages are useless in a consolidation** — draw your own horizontal support/resistance instead. They only function as support/resistance in a clear trend.
15. **Map supports across daily, weekly AND monthly charts** — each timeframe reveals different support levels. Three or four buy levels is enough.
16. **Always buy in 3–4 tranches, never a full position at once.** This produces a good average cost without requiring you to call the exact bottom, and lets you exploit deeper corrections rather than fear them.
17. **Speculative growth names are a deliberate, sized-down 5% exception** — revenue growing 50–100%, profit/cash flow only just turned positive, strong moat required, and **half the normal position size**. Upgraded to a normal holding only after 5 years of consistency.

📋 THE VMI 7-STEP CHECKLIST (One Page — Apply to Any Stock)

Q1 — IS THIS A GREAT BUSINESS? (Steps 1–5)

STEP 1 — Consistent Track Record *(StockOracle → Financials → Financial Trends; Income Statement; Statement of Cash Flows; Company Metrics)*
- [ ] Sales revenue rising consistently for **5–10 years** (most important metric)
- [ ] Cash flow from operations rising consistently for **5–10 years** (second most important)
- [ ] Net income rising consistently — **OR**, if not, **operating income** rising consistently
- [ ] 1–2 down years acceptable; overall **uptrend** required
- [ ] Revenue and cash flow kept growing **through past recessions/crises** (resilience test)
- [ ] Gross margin **consistent or increasing** over 5 years
- [ ] Net margin **consistent or increasing** over 5 years
- [ ] Current margins **≥ close to the 5-year average**
- [ ] ❌ **REJECT if margins have fallen for 5 straight years**
- [ ] ❌ **REJECT if revenue/profit/cash flow is cyclical up-down-up-down** (oil & gas, airlines, commodities)
- **Exceptions:** Banks / property developers / commodity companies → **ignore cash flow from operations**, use revenue + net income only.
- **Speculative growth exception:** revenue growing **50–100%**, not yet profitable → require **net income and CFO turned positive in the last 1–2 quarters** (quarterly view) + strong moat → buy at **HALF the normal allocation**; upgrade to normal after 5 years of consistency.

STEP 2 — Positive Projected Growth Rate *(StockOracle → Company Metrics → Projected Growth; also finviz.com, Yahoo Finance, zacks.com; StockOracle → Earnings → transcripts; StockOracle → AI Insights → Growth Drivers)*
- [ ] Projected earnings growth rate for the next **3–5 years is POSITIVE** → PASS (higher is better)
- [ ] Long-term projected growth rate is **POSITIVE**

- [ ] Identify the actual **growth catalysts** (new products, new markets, new outlets, more customers) from annual/quarterly reports and earnings-call transcripts
- [ ] ⭐ **Bonus:** the company sits in one of the **6 high-growth industries** — AI & robotics · AI energy infrastructure · healthcare (aging population) · digital payments · cybersecurity · new space economy

STEP 3 — Sustainable Competitive Advantage / Economic Moat ⭐ MOST IMPORTANT *(StockOracle → Oracle IQ → Oracle Moat / Community Moat / AI Moat; plus 10-year price chart)*
- [ ] **≥ 3 of the 5 moat sources present:**
- [ ] 1. Strong **brand / brand monopoly** (top-of-mind recall)
- [ ] 2. High **customer switching costs** (expensive/inconvenient to leave; usually subscription)
- [ ] 3. **Network effect** (more users → better product → more users)
- [ ] 4. High **barriers to entry** (patents [20 yrs], regulatory, IP, trademarks)
- [ ] 5. **Economies of scale**
- [ ] **Pricing-power test:** can it raise prices every year without losing customers/market share? *(Yes = wide moat; No/must discount = narrow at best)*
- [ ] Moat grade = **WIDE** (preferred — 70–80% of buys) or **NARROW** (acceptable)
- [ ] ❌ **NEVER BUY A NO-MOAT COMPANY** — many competitors, commodity product, customers buy on price alone
- [ ] **StockOracle AI Moat score ≥ 7/10** (9/10 = very wide; **6/10 = "not strong enough," usually skip**; 2/10 = no moat, most dangerous)
- [ ] **10-year chart shortcut:** clear 10-year uptrend = moat present; 10-year downtrend = no competitive advantage
- [ ] ❌ Avoid the structurally no-moat industries: commodities, airlines, auto manufacturers, property developers, construction, shipping, oil & gas

STEP 4 — Profitable and Operationally Efficient *(StockOracle → Company Metrics → Profitability Ratios [free]; Morningstar → Key Metrics → Profitability & Efficiency; TradingView → CCC indicator + "Total Revenue vs Accounts Receivables" Piranha Profit script)*
- [ ] **ROE ≥ 12%** (very profitable) — **> 15% = super exceptional**
- [ ] **ROIC ≥ 12%** (good) — **≥ 15% = excellent**
- [ ] ROE/ROIC trend **consistent or rising**, and **current ≥ 5-year average** (consistently declining = warning)
- [ ] **Sales revenue growing FASTER than accounts receivable**
- [ ] **Cash conversion cycle consistent or declining** (not sharply rising); **negative CCC = fantastic**
- **Exception: banks, insurance companies, regulated utilities → use ROE only, skip ROIC** (they would all fail it)

STEP 5 — Conservative Debt Structure *(StockOracle → Company Metrics → Debt & Liquidity Ratios [free]; StockOracle → Overview → Financial Strength for CET1)*
- [ ] **Current ratio > 1** *(exception: upfront-collecting subscription models — ServiceNow, Apple — where deferred revenue inflates current liabilities; <1 is OK there)*
- [ ] **Debt / EBITDA ≤ 3** (all debt repayable from profits within 3 years)
- [ ] **Debt servicing ratio < 30%** = interest expense ÷ cash flow from operations × 100%
- [ ] **Interest coverage ratio > 3 (OK) / > 5 (very safe)**
- **Exceptions: Banks → CET1 ratio > 10%. REITs → gearing ratio (total debt ÷ total assets) ≤ 45%, ideally < 40%.**

> ✅ **Passed Steps 1–5?** → It's a great business → **ADD TO WATCH LIST. DO NOT BUY YET.**

Q2 — IS THIS A GREAT PRICE? (Step 6)

STEP 6 — Undervalued vs Intrinsic Value *(StockOracle → Intrinsic Value tab; StockOracle Pro → Oracle Value [normalizes capex]; manual intrinsic value calculators)*
- [ ] **Share price < intrinsic value** → the bigger the gap, the bigger the margin of safety
- [ ] Use the **right method for the business type**:
- [ ] **Normal company → DCF** (project 20 years of cash, discount to present value). *Requires consistently rising CFO.*
- [ ] **CFO inconsistent, earnings consistent → Discounted Net Income.** ***Requires net income rising consistently for ≥5 years.***
- [ ] Both inconsistent → DO NOT BUY **(it should already have failed Step 1)**
- [ ] Bank → 5-year average P/B × current book value per share
- [ ] REIT → 5- or 10-year average P/NAV × current NAV per share**; plus** ≥5% dividend yield **for Singapore-listed REITs**
- [ ] Speculative growth → 5-year average P/S × current revenue per share**, or** P/S ÷ projected revenue growth: <0.2 undervalued · 0.2–0.3 fair · >0.3 overvalued
- [ ] ❌ Do not buy a great business at an overvalued price** — break-even can take years

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Q3 — IS THIS A GREAT POINT OF ENTRY? (Step 7)

STEP 7 — Buy the Wave Down into Support, in Tranches *(TradingView charts — daily, weekly, monthly)*
- [ ] ❌ **NEVER buy during a wave up. Never chase.** Wait for the stock to come to you.
- [ ] Identify which scenario the stock is in and use the matching trigger:
- [ ] **A. Dip on an uptrend** — uptrend = higher highs & higher lows, 50 MA > 150 MA both sloping up, price above rising 200 MA. **Buy on the wave down into the 50 / 100 / 150 / 200 MA** (whichever MA that specific stock historically respects).
- [ ] **B. Support of a consolidation** — **ignore moving averages, they are useless sideways.** Draw horizontal support/resistance from the pattern and **buy each time price returns to support.**
- [ ] **C. Reversal of a downtrend** — buy when **50 MA crosses above 150 MA** and both slope up, **OR** price rises above a rising **200 MA**, **OR** price action makes a **higher high AND higher low**. *(Pre-emptive entry allowed once higher highs/lows appear and price crosses a flattening 50 MA.)*
- [ ] **Buying into a still-falling downtrend is allowed ONLY IF BOTH:** (1) price is at the **low of the downtrend after a sharp ~90-degree drop**, AND (2) price is at a **strong support on the weekly or monthly chart**
- [ ] **Map supports on ALL THREE timeframes — daily, weekly, monthly.** Support = a moving average **or** a line you draw from previous swing lows.
- [ ] **Set 3–4 buy levels** (that's enough)
- [ ] **Buy in 3–4 tranches** — e.g. planned 100 shares → **25 shares at each level**; the lower it goes, the more you buy; **stop once the full allocation is filled**
- [ ] Accept **averaging down** (preferred: lower average cost) **or averaging up** (also fine while still undervalued)
- [ ] Goal is **near the bottom, not the exact bottom** — and above all, **not near the top**

Related follow-on lessons: 54 – VMI Case Study (GOOGL); 55/56 – VMI Case Study (MSFT); 57/58 – VMI Case Study (AMZN). Adam also notes that subscribers to the UIP (Ultimate Investors Playbook) receive a monthly deep-dive applying the seven steps to two great companies.

Part VII — Worked Case Studies: GOOGL, MSFT, AMZN

07 — VMI Case Studies (Full Worked Applications of the 7-Step Formula)

Source lessons:
- Lesson 54 — VMI Case Study (GOOGL) — 37:54
- Lesson 55 — VMI Case Study Part 1 (MSFT) — 23:24
- Lesson 56 — VMI Case Study Part 2 (MSFT) — 33:16
- Lesson 57 — VMI Case Study Part 1 (AMZN) — 21:34
- Lesson 58 — VMI Case Study Part 2 (AMZN) — 24:46

> **Vintage warning.** The three case studies were recorded at different times with different toolsets. The GOOGL study is the newest (StockOracle-era, AI Insights / Oracle IQ, revenue base ~$402B, post-Trump-trade-war chart). The MSFT and AMZN studies are older (Yahoo Finance / GuruFocus / Morningstar / FinViz / Zacks / Simply Wall St era; MSFT priced as of **15 January 2022**). Numbers are quoted exactly as spoken in each lesson and are not reconciled across lessons.

The 7-Step Framework Being Applied

Every case study runs the same checklist. Steps 1–5 answer *"is it a great business?"*; step 6 answers *"is it at a great price?"*; step 7 answers *"is this a great point of entry?"*

Step	Question	Evidence used
1	Consistently increasing sales revenue, net income, cash flow from operations (5–10 yrs)? Plus stable/rising gross & net (or operating) margins	Income statement, cash flow statement, margin history
2	Positive projected growth rate + identifiable growth catalysts?	2–3 independent growth projections + qualitative growth drivers
3	**Sustainable competitive advantage (economic moat)?** — Adam repeatedly calls this "the most important step of all"	Brand monopoly, switching costs, barriers to entry, economies of scale, network effect
4	Profitable and operationally efficient?	ROE, ROIC, revenue growth vs accounts-receivable growth, cash conversion cycle
5	Conservative debt?	Current ratio, debt/EBITDA, debt servicing ratio (and interest coverage)
6	Is the price below intrinsic value (margin of safety)?	DCF (cash from operations) or discounted **free** cash flow; a valuation *range*, not one number
7	Is it a great point of entry?	Charts: 10-yr monthly → 5-yr weekly → 1-yr daily; identify 3–5 support levels **below** intrinsic value; add in tranches

Standing rules Adam repeats across all three:
- Only invest in businesses whose products/services you personally use — it is the only way to genuinely judge the moat (Step 3 has no number attached to it).

- Passing steps 1–5 ⇒ the stock goes on the **watch list**, not into the portfolio. You then "watch it, watch it, watch it" for price.
- Valuation is always a **range** (optimistic / base / pessimistic), never a single point.
- Do the full 5-step business analysis **once a year at most** — "once you realise a business is a great business, it doesn't change that easily… it would remain a great business for many, many years, for the next five, six, seven years."
- Buying a stock = buying a business. Compare the diligence to buying a car, a TV, a fridge — or dating someone before you marry them.

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1. ALPHABET (GOOGL) — Lesson 54

1.0 Idea sourcing and share class

Adam's origination logic: start from products you use daily. He lists YouTube (he creates videos), Google Search, **Gemini**, Google Maps, Gmail — therefore analyse Alphabet.

Two tickers, one company:

Ticker	Class	Voting rights
GOOGL	Class A	Yes
GOOG	Class C	No

"For most of us, we don't really bother voting, so it really doesn't matter which one you really buy… their share prices are very close anyway." Case study uses the **Class A (GOOGL)** shares.

Primary tool for this case: **StockOracle** (data provider: **FactSet**), plus FinViz, Zacks, and **TradingView** for charts.

1.1 Step 1 — Consistent revenue / net income / operating cash flow

Annual figures, in USD billions, oldest → newest (5 years):

Metric	Yr1	Yr2	Yr3	Yr4	Yr5	Verdict
Revenue	257	280	307	349	402	Consistently increasing ✔
Net income	76	59	73	100	132	General uptrend (one down year) ✔
Net operating cash flow	65	91	91	101	125 → 164*	Generally uptrending ✔

* Adam reads the operating cash-flow line as "$65B, $91B, $91B — flat this year — $101, increasing $125, $164." The latest annual operating cash flow used later in the DCF is **$164,713M**.

Key teaching point: **net income does not have to rise every single year.** "There's a general uptrend, although there was a year it dropped, doesn't have to go up every single year." Fallback rule: if net income isn't consistent, check that **operating income** is consistent (it is).

Margins (Step 1, part 2): checked under StockOracle → *Company Metrics*. Over the last 5 years, **gross margins have been more or less quite stable** and **operating margins have also been pretty stable**. ✔

→ **Step 1 PASSED.**

1.2 Step 2 — Growth rate and growth catalysts

Quantitative — three independent projections gathered:

Source	Projected growth
StockOracle / FactSet — 3-to-5-year earnings growth	**15.3%**
StockOracle / FactSet — long-term growth (beyond 5 yrs)	**10.13%**
StockOracle / FactSet — cash flow growth rate	17% (noted but *not used* — Adam takes the two above "to be more conservative")
FinViz — next 5 years	**12.88%**
Zacks — projected growth	**14.72%**

Average of the three 5-yr figures (15.3 + 12.88 + 14.72) ÷ 3 = **14.3%** — used later as the conservative case.

Navigation note: StockOracle → *Company Metrics* → scroll to *Projected Growth*. Adam keeps these in his 10 "favorites" (customisable via three-dots → Edit Favorites; **479 metrics** available).

Qualitative — growth drivers pulled from StockOracle's **AI Insights**:
1. Explosive **AI and cloud infrastructure** expansion
2. Dominance in **digital advertising and search**, with AI monetisation
3. **Enterprise AI adoption** via the Google Cloud backlog
4. **Subscription and YouTube ecosystem** expansion
5. Strong **free cash flow and financial firepower**
6. Global reach and **ecosystem integration**
7. **Waymo** (autonomous-driving market leader) and "other bets" — long-term optionality

Rule stated: "As long as it has got a positive growth rate, it passes this step." All figures are positive and double-digit.

→ **Step 2 PASSED.**

1.3 Step 3 — Economic moat (the most important step)

Tool: StockOracle → **Oracle IQ** → *AI Moat Analysis*. Standard: invest in **wide moat** or **narrow moat**; **avoid no moat**. Adam's caveat: "You may or may not agree with it, you can always do your own research, but this kind of summarizes and brings together all the research that you can find on your own."

Alphabet moat score: 9 / 10

Moat source	Score
Brand power & pricing power	9 / 10
Barriers to entry	10 / 10
Switching costs	8 / 10
Network effect	10 / 10
Economies of scale	9 / 10

Reasoning quoted from the analysis and Adam's commentary:
- **Search**: Google holds >90% global market share, "dwarfing rivals like Microsoft Bing at 4%" and Yandex / DuckDuckGo at roughly 1%.

- **Brand loyalty**: "Google this, Google that" — Google *is* the verb for search.
- **Pricing power**: advertisers "have to pay top dollar" to advertise on Google and YouTube.
- **Barriers to entry**: "It's not easy that anyone else can start a website and compete with Google search, with their kind of history and their data."
- **Cloud**: Google Cloud is a **fast-growing but distant third** behind AWS and Microsoft Azure.
- **AI**: Gemini competes with OpenAI's ChatGPT and others, "supported by massive R&D and infrastructure investments."
- Moat root cause: a dominant global advertising platform anchored by Search + YouTube, which together generate unparalleled **user intent data** and advertiser reach.

→ **Step 3 PASSED.**

1.4 Step 4 — Profitability and operational efficiency

Benchmarks restated: ROE/ROIC ≥ **12% = great business**; **>15% = excellent business**; want them stable or increasing.

Metric	Latest	5-year average
Return on equity (ROE)	**35.7%**	30%
Return on invested capital (ROIC)	**32%**	27%

Both are latest > 5-yr average → improving. "High ROE, high ROIC."

Revenue vs accounts receivable (StockOracle → Financials chart): revenue (blue) growing **much faster** than accounts receivable ✔. Red-flag rule stated explicitly: "If it was the other way around, where accounts receivables are growing faster than revenue, that's a red flag."

Cash conversion cycle: Adam flags this as **not applicable** to Alphabet — "Alphabet doesn't sell a physical product." He looks at it anyway; it has been "more or less pretty consistent over the last five to ten years."

→ **Step 4 PASSED.**

1.5 Step 5 — Conservative debt

StockOracle → *Company Metrics* → *Debt and Liquidity Ratios*.

Ratio	Alphabet	Threshold
Current ratio	**2.0**	> 1
Debt servicing ratio	**0%**	< 30%
Debt / EBITDA	**0.33**	≤ 3
Interest coverage	**112×**	at least 3–5×

"On all metrics, Alphabet is a very, very financially strong business."

→ **Step 5 PASSED.** Conclusion: great business, high-quality company → goes on the watch list.

1.6 Step 6 — Intrinsic value (Discounted Free Cash Flow, with normalised capex)

Method choice. Default method = **DCF**. Precondition: cash from operations must be increasing consistently — it is, so DCF is permitted. Adam runs the **discounted free cash flow** variant using StockOracle's automated calculator

(StockOracle → *Intrinsic Value* → scroll to automated calculator), which auto-pulls latest FCF, total debt, cash & investments, discount rate, shares outstanding, and projected growth rates.

Raw output first: the calculator's untouched answer is **$190.89** intrinsic value.

Why Adam refuses the raw number — the capex normalisation adjustment (the key lesson of this case study):

- Free cash flow = **cash flow from operations − capex**.
- The calculator uses the *latest* free cash flow with no adjustment.
- The hyperscalers — "Meta, Microsoft, Amazon, and Google" — have recently poured money into **AI infrastructure**, so capex spiked and free cash flow collapsed.
- "But will this last forever? No, it's temporary." Taking the depressed latest FCF "will underestimate the intrinsic value."
- Fix: **take the average capex of the last five years** to normalise.

Latest-year derivation (StockOracle → Financials → Cash Flow Statement → Annual):

Item	USD millions
Cash from operations (latest)	164,713
Capex (latest)	91,447
Latest free cash flow	**73,266** ← what the calculator used

Five-year capex, oldest → newest (USD millions): **24,640 → 31,485 → 32,251 → 52,535 → 91,447**. Adam highlights "a big jump over here" between years 4 and 5, plus the earlier step-up at 52,535.

Average capex = (24,640 + 31,485 + 32,251 + 52,535 + 91,447) ÷ 5 = **46,471.6**

Normalised free cash flow = 164,713 − 46,471.6 = **118,241.4** (USD millions)

Base case result. Plugging 118,241.4 into the calculator and recalculating:

> **Base case intrinsic value = $304.13**

Growth inputs behind the base case: **15.3%** (first 5 yrs) / **10.13%** (next period) / **4%** (terminal).

Conservative case. Replace the 15.3% FactSet figure with the **14.3%** three-source average (FactSet 15.3, FinViz 12.88, Zacks 14.72), leaving everything else identical:

> **Conservative intrinsic value = $292.41**

Valuation range: $292 – $304.

The lazy alternative. "If you're very lazy, then just look at the Oracle Value." StockOracle's own **Oracle Value = $294**, which Adam notes falls as "kind of like a median between this range that I just did for you." So a user can either do the manual capex normalisation or simply accept the Oracle Value.

Scenario	5-yr growth input	Intrinsic value
Calculator raw (unadjusted FCF of 73,266)	15.3%	$190.89
Base case (normalised FCF 118,241.4)	15.3%	**$304.13**
Conservative (normalised FCF 118,241.4)	14.3%	**$292.41**

Scenario	5-yr growth input	Intrinsic value
StockOracle "Oracle Value" shortcut	—	$294

1.7 Step 7 — Technical analysis and entry plan

Charting software: **TradingView**. Workflow is always **10-yr monthly → 5-yr weekly → 1-yr daily**.

Chart hygiene notes:
- Click **L** for logarithmic scale; **A** to auto-readjust.
- **Do NOT enable ADJ (adjust for dividends)** when drawing support levels. "I only adjust for dividends when I'm comparing the performance of a few stocks, but if I'm looking at just one stock, I do not adjust for dividends."

(a) 10-year monthly. Requirement: "If it's a great company, you should see a long-term up trend. If you see a long-term down trend, then think twice." Alphabet shows a clear long-term uptrend of wave-up/wave-down. The big drawdowns identified on the chart: the **COVID crash**, the **2022 bear market**, and "the trade war unleashed by Trump," each followed by a wave up.

First action: draw the intrinsic value as a **horizontal ray, dotted, green** at the conservative figure **292**. Rule: "I want to make sure that I only buy when the price goes below this green line."

Supports identified on the monthly:
- **50 MA (blue line) = $163.94** — historically a very strong support during bear markets/crashes. Caveat stated plainly: "Am I predicting that the price is gonna drop to 163.94? Of course not… In fact, the price may never drop to that level." The reason to draw it is probability, not prediction.
- **20 EMA (red line)** — acts as support during *normal* wave-downs ("toing, toing, toing").

(b) 5-year weekly. Structure read: uptrend → downtrend (2022 bear market) → uptrend → sharp correction → uptrend again.
- The **50 MA on weekly** provides support and **coincides with the 20 EMA on monthly** — "the 50 moving average on the weekly candles is equivalent to the 20 EMA on the monthly candles. They actually coincide." Double confirmation of that level.
- The **100 MA on weekly (orange line) ≈ $207** has previously acted as support **and** coincides with a previous swing high → two points of confirmation. Support drawn at **207**.

Caveat: "Support levels can be subjective. You can have two different investors looking at the charts and they identify different support levels."

(c) 1-year daily. The stock had been in a sustained uptrend, but "recently you can see that it waved down and then waved up and waved down and waved up. So it looks like it's at the **early stages of a short-term downtrend**." No one can predict how long a downtrend lasts.

Adam's own position and behaviour: he is already invested in Alphabet, has been "for quite a number of years." Is the short-term downtrend a reason to sell? "No, of course not… It's a great company. In the long run, it will always go higher… In the short term, it can go on a short-term downtrend, but eventually the trend will reverse back up." He doesn't sell — he adds.

The nearest support that he rejects. There is a visible support level on the daily chart where price has bounced repeatedly, but it sits **above** the $292 intrinsic value. He will not buy there: "while it's a support level and it could bounce off, this support level is **not below the intrinsic value**. So I'm not buying when it's undervalued with a margin of safety, I'm paying the fair price… there's nothing wrong with that, but for me, I'm very conservative. I like to buy at least below the intrinsic value." He therefore pulls the level down to the next swing low.

GOOGL final verdict and entry ladder

- **Business verdict:** great business, passes all 5 quality steps.
- **Value verdict:** worth **$292 – $304**; only buy **below the green $292 line**.
- **Entry plan:** **~$270** (next swing low below intrinsic value), then the mentioned **$237** level, then the weekly-100MA/swing-high level at **$207**, then the monthly-50MA level at **$163.94**.
- **Doctrine:** "For every stock that I think is a great company, I identify the intrinsic value, and then below that, I will add at least **four to five support levels**. And this is where I will add shares if the price drops to that level. And I'll keep adding until I have got a **full position**. Once I've got a full position, I won't add more. But I'll let the position grow organically."
- Full-position sizing / portfolio allocation is deferred to the portfolio-construction lesson.

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2. MICROSOFT (MSFT) — Lessons 55 & 56

Listed: **NASDAQ**. Sector: **Technology**. Industry: **Software**.

2.0 Investment thesis first

Before analysing anything, define **why** you are buying: growth, predictability, defensiveness, dividends, etc. (There are eight investment categories, covered in a later lesson.)

> **MSFT thesis: growth AND predictability.** "It is both a growth stock, it grows at high double digits, at the same time it is a very predictable company. It's **recession-proof and pandemic-proof**."

Understand the business by being a customer: Adam uses Microsoft Office for home and office computing, LinkedIn, Microsoft Teams ("a more sophisticated version of Skype") at his office, and plays Xbox with his kids. Supplement with the annual report, the company website, and an online business description.

2.1 Business overview and segments

Microsoft "develops, licenses, and supports software, services and solutions worldwide" — productivity software, cloud computing, advertising campaigns.

Three reporting segments:

Segment	Contents	Competitive position
Intelligent Cloud	Azure, Visual Studio, Windows Server	**#2** in enterprise cloud behind Amazon; "Azure is fast catching up"
Productivity & Business Processes	Office 365 (Excel, PowerPoint, Word), LinkedIn, Microsoft Teams	Dominant; enterprises use Teams over Zoom
More Personal Computing	Windows, Xbox, HoloLens (VR/AR — "huge potential in the upcoming metaverse")	Windows ~86% desktop OS share

Revenue breakdown:

Segment	Share of revenue
Personal computing	~34%

Segment	Share of revenue
Cloud (enterprise)	~30%
Enterprise productivity & business processes	~35%

2.2 Step 1 — Consistent revenue / net income / operating cash flow

Tools: **Yahoo Finance** (4 years + TTM), then **GuruFocus** for the longer history, then **Morningstar** for margins.

Yahoo Finance annual figures (FY2018, 2019, 2020, 2021, plus TTM), USD billions:

Metric	2018	2019	2020	2021	TTM
Revenue	110	125	143	168	176
Net income	16	39	44	61	67
Cash flow from operations	43	52	60	76	**81** (81,945 M)
Free cash flow	positive	positive	positive	positive	positive ✔

All three lines increase consistently. Free cash flow is "positive all the way."

GuruFocus long-history check. Revenue grows nicely long-run; net income also grows long-run with one dip ~7 years back. Adam's method for a dip: **go to the annual report and read the income statement to find the cause.** In this case the drop was due to **changes in taxation** — "which is fine because it is not a core business expense that caused the drop, it is a non-core operating expense. So the actual business itself was still growing." Operating cash flow (grey) rises consistently; free cash flow (green) positive.

Margins (Morningstar → Operating Performance):

Margin	5-yr average	Recent years
Gross profit margin	**65%**	65 → 67 → 68 → 68 (recent > 5-yr avg ✔)
Net profit margin	**26%**	30 → 36 → 38 (rising ✔)

Rule restated: "As long as it's consistent or rising, that's great. As long as it's not falling."

→ **Step 1 PASSED.**

2.3 Step 2 — Growth projections

Source	Projection
FinViz — past 5 years actual	**25.7% p.a.**
FinViz — next 5 years	**16.53%**
Zacks — next 3–5 years	**12.04%**
Simply Wall St	**10.8%**
Average of the three forward figures	**13.12%** ("13%")

Adam notes FinViz also confirms the sector (Technology) and industry (Software).

Outlier rule: "Sometimes if one website gives a very different figure — for example 16% and 14% and 3%, and 3% is like way out there — then I would ignore it. So either it's ridiculously low or ridiculously high, I'll ignore that. It's an outlier. I'll take the average of the different websites [that] have more or less the same growth projections."

Pass criterion is simply that growth is **positive**. Optimistic → take the highest; pessimistic → take the lowest; unsure → average.

→ **Step 2 PASSED.**

2.4 Step 3 — Economic moat (qualitative only)

"Step three is more qualitative. In other words, there's no number to look at." The problem with owning stocks whose products you don't use: "it's really hard to understand the company's competitive advantage."

Checklist of moat sources run against Microsoft:

Moat source	Microsoft?	Evidence
Brand monopoly	✔ Yes	Top-of-mind recall: word processing → Word; spreadsheets → Excel; presentations → PowerPoint; operating system → Windows
High switching costs	✔ Yes	*Individuals*: "I'm so used to using PowerPoint, Excel, and Microsoft Word… I've learned it all my life since I was a kid." *Companies*: switching off Azure to another cloud is "costly and expensive and frustrating… once you're locked in"
Network effect	✔ Yes	Adam works in Keynote on a Mac but "still have to convert to PowerPoint to share with my colleagues… because everyone is using it, you gotta use it." Also **LinkedIn**
Barriers to entry	(listed as a source; not separately scored)	—
Economies of scale	(listed as a source; not separately scored)	—

Verdict: "Microsoft definitely has three of the economic moat sources… it definitely has a sustainable competitive advantage."

The chart-as-moat-proxy shortcut: "Usually a company with a wide economic moat has a long-term uptrend in the stock price… if you see that the price goes up like that in 10 years, it usually has a wide economic moat because **only companies that have got no close competition are able to perform so well**. So that's a clue." Microsoft's 10-year monthly chart is a clear uptrend.

Competitor mapping (search "Microsoft's competitors" online — and note that published lists can be stale; the article Adam opens omits Amazon and still lists IBM):

Segment	Competitors
Cloud	Amazon (AWS), Alphabet (Google Cloud)
Enterprise IT solutions	Oracle, SAP
Personal / application software, desktop OS	Apple — but **Apple has 14% desktop OS share vs Windows 86%**

Relative-strength test on TradingView (5-year comparison overlay):
- vs **Oracle and SAP** — "Microsoft of course performs a lot better." Principle: "obviously you want to invest in the strongest performer."

- vs **Amazon and Alphabet** — not a clean comparison (Amazon also does e-commerce; Alphabet also does search advertising and YouTube), "they compete in only one part of the business, but not in all parts." Even so, **Microsoft outperformed both over the last 5 years.** Adam adds: "I own all three companies anyway. So I don't care which one wins, because I own all three."
- vs **Apple** — "pretty much neck to neck, very, very close, with Apple slightly above Microsoft." Not a direct competitor across most segments since "Microsoft doesn't really make much hardware."

→ **Step 3 PASSED.**

2.5 Step 4 — Profitability and operational efficiency

Four metrics: ROE, ROIC, revenue vs accounts receivable, cash conversion cycle. ROE/ROIC source: **Morningstar → Operating Performance**. Threshold: **above 12%**.

Metric	Last 12 months	5-year average
ROE	**49%** ("wow")	34%
ROIC	**33%** ("double wow")	20%

Revenue vs accounts receivable — TradingView build instructions (given step by step):
1. Financials → Revenue → **Total Revenue, Annual**
2. Indicators → search **Accounts Receivables** → **Annual**
3. On one pane, click the three dots (*More*) → **Move to existing pane above** (merges the two panes)
4. Right-click the price axis (or two-finger click on a Mac trackpad) → **Merge all scales into one on the right**

Result: revenue (blue) rising strongly, accounts receivable (magenta) rising more slowly ✔.

Cash conversion cycle: "mainly applies to retail companies that sell physical goods, that have physical inventory. So it doesn't really apply to Microsoft that much, because Microsoft sells software and services" — checked anyway "for practice purposes." It is **declining** ✔. Rule: want it declining or consistent; "the shorter the cash conversion cycle, the faster the company makes its money back. But if it is rising, then that's a bad sign."

→ **Step 4 PASSED.**

2.6 Step 5 — Conservative debt

Ratio	Microsoft	Threshold	Source
Current ratio	**2.16**	> 1	GuruFocus
Debt / EBITDA	**0.7**	≤ 3	GuruFocus
Debt servicing ratio	**2.8%**	< 30%	calculated manually

Debt servicing ratio — worked by hand (Adam: "something which you have to calculate yourself because I don't find it on any website"):

```
Debt servicing ratio = (Net interest expenses ÷ Cash flow from operations) × 100%
```

- Net interest expense (income statement, annual/TTM): **$2,296 million** (displayed as 2,296,000 in thousands)
- Cash from operations (cash flow statement, TTM): **$81,945 million**
- 2,296 ÷ 81,945 × 100 = **2.8%** → far below the 30% ceiling ✔

→ **Step 5 PASSED.** Great business → watch list.

2.7 Step 6 — Intrinsic value (Discounted Cash Flow from operations)

Method choice: primary method = **discounted cash flow** using **cash flow from operations**. Precondition — operating cash flow must be increasing consistently: 43 → 52 → 60 → 76 → 81 billion. Yes. Method permitted.

Calculator hygiene: key in only the **white boxes**; shaded/coloured boxes contain built-in formulas — "if you change the coloured boxes, it will change the formula."

Full input set

Input	Value	Where it comes from
Name / symbol	Microsoft / MSFT	—
Valuation method	**Discounted cash flow** (from operations)	—
Statement currency	USD	Financial statements
Stock listing currency	USD	Exchange listing
Exchange rate	1.00	USD:USD
Most recent cash flow from operations	**$81,945 million**	Cash flow statement, TTM
Total debt (short-term + long-term)	**$53,288 million**	Balance sheet, **latest quarter**: current debt 3,249 + long-term debt 50,039
Cash and short-term investments	**$130,584 million**	Balance sheet, latest quarter: cash & equivalents + short-term investments
Growth rate, first 5 years	**13.12%**	Average of FinViz 16.53 / Zacks 12.04 / Simply Wall St 10.8
Growth rate, next 5 years	**13.12%**	Same as first 5 years, **subject to a maximum of 15%**
Growth rate, final 10 years (terminal)	**4%**	US average GDP growth + 1% ("last I checked it's 4.18%… just fix it at 4%, it's fine")
Shares outstanding	**7,510 million** (7.51 billion)	FinViz — note the calculator wants **millions**, so 7.51 bn × 1,000
Beta	**0.88**	FinViz
Discount rate	**5%**	Discount-rate table keyed off beta (see below)

Currency discipline (a general lesson taught inside the MSFT case): the statement currency and the stock listing currency can differ. For **Alibaba**, statements are in **RMB**; the listing currency is **USD** if bought on the US exchange or **HKD** if bought in Hong Kong — you must then enter the correct exchange rate. Covered further in an Alibaba/Tencent case study.

Cash-flow normalisation rules taught here (applied later to AMZN and GOOGL):
- Check that the latest figure "is similar or higher than previous years" — you want a gradual increase to project forward.
- **Sudden big jump** in the last 12 months → do **not** use it; it may not be sustainable. Use the prior year, or an average of the last 3–4 years.
- **Sudden big plunge** (e.g. pandemic-driven) → do **not** use it either; it won't last. Use a 3–4-year average for a "more

normal cash flow."
- Microsoft grows "very, very consistently," so the latest figure is used as-is.

Debt definition: "Total debt is **not** total liabilities. It is short-term debt plus long-term debt." Current debt = short-term debt (same thing). Balance-sheet items are always taken from the **latest quarter**, never annual.

Discount rate table: driven by **beta**; separate tables exist for US stocks and China stocks. Beta 0.88 sits between the "<0.8 → 4.6%" and "1.0 → 5.6%" rows. Adam first says 5.6%, then reasons "0.88 is closer to 0.8, so why don't you use 5" and settles on **5%** — "Doesn't have to be exact."

Results

> **Base case intrinsic value = $398** (growth 13.12%)

Current share price as of **15 January 2022 = $310** → the stock is **undervalued by 22%**.

Pessimistic scenario 1 — use the lowest growth projection. Replace 13.12% with Simply Wall St's **10.8%**:

> **Pessimistic value = $337** → at $310, still an **8% discount** ("which is still not bad")

Pessimistic scenario 2 — halve the growth after year 5. Keep the first 5 years at 13.12% but set the next-5-year rate to **6.56%** (13.12 ÷ 2):

> **Pessimistic value = $324.99 ≈ $325** → still undervalued

Scenario	1st-5-yr growth	Next-5-yr growth	Intrinsic value
Base case	13.12%	13.12%	**$398**
Pessimistic 1 (lowest projection)	10.8%	10.8%	**$337**
Pessimistic 2 (growth halves after yr 5)	13.12%	6.56%	**$325**

Valuation range: $325 – $398, against a share price of **$310**.

> "Do you feel pretty safe buying at this price knowing that the company's worth anywhere between $325 to $398? Yeah, I am."

Principle: "If the current share price is below **even the pessimistic one**, that gives you a great margin of safety."

2.8 Step 7 — Technical analysis and entry plan

Entry triggers (stated as the three valid setups): buy on **a dip within an uptrend**, at the **support level of a consolidation**, or when price **reverses into a new uptrend**.

The **green intrinsic-value line is drawn at $325** — i.e. deliberately at the *most pessimistic* value, "but of course you can put the more optimistic one as well."

(a) 10-year monthly. "I want to see a clear uptrend like that. If I don't get a clear uptrend, I'm not interested in the stock in the first place." ✔ Then: where does price land whenever there's a crash? Repeatedly at the **20 MA (red dotted line)** — multiple historical touches.
→ Support drawn at **$262**. "If Microsoft crashes because of whatever reason — bear market or nuclear war or whatever it is — then you can be sure that it's going to hit this support level and you want to be a buy at 262."

Important qualifier: this does **not** mean waiting only for 262. "We can start buying now because it's already undervalued… you never know when it's going to hit that. It may hit that this year, next year, three years from now… So if you want to wait only when it reaches there, then you may buy maybe once every few years."

(b) 5-year weekly. Price hit the **50 MA (blue line)** — a strong historical support that also coincides with a **previous swing low**.
→ Support drawn at **$283**. Also, the **20 MA on weekly** is a strong support.
→ Support drawn at **$305**.

(c) 1-year daily. Clear wave pattern; price is currently waving down and has found support at the **green line = 150 MA**, which coincides with **previous resistance turning into support**.
→ Support drawn at **$316**. "Resistance becomes a support. Beautiful, right?"

MSFT final verdict and entry ladder

Support level	Where it comes from
$316	Daily 150 MA + previous resistance turned support
$305	Weekly 20 MA
$283	Weekly 50 MA + previous swing low
$262	Monthly 20 MA (crash-level support)

Green intrinsic-value line: **$325**. Price: **$310** (already below the pessimistic value).

Tranche execution as described: "After this wave up, it's now wave down, it hit the first level, and it's undervalued — I could have bought **one quarter of my planned allocation** here. Now it hit the second level, I buy another one quarter, so I could have bought **half** of the intended Microsoft shares I want to buy in my portfolio… and of course if it drops even further, then I would add even more until I have a **full position**." If it never reaches the lower levels and instead waves back up and down again, you simply buy the next time it returns to a level.

Verdict: great business, undervalued, and at a valid entry point — buy in quarters at $316 / $305 / $283 / $262.

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3. AMAZON (AMZN) — Lessons 57 & 58

Listed: **NASDAQ**.

3.0 Thesis and classification

> **Thesis: large-cap GROWTH.** "When a company grows more than 20%, we classify it as a growth company."

Sector correction Adam insists on: "Some people think that Amazon's a technology stock. **No it's not.** It is a **retail company** that is within the **consumer discretionary sector**," in the **retail industry**.

Understand the business as a customer — Adam is an Amazon Prime member.

3.1 Business overview, segments and competitors

Business	Description	Competitors
Retail / e-commerce (Amazon, Amazon Go)	"One of the highest-grossing e-commerce retailers in the world" — deliberately *not* "the highest": **Alibaba is the biggest by gross merchandise value.** Amazon is "smaller than Alibaba in terms of GMV" but "a lot more profitable… much higher margins because of its cloud business"	Alibaba, Walmart, Target, eBay
Whole Foods	Supermarket, healthy foods	(grocery)
Cloud — AWS	**The most profitable business, not retail.** "In cloud, Amazon is the market leader" — **33% market share of the cloud market as of Q2 2020**	Microsoft Azure, Alibaba Cloud, Google Cloud
Game streaming — Twitch		YouTube Gaming, Mixer
Amazon Prime Music		Spotify, Apple Music
Amazon Alexa		Google Home
Amazon Logistics (smaller)		FedEx
Amazon Pay (smaller)		Stripe, Apple Pay, PayPal, Google Pay
Kindle, Audible, Amazon Prime video (smaller)		Netflix

Key structural insight: **retail is not the profit engine — cloud is.**

3.2 Step 1 — Consistent revenue / net income / operating cash flow (and the negative FCF investigation)

Yahoo Finance, annual + TTM, USD billions (only ~3 years + TTM available on Yahoo):

Metric	Yr1	Yr2	Yr3	TTM
Revenue	232	280	386	**457**
Net income	10	11	21	**26**
Cash flow from operations	30	38	66	**54** (a drop)
Free cash flow	17	21	**25.924**	**NEGATIVE**

Revenue ✔ consistently up. Net income ✔ consistently up. Operating cash flow: "generally we have an uptrend" despite the TTM dip.

The negative free cash flow investigation (the signature lesson of this case)

Free cash flow turned **negative** in the last 12 months. Adam's diagnostic logic:

```
Free cash flow = Cash from operations − Capital expenditure (PP&E)
```

"If free cash flow turns negative, it's because of two reasons: either the operational cash flow is negative, or the capex is very big. But in this case, the operational cash flow is still very healthy… So the only reason free cash flow turned negative is because of **huge amounts of capital expenditure**."

Where to look: capex sits **under investing cash flow** on the cash flow statement, as **PP&E**. It shows "a huge jump from previous years."

Research finding (Adam's own answer posted to his student chat groups): In the last 12 months Amazon spent **a whopping $45 billion on capital expenditures — more than twice** what it normally spends. Reasons:
- New **Amazon Fresh** supermarkets
- **Cloud infrastructure**
- **Voice-activated technology**
- **Logistics — "the biggest driver of this growth in CapEx"**

Is it good or bad? "To me, the answer is **good**. Why? Because they're making a lot of investments that may cause the free cash flow to turn negative temporarily for one year, but by investing heavily in all these new businesses like cloud infrastructure, logistics, it will lead to higher growth and higher profits in the future."

Boundary condition: "It is fine for **one year** of negative free cash flow, but of course if this persists for **more than one year — like two or three years — then you gotta be careful.**"

Method note: "When you have got something out of the ordinary, you need to do more research. You need to go online and research… read their annual reports to find out is it a good thing."

GuruFocus long-run view: net income (blue) increasing consistently over five years; operating cash flow (grey) increasing consistently; free cash flow (green) consistently positive except the last 12 months. Net income looks tiny beside revenue on the same axis — use the separate chart to see it clearly.

The anomaly flagged for later: "Their operating cash flow is much higher than their net profits and net income… I'll talk more about this later on in the intrinsic value calculation." (This is what changes the valuation method — see §3.7.)

Margins (Morningstar → Operating Performance):

Margin	2018	2019	2020	TTM	vs 5-yr average
Gross margin	26%	25%	22%*	**25%**	**above** 5-yr avg ✔
Net profit margin	4%	4%	5% (from 1% earlier)	**5.7%**	**above** 5-yr avg ✔

* Adam reads the gross margin row as "22%, 25%, 26%, 24% — so it's more or less consistent, it's pretty okay," and net margins as "1%, ooh, that's really low, 4%, 4%, 5% — so that's growing and that's consistent as well," with TTM at gross 25% / net 5.7%. Both TTM figures exceed the 5-year averages → "profit margins look healthy."

→ **Step 1 PASSED.**

3.3 Step 2 — Growth projections and catalysts

Qualitative catalysts: the business is primarily **e-commerce + cloud computing**, "and they are both in huge growth industries." E-commerce is a **long-term secular growth industry** ("more and more people are spending online") and cloud computing "is also a secular growth industry."

Source	Projection
FinViz — past 5 years actual	**101% p.a.** — "this 101% is on average **per year** growth, not total five years"

Source	Projection
FinViz — next 5 years	**36%** ("expected to moderate")
Zacks — next 3–5 years	**24.68%**
Simply Wall St	**29.8%**
Average of the three forward figures	**≈30%**

"They are all pretty similar." (36 + 24.68 + 29.8) ÷ 3 ≈ **30%**.

→ **Step 2 PASSED.**

3.4 Step 3 — Economic moat

Moat questions run against Amazon:

Moat source	Amazon?	Reasoning
Brand monopoly	✔ Yes	Top-of-mind for e-commerce; also for cloud infrastructure
High barriers to entry	✔ Yes	Amazon Prime same-day free delivery — "because of their scale, their size, they can do it faster and cheaper than anyone else"
Huge economies of scale	✔ Yes ("you bet")	Scale of fulfilment network
Network effect	✘ "Maybe not"	Adam explicitly declines to claim it
High switching costs	⚠️ **Split by segment**	*E-commerce/individual:* **No** — "if you want to buy on Amazon today, you can always buy on Alibaba tomorrow, you can buy on eBay the next day, so you could switch easily as an individual." *Cloud/enterprise:* **Yes** — "once a company hosts their stuff on Amazon Web Services, it's gonna be expensive and difficult to switch to Microsoft Azure"

Market-share anchor: **AWS holds 33% of the cloud market as of Q2 2020 — clear market leader.**

Verdict: "Amazon definitely does have at least **two or three** of these sources of a wide moat."

→ **Step 3 PASSED.** (Note this is a *weaker* moat verdict than GOOGL's 9/10 or MSFT's clean three-source pass — Adam is explicit that two of the five sources fail or are segment-limited.)

3.5 Step 4 — Profitability and operational efficiency

Morningstar → Operating Performance:

Year	ROE	ROIC
2017	12%	14%
2018	28%	11%
2019	21%	14%
2020	27%	13%
TTM	**25%**	**above 12%**

Both TTM figures are **above 12%** and **above their 5-year averages**. Adam acknowledges ROIC "has been below 12% in 2017 and 2019 slightly, but last 12 months it's way above 12% and above the five-year average, **so that's fine for me.**" (A tolerance ruling: occasional dips below threshold are acceptable if the latest reading and trend are strong.)

Revenue vs accounts receivable — same TradingView merge procedure as MSFT (annual revenue + annual accounts receivables → *Move to existing pane above* → right-click axis → *Merge all scales into one on the right*). Over 5 years, revenue grows "a lot more than accounts receivables" ✔.

Cash conversion cycle — TradingView → Indicators → Financials → Cash Conversion (viewable quarterly), last 5 years: **declining** ✔. "The shorter the cash conversion cycle, the more efficient the company is at turning around its cash." Note: unlike MSFT and GOOGL, this metric is genuinely applicable to Amazon because it is a physical-goods retailer.

→ **Step 4 PASSED.**

3.6 Step 5 — Conservative debt

Ratio	Amazon	Threshold	Source
Current ratio	**1.12**	> 1	GuruFocus
Debt / EBITDA	**1.76**	≤ 3	GuruFocus
Debt servicing ratio	**2.4%**	< 30%	calculated manually

Debt servicing ratio worked by hand — note the *net* interest treatment:

From the income statement (quarterly view):
- Interest **income**: **$430 million**
- Interest **expense**: **$1.7 billion**
- **Net interest expense = $1,311 million**

Cash from operations (TTM): **$54,672 million**

1,311 ÷ 54,672 × 100 = **2.4%** → "only 2.4% of the company's cash from operations is used to pay its interest expenses, which is very, very low." ✔

(Contrast with MSFT, where Adam used the gross interest-expense line directly. Here he explicitly nets interest income against interest expense.)

→ **Step 5 PASSED.** Great business → watch list.

3.7 Step 6 — Intrinsic value: WHY AMAZON GETS A DIFFERENT METHOD

This is the most important methodological difference across the three case studies.

Adam's default is **discounted cash flow from operations**. He does **not** use it for Amazon. The rule:

> **When cash from operations is more than ~1.5× net income, switch to the discounted FREE cash flow method — it is more conservative.**

Evidence for Amazon:

Year	Net income (USD bn)	Cash from operations (USD bn)	Ratio
2018	10	30	3.0×

Year	Net income (USD bn)	Cash from operations (USD bn)	Ratio
2019	11	38	3.5×
2020	21	66	3.1×
TTM	26	54	2.1×

Why the gap exists — the accounting explanation Adam gives:
- **Depreciation and amortisation is a non-cash expense.**
- Net income is calculated **after deducting** D&A.
- Cash from operations **adds D&A back**.
- Amazon's D&A is enormous — "you can see that depreciation over here, **31 billion, 24 billion**, makes up a huge chunk of the expenses."
- Therefore operating cash flow balloons relative to net income.
- Conclusion: "This tells us that this company has **high capital expenditures or high depreciation costs**. So for these kinds of companies it makes sense to use free cash flow instead, because free cash flow would be the cash from operations after deducting capital expenditure."

Which free cash flow figure to use — normalisation, in the opposite direction from GOOGL.

Amazon's FCF history: 17bn → 21bn → **25.924bn** → **negative (TTM)**.

"Does it make sense to take this number? It doesn't make sense, right? Because this is exceptionally low because of a one-time increase in capex… **whenever you see a big change in the last year or so, you can ignore that and take the year before that, before this skew happened** — that it's more of a normalized free cash flow."

> **FCF input used = $25,924 million** (the FY2020 figure, i.e. the year *before* the capex spike).

(Compare: for GOOGL, faced with the same hyperscaler-capex problem, Adam instead averaged five years of capex and subtracted that from the latest operating cash flow. Two different normalisation techniques for the same underlying distortion.)

Full input set

Input	Value	Source / note
Name / symbol	Amazon / AMZN	—
Valuation method	**Discounted FREE cash flow**	Because CFO > 1.5× net income
Normalised free cash flow	**$25,924 million**	FY2020 (pre-capex-spike)
Total debt	**$50,055 million**	Balance sheet, **latest quarter**. Adam expands the liabilities tree and finds **no short-term debt line at all** — "you don't see any short-term debt, do you? Nope, so there isn't any." Long-term debt only
Cash and short-term investments	**$78,988 million**	Balance sheet, latest quarter: cash & equivalents **+** short-term investments added together
Growth, first 5 years	**30%**	Average of FinViz 36 / Zacks 24.68 / Simply Wall St 29.8
Growth, next 5 years	**15%**	Same-as-first-5-years rule, **capped at the 15% maximum** — 30% exceeds the cap
Growth, final 10 years	**4%**	US GDP growth + 1%

Input	Value	Source / note
Shares outstanding	**507 million**	FinViz
Beta	**1.1**	FinViz
Discount rate	**6.3%**	From the (updated) beta table
Statement currency / listing currency	USD / USD, rate 1:1	Final currency check

Discount-rate table update — explained here: the table Adam uses for Amazon "is slightly different from the table which I used in calculating Microsoft… there's been a recent update on this table." The underlying formula:

```
Discount rate = Risk-free rate + (Beta × Market risk premium)
```

where the **risk-free rate is the 5-year average of the 10-year Treasury bond rate** and the **market risk premium is also a 5-year average**. Because those averages move, the table is periodically refreshed and posted to the chat groups and course website. "Always ensure that you're using the latest calculator." He explicitly tells MSFT-case viewers to redo their number with the newer table.

Results

> **Base case intrinsic value = $3,312** (growth 30%)

Current share price (Friday's close): **$3,242.76** → Amazon is **undervalued by 2.1%**.

Optimistic scenario — use FinViz's 36% growth, all else unchanged:

> **Optimistic intrinsic value ≈ $4,071**

Scenario	1st-5-yr growth	Intrinsic value
Base / conservative	30% (3-source average)	**$3,312**
Optimistic	36% (FinViz)	**$4,071**

Valuation range: ~$3,312 – $4,071, against a price of **$3,242.76** — below even the low end of the range.

> "Notice that intrinsic value is never one number. It's always kind of like a range. Why? Because we can't be absolutely sure about what the growth of the company is going to be. We can only do projections."

Third-party cross-check. Morningstar Premium (paid) publishes its own **fair value ≈ $4,001**. Adam's comparison logic: "For Morningstar, their valuation most of the time tends to be higher than my valuation, because they tend to be more optimistic — or rather, **I tend to be more pessimistic**. But I like to look at their valuation as well to kind of get a sense of where my valuation is relative to theirs." He concludes he is "really, really comfortable" adding shares at the current price given the independent confirmation.

(Transcript note: the spoken sentence "way above even my optimistic valuation of 4,070" is internally inconsistent with a $4,001 Morningstar figure — the ASR is unreliable on these two numbers; the intent is clearly that Morningstar's fair value sits at or above Adam's optimistic case and far above the market price.)

3.8 Step 7 — Technical analysis and entry plan

Entry doctrine restated: "Prices move in wave patterns. Wave up, wave down… we always want to wait for a wave down to a support level. We buy either at **the dip of an uptrend**, at **the support level of a consolidation**, or when **price reverses into a new uptrend**." Timeframes: daily, weekly and monthly, adding **in tranches**.

(a) 10-year monthly. Clear long-term uptrend ✔. Historical wave-downs repeatedly find support at the **20 EMA (red dotted line)**.
→ Support drawn at **$3,116**. Adam notes this level also lines up with the **100 MA on weekly** — "that's a pretty significant level."

(b) 5-year weekly. Strong support at the **50 MA (blue line)**.
→ Support drawn at **$3,351**.
Also: **100 MA (orange line)** has acted as strong support, coinciding with the $3,116 level.
An additional previously-tested level from repeated swing lows → support at **$2,881** ("swing low, swing low, swing low… this is a pretty strong support").

(c) 1-year daily. Price is in a **sideways consolidation**, not a trend. Rule for consolidations: "When it's moving sideways, you want to buy **near the support of the consolidation**." The consolidation support is visible with three touches ("support, support, support"), and price "just waved down to this support level here at **$3,198**."

AMZN support ladder

Support level	Basis
$3,351	Weekly 50 MA
$3,198	Daily consolidation support (current touch)
$3,116	Monthly 20 EMA + weekly 100 MA (double confirmation)
$2,881	Repeated weekly swing lows

AMZN verdict

> "So right now it just waved down to this support level here at 3,198. So yeah, **I think this is a reasonable point to add shares of Amazon**."

Great business ✔ · below the pessimistic end of the valuation range ✔ · sitting on consolidation support ✔ → **BUY, in tranches.**

Position-sizing note unique to AMZN (a ~$3,242 share price):
- Four support levels → buy the planned allocation in **four tranches**.
- For small accounts, one share costs $3,242. **Fractional shares** solve this — "with certain brokers, like Interactive Brokers, you can buy fractional shares… you could actually buy **0.25 shares at a time**," i.e. roughly **$800 per tranche**, accumulating to one full share across the four levels.
- Larger accounts simply scale the same ladder — "one share here, one share here… or three shares, three shares. It's up to you how you want to divide it."
- Since price is already below the first identified support, an investor with **no existing position** could enter in **thirds** across the three remaining levels.

Cross-Case Lessons

A. The formula is invariant; the *inputs and adjustments* are judgement calls

All three names run the identical seven steps in the identical order, and in all three, steps 1–5 only earn a place on the **watch list**. What differs is never the framework — it's how Adam handles anomalies inside it. The skill being taught is not the checklist; it's knowing when a reported number is lying to you.

B. Three ways the same distortion was handled

The hyperscaler AI-capex problem hit both GOOGL and AMZN, and Adam used **two different fixes**, which tells you the principle matters more than the recipe:

Company	Distortion	Fix
GOOGL	Latest capex spiked to $91.4bn (vs $24.6bn five years earlier), crushing FCF to $73.3bn	**Average the last 5 years of capex** ($46,471.6M) and subtract from the latest operating cash flow → normalised FCF $118,241.4M. Raw calculator value $190.89 → adjusted **$304.13** (a **59% understatement** if left unadjusted)
AMZN	$45bn capex year drove FCF **negative**	**Skip the distorted year entirely**; use the prior year's clean FCF of $25,924M
MSFT	None — "growing very, very consistently"	Use the latest figure as reported

Generalised rule stated in the MSFT lesson and applied in both others: **a sudden spike or a sudden plunge in the last 12 months is never the number you project from.** Use the prior year, or a multi-year average.

C. Valuation method is chosen by the *shape of the financials*, not by preference

Company	Method	Trigger
MSFT	Discounted cash flow **from operations** (the default)	Operating cash flow rising consistently; no CFO/NI distortion
GOOGL	Discounted **free** cash flow (with normalised capex)	Operating cash flow rising consistently; capex temporarily inflated
AMZN	Discounted **free** cash flow (prior-year figure)	**Cash from operations > 1.5× net income** → the D&A-driven gap means CFO flatters the business; FCF is the conservative choice

The AMZN case is the explicit teaching moment: a huge CFO-to-net-income gap is a *signal about the business model* (capital-intensive, high depreciation), and the valuation method must respond to it.

D. Intrinsic value is always a range — and the range is built by flexing growth

Every case produces at least two numbers, produced by changing the growth assumption and nothing else:

Company	Low end	High end	Price at time	Position vs range
GOOGL	$292.41 (14.3% avg growth)	$304.13 (15.3% FactSet)	not stated; supports drawn at 270 / 237 / 207 / 163.94	Trading **above** the range — wait
MSFT	$325 (growth halves after yr 5) / $337 (10.8% growth)	$398 (13.12% growth)	**$310**	**Below** even the pessimistic value → 22% undervalued vs base case

Company	Low end	High end	Price at time	Position vs range
AMZN	$3,312 (30% growth)	$4,071 (36% growth)	**$3,242.76**	**Below** the low end → 2.1% undervalued

"The main thing that can affect your valuation is your growth projections." Three ways to build the pessimistic case are demonstrated: (1) take the **lowest** of the three source projections; (2) **halve** the growth rate after year 5; (3) take the **average** of all sources rather than the highest.

E. Growth rates are triangulated from independent sources, never taken from one

Every case pulls 2–3 forward growth projections and reconciles them:

Company	Sources	Spread	Average used
GOOGL	FactSet 15.3 / FinViz 12.88 / Zacks 14.72	2.4 pts	14.3%
MSFT	FinViz 16.53 / Zacks 12.04 / Simply Wall St 10.8	5.7 pts	13.12%
AMZN	FinViz 36 / Zacks 24.68 / Simply Wall St 29.8	11.3 pts	~30%

Outlier discipline: discard any source that is "way out there" (Adam's example: 16%, 14%, 3% → drop the 3%). Optimist → highest; pessimist → lowest; unsure → average.

Standing DCF conventions across all three:
- Next-5-year growth = first-5-year growth, **capped at 15%** (binding for AMZN at 30% → 15%; non-binding for MSFT at 13.12% and GOOGL at 15.3%).
- Terminal 10-year growth = **US GDP growth + 1% ≈ 4%**.
- Discount rate = risk-free rate + (beta × market risk premium), read off a beta table that is periodically updated; precision is unnecessary — "doesn't have to be exact."
- Balance sheet items (total debt, cash & short-term investments) always from the **latest quarter**; total debt = short-term + long-term debt, **not** total liabilities.
- Always verify **statement currency vs listing currency** and the exchange rate.

F. Step 3 (moat) is the one step with no number — and the verdicts genuinely differ

Company	Moat verdict	Weak spots Adam concedes
GOOGL	**9/10** — near-perfect across all five sources (network effect and barriers to entry both 10/10)	Google Cloud is a "distant third" behind AWS and Azure
MSFT	Clean pass — **three** of the five sources demonstrated (brand monopoly, switching costs, network effect)	Barriers to entry and economies of scale not separately argued; Apple neck-and-neck on total return
AMZN	Weakest of the three — "at least **two or three**" sources	**Network effect: "maybe not." Switching costs: none for individual e-commerce shoppers** (you can switch to Alibaba or eBay tomorrow); only present in the cloud/enterprise segment

Two proxies for moat are taught: (i) **a 10-year uptrend on monthly candles is a clue to a wide moat**, "because only companies that have got no close competition are able to perform so well"; (ii) **relative-strength comparison against named competitors on TradingView** — "obviously you want to invest in the strongest performer."

The prerequisite for judging a moat at all is being a customer: the case studies all open with Adam listing the products he personally uses. "When you invest in a stock where you don't use the product or the service, it's really hard to understand the company's competitive advantage."

G. Metrics get waived when the business model doesn't support them

- **Cash conversion cycle** is checked for all three but is only *meaningful* for Amazon. For GOOGL ("doesn't sell a physical product") and MSFT ("sells software and services") Adam runs it "for practice purposes" and gives it no weight.
- **ROE/ROIC dips are tolerated** if the latest reading and trend are strong — Amazon's ROIC fell below 12% in 2017 and 2019, "but last 12 months it's way above 12% and above the five-year average, so that's fine for me."
- **A down year in net income is not disqualifying.** GOOGL: net income fell from 76 to 59 mid-window — "doesn't have to go up every single year." MSFT: a net income dip ~7 years back was traced to **taxation changes**, "not a core business expense… the actual business itself was still growing." The lesson is to **go read the annual report and find the cause** rather than react to the shape of the chart.

H. Step 7 is identical in structure every time

The chart routine never varies: **10-year monthly → 5-year weekly → 1-year daily**, log scale on, dividend adjustment off (that's only for comparing multiple stocks).

1. Draw the **intrinsic value as a green horizontal line** — usually the *conservative* end of the range (GOOGL at 292, MSFT at 325).
2. Identify supports where price has **historically** bounced: moving averages (monthly 20 EMA / 50 MA; weekly 20 / 50 / 100 MA; daily 150 MA), previous swing lows, and previous resistance turned support. Confluence of two signals = a stronger level.
3. Keep only supports that sit **below the green line**. This is the hard rule and it is what makes GOOGL a *wait* rather than a *buy*: there was a perfectly valid daily support level, but it was above intrinsic value, so Adam refused it — "I'm not buying when it's undervalued with a margin of safety, I'm paying the fair price… I like to buy at least below the intrinsic value."
4. Target **4–5 support levels** per name and **add in tranches** (quarters for MSFT, quarters or thirds for AMZN) until a **full position** is reached, then stop adding and "let the position grow organically."

Supports found across the three:

Company	Support ladder
GOOGL	270 (swing low) · 237 · 207 (weekly 100 MA + swing high) · 163.94 (monthly 50 MA)
MSFT	316 (daily 150 MA + old resistance) · 305 (weekly 20 MA) · 283 (weekly 50 MA + swing low) · 262 (monthly 20 MA)
AMZN	3,351 (weekly 50 MA) · 3,198 (consolidation support) · 3,116 (monthly 20 EMA + weekly 100 MA) · 2,881 (swing lows)

I. Supports are probabilities, not predictions — and are subjective

Stated twice, in near-identical language:
- On GOOGL's $163.94 monthly-50MA level: "Am I predicting that the price is gonna drop to 163.94? Of course not. I can't predict where the price will go in the short term. In fact, the price may never drop to that level." The reason to draw it is that "there's a **high probability** that it would tend to bounce off," because that has been the stock's technical pattern.
- On MSFT's $262 level: waiting *only* for the deepest support "may [mean you] buy maybe once every few years."

Since the stock was already undervalued at $310, "we can start buying now."
- And explicitly: "**Support levels can be subjective.** You can have two different investors looking at the charts and they identify different support levels."

J. A short-term downtrend in a great company is not a sell signal

The GOOGL case ends with the stock entering "the early stages of a short-term downtrend" while Adam holds a multi-year position. "Is this a reason for me to sell Alphabet? No, of course not. Why? It's a great company. In the long run, it will always go higher, because it doesn't go up in a straight line." The response to weakness in a business that still passes steps 1–5 is to **add at the next support below intrinsic value**, not to exit.

K. Effort calibration

- The 5-step business analysis is a **once-a-year** job at most — quality does not decay quickly.
- Tooling can substitute for effort at a cost in precision: StockOracle's **Oracle Value ($294 for GOOGL)** lands close to the midpoint of the manually-derived $292–$304 range. "If you're very lazy, then just look at the Oracle Value."
- Third-party valuations (Morningstar's fair value) are worth consulting as a **sanity check on your own conservatism**, not as a substitute — Adam's read is that Morningstar runs optimistic relative to him, so agreement from them on a cheap price increases his comfort.
- The justification for the whole exercise: "You are buying a piece of the business… Because once you buy a great business, you have no more worries. In the short term, the price may go up, it may go down. You don't care, because over time the business will keep growing in value, and the share price will be a lot higher over time."

Part VIII — Portfolio Construction

Lesson 8 — Portfolio Construction (Building a Winning Portfolio)

Source videos: 60 (How to Diversify), 61 (Favourite Sectors), 62 (Dividend & Predictable Stocks), 63 (Growth Stocks & More), 64 (Maximizing Portfolio Returns)

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1. Where This Lesson Sits in the VMI Process

The three decisions of investing, in order:

1. **What to buy** — only stocks of great companies (5 steps of a great business).
2. **When to buy** — buy when price is below intrinsic value AND price has retraced to a level of support.
3. **How many shares to buy for each stock** — position sizing / portfolio construction. Described as "even more important at times" than the first two.

This lesson is entirely about #3.

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2. Why Diversify — The Core Argument

2.1 The "shit can happen" principle

- No matter how great the business or how confident you are, **never put all your eggs in one basket**. Events occur that cannot be anticipated or predicted.
- **Never go 100% into one stock**, no matter how good the analysis.

2.2 Case study — Boeing

- Boeing was a high-quality company passing all criteria of a great business.
- For years: revenue up, profits up, free cash flow up; effectively a **duopoly with Airbus**.
- 2020 broke it:
- Two air crashes → the 787 fleet grounded.
- Global pandemic → borders closed, airlines couldn't fly, orders postponed/cancelled.
- Result: Boeing's profits went **negative for the first time in its history**; share price dropped **80%**.
- Adam owned Boeing before the pandemic and could not predict it. His Boeing position fell 80%.

2.3 The payoff of diversification — actual portfolio results

Year	Damaged holdings owned	Portfolio 1 return	Portfolio 2 return
2020	Boeing (−80%)	**+44%**	**+41%**
2023	Estee Lauder, Boston Beer, Disney, Alibaba (all "whacked")	**+48%**	**+43%**

- 2023 total gain across the two portfolios: **US$2 million in profit**, despite owning Estee Lauder, Disney, Boston Beer and Alibaba, all of which dropped hard.

- Contrast: people who bought Alibaba, Disney or PayPal **all-in** lost heavily on the exact same stocks. The difference was diversification, not stock selection.
- 2023 also cited in Lesson 61 as **+45% portfolio vs. ~+24% S&P 500**; last 5 years **+145% portfolio vs. +83% S&P 500**.

2.4 The eight-engine aircraft analogy

- Would you fly on a plane with **one** engine? No — even the highest quality Rolls-Royce engine can be knocked out by an unpredictable event (bird strike).
- A plane with **eight Rolls-Royce engines**: the probability of all eight failing simultaneously is "almost zero." Lose one or two, six remain to keep the plane flying.
- Portfolio equivalent: hold several great companies so one or two malfunctioning temporarily doesn't ground the portfolio.

2.5 The objective is capital preservation, not maximum gain

- The most important thing is **not** making the greatest gains — it is **never losing money** in the portfolio.
- Short term the portfolio will go up and down; over 2, 3, 4 years you never want to be down.
- Buffett-sourced principle stated in the lesson: if you buy a **group of great companies** and buy them **for less than what they are worth**, it is **impossible to lose money** over a period of time.
- Goal restated: double-digit returns achieved in a **very low risk** way — described as effectively "zero risk" when done collectively over time.
- Formal aim of diversification: **low volatility AND high returns**.

2.6 There is no sure-win stock

- Even Warren Buffett, running Berkshire Hathaway for 58 years, made bad investments:
- **Airline stocks** — lost money.
- **Occidental Petroleum** — still losing money.
- **IBM** — lost money.
- But he also owned **Apple, MasterCard, McDonald's, Coca-Cola**, which more than made up for the mistakes.
- **The key to a winning portfolio is not being right all the time** — it is to **win 80–90% of the time** and minimise the losses from the **10–20%** of investments that turn bad.

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3. Diversification Rules (Hard Numbers)

3.1 Number of stocks

Rule	Number
Absolute minimum stocks in portfolio	**8**
Comfortable working range ("concentrated diversification")	**8 to 30**
Recommended cap for non-full-time investors	**30**
Adam's own portfolio (full-time investor)	**~40 stocks**

Rule	Number
Absolute rule-of-thumb ceiling	**60**

- The 8-stock minimum applies **regardless of capital size**: $1,000, $10,000, $100,000 or $10 million — all get split across at least 8 stocks.
- **Why not more than 60?** There are roughly **6,000 stocks in the US market**. VMI only invests in the **top 1%** of highest-quality companies. 1% × 6,000 = **60**. Going beyond 60 means you have left the top 1% quality band.
- **Broad diversification** (100 or 1,000 stocks) is not "wrong", but has two problems:
  1. You cannot monitor or keep track of them → you won't do as well.
  2. Too many stocks inevitably forces **lower-quality stocks** into the portfolio.
- Adam can hold 40 only because he is a full-time investor whose only daily job is watching his stocks.

3.2 Diversify across sectors, industries and markets

- The 8–30 stocks **must not all be from the same industry or sector**.
- Also diversify across **different markets** where appropriate.

Adam's actual sector allocation (capital-gain portfolio):

Sector	Allocation
Technology	**27%**
Consumer Discretionary	**22%**
Healthcare	**15%**
Financials	**13%**
Communication Services	**12%**
Consumer Staples	**7%**
Industrials	**2%**
(China ETFs, stated separately in Lesson 61)	**3%**

Geographic allocation (capital-gain portfolio):

Market	Allocation
US-listed	**~90%**
China / Hong Kong	**3%**
Europe	balance (remainder)

- **Separate dividend portfolio**: 100% **Singapore-listed** — only the banks and core REITs — for the sole purpose of collecting dividends.

3.3 Correlation / clustering considerations (as taught)

- Own stocks from **different sectors and industries** so a sector-specific shock (e.g. China regulation on Alibaba and other China stocks) can't take out the portfolio.
- **Defensive stocks** are deliberately held as ballast: they hold up or rise during recessions/bear markets when growth and cyclical stocks fall — i.e. deliberately holding negatively/less-correlated buckets.
- **Cyclical stocks** rise first in a boom and "drop the most like flies" when a recession is expected.
- Owning both defensive and growth exposure is the mechanism for "low volatility with high returns."

4. Position Sizing — The Equal-Weight Framework

4.1 Core rule

> Divide capital **as equally as possible** among core long-term investments.

Full allocation per stock = 100% ÷ (number of stocks you plan to own)

Planned portfolio size	Full allocation per stock
10 stocks	**10%**
20 stocks	**5%**
30 stocks	**3.33%**
40 stocks (Adam's)	**2.5%**

- This per-stock % is called a **"full allocation"** or **"full position."**
- Worked example: $10,000 capital ÷ 10 stocks = **$1,000 per stock = 10% each**.
- Each year new capital added (e.g. another $10,000) is divided again across the same target number of stocks and topped up accordingly.

4.2 Reduced size for higher-risk categories

- For **speculative growth companies** and **turnaround companies**: invest only **half or one-third of the normal position**.
- If normal allocation is 10% → buy **5%**.
- If normal allocation is 5% → buy **2.5%**.
- If normal allocation is 2% → buy **1%**.
- Rationale (worked through in Lesson 63): if a 1% speculative position goes to zero you lose 1% of the portfolio. In practice failures usually lose about half, so realistic damage is **~0.5%**. Winners in this category can go up **5–10×**.

4.3 Fractional share math example ($10,000 / 10 stocks / $1,000 per stock)

Stock	Share price	Shares bought	Approx. value
Amazon	$150	**6**	~$900 (7 shares would be ~$1,050)
Alphabet	—	**7**	~$1,000
Meta	—	**3**	~$1,000
NVIDIA	$350	**3**	~$1,050

- You can't hit exactly $1,000 — slightly under or slightly over is fine; it is "roughly a 10% allocation."

4.4 The three conditions for ADDING shares

Add shares only if **all three** are true:
1. The stock is **undervalued** — share price below calculated intrinsic value.
2. The share price has **waved down / retraced to a level of support**.
3. You **do not yet have a full position** in that stock.

> "Once a particular stock has a full allocation, I will not add more shares... Never go more than your full allocation." — the discipline rule.

4.5 Buying in tranches

- Never buy a full position in one go. Buy in **3 to 4 tranches** (typically **quarter positions**).
- Mechanic: price becomes undervalued and hits **support level 1** → buy ¼. Drops to **support level 2** → buy another ¼. Continue until the full allocation is reached.
- Applied to the 6-share Amazon example: buy 1–2 shares at the first support level, another 2 at the second, etc., until 6 shares are held.
- **Apple worked example (real buy levels from his chart):**
- Intrinsic value line ≈ **$170**
- Buy levels: **$165, $157, $150, $141**
- Apple only reached the **first** buy level → only **one quarter** bought → it then flew up, so the position stayed under-filled.

4.6 The portfolio is NEVER perfectly balanced — and that's correct

- At any point in time you will **not** have exactly the target weight in every stock, because not all stocks are at a buy point simultaneously.
- Illustration: NVIDIA overvalued → **0% allocation** held; Amazon dropped all the way → **full allocation**; Apple dropped to only the 1st/2nd buy level → **half allocation (5% instead of 10%)**, the rest bought only if it falls further.
- Real snapshot from Adam's ~40-stock portfolio (full allocation = 2.5%):
Holding	Actual weight	Reason
Apple	**1.7%** (also stated as 1.77%)	Only hit the 1st buy level; bought ¼ then it rallied away
ServiceNow	**~2.5%**	Full allocation achieved
Meta	**6.9%**	Grew **organically** from 2.5% as the share price rose
(new position, recently bought)	**0.96%**	Bought a quarter position; slightly above ¼ because it rose after purchase

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5. Rebalancing Rules (Both Directions = NO)

Situation	Action	Reasoning
Position **grows** above full allocation (e.g. Meta 2.5% → 6.9%; NVIDIA 5% → 10%)	**Do NOT sell to rebalance down**	As long as the company is still great, still growing, and **not crazily overvalued**, let it run. "Let it grow to its fullest potential. **Do not cut your flowers.**" This is what Buffett and Peter Lynch do.
Position **falls** below full allocation (e.g. 2.5% → 1.25%, or 5% → 2.5%) purely from price decline	**Do NOT top up to restore the weight**	Once the full allocation has been bought, that's it — the position "either grows organically or drops organically."
Buying more before full allocation is reached	**Allowed**, but only if undervalued + at support + not yet full	The tranche-building rule (§4.4)

Situation	Action	Reasoning
Adding new annual capital	Divide the new money across the target stock count again and top up per the buy rules	

- The classic mistake called out explicitly: selling the stocks that went up a lot and using the money to buy the ones that went down — **"cutting your flowers and watering the weeds in your garden."**
- Fund managers do this. VMI investors do not.

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6. The Three Broad Stock TYPES (by economic behaviour)

Type	Product/service	Growth profile	Volatility	Outperforms when	Underperforms when
Defensive	Low-priced necessities (detergent, toothpaste, sanitary pads, batteries, shaving cream/razors, potato chips)	Low to moderate, **single digits 5–10%**, very consistent	Low, even in recessions	Recessions, recession fears, bear markets	Booming economy (money rotates out into cyclicals/growth)
Cyclical ("moody" stocks)	Higher-priced / luxury / economy-dependent (cars, houses, construction & renovation, industrial equipment, bank borrowing)	High profits in expansions, lower profits or **losses** in recessions	High	Economic boom — first to fly up	Expected recession / bear market — "drop the most like flies"
Growth (moderately cyclical)	Tech, digital, secular growth industries	**High growth**, revenues/profits more consistent through cycles	Moderate–high	Most of the time; core engine of the portfolio	Sharp risk-off periods

- Defensive stocks are also called **"boring businesses."** Boring ≠ bad: **Warren Buffett loves boring companies** because boring = **predictable**, consistent sales and profits, and few new competitors want to enter a boring industry.
- Tactic: **buy defensive stocks when they are down, cheap and unwanted** — then they protect the portfolio in the next recession/bear market.

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7. The 11 Sectors — Full Sector-by-Sector Breakdown

Classification: 3 defensive + 5 cyclical + 3 growth/moderately-cyclical.

7.1 DEFENSIVE SECTORS (3)

Healthcare — ETF **XLV** — *one of his favourite sectors*

- **Why:** it is **both defensive AND has good growth potential**; sits in a long-term **mega-trend: the global aging population**.
- Performance: **on par with the S&P 500** over the long run.
- Sub-industries:
  | Sub-industry | Verdict | Names given |

|---|---|---|
Biotechnology	**Avoid for long-term investing.** Most don't make money or lack consistent profits; one or two make a lot. Super high risk / super high return — "great for speculation" only	—
Pharmaceuticals	**Normally avoid.** Most lack consistent profits. Patents last only ~**20 years**; once expired they need new drugs in the pipeline → **limited economic moat**	Only two with consistent profits: **Eli Lilly** and **Novo Nordisk**. He admits breaking his own rule buying Novo Nordisk cheap last year — still below his average cost, "punished by the market gods"
Hospitals	**Loves them** — very, very predictable	**HCA Healthcare** — "has done very well for me"
Medical equipment / supplies	**Likes** — more predictable, no 20-year patent cliff	**IDEXX Laboratories**, **Thermo Fisher**
Health insurance / managed care	Positive long term; recent regulatory hits judged short-term	Market leader **UnitedHealth Group**
(Other healthcare names in his portfolio)		**Abbott Laboratories**

- Industry ETF he owns: **IHI** (healthcare medical devices).

Consumer Staples — ETF **XLP**

- **Definition:** manufacturing of food, beverages, household and personal products; also includes education and training services companies.
- Essentials bought regularly at the grocery store / supermarket; **low-priced items** — toiletries, soft drinks, energy drinks, snack foods.
- **Growth: very, very slow. Underperforms the S&P 500** over the long run.
- **How he trades them (different from growth stocks):** buy when **very, very undervalued**; **sell when overvalued**; buy back when undervalued again. He does **not** hold consumer staples long term. (Growth stocks, by contrast, he holds through ups and downs and does not trade in/out.)
- **Why hold them at all:** in a recession staples **hold their price** and protect the portfolio.
- Names: **Monster Beverage** (high growth exception), **Celsius** (up-and-coming energy drink in the health niche — an exception with high growth), **PepsiCo** (very low growth), **Coca-Cola** (very low growth), **Walmart**, **Costco**, **Hershey's**, **McDonald's**.

Utilities — ETF **XLU**

- Electricity, gas and utilities to residential homes and organisations.
- **Generally he does not touch utilities.** Demand for electricity grows **less than 2% per year** — very low growth. People buy them for **dividends, not growth**. Fairly defensive.
- **Underperforms the S&P 500** long run → avoided.
- **AI-driven exception:** demand for energy to drive **data centres** means some utilities are now growing **double digits** — specifically **IPPs (Independent Power Producers)** which can set their own market price.
- IPP names: **Constellation Energy, Vistra, Talen Energy, NRG Energy.**
- Problem: at time of recording many are **quite overvalued** and lack a consistent profit track record → **he doesn't own them individually; he owns them via an ETF, ticker AIPO.**
- **Regulated utilities** — price set by the state government → very low growth → **avoids these**: **NextEra Energy, Southern Company.**

Defensive summary: buys a lot of **healthcare**, some **consumer staples**, generally **no utilities**.

7.2 CYCLICAL SECTORS (5)

Basic Materials / Commodities

- Chemicals, building materials, paper; commodity exploration and processing — gold, silver, steel, aluminium, fertilizer.
- Names: **Steel Dynamics, Nucor, Alcoa, DuPont, Freeport, Linde.**
- **Avoids the sector.** Reasons:
  1. Revenue and profits are **tied to commodity prices** → very unpredictable, very cyclical. He wants **consistency and predictability**.
  2. **Very low profit margins** (heavy competition).
  3. **Lower return on equity.**
  4. **High debt / highly leveraged.**
- Underperforms the S&P long run.
- **Exception he owns: Linde PLC (LIN)** — industrial gas. They **do not mine** the gas; they **produce industrial gas in a factory**, so the price is **not affected by normal commodity cyclicality** → the company is predictable despite the sector label.

Energy

- Oil and gas, oil field services and equipment, coal.
- Names: **ConocoPhillips, Chevron, British Petroleum, ExxonMobil.**
- **Stays away.** Share prices are very up and down over the long run and very unpredictable — e.g. big years when Middle East conflict pushes oil prices up; collapse when oil prices fall (COVID).
- Verdict: **good for speculation / swing trading / short-term momentum trading — not for long-term investing.** Underperforms S&P long run.

Financials — *the one cyclical sector he focuses on*

- Includes banks, savings & loans, asset management, credit services, brokerage firms, financial ratings and analytics firms, payment technology companies, insurance companies.
- Sector is cyclical overall; growth generally **high single digits**, not double digits — **but** some **fintech** names do have high growth rates, and some companies within it are far less cyclical.
- **Banks — be careful.** Banks are **very highly leveraged** and **could easily go to zero** if they carry toxic assets on the balance sheet, which is very hard to detect unless you are a professional bank analyst.
- Avoids **regional banks**.
- Would only consider **top global banks** at a good price: **JP Morgan, Goldman Sachs**.
- **Big fan of the three Singapore banks** — very conservative, very good dividends: **DBS, UOB, OCBC**.
- **Asset managers:** "okay, not bad investments" — high single digit, worth buying when undervalued. Two main ones: **Blackstone** and **BlackRock**. He used to own them; **currently owns neither.**
- **His favourite financial sub-group: financial ratings & analytics companies and companies that own stock exchanges** — these are **"toll-booth" companies**, monopolistic, little or no competition in their region.
- Favourites: **S&P Global**, **MSCI**.
- **Payment technology favourites: Visa and MasterCard.**
- **Insurance:** likes the strong ones. One of the strongest insurers in the world: **Chubb**. Also **Arch Capital (ACGL)**.

Real Estate — *avoids the entire sector*

- Includes mortgage companies, property management companies, and **REITs**.
- Why avoided: **high debt, low profit margins, no strong economic moats, weak return on equity** (very competitive sector). Worst long-run underperformer versus S&P among the sectors shown.
- Names: **Regency Centers Corp, Extra Space Storage, Iron Mountain, SL Green/"SX" Property Trust.**
- **REIT exception — dividends only, never capital gain**, and **only Singapore-listed REITs**:
- Singapore REITs have among the **highest dividend yields in the world** and the **dividend is tax free**.
- **US-listed REITs: 30% withholding tax on dividends** → doesn't make sense after tax; they also carry much **higher leverage/debt**.

Industrials

- Machinery, tools, industrial products; includes **aerospace & defense**, transportation, logistics services.
- Generally avoided/underweighted: **higher debt, lower profit margins, weaker economic moats.** Only **2%** of his portfolio.
- Big names, but too cyclical with inconsistent profits → he doesn't like to hold: **Deere (DE), Honeywell (HON), Caterpillar (CAT).**
- **The rare industrials he DOES own:**
- **Waste Management** — clears waste; very predictable, very consistent profits and revenues.
- **Pool Corporation** — largest manufacturer and wholesale distributor of swimming pools and pool maintenance in the US.
- **Copart (CPRT)** — leading vehicle auction / salvage company.
- **Lockheed Martin** named as a leading military defense company (also **Raytheon**, **General Dynamics**) — but **he does not own them right now**.

Cyclical summary: of the 5 cyclical sectors he focuses mainly on **Financials**; buys a bit of the rest, but not much.

7.3 GROWTH / MODERATELY CYCLICAL SECTORS (3) — the majority of his buying

> "The majority of stocks that I buy are not from the defensive sector, not from the cyclical sector, but from the growth sectors." Revenues and profits stay more consistent whether in recession or boom.

Technology — ETF **XLK** — **highest long-run growth of all sectors; his biggest holding at 27%**

- Design, development and support of computer applications; manufacture of computer equipment and components.
- Names he owns: **Apple, Microsoft, ServiceNow, Palo Alto Networks** (cybersecurity), **NVIDIA** (semiconductors), **Palantir** (AI enterprise), **Salesforce**, **Veeva Systems**, "and a couple more."
- Industry ETFs owned: **SOXX** (semiconductors); cybersecurity ETFs named: **IHAK**, **HACK**.

Consumer Discretionary — ETF **XLY** — **second-highest long-run growth; 22% of portfolio**

- Retail stores, auto and auto-parts manufacturers, residential construction (**this part he does not buy**), lodging facilities, restaurants, entertainment companies.
- Definition vs. staples: **higher-priced branded goods you have to think about before buying**, versus staples which are low-priced and bought without thinking.

- Favourites owned: **Booking.com** (world's biggest online travel agent), **MercadoLibre** ("the Amazon of Latin America"), **Amazon** (biggest e-commerce company in the US and the world — note Amazon is also big in **digital advertising** and **AWS**, which is really technology), **Lowe's**, **Home Depot**, **Nike**.

Communication Services — 12% of portfolio

- Old-style telcos (**AT&T**, phone lines) — **does not invest** in these.
- What he does invest in: **social media**.
- **Meta Platforms** — owns Facebook, WhatsApp, Instagram, Messenger.
- **Alphabet / Google** — owns YouTube.
- **Netflix** — added recently; he never owned it in the past and regrets it, so bought once it got cheap again.

7.4 Long-run sector performance vs. S&P 500 (as presented)

Rank / verdict	Sectors
Beat the S&P clearly	**Technology** (highest), then **Consumer Discretionary (XLY)**
On par with the S&P	**Healthcare**
Underperform — avoided	**Real Estate, Energy, Utilities, Materials**
Underperforms but held anyway	**Consumer Staples** — kept because they are defensive and useful in a bear market/recession

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8. The 8 Stock Investment Categories (Investment Thesis Framework)

> Before you buy any stock you must state **why** you are buying it. It has to fit into one of these eight categories. **If it doesn't fit, don't buy it.**

Balanced-diet analogy:
- **Protein / meat** = growth stocks (build the portfolio, great when younger; too much when older can be unhealthy).
- **Carbohydrates** = predictable stocks (balance the growth stocks).
- **Spice / chilli** = speculative growth stocks (great in small amounts; too much = indigestion).
- **Vegetables / fibre** = dividend stocks (consistent income).

Category comparison table

#	Category	Mascot / label	Defining criteria	Growth / return profile	Risk	Position size
1	**Dividend Cash Cows (Income stocks)**	—	Yield ≥5% (SG), rising DPS 5 yrs, defensive sector	Low capital appreciation; income	Low	Separate portfolio
2	**Predictable Companies**	Loyal **dog**	Wide moat, predictable earnings & CFO, product never obsolete	**5–10%** (single/high-single digit); can still deliver double-digit EPS via buybacks	Low	Full allocation
3	**Large Growth Companies**	**Cheetah**	Big companies still growing **above 20–25% p.a.**; profitable ≥5 years	Highest long-run compounding	Moderate	Full allocation (narrow-moat → smaller, esp. initially)

#	Category	Mascot / label	Defining criteria	Growth / return profile	Risk	Position size
4	**Speculative Growth**	—	Revenue growth often **>100%**; **not yet consistently profitable**	5–10× potential; can also lose big	**Very high**	**½ or ⅓ of normal**
5	**Deep Cyclicals**	—	Capital-intensive, highly economy-sensitive, can't respond to demand fast	Roller coaster; must trade the cycle	High	Rarely/never used by Adam
6	**Deep Value ("cigar butts")**	Cigar butt	Price below net-net liquidation value; lousy companies	Slow, multi-year	High + long dead money	Not used by Adam
7	**Turnaround Companies**	—	Great company, profits **−25% in one year** OR **declining 2 consecutive years**, moat intact	Stock can be **40–50% below intrinsic value**	Higher than growth/ predictable	**Smaller than usual**
8	**ETFs**	—	Basket — index, country, sector or industry	Market-like	Low (no company-specific risk)	—

8.1 Category 1 — Dividend Cash Cows / Income Stocks

Who it's for: older investors, near-retirees, retirees. Younger investors may skip the category entirely — dividend stocks "don't really grow your wealth very much" (low capital appreciation). Adam (age 50 at recording) keeps a **separate portfolio just for dividends.**

Selection criteria — same "great company" criteria PLUS these additional metrics:
1. **Dividend yield ≥ 5%** (rule of thumb given today's interest-rate environment; may change over time).
- `Dividend yield = dividend per share (trailing 12 months) ÷ current share price`
- Example: $2.50 DPS ÷ $50 share price = **5% yield**.
2. **Consistently INCREASING dividend per share for at least the last 5 years.**
3. **Consistently increasing net income AND cash flow from operations for the last 5 years.**
4. **Stable share price over the last 5 years** — either trending upward or ranging. It must not be consistently declining. (Sideways is fine; you're collecting 5%.)
5. **Avoid very cyclical companies** — airlines, commodity companies, shipping companies, property companies — because in a downturn their profits drop and **they cut dividends**. Buy **defensive** companies that can keep increasing dividends through recessions.
6. **Conservative debt:** debt servicing ratio **< 30%**; current ratio **> 1**.
7. **Dividend payout ratio between 20% and 100% — never above 100%.**
- `Payout ratio = dividend per share ÷ earnings per share`
- $1.00 EPS paying $0.50 dividends = **50% payout — very good.**
- $1.00 EPS paying $1.20 dividends = **>100% — bad**: the dividend isn't coming from profit, it's coming from somewhere else and is **not sustainable**.

The yield trap warning: A high dividend yield can coexist with a **falling** dividend per share — this happens when DPS drops but the **share price drops even more**. Never look at yield alone; DPS must be rising consistently for 5+ years.

Market choice — why Singapore, not the US, for dividends:
| | US-listed | Singapore-listed |

|---|---|---|
Typical high-quality yield	Very low; **3% is already considered pretty good**; you cannot get 5% on quality names	**4%, 5%, 6%**
Dividend tax (for foreigners/Singaporeans)	**30% withholding tax**	**Tax free**
Capital gains tax	Tax free (for Singaporeans)	—
Verdict	Best market for **capital gain**, not dividends	Used exclusively for dividends

- His Singapore dividend universe is deliberately narrow: **the 3 banks (DBS, UOB, OCBC)** plus **selected high-quality REITs**.

Screening for US dividend stocks (Finviz — free screener), if you want them anyway:
- *Descriptive* tab → **Dividend yield over 3%**
- *Descriptive* tab → market cap **mid to large** (companies **over $2 billion**)
- *Fundamental* tab → **EPS growth positive past 5 years**
- *Fundamental* tab → **Sales growth positive past 5 years**
- *Fundamental* tab → **EPS growth next 5 years ≥ 5%**
- *Fundamental* tab → **Current ratio over 1**
- Result: out of **~6,000 US stocks, only 49 stocks** pass.
- Then manually **exclude the very cyclical sectors** — real estate, commodities, energy.
- Names surfaced by the screen in the demo: **Bank of America, Cisco, Gilead Sciences.**
- **Yahoo Finance → Statistics tab** shows dividend info; Cisco example: **trailing dividend yield 3%**, **forward dividend yield 3%**.
- **Stock Oracle** (his tool) → Overview page for standard dividend info; dedicated **Dividends tab** shows payout ratio, current & forward yield, **dividend growth rate over 3, 5 and 10 years**, next pay date and ex-dividend date, charts of yield history over 10 years, DPS (purple line), EPS supporting the rising dividend, payout ratio and full dividend history table.
- PepsiCo example shown: next pay date **31 March 2026**, ex-dividend date **6 March 2026**. Rule: hold the shares **before the ex-dividend date (6 Mar)** to receive the dividend on the pay date (31 Mar).

Higher-yield alternatives beyond dividend stocks (for income investors):
Instrument	Yield stated
Corporate bonds	**~4.4%**
Treasury bonds	**~4%**
High-yield bond funds	**5.77%** quoted; the one he just bought yields **~8–9%**
Private credit BDCs (Business Development Companies)	**8–9%** normally; **~12% at time of recording** due to bad news in the private credit market
Other income funds	—

- **BDCs are the ONLY kind of US-listed security he buys for dividends.**
- They are **not analysed like normal companies** — unique structure.
- **Legally required to pay out 90%** of profits / net investment income as dividends.
- Because of that structure they are **not subject to the 30% withholding tax**. (Interactive Brokers does withhold 30% but **refunds it the following year**.)
- He buys only the **highest-quality private credit BDCs** — that's how he gets very high yield with long-term safety.

8.2 Category 2 — Predictable Companies (mascot: the loyal dog)

Definition: companies selling a product or service **you use regularly**, so revenue and profits are very predictable.

Criteria:
- **Large** companies with **wide economic moats**.
- **Predictable earnings and cash flow from operations.**
- Sell a product/service that **never goes obsolete** — hamburgers, soft drinks, chocolate.
- **Very low uncertainty** in net income and cash flow.
- **Slow but steady capital appreciation.**

Growth profile: NOT 20–30%. Typically **single digit to high single digit — 5% to 10%.**

Two sub-types:

Sub-type	Sectors	Examples given
Defensive predictables	Healthcare, Consumer Staples	**PepsiCo (PEP)** — beverage & snacks; **McDonald's** (fairly defensive); **Waste Management** (very defensive); **Monster Beverage**; **Thermo Fisher** (healthcare medical devices & instruments)
Moderately cyclical predictables	Finance, Consumer Discretionary (also predictable Technology)	**Chubb** (insurance); **S&P Global** (global ratings & data analytics); **MasterCard** (payment technology); **Visa**; **Apple** (predictable technology)

Why they still work as investments despite low revenue growth:
- Heavy **share buybacks** let them grow **EPS by double digits** even with low revenue growth.
- You can still get **high double-digit returns and beat the market** by buying them **at big discounts during temporary crises**.
- When the price gets **way overextended or overvalued**, he combines the holding with **selling covered call options** to create extra income and boost returns (covered in Lesson 12 — "turbocharging my portfolio using options").

8.3 Category 3 — Large Growth Companies (mascot: the cheetah) — *most of his stocks*

Criteria:
- **Big companies that still have huge growth potential — growing above 20% to 25% a year.**
- **Profitable for at least 5 years** (this is the dividing line vs. speculative growth).
- Majority are **technology-related**, plus companies in **secular / long-term growth industries**: **cybersecurity, e-commerce, software**.
- Usually the stock is **overvalued** — you must be **patient**; they become undervalued during **corrections, pullbacks and bear markets**, and those are the times to **build a strong position**.

Two moat grades:

Moat	Definition	Position sizing	Names
Wide moat	Brand monopoly, high switching cost, network effect lasting **at least 20 years**	Full allocation	**Microsoft, NVIDIA, Google/Alphabet, Meta Platforms, Amazon, Adobe, MercadoLibre, Veeva Systems, Salesforce, ServiceNow**

| **Narrow moat** | Competitive advantage lasting **at least 10 years** | **Smaller positions, especially initially** — a bit more risky | **Booking.com, IDEXX, Tesla** |

- **Tesla is explicitly NOT owned** despite being a high-quality narrow-moat growth company: they had to **cut prices to fend off competition**, which signals **damaged pricing power — a red flag**.

Capital-return behaviour: in early growth stages they usually **don't pay dividends** and may **not do buybacks** — in fact they may **issue more shares** and hand out heavy **stock-based compensation**. That's acceptable because high growth in revenue, profit and free cash flow outruns the share-count dilution, so **FCF per share and EPS still grow double digits**. At maturity (e.g. Microsoft) they start buybacks and dividends.

8.4 Category 4 — Speculative Growth Stocks

Criteria:
- Relatively **new** growth companies with the potential to become **wide-moat industry dominators**, but **currently only a narrow moat — or no moat yet** (still building it).
- **Revenue growing at very high rates, usually over 100%.**
- **Not yet generating consistent net profit and cash flow from operations** — often still loss-making due to high investment costs and lack of economies of scale.
- **Difference vs. large growth stocks:** large growth stocks are **already profitable for at least 5 years**; these are not. Adam's own entry rule: **buy them once they can turn a profit for at least the first year** (he does not require the 5-year track record) — but acknowledges they are **a lot more risky**.
- **They do not yet pass all the steps of a great business.**

Valuation method: since there's no consistent profit/cash flow, use the **Price-to-Sales-Growth (PSG) ratio**:

> **PSG = Price-to-Sales ratio ÷ projected revenue growth rate**
> - **0.2 to 0.3 = fair value.**
> - **Above 0.3 = overvalued.**
> - **Below 0.3, and especially below 0.2 = really undervalued.**
> - Pre-calculated on Stock Oracle, or compute manually.
> - **Worked example — AST SpaceMobile:** projected revenue growth **127.66%**; price-to-sales ratio **324**; 324 ÷ 127.66 ≈ **PSG 2.54 → way overvalued** → he has **not entered a position**.

Position sizing (critical): half or one third of the usual position. 5% normal → **2.5%**; 2% normal → **1%**.
- **Downside math:** worst case (goes to zero) on a 1% position = **−1% of portfolio**. Realistically, failures lose about **half** and never come back → **−0.5% of portfolio**, which is acceptable.
- **Upside math:** winners can go up **5× to 10×** the money invested.
- Verdict: **super high risk, very high return → keep a small allocation.**

Cautionary base rate: *"For every one Amazon that made it, 99 others didn't make it."*
- Amazon itself was speculative: it **did not make any money for its first 7 years** after listing, but revenue grew like crazy and the **share price rose 15-fold** during that period.

His actual speculative positions:
Stock	Story
Palantir	Bought **2022**, before it had one year of net profits or free cash flow — entered **the moment it had one quarter of profits**. Bought before ChatGPT/AI hype; one of the first **AI enterprise** companies transforming other

businesses (e.g. Airbus). Invested **just 1% of portfolio**. After ~4 years it went up **~20×** (also stated as "24"). He **sold some** when it got really overvalued but still holds a sizable position. |
Celsius	Bought about a year ago, before the 5-year consistent profit/cash-flow track record. Up "quite a bit" but not yet double/triple; he expects it will be.
Uber	Speculative growth stock he **did NOT buy** — lacked the same conviction, though he thinks it's not a bad company.
AST SpaceMobile	Interesting; potential huge winner in the **new space economy** — **not bought: way overvalued** on PSG (2.54). Will consider if fairly priced or undervalued.
Rocket Lab	Same as above — new space economy, **not bought yet, overvalued**.

8.5 Category 5 — Deep Cyclical Stocks

Criteria: capital-intensive industries — **shipping, real estate, property development, commodities, oil and gas** (also offshore & marine, industrial companies). Highly sensitive to the economic cycle and **unable to respond to demand changes quickly**.
- Big profits in booms; smaller profits or **losses** in recessions. Revenue and profit charts zig-zag; share price is a roller coaster.
- **Every drop can be as much as 50%.**
- **He very rarely invests here** and avoids them — but covers it because many investors focus on them.
- **You cannot hold these long term.** Holding through the ups and downs produces poor returns because of the size of the drawdowns. You must **get in and get out fast** — the opposite of predictable/growth companies which he holds for the long run to compound.

How to trade them (the inverted rules):

	Deep cyclicals	Normal VMI stocks
Buy when	Profits are **low or in temporary loss**, and the stock is **very undervalued**	Profits growing, price below intrinsic value
Sell when	They are **making a lot of money** and are **overvalued**	Only on exit rules
Chart timing	Use **weekly and monthly** charts; draw support and resistance; buy near **support**, sell near **resistance**. Buy after a **40–50% drop** into a very strong support level — in the Caterpillar example, the **200 moving average on monthly candles**	Daily/weekly support levels

- **Example used: Caterpillar (CAT)** — revenue and profit visibly cycle down/up/down/up, share price mirrors it, and every decline can be ~50%.

8.6 Category 6 — Deep Value Stocks ("cigar butts")

Origin: made popular by **Benjamin Graham**, the father of value investing, and used by **Warren Buffett when he first started.**
Adam does not use this style and owns nothing in this category — included only for completeness.

The concept: deliberately buying **mediocre or lousy companies** — but so cheap that the **share price is below the liquidation value of the company**. You are effectively hoping the company liquidates and what's left over exceeds what you paid. Often **distressed assets**. Buying below net cash means **you get the business for free**.

The core metric:

> **(Current assets − Current liabilities − Long-term debt) ÷ Shares outstanding**
>
> (Current assets − current liabilities = **net working capital**; then subtract long-term debt.)
>
> **This figure must be HIGHER than the current share price.**

Additional criteria if you do use it:
- **Positive cash flow from operations for the last 12 months** — still profitable.
- **Low debt: current ratio > 1**, **debt-to-equity ratio < 1**, debt servicing ratio (stated in transcript as "more than 30%" — read alongside the standard VMI rule of **< 30%**).
- Balance sheet **not deteriorating**.
- **Avoid sunset industries** and companies whose products are becoming obsolete, no matter how cheap — the value of the company will keep eroding.
- Note that most deep value companies have **no economic moat at all**.

Entry / exit timing:
- **Entry:** buy at a **support level in a consolidation**, or when it is **reversing into a new uptrend** — defined as the **50 MA above the 150 MA**. Never buy in a downtrend.
- **Exit:** when the **50 MA crosses back below the 150 MA**, or when the company **starts to report a loss**.

Why he dislikes it: recovery is not immediate; it requires **long-term patience** and may take **several years** for the company and the share price to recover — money is tied up too long.

8.7 Category 7 — Turnaround Companies

Definition: great businesses hit by **temporary bad news** — a lawsuit, boycott, scandal, mismanagement, recession — that causes profits to fall temporarily.

The formal trigger (either one qualifies):
- **Profits drop more than 25% in one year**, **OR**
- **Profits decline for two consecutive years.**

Key requirement: the **economic moat / competitive advantage must remain intact.**

Opportunity size: in such cases the stock price can fall **40% to 50% below intrinsic value** — the time to buy the great company.

Risk warning: turnarounds are **more risky than usual growth or predictable stocks** because:
- Some take a **long time** to turn around.
- Some **never** turn around — the short-term problem becomes a long-term problem, or the moat deteriorates because of the problem.

Position sizing: smaller than usual allocation (grouped with speculative growth as ½ or ⅓ of a normal position).

Case studies:

Stock	Problem	Outcome
Meta Platforms	2021–2022: Apple's **privacy changes** broke ad targeting + heavy **metaverse** spending → **profits dropped ~40%**, panic, **share price dropped 70%**	Bought more at the discount, roughly **$70–$80 per share**; now near **$400 per share**. **Now one of his biggest positions and his most profitable stock.** Successful turnaround; reclassified back to a growth stock
Boeing	2020: plane crashes, groundings, pandemic order cancellations, profits went negative; stock **−70/−80%**	

Already owned it, **added more**. Sold on the rebound for a **small profit** and **will not buy it back**: since the crashes there have been more problems (doors falling off), the **brand image has been punished by execution problems** and **the moat is deteriorating → no longer a great company** |
Estee Lauder	COVID **supply chain** problems + **weak Chinese consumer**; **profits down two years in a row**	**Still holding**; has **not yet turned around**; expects a turn in **1–2 years**
Boston Beer	Over-projected demand for **hard seltzer**, built too much capacity, demand dropped → short-term profit losses; **share price −60%+**	**Still holding**; believes it is **beginning to turn around now**
Disney	Struggling transition from **cable TV to streaming TV**	Judged a **long-term structural problem, not short-term → sold at a loss**. Down 26% in the 10-stock sample

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8.8 Category 8 — ETFs

Types:
- **Index ETFs** tracking an entire market — e.g. **S&P 500 ETF**, **China ETF**.
- **Sector ETFs** — **XLV** (healthcare), **XLK** (technology), **XLY** (consumer discretionary), **XLP** (consumer staples), **XLU** (utilities).
- **Industry ETFs** — cybersecurity: **IHAK**, **HACK**; medical devices: **IHI**; semiconductors: **SOXX**; AI-power/IPP: **AIPO**.
- **Bond ETFs** and **commodity ETFs** — commodity ETFs are for **short-term speculative trades only, never long-term investments**.

Why they're useful:
- You buy a **basket of many companies → no need to calculate intrinsic value** and **no company-specific risk**.

How to buy them: purely on technical analysis — trends, moving averages, support and resistance.
- Some ETFs publish a P/E and growth rate, giving a rough valuation: **PEG = P/E ÷ earnings growth rate**; **PEG below about 2 = reasonably priced.**

ETFs he owns: IHI (healthcare medical devices), **SOXX** (semiconductors), **AIPO** (AI/IPP power), **China ETFs (3% of portfolio)**.

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9. Category Migration — Stocks Move Between Buckets

A stock can transition from one category to another; the thesis must be updated.

Stock	From → To	Trigger
Apple	Growth → **Predictable**	Grew >20% p.a. for years; growth has now slowed below 20%, but sales/profits became more predictable thanks to **recurring revenue from the services business**
Meta	Growth → **Turnaround** (2021–22) → **Growth again**	Temporary bad news, then problems solved
Palantir	**Speculative growth** → will upgrade to **Growth**	Once it achieves **5 years of consistent sales and profits**

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10. The Pre-Purchase Checklist (Four Questions)

Different investors have different **account sizes, risk profiles, objectives and portfolio strategies** — so run every candidate through this:

1. **Is this a great business?** Put it through the fundamental criteria — must pass the **5 steps of a great business**.
2. **Is the stock undervalued?** Share price **below intrinsic value** AND price has **retraced to a level of support**.
3. **What type of stock is this — which of the 8 investment categories, and which sector?** Then ask: **is this the kind of stock I want in my portfolio**, given my strategy? (If you only want growth, you don't buy DBS Bank. If you want dividends, you don't buy NVIDIA or ServiceNow — they don't pay dividends.)
4. **Do I already have a full allocation for this stock?** If you plan 20 stocks, the maximum is **5%**. Once 5% is bought — **do not buy more.**

The comparison rule: never compare across categories. *"You can't compare a steak with an apple."*
- NVIDIA (growth) must be compared to **ServiceNow, Salesforce, Microsoft** — other growth stocks.
- DBS (dividend) must be compared to **OCBC, UOB, CapitaLand Ascendas Trust** and other high-dividend stocks.

Mixed strategies are fine — e.g. "50% growth, 20% dividend, 20% [other]" — but you must **know your strategy first**, then pick stocks that fit it.

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11. Adam's Own Category Allocation

Category	His holding
Large growth	**Almost half of his stocks** — the reason the portfolio beats the S&P 500
Predictable	"Quite a number" — both defensive (healthcare, consumer staples) and moderately cyclical (predictable tech like **Apple**; financials like **Visa, MasterCard**). Held because they are **resilient in a recession/bear market and balance the portfolio**
ETFs	Some industry ETFs — **IHI** (healthcare medical devices), **SOXX** (semiconductors), plus **3% China ETFs**
Speculative growth	**Only one — Palantir**
Turnaround	"A few" — **Estee Lauder, Boston Beer**. Deliberately keeps these few because they are riskier
Deep value	**None**
Deep cyclical	**None**
Dividend	**Entirely separate portfolio** — Singapore-listed banks + high-quality REITs, purely for dividend collection

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12. Maximizing Portfolio Returns — The 5% / 95% Rule

> **Over time, 95% of your profits will come from only 5% of the companies that you buy.**

12.1 Buffett's evidence

- Berkshire Hathaway run for **over 58 years**.
- Berkshire grew **170,000%** vs. the **S&P 500's 3,900%** — **outperformance of 43×**.

- From his shareholder letter, he describes his collection of businesses as: **a few with truly extraordinary economics; many with very good economic characteristics; and a large group that are marginal.**
- Translation: most of what he owns is so-so; **the few account for all the profits.**
- Over 58 years Buffett bought and owned **over 210 stocks and businesses**. Only **5% — "about a dozen" (12 of 200+) — contributed 90% of all his gains.**

Buffett's 5-year holding performance as presented (Apple, HPQ, AXP, CVX, Bank of America, Coca-Cola, Occidental Petroleum, NU, KHC, PARA):
- Only **Apple** did very well.
- **HPQ** and **AXP** did slightly better than the S&P 500.
- **CVX, Bank of America, Coca-Cola, Occidental** made positive gains but **underperformed** the S&P 500.
- **Three stocks made negative returns: −30%, −35%, −69%.**

12.2 The same distribution in Adam's own portfolio

Expected outcome per 10 stocks bought:

Bucket	Count out of 10
Exceptional outperformers	**2 to 3**
Do well	**4 to 6**
Suck (at least in the short term)	**2 to 3**

Real 5-year sample of 10 of his ~40 holdings:

Stock	5-year return	Bucket
NVIDIA	**+700%**	Way outperformed S&P
ASML	**+281%**	Way outperformed S&P
Apple	**+257%**	Way outperformed S&P
Microsoft	**+237%**	Way outperformed S&P
Lowe's	pretty good, slightly above S&P	Did well
Meta	pretty good, slightly above S&P	Did well
Amazon	just slightly below S&P	Did well
Estee Lauder	**−15%**	Underperformer
Disney	**−26%** (already sold)	Underperformer
Alibaba	**−41%**	Underperformer

12.3 The operating conclusion

- **When you buy the 10 stocks you will never know which ones will outperform** and which will underperform. You only know after it happens — often after 5 years.
- Therefore: **keep all of them**, so the **massive gains from the winners more than offset the temporary drops from the losers.**
- Net-net the portfolio still grows and beats the S&P 500.
- This is the mathematical justification for **not rebalancing**, **not cutting winners**, and **not averaging into losers**.

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13. Key Takeaways

1. **Never go all-in on one stock** — no matter how great the business. Boeing was a great company and lost 80% on events nobody could predict.
2. **Minimum 8 stocks, comfortably 8–30, hard ceiling 60** (top 1% of 6,000 US stocks). Adam runs ~40 only because he invests full time.
3. **Full allocation = 100% ÷ number of stocks.** 10 stocks → 10%; 20 → 5%; 30 → 3.33%; 40 → 2.5%.
4. **Speculative growth and turnaround positions get ½ or ⅓ of a normal position** — 1% instead of 2%, 2.5% instead of 5%.
5. **Three conditions to add shares:** undervalued + at support + not yet at full allocation. Once full, stop buying.
6. **Buy in 3–4 tranches** (quarters) at successive support levels, never all at once.
7. **Never rebalance in either direction.** Don't trim winners that grew organically (Meta 2.5% → 6.9% stays); don't top up losers back to target weight. *"Do not cut your flowers… don't water the weeds."*
8. **Diversify across sectors, industries and markets**, not just across tickers. ~90% US, 3% China/HK, remainder Europe.
9. **The objective is never losing money over 2–4 years**, not maximum gain. A portfolio of undervalued great companies held over time makes losing money effectively impossible.
10. **Win rate matters more than perfection:** aim to be right **80–90%** of the time and cap losses on the failing 10–20%.
11. **Three stock types:** defensive (5–10% growth, protects in recessions), cyclical (boom/bust), growth/moderately cyclical (the core engine).
12. **Favourite sectors: Technology, Consumer Discretionary, Healthcare, and Financials** (the one cyclical sector he leans into, via toll-booth companies, payment networks and strong insurers).
13. **Avoided sectors: Real Estate, Energy, Basic Materials, Utilities** — all underperform the S&P long run, with high debt, low margins, low ROE and unpredictable profits. Consumer staples underperform too but are kept as recession ballast.
14. **Every purchase needs a thesis** that fits one of the **8 investment categories**. If it doesn't fit, don't buy it. Only ever compare stocks within the same category.
15. **Dividends: Singapore, not the US.** US dividends are taxed **30%** for foreigners and quality yields max out near 3%; Singapore yields 4–6% and are **tax free**. The exception is **BDCs**, the only US-listed income vehicle he buys (8–12% yields, exempt from the 30% withholding).
16. **Dividend selection metrics:** yield ≥5% (SG), rising DPS 5 years, rising net income and CFO 5 years, stable/rising price, payout ratio 20–100%, current ratio >1, debt servicing <30%, avoid cyclicals.
17. **Speculative growth valuation: PSG = P/S ÷ projected revenue growth. 0.2–0.3 = fair; >0.3 = overvalued.**
18. **Turnaround trigger: profits −25% in one year, or two consecutive years of decline**, with the moat intact; stock can be 40–50% below intrinsic value.
19. **The 5%/95% rule:** 95% of profits come from 5% of holdings. Buffett: 12 of 210+ stocks produced 90% of gains over 58 years. Expect 2–3 winners, 4–6 decent, 2–3 duds per 10 stocks — and you can't know in advance which is which, so hold them all.
20. **Categories are not permanent** — Apple went growth → predictable, Meta went growth → turnaround → growth, Palantir will go speculative → growth once it has 5 years of consistent profits.

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14. Model Portfolio Blueprint

14.1 Structure

Parameter	Value
Minimum number of holdings	**8**
Recommended range (part-time investor)	**8 – 30**
Maximum (full-time investor)	**~40**
Absolute ceiling (top 1% of 6,000 US stocks)	**60**
Full position size	**100% ÷ number of holdings**
Full position at 10 / 20 / 30 / 40 holdings	**10% / 5% / 3.33% / 2.5%**
Position size for speculative growth & turnarounds	**½ or ⅓ of full** (e.g. 5% → 2.5%; 2% → 1%)
Entry tranches per position	**3–4** (quarter positions at successive support levels)
Maximum position ever purchased	**The full allocation — never more**
Rebalancing (up or down)	**None**
Number of separate portfolios	**2** — a capital-gain portfolio and a **separate dividend portfolio**

14.2 Sector allocation target (Adam's live capital-gain portfolio)

Sector	Weight	Role
Technology	**27%**	Highest long-run growth; biggest holding
Consumer Discretionary	**22%**	2nd highest long-run growth
Healthcare	**15%**	Growth + defensive; aging-population mega-trend
Financials	**13%**	The favoured cyclical sector (toll-booths, payments, insurance)
Communication Services	**12%**	Social media (Meta, Alphabet, Netflix)
Consumer Staples	**7%**	Recession ballast despite underperforming the S&P
China ETFs	**3%**	Geographic diversification
Industrials	**2%**	Minimal — only the predictable exceptions
Utilities / Energy / Real Estate / Basic Materials	**~0%**	Avoided — underperform, high debt, low margins, unpredictable

14.3 Geographic allocation

Market	Weight
US-listed	**~90%**
China / Hong Kong	**3%**
Europe	remainder (~7%)
Dividend portfolio (separate)	**100% Singapore-listed** — DBS, UOB, OCBC + selected high-quality REITs

14.4 Category allocation

Category	Target weight / count
Large growth	**~50% of holdings** (almost half) — the return engine
Predictable	"Quite a number" — defensive + moderately cyclical ballast
ETFs	Small satellite positions (IHI, SOXX, AIPO, China ETFs ~3%)
Speculative growth	**Rare — currently 1 name (Palantir)**, at ½–⅓ position size
Turnaround	**A few, deliberately limited** (Estee Lauder, Boston Beer), at reduced size
Deep value	**0%**
Deep cyclical	**0%**
Dividend	**0% in the capital-gain portfolio** — housed in a fully separate SG portfolio

14.5 Expected outcome (per 10 holdings)

Outcome	Count
Exceptional outperformers	**2–3**
Solid performers	**4–6**
Losers (at least short term)	**2–3**
Target win rate	**80–90%**
Share of profits from the top holdings	**95% of profits from 5% of stocks**

14.6 Track record this blueprint produced

Period	Portfolio	S&P 500
2020	**+44%** / **+41%** (two portfolios)	—
2023	**+48%** / **+43%** (also quoted as +45%) — **US$2m profit**	**~+24%**
Last 5 years	**+145%**	**+83%**

Next lesson: portfolio management — exit rules and how to consistently monitor the portfolio to beat the market.

Part IX — Finding Stocks, Monitoring, Drawdowns & Exit Rules

Section 09 — Finding Stocks, Monitoring, Drawdowns & Exit Rules

Source: Whale Investor / Value Momentum Investing (VMI), Lesson 9 — videos 66–72
(66 Finding Great Stocks · 67 Stock Screeners · 68 Monitor Your Portfolio · 69 Dealing with Drawdowns · 70 When to Sell Pt.1 · 71 When to Sell Pt.2 · 72 Reasons Not to Sell)

Lesson 9 has three parts: (1) where to source investment ideas, (2) portfolio management / monitoring, (3) exit rules. Everything sourced here still has to be pushed through the **7-step criteria** before it can be bought — sourcing only generates candidates, it never generates a buy.

PART 1 — SOURCING: WHERE GREAT STOCKS COME FROM

1.1 The core principle — buy what you use ("consumer monopolies")

The best-performing stocks he owns did **not** come from insider tips or expensive paid reports. They came from **businesses he patronises himself** — where he uses the products and services regularly and pays for them all the time.

Guiding question: *"What products or services do I use regularly, and what do I pay for all the time?"* Companies that answer that question repeatedly across millions of people are **consumer monopolies**.

His own portfolio, mapped to his own usage:

Stock owned	Why (personal usage)
Microsoft	Uses Excel and Word every single day
Meta	Uses Facebook, Instagram, WhatsApp every day
Amazon	Orders on Amazon, watches Prime Video
Alphabet	Google search, Google Maps, Google Docs, Gmail, YouTube — "can't live without Google"
Visa & Mastercard	Uses both cards every day when buying things
Adobe	Uses Adobe heavily for design; daughter uses it in college for design
Apple	iPhone, Apple Watch, whole family on MacBooks and iPads

The Buffett precedent. He learned this from a Buffett interview many years ago. Asked how he found his greatest investments, Buffett's answer: he buys businesses he patronises all the time.
- **Coca-Cola** — Buffett loves drinking Coke; admits to **4–6 cans a day**.
- **Gillette** — he shaves with Gillette blades.
- **American Express** — he uses their credit cards and used to use their travellers' cheques.
- **Apple** — Buffett's biggest investment, and the trigger was not an analyst research paper. His **granddaughter** told him she couldn't live without her iPhone; Buffett realised how many people can't live without Apple products.
- Buffett's fortune also built on Coca-Cola, American Express, Visa, Mastercard, McDonald's, Bank of America.

Buffett's logic, as restated: *if I use these products every day, so do millions of people worldwide* → proves **brand monopoly** and **market leadership** → these are the companies you want to own → they have **very predictable future revenue and cash flow**.

1.2 The daily-life inventory exercise

Walk through your own day from waking to sleeping and list every listed company you touch:
1. You wake up and bathe — **which soap? which shampoo?** Research those brands if listed.
2. **What toothpaste** do you use?
3. Breakfast — **where does it come from? Where do you buy your groceries?**
4. **What car do you drive?**
5. Lunch and dinner — McDonald's, KFC, a restaurant chain — many are listed companies.
6. **What tools/software do you use every day** for work — Microsoft, Google, other software?

That inventory *is* your candidate list to analyse.

1.3 Critical caveat — usage ≠ automatic buy

Using a product does **not** make it a great company. It still has to pass the **7-step criteria**.

Netflix counter-example. He watches Netflix occasionally and his family subscribes, but he does **not** own the stock. After the 7-step research he concluded Netflix is a *good* business, **not a great** one. Reasons:
- Netflix must spend **hundreds of millions of dollars generating its own content** → huge **maintenance capex**, continuous spend just to maintain content and retain customers.
- Content is a **hit-and-miss** business: some movies are hits, some flop and lose millions → **not predictable**.
- He prefers **Alphabet (YouTube)** instead: YouTube does **not** generate its own content — users generate it → **very low maintenance capex**.

1.4 Source #2 — Great companies hit by temporary bad news

No matter how great a stock is, bad news will at some point knock the share price down **20%, 30%, 40%, 50%** — and that can be a chance to buy a great company at a big discount.

Two sub-cases:
- Bad news that **genuinely dents short-term profits** for one, two, three quarters or even a year.
- Bad news where **profits and revenue don't drop at all** — only sentiment/emotion drops. *This is the best kind*, because the business is untouched and the fall is purely irrational selling.

The test that must be satisfied: the bad news must be **temporary**. The **business model** and the **competitive advantage / economic moat must remain intact**, so that over the medium-to-long term the company keeps growing revenue and profits and makes back the short-term damage.

Example A — Meta (2022), stock down ~70%. Two pieces of bad news:
1. Zuckerberg started spending on the **metaverse**, not yet profitable → expenses up, profits temporarily down.
2. **Apple changed its privacy settings** (app tracking opt-out) → blocked Facebook's ad targeting → advertising ROI dropped.

He held his shares and **bought more at a 60–70% discount**. His confidence came from:
- Research showing Meta still had its competitive advantage and was building an **AI engine to circumvent Apple's privacy changes** — i.e. a short-term problem.
- **He was still using Facebook/Instagram/WhatsApp just as much** — no reduction in his own usage.
- **He was himself an advertiser** — his company spends **millions of dollars a year** on Facebook ads. He asked: is there

an alternative? *Not really* — Facebook gave the best return on ad dollars. Was he planning to spend less next year? *No — more.*

Because he was a customer AND an advertiser, he understood the business well enough to keep adding shares while others panic-sold. Meta is now one of his biggest positions and one of his highest-returning.

> Lesson: **your biggest positions should be the businesses you understand best — and you only truly understand a business if you are a customer of it.** Buy what you know.

Example B — Ozempic scare (2023). News about the weight-loss/diabetes injectable (publicised via Kim Kardashian and Elon Musk) triggered irrational sector-wide selling:
- **Food stocks −20% to −30%**: Pepsi, Hershey's, McDonald's, Yum Brands — on the fear people would eat fewer chocolates, cookies, potato chips.
- **Healthcare stocks −10% to −30%**: Thermo Fisher, HCA Healthcare, Abbott Laboratories — on the fear that thinner people get less heart disease and fewer medical problems.
- Revenue and profits **did not actually drop**. He used it to buy more **McDonald's**, plus **Hershey's** and **Pepsi**, and the marked-down healthcare names.

Example C — Booking Holdings / Booking.com (2020 COVID). Largest online travel agent in the world (owns Priceline, Agoda and other travel brands). Global borders shut, **revenue dropped ~90%, profits ~90%**, share price plunged. He **bought aggressively at huge discounts** while everyone panicked, because (a) it passed all his criteria, (b) the profit collapse was from an external cause, not the company's fault, and (c) the pandemic would not last forever.
- Note what he did **not** buy: hotels, airlines, cruise ships — they **did not meet his investment criteria**. Booking did, because it is an **online business with a very strong balance sheet**.

Rules attached to the bad-news source:
1. Do **not** buy blindly just because you read bad news — the company may not be a good business in the first place. It must pass the investment criteria.
2. The **news only points you to the opportunity**. The actual buy decision comes from **fundamentals + technicals**: Is it undervalued? Has price retraced to a strong support level? Is there a reversal pattern? Entry criteria must be met before pulling the trigger.
3. **Exception — financial fraud or financial irregularity: don't touch it.** "Where there's smoke, there's a big fire." Every other type of short-term bad news is an opportunity.

1.5 Source #3 — Stock screeners

- Best **free** screener: **finviz.com**
- Best **paid** screener: **gurufocus.com**
  (Full setups in Part 2 below.)

1.6 Source #4 — Following super investors (13F filings)

Website: dataroma.com — free; tracks the **13F filings** of well-known US hedge-fund managers and super investors managing billions.

What you can see there:
- A list of tracked **super investors**: Bill Ackman, Warren Buffett, Li Lu, Chase Coleman, and others.
- Each investor's **top-10 holdings** and their **most recent purchases**.
- Example cited — **Bill Ackman, Q3 2023**: added **Google, Hilton, Howard Hughes**. (He describes Ackman as one of the few fund managers he actually respects.)
- Example cited — **Warren Buffett / Berkshire Hathaway**: largest holding **Apple at ~50% of the portfolio**; Q3 buys

included **LLYVK (Liberty Media), Sirius holdings, Atlanta Braves Holdings**; prior quarter **Capital One Financial, Occidental Petroleum**.

More useful than any single investor: the aggregate. Look at the stocks that many super investors are buying — the "most popular stocks among super investors" — on the Dataroma home page, right-hand side:
- **Top 10 most-owned stocks by super investors** (as shown): Google, Microsoft, Meta Platforms, Amazon, Visa, Berkshire Hathaway, Wells Fargo, UnitedHealth, Mastercard. *He owns 8 of the 10* — but bought them from his own analysis, not because they did.
- **Top 10 buys last quarter** (as shown): Alphabet, Meta, Amazon, Disney, Microsoft, UnitedHealth, CVS Health, Thermo Fisher, Dollar General, Charter Communications.

Hard warnings on this source:
1. **Never follow blindly.** Respecting an investor doesn't mean copying their picks.
2. He respects Buffett tremendously but would **never buy Occidental Petroleum (OXY)** — doesn't meet his criteria and he doesn't like energy stocks; it fell significantly after Buffett bought.
3. Buffett famously bought **airline stocks in 2019**, then cut the loss in the 2020 COVID crash. Airlines are something Adam would **never** buy.
4. Even the aggregate can be wrong — **they could all be wrong together**.
5. Illustrative divergence: **Disney** was one of the most-bought stocks by super investors that quarter — and he **sold his Disney shares** in the same period because it no longer met his criteria. Sometimes he sells what they buy, buys what they don't.
6. Every idea from this source must still go through the **7-step analysis**.

1.7 Source #5 — Subscription research (UIP)

Ultimate Investor's Playbook (UIP), managed by Adam Khoo — subscribe at **insight.piranaprofits.com**. What it provides:
- **Two stocks presented every month**: one picked by him personally, one **voted on by the community**.
- Each gets an **in-depth 7-step deep dive**: investment thesis (market potential, why it beats the S&P 500), business overview (products/services), revenue growth history, net profit, cash from operations, gross profit margins, ROE, ROIC, revenue rising faster than accounts receivable, economic moat (wide vs narrow) and its sources (market leadership, network effect, high switching costs), direct competitor study with 10-year and 5-year relative performance, growth drivers, debt structure, intrinsic value via the intrinsic value calculator, and investment risks.
- *Example given*: **Intuit**, presented May 2022, competitor-compared against **Sage Group, Xero, H&R Block**; presented when price dropped to a strong support level inside the buy zone; later at **$594**.
- **Monthly updates of intrinsic value and buy levels** for every stock.
- *Example given*: **Fortinet (FTNT)** — intrinsic value range **$73 conservative to $90 base case**, share price **$55** (still quite undervalued), with **four buy levels: $61, $56, $52, $47**, updated monthly off strong support levels.
- A **Discord notification channel** where he screenshots every transaction the day he makes it.
- *Examples*: 24 Oct — added **Capital(and) DC REIT at $1.80, 5,600 shares**; 25 Oct — added **Thermo Fisher at $427**, showing the allocation increase and transaction screenshots.

PART 2 — SCREENERS: EXACT SETUPS

2.1 Finviz (free) — finviz.com → Screener

Universe: **~9,500 stocks and ETFs** in the database (not all US-listed; some non-US names included). Three tabs of criteria: **Descriptive, Fundamental, Technical**.

On P/E — deliberately left as "Any". P/E by itself is completely meaningless. A high-P/E stock can be cheap and good; a low-P/E stock can be expensive and lousy. **P/E and Forward P/E → Any** (not used).

Fundamental filters, in the order he sets them:

#	Filter	Setting used	Notes / rationale
1	P/E	**Any**	P/E alone is meaningless
2	Forward P/E	**Any**	same
3	EPS growth next 5 years	**Over 10%** (demonstrated **Over 20%** first → cut 9,000 to **492 stocks**; then relaxed to **Over 10%**)	Use **Over 20%** if you specifically want high-growth names; lower if you want steadier/predictable names
4	Return on Equity (ROE)	**Over 10%**	He wants **12% or higher**, but Finviz only offers Over 10 / Over 15 → picks **Over 10**. After criteria 3+4: **550 stocks**
5	Sales growth past 5 years	**Over 5%**	Wants a consistent revenue track record; can use Over 10% or Over 20% for stricter growth screens
6	PEG ratio	**Under 2**	PEG is only a rough valuation gauge; theoretically >1 = overvalued, but many great companies run above 1, so he is deliberately lenient at **under 2**
7	Current ratio	**Over 1**	
8	EPS growth next year	**Over 5%**	Can raise to Over 10% depending on the stock type sought
9	EPS growth past 5 years	**Over 5%**, then tightened to **Over 10%**	
10	Return on Investment (ROI)	**Over 10%**	Similar to ROIC

Result counts as demonstrated: all fundamental criteria at the looser settings → **69 stocks**. Tightening EPS growth past 5 years to **Over 10%** → **55 stocks**.

Names that surfaced in the demo: **Applied Materials, Arista Networks, Copart, Alphabet, Mastercard, MercadoLibre, Netflix, Winnebago (mid-cap), ZTO Express**; in chart view also **Booking.com, JBL**.

Technical filter added:

#	Filter	Setting	Effect
11	Price vs 200-day SMA	**Price above SMA200**	Confirms overall trend is up. **55 → 37 stocks**

Descriptive filters added:

#	Filter	Setting	Effect
12	Market cap	**Large (over $10bn)** — optional	**37 → 15 stocks**. Leave open (small-cap and above) to keep the full **37**
13	Dividend yield	**Over 2%** — only if hunting dividends	Collapsed the list to **2 stocks**; US stocks are not known for dividends, so normally leave it **Any**

Operating principle: the stricter you are, the fewer candidates; the more lenient, the more candidates. Set strictness according to whether you want **growth stocks** or **predictable stocks**.
Output use: take the surviving list and run each name through the **7-step investment analysis**. The screener does

not qualify a company — it only shortlists.
Extra view: the "Charts" view lets you eyeball all survivors' charts quickly.

Finviz limitations (why he also pays for GuruFocus):
- **US-centric** — you can't sort for India, China or European stocks.
- **You cannot set exact thresholds** — e.g. you cannot ask for ROE of exactly 12%, only the preset buckets.

2.2 GuruFocus (paid) — gurufocus.com → Screener → All-in-One Screener

GuruFocus ships many **preset screeners** (e.g. Ben Graham's lost-formula / deep-value screen, fast growers, high-quality companies). But he builds his own custom screen, saved as **"Adam VMI"**.

Step 1 — Region. Select the regions to hunt in: US only (uncheck Asia, Europe, etc.), or check **US + Asia + Europe** together; Canada, UK, Oceania also available.

Step 2 — The 12 criteria (11 fundamental + 1 valuation). He deliberately keeps it simple:

Fundamental (11):
1. **Current ratio ≥ 1**
2. **Interest coverage ratio ≥ 3**
3. **Debt / EBITDA ≤ 3**
4. **Return on Equity (ROE) ≥ 12%**
5. **Return on Invested Capital (ROIC) ≥ 12%**
6. **5-year average ROIC ≥ 12%**
7. **Share buyback ratio ≥ 1%** — focuses on companies buying back shares. *Not a must, but a great-to-have.*
8. **Financial Strength rank ≥ 7** (GuruFocus proprietary 1–10 ranking)
9. **Profitability rank ≥ 7** (GuruFocus proprietary 1–10 ranking)
10. **Predictability rank = 4 to 5 stars** (GuruFocus ranks predictability from past revenue, profits and cash flow)
11. *(the above 8–10 are GuruFocus's own internal ranking metrics, which he deliberately piggybacks on)*

Valuation ratio (1):
12. **Price-to-GuruFocus-Value ratio: 0 to 1** — i.e. only stocks **not overvalued** on GuruFocus's own valuation.
- Caveat he states: he doesn't always agree with GuruFocus's intrinsic value — he finds them **sometimes too optimistic / too aggressive**, and **his own intrinsic value is always below theirs**. He uses it for screening only and **still does his own valuation** to confirm.

Result: these 12 criteria filtered down to **21 stocks**. Names shown: **Alphabet, Visa, Johnson & Johnson, ASML, Cisco, Nike, Infosys, Agilent, Wipro**, and (transcript unclear) a life-sciences name — many of which he already owned.

Screening cadence: he screens at least ONCE A MONTH, because new stocks appear and old ones drop out.

Screener output is never a buy signal. Everything that survives still goes through the full **7-step analysis**.

PART 3 — MONITORING: WATCH LIST AND PORTFOLIO ROUTINE

3.1 The watch list

When a stock goes on the watch list: once it passes the **first five steps** of the 7-step analysis — i.e. it is confirmed a great business but the *price* is not yet right.

Watch list size: roughly 20–30 great companies.

Daily routine: spend ~5 minutes a day running through the watch list, waiting for the right entry.

The entry condition being waited for (both must be true):
1. Price drops **below intrinsic value** (undervalued), **AND**
2. Price waves down to a **level of support you have already identified**.

The whole point is **patience** — wait for the right point to pull the trigger.

3.2 Platforms for building watch lists

TradingView.com (his charting software)
- Right-hand panel → open the watch-list icon → "Create new list" → name it → save → click **+** to add tickers.
- You can run **multiple watch lists**, either by market (US Stocks Watchlist / Asian Stocks Watchlist / European Stocks Watchlist) or by **investment category** (dividend stocks / growth stocks / more speculative growth stocks).
- Demo tickers added: **Abbott Laboratories, BlackRock, Accenture, Pepsi, Procter & Gamble, Fortinet, Microsoft** — deliberately a mix of technology, consumer staples and financials.

Thinkorswim (TD Ameritrade) — both a broker and a charting platform. His watch list sits on the left.

Interactive Brokers — New Window → Watch List; the panel can be dragged/docked anywhere on the workstation (he keeps it on the right, chart on the left, then locks the layout).
- **Free TradingView charts inside IBKR**: New Window → Chart → **Advanced Chart**. Functions are somewhat limited versus standalone TradingView, **but you can add all the moving averages for free** — whereas standalone TradingView requires a paid subscription for many of those moving averages.

3.3 How the chart is marked up (the actual monitoring template)

For every stock on the list he draws:
- **A green line at the intrinsic value.**
- **Three to four buy levels per stock**, each placed at a **significant support level below the intrinsic value**.

Worked examples:
- **Abbott Laboratories** — intrinsic value **$111**; buy levels **$104, $101, $96, $88**. Price recently dropped through level 1, level 2, level 3 and almost to level 4 → shares could have been added at each; it then reversed back up. Still undervalued at the time of recording.
- **Accenture** — intrinsic value **$314**; buy levels **$301, $282, $262, $242**. Price fell all the way to **$242** (the last buy level) last year — a great time to add.

How the levels are used: buy **in tranches**, roughly a quarter at each level; by the last support level you should hold a **full position**. Never buy all at once — average into the position. A great business will always rebound once market sentiment shifts.

Once bought, the stock moves from watch list → portfolio. His own portfolio holds **40 stocks**; a normal target is **10–40 stocks**.

3.4 The daily portfolio check

- **Yes, monitor regularly — but under 10 minutes a day**, when the market is open.
- **What you are looking for:** whether any price has retraced into a **buy zone** — i.e. **below intrinsic value AND at/near an identified support level**. If you don't yet hold a full position, add more shares.
- Once the intrinsic value is exceeded, you can't buy any more until either the intrinsic value rises on revaluation, or price drops back below it.

What NOT to do while checking: do not be concerned about **daily fluctuations**. That is volatility, and volatility is normal. Don't freak out if a great company drops **10%, 20% or 30%** — most of the time a price drop has **nothing to do with the value of the underlying business**. "Be chill. Be zen."

Mr. Market framing. The market behaves like a drunk, psychotic person with bipolar disorder — very happy one day, very sad the next. Daily prices are driven by **emotions, sentiment and manipulation**, not logic. On fearful red days you have exactly **two** legitimate options:
1. **Take advantage of him** — buy more shares of great companies at depressed prices (if you don't yet hold a full position).
2. **Ignore him** — if you already hold a full position, go fishing, don't even look.

What you must **never** do is be **influenced** by him — getting scared when he's scared, excited when he's excited. That is the road to ruin.

Prices never go up in a straight line; they move in **wave patterns** — wave up, wave down, wave up. "After breathing out, you have to breathe in."

3.5 The 10 reasons a stock price drops (the diagnostic checklist)

When price drops, classify the cause. **Reasons 1–9 are ~99% of cases, are all temporary, and are all buying opportunities (or things to ignore). Only reason 10 justifies concern.**

Non-company-specific (macro/market):
1. **Macro or geopolitical news** — a Middle East war, the Ukraine war, recession fears, the Fed raising interest rates. Hits all companies, not just yours.
2. **Bear market** — happens on average **once every six years**; the whole market turns negative and good and bad companies drop together for purely emotional reasons.
3. **Correction** — happens on average **once a year**; market drops about **10–15%**; good companies drop too.
4. **Sector rotation** — money rotates between sectors. In 2023 institutions rotated **out of healthcare into technology**, so all healthcare names fell regardless of quality and most tech names rose regardless of quality. Nothing wrong with your company. Buy the discount; rotation eventually reverses (he suspected a rotation back into healthcare in 2024).
5. **Industry-specific news** — e.g. the 2023 Ozempic story dragging the whole food industry down (McDonald's, Pepsi, Yum Brands, Hershey's).
6. **Manipulation by market makers and algo traders.**

Company-specific but still temporary:
7. **Missed earnings estimates or lowered forward guidance.** Companies report **four times a year**; analysts publish a projection beforehand (e.g. "McDonald's will earn $5 per share"). Beat → price usually jumps; miss → price gaps down.
- **Can you predict a beat or a miss? No — it's a coin toss, 50/50.**
- Don't be upset by a miss and don't be too happy about a beat: **a great company will have bad quarters and good quarters.** A miss + panic selling = an opportunity to add.
- **Lowered forward guidance** can knock **5–10%** off even after a beat. *Example*: Adobe announced better-than-expected earnings but guided next-year revenue below projection → **stock dropped 6%**. Not a concern; it's a buying opportunity.
8. **Cyclical slowdown in demand.** Even great businesses aren't linear. Semiconductors (ASML, Texas Instruments, Nvidia, AMD) trend up long term but dip in between. Online advertising grows long term but ad spend dips in slowdowns — in **2022** ad-spend fear of recession dented **Alphabet and Meta** revenue, which **re-accelerated the following year**. Semis did the same.
9. **Other temporary bad news** — a **product recall** (e.g. Tesla recalling cars over a manufacturing fault), a **lawsuit**, a

sex scandal, or a **forced seller**: a major shareholder dumping stock for personal reasons unrelated to the business. All temporary; all buying opportunities.
- **The one exception inside this bucket: accusation of financial fraud → sell immediately, do not hold.**

The only genuinely dangerous one:
10. **A structural / long-term problem** — permanent damage to the business model; revenue will never grow again, profits will never re-accelerate, or the **economic moat has permanently deteriorated** so the company can no longer fend off competition. **Rare** if you bought a wide-moat company that met all the criteria — but possible, so watch for it.
- Historical examples: **Nokia** (90% market share, disrupted by iPhone touchscreen and BlackBerry); **Blockbuster Video** (disrupted by Netflix streaming); **Kodak** (disrupted by digital photography); **Yahoo** (the #1 search engine, disrupted by Google); **AMC** movie theatres (disrupted by streaming + permanent change in consumer behaviour). None of them ever recovered.
- Root causes: **moat deterioration, permanent disruption of the business model, permanent change in consumer behaviour.**
- If unsure whether a problem is short-term or permanent, raise it in the **Discord private community chat** for feedback.

3.6 Earnings reports — the quarterly routine

- Companies report **four times a year**; earnings seasons cluster in **April, July, October and January**.
- **Finding the next earnings date:** on the **TradingView** chart (also on Thinkorswim), the letters under the chart mark events — **"E" = earnings**. Past E's show historical reports; the **purple** upcoming E shows the next one. Clicking it shows the date and the **analyst EPS estimate**.
- *Example*: Apple's next earnings **25 January 2024**, analysts projecting **$2.08 per share** for the quarter. Beat → price usually jumps; miss → price usually drops.
- **"D" = dividends** — marks the **ex-dividend date**. Buy before the ex-dividend date to receive the dividend.
- **Earnings calendar:** `finance.yahoo.com/calendar/earnings` — see which stocks report in coming days, weeks or months, and search for a specific company's next earnings date.

What to actually read after a report:
1. **The quarterly report itself** — download from the company's own website under **Investor Relations** (e.g. apple.com, adobe.com): latest quarterly financial reports and annual reports.
2. **The conference call** — after results, the CEO and management take analyst questions. Listen to the audio or read the transcript. Find it by Google-searching e.g. "Adobe conference call"; transcripts show in the results.
3. **seekingalpha.com** — search for conference calls and earnings results for all major stocks.

What you are reading for: how **management is executing on its plans**, and — if revenue or profits dipped — **management's explanation of what happened**.

Framing: don't be overly concerned about short-term beats or misses. You are only concerned with **long-term structural damage to the competitive advantage or business model**. That doesn't happen overnight — **it takes years for a company to lose its competitive advantage**, so you will see it coming.

3.7 Corporate announcements — how to read each one

Share buybacks — usually bullish, but conditional. Fewer shares outstanding → higher value per share. **BUT** a buyback is only good if:
- The shares are **undervalued** when repurchased. Buying back at an **all-time high / expensive price is NOT acting in shareholders' interest**; and

- The **company itself is doing well** — revenue and profits growing.
If the company is doing badly (falling revenue, falling profits, eroding competitive advantage), it should be spending cash on **improving the business, products, services and hiring better people**, not on buybacks.
- *Cautionary example*: years ago **IBM's** business was deteriorating and the CEO kept buying back shares to **artificially inflate EPS** instead of fixing the business — long-term detrimental.

New share issuance / rights issue — bearish to neutral. Companies sell new shares when they need to raise money. Most companies he owns hold lots of cash and never need to. If one of your holdings issues shares, **find out what the money is for**:
- To **pay off debt** → not a good sign.
- Because they're **short of money for working capital** → not a good sign.
- To fund an **acquisition** → check the target: is it a good company? Are they overpaying?
Generally he dislikes it, because issuance **dilutes existing shareholders**. In a **rights issue**, existing holders get first right to buy new shares **at a discount**; you can exercise the rights (put in more money), **sell the rights** for cash, or **let them expire**.

Stock split — bullish (mildly), but changes nothing fundamental. A split does **not** change the value of the business, better or worse. It only makes shares look more affordable.
- *Worked example*: Amazon at **$3,000/share**; you own 1 share = $3,000. A **20-for-1 split** gives you **20 shares at $150** = still **$3,000**. Like cutting a pizza into 20 smaller pieces.
- **The intrinsic value splits proportionally.** His pre-split Amazon intrinsic value was about **$3,500**; after the 20-for-1 split the new intrinsic value is **$175**.
- Why it's mildly bullish: at $3,000 few retail investors can buy a share; at $150 many can → more demand → can temporarily lift the price.

Reverse stock split — normally bearish. The company combines e.g. **10 shares into 1** because the share price is too low and they want it to look more respectable.

Cash dividends. Announcement carries an **ex-dividend date** (buy before it to receive the dividend; buy after and you don't) and a **payout date** (when it is credited to your account).

Where to read announcements:
- **Finviz** — enter the ticker (e.g. AAPL) and scroll down for recent news and public announcements.
- **Yahoo Finance** — the ticker page, scroll to **Press Releases**; also **SEC filings**, which can be sorted by **financial reports, corporate changes, tender offer / acquisition reports, proxy statements**. Splits, dividend issues and results all get filed with the SEC.

PART 4 — DEALING WITH DRAWDOWNS

4.1 Psychology is the majority of the game

- Many very smart people with finance degrees and CFAs are **terrible investors** because they can't control their psychology and emotions.
- **"70% of investing success comes down to managing your psychology."**
- A **drawdown** = a temporary drop in the price of the stocks you own and in the market value of your portfolio.
- **Drawdowns are inevitable** — no portfolio goes up in a straight line, no matter how good the investor. Analogy: no matter how good a pilot you are, you cannot fly without turbulence. Be prepared for it, don't panic, ride the wave, and use it to build positions in great companies.

4.2 The evidence — real drawdown numbers

His own portfolio, last 5 years (a period with **two bear markets in five years** — rare — plus the 2020 recession, the Ukraine war, very high interest rates and very high inflation):
- Portfolio **+141%** over 5 years, beating the S&P 500 by a wide margin.
- **2020 drawdown: −25%.**
- **2022 drawdown: −30%.**

Warren Buffett / Berkshire Hathaway, last 10 years (58 years of investing):
- **+312%**, beating the S&P 500; **~12% compounded annual return**.
- Drawdowns along the way: **−18%, −16%, −30%, −27%, −9%, −11%.** Even the greatest investor in the world takes drawdowns.

Peter Lynch / Magellan Fund, 1977–1990 (13 years):
- Fund **+604%** vs S&P 500 **+223%** — beat the index by over **3x**; **~29% compounded annual return**.
- Drawdowns during that run: **−56%, −27%, −42%, −32%.**
- **The kicker: more than 50% of the clients in his fund LOST money.** The best fund in the world, 29% a year, and a majority of investors lost.
- **Why:** investors bought **after** strong runs (when excited), then panic-sold into the −56%, −27%, −42% drawdowns — repeatedly buying high and selling low.
- **Conclusion:** you can own the best fund or the best stocks in the world and still lose money if you buy when everyone is excited and sell when everyone panics.

4.3 The behavioural rules for drawdowns

1. **Expect them and be psychologically prepared before they happen.** If you're not prepared, you'll freak out and do "dumb, stupid things" — selling in fear and being worse off after the rebound.
2. **Don't try to predict them.** There is **no way to predict** when a bear market or correction will hit.
3. **Roller-coaster rule: the only people who get hurt on a roller coaster are those who jump off halfway.** Keep the seatbelt fastened through the ups and downs and you always reach a destination far higher than the start.
4. **On panic days, two actions only:** add shares if you don't hold a full position (price below intrinsic value, at a support level), or **ignore the market entirely** if you do. Never be *influenced*.
5. **Buy when others panic; don't buy more when everyone is excited.** "You've got to buy great companies when it has collapsed, when people are panicking."
6. Alternatively, when everything is euphoric you may **sell a bit to take profits**, or simply hold through the ups and downs — because eventually it goes higher.
7. **Drawdowns don't matter to the end result** as long as you hold great companies — the compounding continues.
8. **Optional hedging (see §6.4)** — covered calls and protective puts during corrections/bear markets. *Nice to have, not a must.*

PART 5 — SELL RULES: THE SIX VALID REASONS TO SELL

Default stance: when you own shares in a great business, **hold as long as possible** — for years — and let it compound **5x, 10x, 50x or even 100x**. There are only **six valid reasons** to sell.

SELL RULE 1 — The business is no longer great: fundamentals have deteriorated PERMANENTLY

Triggered by permanently changed consumer preferences, disruption by technology or competition, or **permanent deterioration of the economic moat**.

Frequency: rare. A **wide-moat** company should sustain its moat for **at least 20 years**; a **narrow-moat** company for **at least 10 years**. And moats **never erode overnight — it takes years**, which means you always have plenty of time to get out.

1a. Disruption by new technology or competition — the company can no longer generate increasing revenue, profits and free cash flow.
- **Singapore Press Holdings** — a monopoly in Singapore media, disrupted by social media, Google and Facebook. He knew first-hand: he used to advertise **100%** in the Singapore newspapers, watched his ad effectiveness diminish as readers moved online, and shifted his **entire ad spend to Google and Facebook** — as did many other advertisers. Share price collapsed; shares eventually delisted.
- **Nokia** — **90% market share** in the 1990s. Apple launched the first touchscreen iPhone; Samsung copied; BlackBerry arrived. Nokia's **strategic blunder**: phones cost $100–$200 at the time, and when Apple priced at ~$500 Nokia declared nobody would pay that and kept making $200 phones. Market share fell from **90% to ~1%**; share price fell from **$42 to ~$2**. The tell came early — *people around him started swapping Nokias for iPhones and BlackBerrys*, **before** the bad financials and **before** the price collapse. Anyone holding at $42 had years to exit.
- **Kodak** — market leader in film. When the digital camera appeared, and then phones that took pictures with no film, the outcome was obvious. Kodak's price was still high and it was still making a lot of money for years, because adoption was gradual — again, **lots of time to get out**.
- **Intel** — every computer once carried "Intel inside". Then Apple moved to its own chips, others followed, and the Intel-inside badge disappeared from more and more machines. Again visible **before** the price collapsed.
- **Blockbuster Video** — a US monopoly for video rental (rent a cassette, return in three days or get fined). Netflix streaming arrived; the shift took many years and the demise was visible in slow motion.

1b. Change in consumer preferences.
- **Kraft Heinz** — canned baked beans, Heinz ketchup, Kraft cheese did very well in an era of processed food. As consumers became health-conscious and shifted to fresh food, the dominant food moat began to erode.

How to detect moat erosion — the signals:
1. **Best signal: observe consumers, including yourself.** Are the people around you — and are *you* — no longer buying that product or service? *"The day I stop eating McDonald's is the day I'm going to sell the stock."* (He eats there ~3 times a week for breakfast.) This is precisely why you must own businesses **where you are the customer** — he does **not** own **Starbucks** despite it possibly being a great company, because he doesn't drink Starbucks (he drinks Singapore local coffee at **$1.50** and refuses to pay more than **$3** for coffee), so he has no informational edge. He does own **McDonald's, Yum Brands, Hershey's** because he consumes them.
2. **Profit margins decline consistently for at least 2–3 years** — and the decline is due to **lower prices**, i.e. the company is forced to **discount** to attract customers → it is **losing pricing power**. A great business doesn't need to discount to keep market share. Forced discounting = heavy competition = deteriorating business.
3. **Margins contracting from higher cost of sales.**
4. **Declining market share** — one quarter is fine, three quarters is fine, one year maybe okay, but **losing market share for two straight years** is a sign the business has changed.
5. **Customers visibly switching to competitor brands.**

His own real case — Manulife US REIT (bought ~2018). A US office REIT, a great business when bought. Then:
- COVID-19 (2020) moved people to work-from-home, and **the behaviour did not revert** after the pandemic (hybrid work stuck).

- Occupancy fell from **95% to 30%**, with no sign of returning; revenue and profits dropped and could not get back to pre-pandemic levels.
- The Fed then raised rates **very fast in 2022**. With rents too low, the properties were **devalued**, which pushed up the **debt-to-asset / gearing ratio**.
- Gearing **exceeded 50%** → **breach of financial covenants** → banks could withdraw loans → risk of bankruptcy.
- The business model had changed permanently and the balance sheet was no longer strong, so **he sold at a loss**.
Because it was only a **small percentage of his dividend portfolio**, the year was still positive overall.

SELL RULE 2 — You made a mistake in your initial analysis

Either the company isn't as great as you thought, or it isn't as cheap as you thought because you were **too optimistic in your valuation**.

His own case — Disney. Bought as a **turnaround**: Disney had grown revenue and profits consistently, then transitioned from cable TV networks (a sunset industry, as viewers moved to streaming) to **Disney+**. During the transition, **profits dropped more than 60%** because streaming wasn't yet profitable and cable was in decline. He judged it cheap and expected profits to return to past highs.
After holding **~2 years** and doing more research he concluded:
- Streaming had become far more competitive (Amazon Prime, Apple TV, Netflix).
- Disney's created content **was not as great as before**.
- Content requires **millions of dollars per title** and is **hit-or-miss** — good movies make a lot, bad movies lose millions.
- Verdict: **not a predictable business** — not the kind he wants to own.
Action taken: sold **all** shares, "took the loss like a man", and **immediately reinvested the proceeds into other great, undervalued businesses** to make back the loss.

SELL RULE 3 — Management does not act in the best interest of shareholders

Red flags in this category:
1. Management **buys back shares at an all-time high / when overpriced**.
2. Management **buys back shares when the company is not doing well** (rather than investing in the business).
3. The company **issues new shares to raise capital when it already has plenty of undeployed cash** on the balance sheet.

Any of these → red flag → sell. *(He notes this has never happened to him — yet.)*

SELL RULE 4 — Still a great company, but grossly overvalued

The threshold:
- **Overvalued by 10%, 20% or 30% → HOLD.** A great company's intrinsic value keeps rising and eventually catches up to the price.
- **Overvalued by MORE THAN 50–100% above intrinsic value → start reducing the position.**

How to sell: never all at once — **sell in tranches**, because you can no more pick the top than you can pick the bottom. Sell roughly a third, then half, then the rest; or, if selling 100 shares, sell **25 shares, then another 25 a few days later, then 25 more** — averaging out the position. Reinvest the proceeds into another business just as great but at a more reasonable valuation.

- *Illustration*: if McDonald's intrinsic value is **$250** and the price runs to **$500** (100% overpriced), start selling — even though McDonald's is still a great company.

Supporting technical trigger — the parabolic move. When a stock is extremely overvalued you will usually also see the price go **parabolic and far from its moving averages**:
- A **sustainable uptrend** advances at roughly a **45-degree angle**, waving up and down, staying near its moving averages — in this context, the **20 EMA** is normally closest to price in an uptrend (and price stays near the 50 MA too).
- A **parabolic move** is a **90-degree**, vertical advance with **very long green candles**, driven by **panic buying** — everyone rushing in. This is usually **the end of the uptrend**, and it's where late retail investors (who heard about the stock from a neighbour or friend) buy right at the top before the collapse.
- Same pattern applies to the **overall index**. Buffett's line, as quoted: a bull market is like sex — it feels best just before it ends.
- **Rule:** if a stock he owns shifts from a 45-degree advance to a **90-degree advance with big long candles, especially while significantly overvalued, he exits** — because it usually collapses.

His own case — Meituan (Hong Kong).
- Bought at about **$70/share**; his calculated intrinsic value was **$220**.
- Price rose through intrinsic value; he held while it was only modestly overvalued.
- It then went **parabolic (90 degrees)**. At **~$330** — i.e. **50% above the $220 intrinsic value** — he sold **one quarter** of his shares.
- It dipped, then went parabolic again to **~$440** — **100% overvalued** — and he **sold all remaining shares** (the remaining half sold above $440).
- Outcome: about a **400% return**. Trade alerts were sent to his **~3,000** UIP subscribers (via Telegram then; Discord now). He also sold **Ping An Insurance** at the time.
- The stock later collapsed. **He did not buy it back**, because **new Chinese government regulations totally changed Meituan's business model** — it was no longer a great business. "No matter how low it goes, I ain't buying it back."

Holding-period nuance by stock type:
- **Secular / large-cap growth stocks** (Meta, Microsoft, Amazon, Nvidia) — he **very rarely sells**; holds many years unless ridiculously overpriced, because they keep compounding.
- **Defensive companies** — also very rarely sold, unless they get crazy expensive.
- **Deep cyclical stocks** — he **very rarely buys** them, and when he does he does **not hold them long** (see Sell Rule 4b).

SELL RULE 4b — the cyclical-stock exit (price-to-book band method).
Deep cyclicals = **banks, insurance, industrial companies, real-estate companies / property developers, shipping companies, commodity companies**. Their profits and share prices swing with the economic cycle — booming economy = big profits, weak economy = small profits or losses.

He only buys them **near or at the bottom of the cycle**, as a **medium-term trade**, and exits **near the top of the cycle**. The tool is the **price-to-book (P/B) ratio** — specifically for **banks, insurance companies and real-estate companies**. On TradingView, add the **price-to-book ratio indicator**.

Worked example — JP Morgan, 10-year P/B history:
1. **10-year high P/B = 1.89**
2. **10-year low P/B = 0.64**
3. **Midpoint = (1.89 + 0.64) / 2 = 1.26** → the median line.
- **Above 1.26 = getting expensive. Below 1.26 = getting cheap. He buys below the midline.**
4. **Sell level = (10-year high + median) / 2 = (1.89 + 1.26) / 2 = 1.56** → the midpoint between the 10-year high and

the 10-year median. **When P/B reaches ~1.56, that's a sell.**
- Accepting imperfection: after you sell it can still go higher — you can never sell exactly at the top.

Caveat to 4b — Singapore bank stocks. He owns a lot of Singapore bank stocks and, although they are cyclical, he **never sells them**. Reason: he buys them **not as medium-term trades but for dividends**, entering when yields are very low-priced/high-yield at about **5–6% dividend yield**, locking in a **tax-free** yield and holding for passive income year after year. **US banks** he *will* sell on the cycle, because their dividends are too small to justify holding.

SELL RULE 5 — You found a better opportunity and need the cash to fund it

You own a great company, but you find another stock that is **just as great or better** and/or **more undervalued**. If you **don't have spare cash**, there is nothing wrong with **swapping** — selling the first to buy the second, or selling **half** of the first and using the proceeds for the second.
Condition: this only applies when you lack the cash. **If you have extra cash, hold the first and buy the second as well.**

SELL RULE 6 — You need the cash for personal expenses

Legitimate: funding a child's college, buying a car, renovating the house.

Framing via founders: when **Jeff Bezos** sells Amazon shares or **Mark Zuckerberg** sells Meta shares, it usually does **not** mean they are bearish — Bezos funds his rocket venture, Zuckerberg funds his charity. Founder selling ≠ bad company.

The attached money-management rule: only invest money you **do not need in the short term**. Money you need in the **next one or two years must not go into the stock market** — put it in **short-term bonds that mature when you need the money, money-market funds, or a fixed/time deposit**. In the short term markets can drop 50% or rise 50%; the worst outcome is being **forced to sell great companies at a 50–60% discount** to raise cash you needed.
Ideal timing: raise personal-expense cash by selling **when stocks are overvalued / at an all-time high**, not when you're forced to.

PART 6 — REASONS *NOT* TO SELL (INVALID TRIGGERS)

6.1 The seven bad reasons and the counter-arguments

BAD REASON 1 — "The price is too high" / "I want to lock in my unrealised gains."
The mistake: the stock is up 20/30/40%, you get nervous about losing the paper profit and sell to "take profit."
Counter: that's a **short-term trading mentality**, not an investor's. To get a **5x, 10x or 50x** stock you must **resist selling at +50%, +100%, +150%, +200%. There is no such thing as "too high"** — high can go higher, especially when the business is growing and **intrinsic value keeps rising**. Stocks can run for years and decades; selling early cuts off the entire compounding tail.

BAD REASON 2 — "I'll sell high and buy back lower" (trading in and out).
Counter: he did exactly this for years — on Apple, Microsoft, Meta — and he **made money doing it**. But had he simply **held Apple, Meta, Microsoft and Amazon for the last 20+ years**, he'd be **five to ten times richer**. Two failure modes he lived through repeatedly:
- He sells near a perceived high; price dips slightly, then **flies up without him**; to re-enter he'd have to pay far more, missing 100–300% of upside.

- He sells, price *does* fall, he plans to re-buy at a trend line — but it **zooms up before reaching the trendline** and takes off without him.
Yes, you can occasionally nail it. But **over the long term you will mistime it often** and end up with far lower returns than simply holding.

Quantified illustration — Amazon, 2018–2020, trading the 50/150 moving-average crossovers (pre-split prices):
- Buy **1,006**; downtrend signal (50 crosses below 150) → sell **1,330** → transcript states a **−17%** leg outcome in the running total;
- Re-buy on uptrend at **1,009**; sell on downtrend at **1,776** → **−6.5%** leg;
- Re-buy at **2,000**; runs to **3,352** → **+62%** leg.
- **Total from trading in and out: +24% over two years** (62 − 6.5 − 17).
- **Simply buying and holding through the ups and downs: +108%.**
- Longer horizons for Amazon: **+610% over 5 years**, **+2,176% over 10 years**. Very few investors capture even a fraction — they freak out on dips or take profits on pops and get thrown off the ride.

The relationship analogy: great compounding companies are **rare — less than 1% of US companies** are great companies to hold (Microsoft, Nvidia, Amazon, Meta). Finding one is like finding a great spouse — beautiful, funny, loyal, intelligent, kind. Once found, you hold on and go through their ups and downs. If you keep running away every time they're in a bad mood, one day you come back and they're gone. **If you leave a great company, it will leave you behind.**

> **"Time in the markets is more important than timing the markets."**

BAD REASON 3 — "A crash / recession is coming" (your own prediction or an expert's).
Counter: nobody can predict the next recession or crash. Case in point **2023**: at the end of 2022 the majority of economists and analysts said a recession was inevitable and most predicted a crash. Investors **sold near the bottom** at end-2022, and some **shorted**. In 2023 the **S&P 500 rose over 20%** and the **NASDAQ over 40%**. The sellers missed it and the shorts took huge losses. **Never sell on an economic or market prediction.**

BAD REASON 4 — Macro or geopolitical bad news.
Inverted yield curve, negative GDP, high interest rates, a new war, declining M2, the Fed doing QT, an earthquake, a virus.
Counter: you will usually regret it — the market keeps rising, or dips slightly and rises without you.
The wall-of-worry evidence: from **March 2009 to December 2019** the S&P 500 gained **495%**. Very few investors captured a fraction of it, because the news scared them out (bad news drives views and eyeballs). The headlines through that 500% bull market included: 663,000 jobs lost in March; Chrysler and GM bankruptcies; the BP oil spill; the flash crash; the most powerful earthquake on record; the S&P downgrade of US debt; the European sovereign debt crisis; a US government shutdown; Ebola contagion fears; Brexit; Charlottesville; the S&P 500 entering a bear market; a second government shutdown; the trade war; yield-curve inversion; Iranian bombing; the emergency Fed injection into the repo market.

> **"There is always something to worry about," and "the bull always climbs a wall of constant worry."**
> Peter Lynch: your success or failure depends on your ability to **ignore the worries of the world long enough** to let your investments succeed.

BAD REASON 5 — Panic during a correction or bear market.
Corrections drop **10–15%**; bear markets drop **20–25%**. People who don't understand what they own start fearing the price will keep dropping forever or go to zero, and sell **near or at the bottom**.

Counter / evidence: many who sold at the **March 2020** bear-market bottom then watched the market rocket **~80% one year later**. Many who panicked and sold in **September/October 2022** — when everyone said it would get worse — watched 2023 rise **20–40%**. Stay invested in great companies and ride the temporary drawdowns.

BAD REASON 6 — Selling a great stock purely to reduce its allocation (mechanical rebalancing).
The practice: fund managers rebalance at year-end to keep equal weights.
Worked example: 10 stocks, **$10,000 invested, $1,000 each = 10% allocations**. One year later one winner (e.g. an Nvidia-type) **doubles** while the rest are flat → the winner is now **~18%** of the portfolio and each of the other nine is **~9%**. The manager sells down the winner and spreads the cash into the other nine to restore 10% each.
Counter: Warren Buffett and Peter Lynch both call this stupid. If stock #1 is still a great business and still growing, **do not sell it just because it's big**. Buffett's Apple started as a normal-sized position and grew to **~50% of his entire public portfolio** — he does **not** trim it; as long as Apple stays great, keeps growing, keeps dominating and isn't too overpriced, it makes sense **not only to hold but possibly to buy more**. This is a key difference between super investors and ordinary fund managers — and part of why **over 90% of fund managers can't beat the market**.

> Peter Lynch: selling your winner to fund the others is **"cutting the flowers and watering the weeds."**

BAD REASON 7 — "The market/my portfolio is at an all-time high, so I should get out before it comes down."
At all-time highs there are always two camps: chasers who want to buy more, and sellers who want out before a fall.
Counter:
- If already invested, **do not buy more** at an all-time high — follow the rules and only add when price drops to undervalued levels **and** to support levels.
- But that does **not** mean sell. He used to sell near perceived tops; sometimes it did dip, but then ran much higher without him; or it dipped and reversed up before he could re-buy. Holding through would have left him **much richer**.
- **The risk of mistiming is asymmetric:** if you're **in** the market and it drops, no problem — you're still in and it comes back. If you're **out** and it runs, **you can't get back in except at a much higher price**.

> **"The risk of staying out of the market over the long run is worse than the risk of staying in the market."**

6.2 The cost of missing the best days (why timing kills)

$10,000 invested in the S&P 500 for 20 years, January 2003 to 2022:

Scenario	20-year return
Fully invested (held through all ups and downs)	**+548%**
Missed the best **10** days	**+197%**
Missed the best **20** days	**+78%**
Missed the best **30** days	**+17%**
Missed the best **40** days	**−19%**
Missed the best **50** days	**−42%**
Missed the best **60** days	**−57%**

- Missing just **10 days out of 20 years cuts returns by more than half**.
- **7 of the 10 best days in the entire 20-year history occurred during bear markets** — precisely when most investors are out of the market, which is why they end up with big losses.

- "But can't I just miss the 10 *worst* days?" — only with a crystal ball. **In trying to dodge the 10 worst days you will unknowingly miss the 10 best days.** Better to stay put and miss none.
- The market averages about **10% a year**, but it does not accrue smoothly — **an entire year's return can arrive in a single day.**
- **His own screenshot evidence** from one such day (two Interactive Brokers portfolios): one portfolio **+10.5%** and the other **+10.83%** in a single day; **+$175k** and **+$231k** respectively — **+$396,000 USD in one day.** Missing that one day would have changed his returns for that year and for the five-year period.

> Peter Lynch: **far more money has been lost by investors preparing for or anticipating corrections than has been lost in the corrections themselves.** People predict a crash, get out or stay out, and lose far more in opportunity cost than they'd have lost by simply riding the short-term swings.

6.3 Quoted words of wisdom he credits with changing his approach

- **Charlie Munger:** the big money is not in the buying and selling, it is in the **waiting** — sitting patiently. Few people can do it.
- **Peter Lynch:** his best stocks delivered their greatest gains in the **third, fourth or even fifth year** of ownership — not the third or fourth *week*. People want money rapidly; it doesn't work that way. You can make small money fast; you will **never make millions with a short-term mentality**.
- **Charlie Munger:** the first rule of compounding is **never interrupt it unnecessarily**.
- The stock market is a **device for transferring money from the impatient to the patient**.

6.4 The third option at market highs — hedging (optional)

Instead of holding blindly (option 1) or trying to sell high and buy back low (option 2, "good luck to you"), there is a **third option**:
- **Hold your shares — don't sell** — and, if a correction or bear market is developing:
- **Sell covered call options** to generate extra income while price corrects and consolidates.
- **Buy protective put options** to generate profits as the market drops, offsetting the drawdown.
- Structures he showed from his own bear-market portfolios: **calendar spreads** producing short-term profits while holdings fell; a **bear put spread combo** producing put-option profits during a bear market. This is **hedging**.
- **Explicitly optional:** "learning to hedge your portfolio is not a must. It is nice to have, but it's not a must." If you don't know how, just hold — you will still do very well simply holding great companies through short-term ups and downs. Options are taught in a separate Piranha Profits options trading course.

KEY TAKEAWAYS

1. **The best ideas come from your own life.** Buy the businesses you personally patronise — consumer monopolies with brand power, market leadership and predictable revenue and cash flow. Do the daily inventory: soap, toothpaste, breakfast, groceries, car, lunch, dinner, software.
2. **Using a product is a lead, not a decision.** Every candidate must pass the **7-step criteria** — Netflix is used in his household and still isn't owned (high maintenance capex, hit-or-miss content, not predictable).
3. **Bad news on great companies is the second-best source of ideas** — but only if the bad news is temporary and the moat is intact. Fraud accusations are the one exclusion.
4. **News points to the opportunity; fundamentals and technicals authorise the buy** — undervalued vs intrinsic value + at a support level + reversal pattern.

5. **Screeners shortlist; they never qualify.** Finviz free (P/E deliberately "Any"), GuruFocus paid for exact thresholds and non-US regions. **Screen at least once a month.**
6. **Follow super investors for ideas, never for orders.** Dataroma 13F data; the aggregate "most owned / most bought" lists are more useful than any one manager — and they can all be wrong (he sold Disney while they were buying it).
7. **Watch list of 20–30 great companies, 5 minutes a day; portfolio review under 10 minutes a day.** Mark the chart with an intrinsic-value line and **3–4 support-based buy levels**; buy a quarter at a time.
8. **Daily price moves are noise.** 9 of the 10 causes of a price drop are temporary and are buying opportunities. Only **permanent structural damage** matters — and it takes years to develop, so you will see it coming.
9. **Read the quarterly report and the conference call transcript** (company IR page, Seeking Alpha, Google search). You're checking execution and management's explanation — not celebrating beats or mourning misses. Beats and misses are a coin toss.
10. **Drawdowns are unavoidable and psychological preparation is 70% of the game.** His own: −25% (2020), −30% (2022), yet +141% over five years. Buffett: −30%, −27%. Lynch: −56%, −42% — while returning 29% a year, and yet **over 50% of his clients lost money** by buying high and selling low.
11. **Only six things justify a sale.** Permanent fundamental deterioration; an analytical mistake; management acting against shareholders; gross overvaluation (>50–100% above intrinsic value); a strictly better opportunity when cash-constrained; a genuine personal cash need.
12. **Sell in tranches, just as you buy in tranches** — you can no more pick the top than the bottom.
13. **The technical confirmation of a sell is the parabolic move**: a healthy uptrend runs ~45° near the 20 EMA; a 90° vertical run on long green candles is panic buying and usually the end.
14. **Cyclicals are the exception to buy-and-hold** — enter below the 10-year median P/B, exit at the midpoint of the median and the 10-year high (JPM: buy < 1.26, sell ≈ 1.56). Unless, like his Singapore banks, you hold them purely for a locked-in 5–6% dividend yield.
15. **The biggest destroyer of returns is being out of the market.** Missing the 10 best days in 20 years cuts +548% to +197%; missing 40 turns it negative — and **7 of the 10 best days happened inside bear markets**.
16. **There is always something to worry about.** The bull climbs a wall of worry; the 2009–2019 500% bull market ran through jobs collapses, bankruptcies, downgrades, a debt crisis, Ebola, Brexit, shutdowns, a trade war and yield-curve inversion.
17. **Never invest money you need within one to two years** — use short-term bonds, money-market funds or fixed deposits, so you are never forced to sell great companies at a 50–60% discount.
18. **Don't cut the flowers to water the weeds.** Mechanical rebalancing out of a winner is why 90%+ of fund managers underperform. Buffett lets Apple sit at ~50% of his public portfolio.

CHEAT SHEET A — SELL CHECKLIST (VALID TRIGGERS)

#	Trigger	Threshold / test	Action
1	**Permanent fundamental deterioration** — moat destroyed by technology, competition or changed consumer preferences	You and the people around you have **stopped buying** the product; profit margins declining **2–3 years straight** because of **forced discounting** (lost pricing power); margins squeezed by higher cost of sales; **market share falling 2 straight years**; customers switching to competitors	**Sell.** Rare — wide moats last 20+ yrs, narrow 10+ yrs — and it never happens overnight, so exit early

#	Trigger	Threshold / test	Action
2	**Accusation of financial fraud or financial irregularity**	Any credible accusation, regardless of company size or fame (Enron, WorldCom both went to zero)	**Sell immediately. Don't look back.**
3	**You made a mistake in the original analysis**	Business isn't as great as you judged, or you were too optimistic in your valuation (Disney: content hit-or-miss, not predictable)	**Sell all, take the loss, immediately redeploy into a great undervalued business**
4	**Management acts against shareholders**	Buybacks at all-time highs / while the business deteriorates; new share issuance despite a cash-rich balance sheet	**Sell**
5	**Grossly overvalued**	**>50–100% above intrinsic value.** (10–30% overvalued = HOLD)	**Start reducing — sell in tranches** (e.g. 25 shares at a time, or 1/3 → 1/2 → rest); redeploy into an equally great, cheaper business
6	**Parabolic price action** (confirming trigger 5)	Trend shifts from ~45° near the 20 EMA to a **90° vertical run with long green candles** = panic buying, usually the end of the uptrend	**Exit**, especially if already overvalued
7	**Cyclical stock reaches the top of its cycle** (banks, insurers, industrials, property, shipping, commodities)	P/B reaches the **midpoint between the 10-yr median and the 10-yr high** — JPM: high 1.89, low 0.64, median 1.26, **sell ≈ 1.56**	**Sell the medium-term trade.** Exception: cyclicals held purely for locked-in 5–6% dividend yield are never sold
8	**A strictly better opportunity and no spare cash**	The alternative is as great or better and more undervalued, and you have no cash to fund it	**Swap** — sell all or half and redeploy. If you *do* have cash, hold both
9	**Genuine personal cash need**	College fees, a car, home renovation	**Sell** — ideally when the holding is overvalued / at an all-time high, never under forced conditions

CHEAT SHEET B — DO NOT SELL BECAUSE...

#	Invalid trigger	Why it's wrong
1	"The price is too high" / "I want to lock in my unrealised gains"	High can go higher; intrinsic value keeps rising. A 5x/10x/50x stock requires refusing to sell at +50%, +100%, +200%. Trader mentality, not investor
2	"I'll sell high and re-buy lower"	Very high risk of mistiming. Amazon 2018–20: trading the crossovers gave **+24%** vs **+108%** for holding. He'd be **5–10x richer** had he held Apple/Meta/Microsoft/Amazon for 20+ years
3	"A crash / recession is coming" (your prediction or an expert's)	Nobody can predict it. End-2022 consensus said recession was inevitable; 2023 gave **S&P +20%, NASDAQ +40%**. Sellers missed it; shorts got crushed
4	Macro or geopolitical bad news (inverted yield curve, negative GDP, high rates, war, QT, virus)	The bull climbs a wall of worry. March 2009–Dec 2019 = **+495%** through jobs collapses, bankruptcies, the US downgrade, the euro debt crisis, Ebola, Brexit, shutdowns, trade war and yield-curve inversion
5	Panic during a correction (−10–15%) or bear market (−20–25%)	Sellers at the March 2020 bottom missed a **~80%** one-year rebound; panic sellers of Sept/Oct 2022 missed **+20–40%** in 2023. And **7 of the 10 best days in 20 years happened inside bear markets**

#	Invalid trigger	Why it's wrong
6	Earnings miss or lowered forward guidance	Beat/miss is a **50/50 coin toss**; great companies have bad quarters. Adobe beat and still fell 6% on guidance — a buying opportunity, not a sell
7	Sector rotation, industry-wide news, market-maker/algo manipulation, a forced seller dumping stock	Nothing changed at the company. Money rotates back; discounts are opportunities (2023: healthcare sold off wholesale; Ozempic knocked 20–30% off food stocks whose revenues never fell)
8	Cyclical slowdown in demand	Semiconductors and advertising dip and re-accelerate. 2022 ad-spend fear dented Alphabet and Meta revenue; it re-accelerated the next year
9	Product recall, lawsuit, scandal	All temporary. (Fraud is the sole exception — see Cheat Sheet A #2)
10	Rebalancing — trimming a winner just because its allocation grew	Buffett and Lynch both call it stupid; Buffett lets Apple sit at **~50%** of his public portfolio. Lynch: **cutting the flowers and watering the weeds**. It's part of why **90%+ of fund managers can't beat the market**
11	The market / your portfolio is at an all-time high	Don't buy more, but don't sell either. **The risk of being out is worse than the risk of being in** — if you're in and it drops, it comes back; if you're out and it runs, you can only re-enter far higher
12	A founder or CEO is selling shares	Bezos funds rockets; Zuckerberg funds charity. Founder selling is usually a personal cash need, not a bearish signal
13	Daily volatility / a 10–20–30% drawdown in a great company	Volatility is normal; drawdowns are inevitable even for Buffett (−30%, −27%) and Lynch (−56%, −42%). The roller coaster only hurts those who jump off halfway

Part X — Investor Psychology (The Most Important Lesson)

Lesson 10: The Psychology of Winning Investors

> Adam Khoo frames this as **the most important lesson in the entire course**: "to achieve success in anything, it takes two factors" — strategy and psychology. **Strategy makes up only 30% of success; psychology/mindset makes up 70%.** Most people focus only on strategy and leave psychology to chance, wrongly assuming they already have "the right psychology."

Part breakdown (content-based, not file-order labels):
- **Part 1 (file 74):** Discipline as the foundation; sticking to rules under pressure; Warren Buffett & the dot-com bubble case study; 2020–2021 hype-stock parallel; circle of competence; never blindly follow experts/analysts — with three detailed case studies (Nobel Group failure, Meituan success, Goldman Sachs short call).
- **Part 2 (file 75):** Patience and long-term thinking; the golf compounding exercise; "buy slowly, sell even slower"; the Fidelity dormant-accounts study; Adam's own Facebook mistake; judging investments by business performance not price; the Washington Post / Warren Buffett interview case study.
- **Part 3 (file 76):** Thinking independently and contrarian psychology; the fear/greed cycle and Fear & Greed Index; "worst day" media pattern; John Templeton quote on bull market phases; why you should never predict the market (with a decade of failed doomsday headlines as evidence); ignoring macroeconomic/geopolitical noise; risk management and capital protection; taking responsibility for mistakes; final 10-rule summary.

Opening Framework: Strategy (30%) vs. Psychology (70%)

- To achieve success in **anything** requires two factors: the right strategy AND the right psychology (emotional discipline, focus, self-motivation to apply what you've learned and follow the rules).
- **Analogy — weight loss:** The strategy for losing weight is simple and freely available (diets, exercise regimes), yet **fewer than 5% of people who attempt to lose weight succeed**, and of that 5%, **3% put the weight back on within a couple of years**. People know *what* to do but lack the discipline to *do* it. "Again, it's all about psychology."
- **Analogy — business:** If strategy alone were sufficient, everyone with an MBA would build a successful business and everyone with a PhD in finance would be a billionaire. Neither is true — many MBAs fail in business; many successful entrepreneurs are dropouts.
- By this point in the course, students already have the strategy (fundamental analysis, technical analysis, identifying great businesses). This lesson covers the missing 70%.

Psychology Rule #1: Absolute Discipline

- **Discipline defined:** You only buy/sell a stock based on **predetermined, objective rules** in your investment plan — rules grounded in fundamental and technical analysis (e.g., buy only businesses with strong fundamentals: consistently increasing sales, earnings, cash flow, low debt, undervalued, on a clear uptrend).
- **What most people actually do instead (the failure modes):**
- Buy/sell based on **rumors** ("I heard this company's going to be taken over").

- Buy/sell based on **emotions** — the two core emotions are **fear and greed**: people buy out of greed and sell based on fear.
- Buy/sell based on **the opinions of other people**.
- **FOMO (Fear Of Missing Out):** buying something you don't understand, outside your plan, because "everyone's buying it" — Adam compares this to **lemmings following each other off a cliff**, "they get slaughtered like lemmings."
- **Panic selling:** selling because of media scare stories (market crash warnings, trade war headlines) — "the moment you sell, the stock keeps going up."
- Doing any of these guarantees you will never make money in the long run.

Case Study: Warren Buffett and the Dot-Com Bubble (1999–2000)

- Buffett is considered the greatest investor not because he's the most intelligent (by his own admission) but because he is **the most disciplined**. Adam's method mirrors his: only invest in profitable, undervalued businesses; avoid unprofitable, overpriced/hyped businesses.
- During the dot-com bubble, unprofitable, overpriced "speculative stocks" (Netscape, AOL, Pets.com) kept rising, while profitable undervalued companies (McDonald's, Procter & Gamble, Coca-Cola) kept falling.
- **1999:** S&P 500 up 18%; Buffett's Berkshire Hathaway **down 23%**. Most investors would have abandoned their plan and chased the hype (FOMO) — Buffett refused.
- He was publicly ridiculed. *Barron's* ran a piece: *"What's Wrong, Warren? After 30 years of unrivaled investment success, Warren Buffett may be losing his magic touch."* Called a dinosaur who didn't understand tech.
- Berkshire fell from ~$80,000/share to ~$54,000/share (23% drop) — yet Buffett stayed unfazed, knowing that **short-term the market is irrational, but eventually prices reflect business fundamentals.**
- **One year later**, the bubble burst: NASDAQ dropped **72%**, S&P 500 lost **28%**. Buffett ultimately outperformed for years. From the $54,000 low, the stock later reached **$443,000**.
- **Lesson:** Stick to your investment plan even when the market temporarily punishes you for it. Short-term the market is irrational/manipulated; long-term it always reflects reality.

Modern Parallel: 2020–2021

- In 2020, unprofitable/overpriced companies (Roku, Zoom, Tesla, Peloton, Fastly) kept rising. Adam refused to buy them and was criticized ("Ed, why aren't you buying these stocks? They're going up.") His reply: "It won't last."
- **2021:** these stocks collapsed 50–60%, while Adam's holdings (Microsoft, Alphabet, McDonald's, Yum Brands) held their value and some kept rising through the correction.
- Reinforces: always stick to the investment plan no matter what others are doing in the short term.

Psychology Rule #2: Invest Only Within Your Circle of Competence; Never Blindly Follow Experts

- **Circle of competence:** only invest in businesses you truly understand and have personally researched. "No matter how smart you are, you can't be an expert in everything… you don't have to win at every game. You only have to win at the game which you understand."
- **Adam's personal examples of businesses he avoids:**
- **Energy companies** — energy prices are too unpredictable.
- **Commodity companies** — commodity prices are too unpredictable.

- **Tesla** — considers it overpriced and the EV market too competitive; despite being told "it's not just a car company, it's a software company, it's a battery company," he still doesn't understand the business well enough to invest, and refuses regardless of criticism.
- He instead focuses on businesses he understands with predictable, consistent cash flow (Facebook, Amazon, Alphabet, Microsoft).
- **Never blindly rely on recommendations/tips from experts** — even from investors you admire.
- **Thought experiment:** if Adam had dinner with Warren Buffett, Carl Icahn, and Bill Ackman and all recommended a stock (e.g., "Micron"), would he buy it? **No** — he'd run it through his own **seven-step investment analysis** first. He only invests if it meets his own criteria, not because admired people invested in it.
- Example: Adam admires Buffett but would never have bought some of Buffett's own picks — General Motors, airlines, Occidental Petroleum, IBM — because they didn't meet his criteria, and these turned out to be some of Buffett's losing investments.
- Always do your own research on every investment, **monitor stocks regularly**, and **keep investment records** — a portfolio tracker and an investment journal. "There's no free lunch in this world... to create wealth, you got to put in the work."
- Analyst reports (from brokers, private banks) can be **leveraged for research** but never blindly followed — always run them through your own checklist.

Case Study: Nobel Group (November 2011) — a failed "buy" recommendation

- A top Singapore broker recommended **buy Nobel Group at $1.56**, target price $2.05. Nobel was a large commodity company, blue chip, part of the Straits Times Index.
- Adam's process: (1) he never buys commodity companies as a rule — automatic disqualifier; (2) still ran it through his seven-step check anyway, and it **failed most steps**:
- **Negative operating cash flow** despite reported increasing profits — a red flag ("why are they generating profits on paper but the cash is not coming in?").
- **Profit margin < 0.5%** — razor thin, vulnerable to any cost increase.
- **Debt-to-EBITDA ratio > 12** (rule of thumb: should never exceed 3).
- **Debt-servicing ratio over 50%.**
- He did not buy. **One week after the recommendation**, price dropped from $1.56 to $1.08 (−31%).
- **Lesson on cutting losses:** if you realize you made a mistake, **cut loss immediately** — "take your loss like a man or like a woman." Most people don't, because "if I sell, I'm gonna lose money... maybe it comes back, I hope." **Quote: "In the markets, hope is for the hopeless. Never have hope."**
- Nobel Group continued: $1.08 → $0.45 → ~$0.05 → **delisted**, company collapsed. Investors "complained they were cheated" (referenced newspaper headline: *"Tales from the collapse of commodity giant Nobel Group"*).
- **Lesson:** (1) if you realize a holding is a mistake, get out; (2) never blindly trust experts/analysts — always do your own research. Many burned investors blamed the broker, the company, or luck instead of taking responsibility — "as long as you keep blaming other people... you will always lose money."

Case Study: Meituan Dianping (September 2019) — ignoring a bearish "avoid" call

- Adam found Meituan Dianping (China stock) as a great company with multi-bagger potential; shared research via YouTube and predicted the stock could double/triple within a year.
- Told his community he was "getting my dry powder ready" as it approached his ideal buy price of $70.
- **10 September:** price hit $70; Adam and students bought in together (students posted their entry prices: $70, $73, etc.).

- Students flagged a **March 2019 CNBC article** where analysts said investors should **avoid** Meituan shares — asked Adam why he was buying against that advice.
- Adam's reply: **"You gotta ignore these analysts, because they may not know what the hell they're doing. You have to trust your own research."**
- **Result: within 17 months, price went from $70 to $440 — up 528%.**
- Sell discipline: started selling at $320 (overvalued vs. his intrinsic value estimate of $240), sold more progressively, sold the majority of the position at $440. (Referenced screenshot from his "Ultimate Investors Playbook" live-alerts channel, showing him near +400% and stating intent to sell half his balance as "quite overvalued.")

Case Study: Goldman Sachs Short Call (21 June 2012)

- Goldman Sachs recommended clients **build short positions on the S&P 500** citing a "weakening domestic economy."
- Adam's rule: **never listen to these recommendations — follow your own investment rules (chart trend).**
- At the time, the chart showed a clear **uptrend** (50-day moving average above the 150-day moving average, and the 150 still sloping upward, even though the 50 had briefly dipped below it).
- **Result:** market didn't fall — it rose **7% over the following months**. People who listened to the "short" call lost money or lost the opportunity for that 7% gain.
- **Takeaway repeated across all three case studies:** never listen to opinions/advice of analysts, bankers, or experts. Follow your investment plan's rules based on fundamentals and chart patterns instead.

Psychology Rule #3: Be Patient and Think Long-Term

- Most people want to get rich quick — "it just doesn't happen. It just doesn't exist." This drives people toward get-rich-quick schemes or speculative hype stocks. You *can* occasionally get lucky (like winning the lottery), "but in the long run, whatever money you make, you will always lose it back."
- Sustainable wealth requires a **long-term timeframe**, patience, and letting investments **compound**.
- **Quotes:**
- "The stock market is a wonderful mechanism that transfers wealth from the impatient to the patient."
- **Charlie Munger:** "The big money is not in buying and selling stocks, the big money is in waiting." (repeated later: "It's in inactivity.")
- Most investors abandon their strategy out of impatience when they don't get rich overnight, or when they see other stocks/crypto rising faster and get "seduced by those quick profits" — jumping into what's hot (which usually crashes) and abandoning good companies (which usually then rise).
- **"Get rich slow, but get rich sure"** — not get rich quick.
- Stock prices/portfolios **do not go up in a straight line** — they cycle through ups and downs even for great companies with strong long-run annualized returns (e.g., a company can compound at 26%/year on average while going flat or down for a full year at a time).
- **Illustrative example:** buy at $100, stock doubles to $200 in 3 years = ~26% annualized return — a great outcome — but this often happens with two flat/sideways years followed by a big move in year three. Most people give up ("lose faith… lose patience… sell the stock") right before the move happens, then watch it double after they sell.

The Golf Compounding Exercise

- Thought exercise: bet 10 cents on hole 1 of golf, **double the bet every hole** for 18 holes. Guess the hole-18 bet before being told.
- Progression shown: Hole 1: $0.10 → Hole 5: $1.60 → Hole 6: $3.20 → Hole 9 (halfway): $25.60 → Hole 10: $51 → Hole 12: $204 → Hole 13: $409 → Hole 15: $1,600 → **Hole 18: $13,000** (starting from just 10 cents).
- **Point:** compounding grows *very slowly* at first ("like watching the grass grow… you get impatient… you're barely say, forget it, I'm gonna do something more exciting") before exploding exponentially later. Between hole 15 and hole 18 (only 4 holes), the value goes from $1,600 to $13,000.
- **Lesson:** starting investing even 1–4 years later than someone else results in a hugely different end outcome over the long run — **start as early as possible** to give money runway to compound.

Psychology Rule #4: Buy Slowly, Sell Even Slower

- Unlike most areas of life where "activity equates to performance," **in investing, less activity often produces more money.**
- **Be very patient before buying.** Common student mistake: after finishing the course, getting "trigger happy" and buying prematurely out of fear of missing the move, before a stock is properly undervalued or at the right buy point.
- **Rule: "If there's no great business at this point of time at a great price, do nothing."** Hold cash rather than invest for the sake of investing.
- **Baseball analogy:** "You only swing when you have the perfect pitch. If there's no perfect pitch, you don't have to swing." Investing is a marathon, not a race.
- Buy aggressively when the market is in fear/correction and great companies are on sale; do NOT buy when markets are high, expensive, and people are euphoric.
- **Be even more patient in selling.** Common mistake: taking profits too fast (e.g., stock up 20%, "let me grab those profits before I lose those profits") — this caps you at small gains and prevents big wins. **"The best stocks are the ones that you hold forever."**
- Clarification on "forever": only sell if you need the cash for something specific (a house, a car). Otherwise let it compound.

Fidelity Dormant Accounts Study

- A study of Fidelity clients' accounts over 30 years found the **best-performing accounts were inactive ones** — accounts the owner forgot about, or where the owner had **died** and no one claimed the stock. Actively-traded accounts performed the worst.
- **Lesson:** "In investing, lower activity actually results in greater returns."

Personal Anecdote: Adam's Facebook Mistake

- Adam bought Facebook years ago at **$19/share** (later ~$350/share at time of filming). A great investment — but he did **not** hold it the whole way, because early in his career he used to jump in and out of stocks (buy, small gain, sell, buy something else).
- Self-admission: "I was an idiot." He estimates that if he had simply held great companies without trading in and out, he'd be **10 times richer** today. He keeps the original Facebook purchase screenshot on his phone as a permanent reminder never to sell a great company prematurely.

The Marriage/Relationship Analogy

- Finding a great company is like finding a great spouse — rare (top companies, like the "top 1%" of people). Once found, you hold on for life.
- When a great spouse (or stock) is in "a bad mood" (temporarily down), you stay and support them — you even "buy more shares at a discount so it compounds even faster." Adding shares during a temporary dip in a great company is the equivalent of being there for a spouse during a rough patch — it strengthens the position/relationship.
- **Why not sell high and rebuy low?** In reality, it's **impossible to time selling at the top and rebuying at the bottom.** More often you sell a good stock and it runs away from you before you can buy it back. Analogy: "your spouse is in a bad mood, so you run away to the pub... by the time you get back, she's run off with your best friend." **Never leave a great company. Never leave a great spouse.**

Psychology Rule #5: Judge Investments by Business Performance, Not Short-Term Stock Price

- Most people wrongly judge investment success by short-term price movement: "stock went up this month = good investment; stock went down = bad investment." This is false.
- **Example:** Alibaba, Tencent, and other fundamentally strong Chinese companies fell due to emotional reactions to bad news (the Chinese regulatory crackdown), leading people to wrongly conclude buying them was "a mistake." It wasn't — if the underlying business is still growing and profitable, short-term price weakness does not equal an investment mistake.
- **Judge success by the business itself**, not short-term price, because short-term price is driven by **emotions and manipulation, not logic or fundamentals** — and is therefore unpredictable.
- If you buy a good company and price falls, **don't feel regret or beat yourself up** — you cannot control short-term price. Short-term weakness in price for a fundamentally sound company is **an opportunity to add shares**, not a reason to doubt yourself.
- **"Let the market serve you and take advantage of the market and not let the market instruct you of what is a good investment."**

Case Study: Warren Buffett and the Washington Post ($10 million → $1.5 billion)

- Buffett bought Washington Post shares in 1973 at ~$20/share when the company was worth ~$100/share — an **80% discount**.
- From the interview footage: Buffett explains he had studied the prospectus in 1971 (meeting Katie Graham around the time of the Pentagon Papers); by 1973 the stock became cheap because its TV licenses were being challenged (tied to Nixon associate "BeBe Rebozo"). Price fell from ~$38 to ~$16–20; Buffett and Munger bought almost 10% (now ~18% after buybacks) when the whole company (4.8 million shares) was priced at only $100 million on the market — while the sum of its parts (Washington Post, Newsweek, four network TV stations including a CBS affiliate in DC and one in Jacksonville, FL) could have fetched roughly **$500 million** if sold piece by piece ("people would have been swimming out in shark-infested waters" to buy the pieces). Buffett: "So what?" the stock kept dropping after he bought it — he was focused on business value, not price.
- Result: the ~$10 million investment became **worth roughly $1.5 billion.**
- **The real lesson Adam draws:** it wasn't just that Buffett found an 80%-undervalued stock — it's that **other analysts and big institutional investors agreed the business was worth ~$100/share (~$500M)** and were still **selling** at $20 because they feared the price would keep falling short-term. Buffett bought because he focused on long-term business value, not short-term price fear.

- **Modern parallel:** Alibaba — most analysts/investors agree it's worth roughly $300/share, but it trades around $100 and few people dare buy it "except a few people like me," out of fear it will keep falling. No one can predict exactly when the price will rebound to reflect true value — could take a month, three months, six months, a year, or three years — but patience with an undervalued, fundamentally strong business pays off. (Buffett's WaPo position also fell further after his purchase, from $20 to $16, before eventually multiplying — reinforcing "you can never catch the bottom, but you don't need to.")

Psychology Rule #6: Think Independently (Contrarian Psychology)

- Many people fail because they are **"creatures of conformity"** — they get excited when others are excited and afraid when others are afraid, following the crowd. This causes buying high (when everyone's excited) and selling low (when everyone's scared) — a losing pattern even if you pick the right companies or buy the index.
- **Rule: train yourself to think and feel opposite of everyone else / opposite of the masses** — "be fearful when others are greedy, and be greedy when others are fearful."
- **"Only you control your emotions. No one can make you feel scared except yourself... You create your emotions based on what you choose to focus on and the meaning you give to the experience."**
- Best way to train this ability: participate in a community/chat group to strengthen the habit of thinking against the crowd.

The "Worst Day" Media Pattern

- Adam's observed pattern: whenever the market has a big drop, the next day financial media runs articles featuring a sad-faced person and headlines using the word **"worst"** (e.g., "Dow craters in worst day..."). Whenever you see the word "worst day" paired with a fearful face, **that's usually the best time to start adding shares** — right before the market bounces back. (Illustrated with a specific example dated 28 October, and repeated pattern examples.)

Fear & Greed Index

- Referenced tool: **moneycnn.com** (CNN Fear & Greed Index) — measures market sentiment, swinging between extreme fear and extreme greed like a pendulum.
- At time of filming (21 December 2021), the index was at **extreme fear** (~level 25); during the COVID crash it fell to as low as **10**. Adam states he had been actively adding shares during that period of fear.
- The market can't be predicted with certainty even from this index, but **extreme fear signals good companies are on discount** — you can only buy a good company at a discount when people are scared.
- Historically: every time people get scared, the market has rebounded.

Market Cycle Emotions

- The market is never a straight line — it moves in cycles ("boom cycle and bust cycle"), each phase carrying a characteristic set of investor emotions:
- **Near the top:** thrill, euphoria ("I'm a genius, we're gonna be rich, I gotta tell everyone to buy") — everyone around you is talking about and buying stocks. This is when you should NOT buy — start taking profits instead.
- **Near the bottom:** anger, panic, depression, disbelief — people telling you "don't touch stocks... we're gonna die." This is when you should buy (good companies).
- **John Templeton quote:** *"Bull markets are born on pessimism, they grow on skepticism, they mature on optimism, and they die on euphoria."*

- Born on pessimism: general public expects the market to crash — that's usually when a bull market begins.
- Grow on skepticism: as the market rises, people doubt it will continue — and it keeps rising anyway.
- Mature on optimism: when people around you start saying "we're gonna do well," the bull market is nearing its end.
- Die on euphoria: when people feel unstoppable/certain of riches, the bull market dies.
- **Corollary: maximum pessimism = best time to buy; maximum optimism = best time to sell.**

Psychology Rule #7: Never Attempt to Predict the Market

- You can assess **probability** via trend (uptrend = higher probability prices rise; downtrend = higher probability prices fall) via chart patterns/moving averages, but you can **never predict short-term price movement with absolute certainty**, because prices are driven by demand/supply, which is driven by emotions, manipulation, and news — and no one can predict tomorrow's news or how millions of people will react to it.
- **Buy/sell decisions should never be based on predictions — always base them on business fundamentals** (good business + undervalued + retraced to a support level → add shares). **Never buy all at once — buy in tranches to average your cost.**
- **Warren Buffett quote:** "We have long felt that the only value of stock forecasters is to make fortune tellers look good."
- **Why predictive mindset is dangerous:** it causes you to abandon your predetermined rules. Example: if you predict a crash, you won't buy an undervalued great company out of fear — and then miss the gains when it rises anyway.
- **Your job as an investor is NOT to predict short-term price direction — it's to buy great companies that thrive under all economic conditions**, building a "bulletproof portfolio" that grows collectively regardless of inflation, interest rates, or recession.
- Your chance of any single prediction being right is essentially **50-50** — "you might as well toss the coin." For every expert predicting up, another predicts down.

Case Study: Dow Jones, October 2009 (contradictory predictions in one week)

- Dow was at 9,800 points. On **6 October**, CNBC headline: "Experts position themselves for Dow 10,000" (bullish). The day before, **5 October**, the same outlet ran: "Dow will fall to 6,300 by year end" (survey of portfolio managers, bearish).
- **"So who the hell do you listen to? You listen to CNBC or CNBC?"** — Answer: don't listen to anyone; follow the trend instead. On 6 October the chart showed an uptrend (50-day MA above 150-day MA).
- **Actual result:** Dow rose to 10,000, then 11,000, then 20,000, eventually 23,000–24,000. Anyone following the bearish "clowns" prediction would have shorted and lost money, or missed the gains.

A Decade of Failed Doomsday Predictions (2008/2010–2019, S&P 500 rose ~262–400%+)

Adam walks through a chronological sequence of major media doomsday headlines that all preceded a ~10-year, ~400%+ bull run in the S&P 500 (SPY roughly 80 → 290+ points, later ~450):
- **2010 (New York Times):** "A market forecast that says, take cover" — quoting forecaster **Robert Prechter**, predicting "a market decline of staggering proportions, perhaps the biggest of the last 300 years."
- **2011 (USA Today):** "S&P lowers its outlook. Could US default on its debt?"
- **2012 (CNBC):** "Europe debt defaults are poised to rattle stocks" (Eurozone sovereign debt crisis fears).
- **2013 (Bloomberg):** "A US default seen as catastrophe, dwarfing Lehman's fall."

- **2013 (unnamed forecaster):** "87% risk of stock crash by year end." (Market rose 20% that year instead.)
- **2015: David Stockman** — "Stocks and bonds will crash soon." (Adam's joke: "Kuh, Stockman, Kuh, Stockman... if you listen to Stockman, you would not have made money, 'cause stockmen say we're gonna die.")
- **2016:** "Here comes the biggest crash in a generation."
- **2018:** "Why markets are still heading for a crash."
- **Pattern/lesson:** financial media predicts a crash **every single year**, because "bad news sells newspapers... sells articles... gets clicks... sells advertisements." Markets DO periodically correct (statistically: down 3–5% about once a year, down 10% about once every 2 years, down 20% about once every 6 years) — but **when it happens is unpredictable**, and when it does happen, "you don't fear them, you take advantage of them." **Rule: stop reading predictions/forecasts; focus on building a watchlist of great companies and add shares during corrections.**
- Reiterated **Peter Lynch quote:** "If you spend more than 13 minutes analyzing economic and market forecasts, you've wasted 10 minutes of your time."

Psychology Rule #8: Pay No Attention to Macroeconomic or Geopolitical Events

- Counterintuitive to most people, who believe successful investing requires tracking GDP, interest rates, yield curves, inflation, etc. Adam calls this "absolute crap and rubbish."
- **Two reasons:**
  1. No one can predict macroeconomic events (interest rate moves, inflation) — at best a 50-50 guess.
  2. Even if you could predict the event, **you can't predict how the market will react to it.**
- Same logic applies to geopolitical events (e.g., Omicron, government shutdowns) — the market reacts instantaneously and unpredictably; there's no way to profit from reading this news in advance.
- Adam admits he *does* discuss interest rates, GDP, and inflation in his newsletter/YouTube videos — but purely for **intellectual and entertainment purposes**; he assigns **zero weight** to any of it in actual buy/sell decisions. All of it is "market noise."
- **Peter Lynch quote:** "Your ultimate success or failure will depend on your ability to ignore the worries of the world long enough to allow your investments to succeed."
- **The only valid reason to sell a stock is that the underlying business itself is no longer a great business** — never sell because of macro or geopolitical reasons.
- Illustrative list of "reasons to sell" that scared people out of the 2009–2019 bull run and caused them to miss ~500% growth: 600,000 jobs lost in March; Chrysler/GM bankruptcy filings; BP oil spill; the Flash Crash; European sovereign debt crisis; a powerful earthquake; S&P downgrading US debt; fiscal cliff talks; US government shutdown; the "taper tantrum"; Ebola contagion fears; Brexit; Charlottesville; "Dow falls to biggest point decline ever"; another government shutdown; S&P entering a bear market; the trade war; the yield curve inverting; an Iranian bombing; an emergency Fed injection. **Every one of these was a reason people sold or avoided buying — and the market kept rising through all of them.**

Psychology Rule #9: Risk Management and Capital Protection

- Most people, before buying, only think about **how much they can make** — they never ask how much they'd lose if wrong. **Professional investors do the opposite: ask "how much can I lose if I'm wrong?" FIRST, then "how much can I make?"**
- **"Always protect your downside and the upside will take care of itself."**

- **Rule #1 of successful investing: do not lose money. Rule #2: how much money can I make.** (Attributed framing, echoing Buffett's famous rules.)
- **Never put all your eggs in one basket** — "there's no such thing as a sure thing." Even a genuinely great business can be hit by unforeseeable news.
- **Example: Boeing.** Adam owned Boeing shares for many years; a great, historically strong business — but the 737 MAX crashes/grounding plus the pandemic caused a **50% share price drop**. If your entire portfolio had been in Boeing, you'd have been down 50%. (It has since recovered.)
- **Solution: diversify across many great companies** so temporary drawdowns in one or two names are buffered by gains in the rest, keeping the overall portfolio growing.
- **Key principle:** "If you buy a portfolio of great companies and you buy them for less than what they are worth, collectively, you cannot lose money."
- **Concrete allocation guidance:** hold a **minimum of 8 companies, ideally 20–30**. Risk only a small % of capital per position — e.g., 10 companies → 10% allocation each; 20 companies → 5% allocation each.

Psychology Rule #10: Take Responsibility and Learn From Mistakes

- You will make mistakes investing — everyone does. **"It's not what happens to us that determines our success or failure. But it is how we choose to respond to what happens that determines the outcome."** You can choose to learn, or make excuses, ignore mistakes, or blame someone else.
- **How do you know if an investment is actually a mistake?**
- **NOT a mistake:** price drops 10/20/30/even 40% after purchase, while the underlying business is still growing and still passes all the fundamental steps. Short-term price is uncontrollable, manipulated, and emotion-driven. If you still have available cash, this can even be an opportunity to **average down** (add shares).
- **IS a mistake:** the underlying business fundamentals have deteriorated (lost competitive advantage / no longer a good business), OR you discover post-purchase that it actually failed one of your investment criteria.
- **When you realize it's a genuine mistake: sell immediately** — regardless of whether the price is above or below your purchase price. "Win, lose, or draw, get out, cut it, take the cash and reinvest it into a great company that can grow your money from there forth." Don't hold out hoping "it's going to come back."
- **The "fire the employee" analogy:** think of yourself as the boss of a company. If you hire a salesperson who brings in no sales and is rude to customers, you fire them immediately rather than paying them more and hoping they change. Treat a lousy stock holding the same way — cut it regardless of sunk cost (salary already paid = losses already taken).
- **"Think logically and not emotionally. Never get emotional about a stock and be emotionally attached to it because the stock doesn't know who you are. The stock doesn't care who you are."** You don't have to "make the money back" from the same stock — you can make it back from other great companies.
- **Never blame anyone for losing money** — not the market, not God, not your broker, not your mother-in-law, your wife, or your kids. **Blaming = giving away control.** "When you blame someone, it means they are controlling your results... As long as people are controlling your results, they are controlling your life, you can't change your life."
- **Take ownership/responsibility** — believing "I am creating my results, I'm creating my life" gives you the power to change your life and your results.
- Closing quote: **"For things to change, I must change first."** Don't wait for the world to get better — focus on making yourself better through constant learning, skill growth, and strengthening your psychology.

Full 10-Rule Summary (as recapped by Adam at the end)

1. **Be disciplined** — follow the rules of your predetermined investment plan.
2. **Invest only in what you understand** — stick to your circle of competence; don't blindly follow experts.
3. **Be patient and think long-term** — wealth compounds slowly, then explosively.
4. **Buy slowly and sell even slower** — patience on entry, extreme patience on exit; hold great companies "forever."
5. **Focus on business performance, not stock price**, to judge investment success.
6. **Think independently** — go against the crowd; be greedy when others are fearful, fearful when others are greedy.
7. **Never attempt to predict the market** — follow the trend and fundamentals instead of forecasts.
8. **Pay no attention to macroeconomic or geopolitical events** — focus purely on the business itself.
9. **Risk management and capital protection** — diversify (8–30 companies), protect downside first.
10. **Take responsibility and learn from your mistakes** — cut real mistakes fast, never blame others.

Key Takeaways

- Investing success is **70% psychology, 30% strategy** — most people over-invest in learning strategy and neglect the psychological/behavioral side entirely.
- The core emotional trap is **fear and greed**: buying out of greed (FOMO, chasing hype) and selling out of fear (panic on bad news) is the default human behavior pattern and the primary destroyer of investment returns.
- Short-term stock prices are driven by **emotion, manipulation, and unpredictable news** — not by business fundamentals. Long-term prices always converge back to reflect business value. This single distinction underlies almost every rule in the lesson (rule #1, #5, #7, #8).
- **Discipline = pre-committing to objective rules before emotion arrives**, then refusing to deviate even under social pressure (public ridicule, underperformance vs. the crowd) or market pressure (drawdowns).
- **Contrarian positioning against crowd emotion** (buy fear, sell greed) is one of the only repeatable edges available to individual investors, because emotional extremes are visible in media tone, sentiment indices, and social conversation.
- **Predictions — economic, geopolitical, or market-direction — are close to worthless** (~50/50 odds) and are actively harmful because they cause you to override your rules. Financial media predicts crashes constantly because fear sells; nearly all such predictions over the past decade were wrong.
- **Inactivity is an edge.** Trading frequency is negatively correlated with returns (Fidelity dormant-account study); the biggest gains come from long, patient holding, not frequent buying/selling.
- **Diversification (8–30 positions, 5–10% max allocation) plus buying only undervalued great businesses is presented as a near risk-free combination** for capital protection: "You cannot lose money if you buy a group of great companies and you buy them for less than what they are worth."
- **A real investment mistake is a change in business fundamentals or a failed criterion — not a price decline.** Confusing these two is one of the most damaging psychological errors an investor can make.
- **Personal responsibility is the foundation that makes every other rule usable.** Blaming external causes (brokers, luck, "the market") removes your ability to learn and improve; only ownership of outcomes creates the power to change results.

Psychology Rules Checklist (Practical)

Before buying:
- [ ] Have I run this through my full seven-step investment analysis myself, regardless of who recommended it?
- [ ] Is this inside my circle of competence — do I genuinely understand the business?
- [ ] Does it meet ALL my predetermined fundamental + technical criteria (profitable, low debt, undervalued, uptrend)?
- [ ] Am I buying because of FOMO, a rumor, or someone else's opinion? (If yes — stop.)
- [ ] Is the crowd euphoric right now? (If yes, be cautious; don't chase.)
- [ ] Is the crowd fearful/panicked right now with "worst day" headlines? (If yes, that's often a buying window.)
- [ ] Am I buying in tranches, not all at once?
- [ ] Have I sized this position so no single stock is more than ~5–10% of my portfolio, with at least 8 (ideally 20–30) holdings?
- [ ] Have I asked "how much can I lose?" before asking "how much can I make?"

While holding:
- [ ] If the price has dropped, has the underlying BUSINESS changed — or is this just short-term price noise? (Only the former is a "mistake.")
- [ ] Am I ignoring today's macroeconomic/geopolitical headlines when making buy/sell decisions?
- [ ] Am I ignoring market forecasts and predictions from analysts, banks, and media?
- [ ] Am I resisting the urge to sell a great company just because it's "up 20%" and I want to lock in quick profits?
- [ ] If a great holding drops temporarily, am I treating it as a chance to add shares rather than a reason to panic?

Before selling:
- [ ] Am I selling because the business fundamentals genuinely deteriorated / it failed a criterion — or because of price movement, macro news, or emotion?
- [ ] If it IS a genuine mistake, am I selling immediately (win, lose, or draw) rather than "hoping it comes back"?
- [ ] Am I emotionally attached to this stock in a way that's clouding judgment? ("The stock doesn't know who you are.")

After a loss or mistake:
- [ ] Am I taking full responsibility instead of blaming the market, broker, or anyone else?
- [ ] Am I extracting the specific lesson (which rule/criterion did I skip or violate)?
- [ ] Am I reinvesting the recovered capital into a great business rather than staying in a lousy one?

General discipline maintenance:
- [ ] Am I thinking and feeling opposite of the crowd (fearful when greedy, greedy when fearful)?
- [ ] Am I thinking in years/decades, not days/months?
- [ ] Am I keeping an investment journal and portfolio tracker?
- [ ] Am I limiting my "activity" — resisting the urge to trade frequently?
- [ ] Am I re-reading these 10 rules regularly to keep them front of mind?

Appendix — Source Video Index

All 55 videos in this folder, in course order (23h 7m total runtime). Every one was transcribed and incorporated into the notes above.

#	Lesson Video	Duration
1	06 - VMI Course Introduction (Must Watch) 2025	11:24
2	07 - Stock Basics 2025 Part 1 (Parts of a Business)	28:25
3	08 - Stock Basics 2025 Part 2 (P E & PEG)	26:41
4	10 - Stock Basics 2025 Part 4 (What Drives Stock Prices)	26:48
5	11 - Stock Basics 2025 Part 5 (How to Make Money in Stocks)	23:55
6	12 - Stock Basics 2025 Part 6 (Stock Index Performance)	32:22
7	14 - Shocking Market Facts	23:43
8	20 - Trend Following Part 1	20:09
9	21 - Trend Following Part 2	36:36
10	22 - Strategy Execution	30:31
11	25 - Part 1 (Keys to Profitable Investing)	27:34
12	26 - Part 2 (Keys to Profitable Investing)	23:03
13	28 - Introduction	18:01
14	29 - Income Statement	32:01
15	30 - Balance Sheet	47:54
16	31 - Statement of Cash Flows	27:45
17	32 - Financial Metrics	22:17
18	33 - Hyflux Case Study	9:46
19	34 - Great Business Models	21:33
20	36 - Introduction & PEG Ratio	16:15
21	37 - Discounted Cash Flow Method Part 1	33:53
22	38 - Discounted Cash Flow Method Part 2	24:38
23	39 - Discounted Net Income Method	20:43
24	40 - Price Book Ratio & Banks	25:02
25	41 - REITs & Price Sales Ratio	20:31
26	43 - Reading Charts	31:27
27	44 - Support & Resistance	32:52
28	45 - Moving Averages	15:44
29	46 - Buying on Uptrends	27:00

#	Lesson Video	Duration
30	47 - Buy Sell on Downtrends	22:23
31	48 - Catching Bottoms	22:21
32	50 - VMI Formula Part 1	30:01
33	51 - VMI Formula Part 2	30:34
34	52 - VMI Formula Part 3	21:00
35	53 - VMI Formula Part 4	28:16
36	54 - VMI Case Study (GOOGL)	37:54
37	55 - VMI Case Study Part 1 (MSFT)	23:24
38	56 - VMI Case Study Part 2 (MSFT)	33:16
39	57 - VMI Case Study Part 1 (AMZN)	21:34
40	58 - VMI Case Study Part 2 (AMZN)	24:46
41	60 - How to Diversify	24:48
42	61 - Favourite Sectors	27:35
43	62 - Dividend & Predictable Stocks	22:33
44	63 - Growth Stocks & More	20:49
45	64 - Maximizing Portfolio returns	23:27
46	66 - Finding Great Stocks	24:14
47	67 - Stock Screeners	14:48
48	68 - Monitor Your Portfolio	29:42
49	69 - Dealing with Drawdowns	15:30
50	70 - When to Sell Part 1	22:40
51	71 - When to Sell Part 2	23:37
52	72 - Reasons Not to Sell	19:10
53	74 - Part 1 (Investor Psychology)	25:18
54	75 - Part 2 (Investor Psychology)	23:18
55	76 - Part 3 (Investor Psychology)	35:50

Compiled from full transcription of all source videos. All figures, thresholds, and examples are as stated by the instructor in the lessons.